California Independent System Operator Corporation

August 22, 2023

The Honorable Kimberly D. Bose
Secretary
Federal Energy Regulatory Commission
888 First Street, NE
Washington, DC 20426

Re: California Independent System Operator Corporation
Docket No. ER23-___-000

Day-Ahead Market Enhancements and Extended Day-Ahead Market

Dear Secretary Bose:

Attached is the submittal by the California Independent System Operator Corporation of a tariff amendment to implement the Day-Ahead Market Enhancements and Extended Day-Ahead Market initiatives. Please contact the undersigned with any questions.

Respectfully submitted,

/s/ John C. Anders

Roger E. Collanton
General Counsel
John C. Anders
Deputy General Counsel
Anthony Ivancovich
Deputy General Counsel
William H. Weaver
Assistant General Counsel
Andrew Ulmer
Assistant General Counsel

David Zlotlow
Lead Counsel
John Spomer
Lead Counsel
Heather Curlee
Senior Counsel
Sarah E. Kozal
Counsel

Counsel for the California Independent System Operator Corporation

I. INTRODUCTION 2

II. EXECUTIVE SUMMARY 6

A. Day-Ahead Market Enhancements 6
1. Addressing Energy Imbalances Today 6
2. Description of Imbalance Reserves and Reliability Capacity 6
3. Bidding for the New Products and Eligibility for Awards 7
4. Impacts on Resource Adequacy Bidding 8
5. Enhancements to the Integrated Forward Market 8
6. Enhancements to the Residual Unit Commitment Process 9
7. Financial Settlements Issues 10
B. Extended Day-Ahead Market 12
1. Pre-Market Timeframe 13
2. Within the Extended Day-Ahead Market 18
3. Post-Market Timeframe 21
4. Offering Access to the Extended Day-Ahead Market 22
a. Onboarding and Costs Related to EDAM 22
b. Historical Revenue Recovery for EDAM Transmission Service Providers 22
C. Severability 23

III. BACKGROUND 27

A. The CAISO's Current Market Structure 27
1. Day-Ahead Market 27
2. Real-Time Market 32
3. Other Noteworthy Market Design Elements 34
B. The Western Energy Imbalance Market Today 37
C. DAME Stakeholder Process 38
D. EDAM Stakeholder Process 40

IV. PROPOSED DAY-AHEAD MARKET ENHANCEMENTS 42

A. Issues with the Current Day-Ahead Market Leading to this Filing 42
1. Drivers of Increased Net Load Imbalances between the Day-Ahead Market and the Real-Time Market 42
2. CAISO Operators Rely on Manual Interventions in RUC to Address Net Load Imbalances 43
3. The Status Quo of Relying on Out-of-Market Operator Interventions is Problematic and Unsustainable 45
B. Description of Key Day-Ahead Market Enhancements 46
1. Overview of Imbalance Reserves 47
2. Overview of Reliability Capacity 48
3. Relationship Between the Products 49
C. New Products Will Address Key Gaps in the Existing Market Structure 51
1. Benefits of the Imbalance Reserves Product 51
2. Benefits of the Reliability Capacity Product 53
3. Why are Both Products Necessary? 54
D. Detailed Description of Major Day-Ahead Market Rule Changes 54
1. Bidding for the New Day-Ahead Products and Limitations on Awards 54
2. Integrated Forward Market – Market Power Mitigation Pass 63
3. Integrated Forward Market – Procurement 67
4. Residual Unit Commitment – Market Power Mitigation Pass 72
5. Residual Unit Commitment – Procurement 74
6. Real-Time Market Obligations from Imbalance Reserves and Reliability Capacity Awards 76

7. Major Financial Settlements Design Elements of the Day-Ahead Market Enhancements 77
E. Additional Design Elements 82
1. Measures to Accommodate Long-Term Contracts 83
2. Real-Time Market Ramp Deviation Settlement 85
3. Congestion Revenue from Deployment Scenarios 89
4. Treatment of Metered Subsystems, Existing Transmission Contracts, and Transmission Ownerships Rights 91
5. Clarifying and Conforming Changes to Existing Tariff Provisions 93
F. Major Issues of Stakeholder Discussion 94
1. Demonstrating the Need for the Imbalance Reserves Product 94
2. Granularity of Procurement and Accounting for Transmission Constraints 95
3. Which Day-Ahead Market Process Should Procure Imbalance Reserves? 96
4. Hybrid Demand Curve 97
5. Limitations on Imbalance Reserves Awards 98
6. Tunable/Configurable Parameters 98
G. Connection to EDAM 99
1. Benefits of Imbalance Reserves in EDAM 99
2. Benefits of Reliability Capacity in EDAM 100

V. THE EXTENDED DAY-AHEAD MARKET 101

A. Participation Framework 102
1. Implementation, Procedures, and Fees 103
2. Transitional Protective Measures 105
3. Participants in Each Balancing Area 107
4. Participation in EDAM Versus the WEIM 107
5. Stakeholder Positions on the EDAM Participation Framework 109
6. EDAM Agreements 111
7. Metering and Telemetry 116
B. Pre-Market Activities 118
1. Ensuring Sufficient Transmission Is Available for the Day-Ahead Market 118
2. Ensuring Resource Sufficiency in the Day-Ahead Time Horizon 141
3. Ensuring Sufficient Resources to Serve Balancing Area Needs 150
C. Day-Ahead Market Activities 151
1. Virtual Bidding 152
2. External Resource Participation 153
3. Greenhouse Gas Accounting 158
4. Locational Marginal Price Formation Changes 174
D. Post-Market Processes and Outputs 177
1. Transmission Revenue Recovery 178
2. Settlements Generally 183
3. Transfer and Congestion Revenue Allocation 183
4. Stakeholder Positions on Transfer and Congestion Revenue Allocation 189
E. Additional General Tariff Provisions 190
1. Creditworthiness 190
2. Dispute Resolution, Liability, and Confidentiality 190
3. Miscellaneous 192
F. Components of the CAISO's Day-Ahead Market Design Not Included in the EDAM Design 192
1. Inter-Scheduling Coordinator Trades 192
2. Ancillary Services Procurement 193
3. Congestion Revenue Rights 194
G. Conforming Tariff Changes for Entities Participating in EDAM 195

VI. EFFECTIVE DATES OF DAME AND EDAM TARIFF REVISIONS, REQUEST FOR WAIVER OF NOTICE REQUIREMENTS, AND REQUEST FOR COMMISSION ORDER 197

VII. FILING 199

A. COMMUNICATIONS 199
B. SERVICE 199
C. CONTENTS OF THIS FILING 200

VIII. CONCLUSION 201

California Independent System Operator Corporation

August 22, 2023

The Honorable Kimberly D. Bose
Secretary
Federal Energy Regulatory Commission
888 First Street, NE
Washington, DC 20426

Re: California Independent System Operator Corporation
Docket No. ER23-___-000

Day-Ahead Market Enhancements and Extended Day-Ahead Market

Dear Secretary Bose:

The California Independent System Operator Corporation (CAISO) submits this tariff amendment to enhance and extend its day-ahead market in the West.[1] This filing implements the most significant market enhancement for the CAISO and Western stakeholders since the CAISO established the Western Energy Imbalance Market (WEIM) in 2014. This market enhancement will increase reliability, decrease costs to ratepayers, optimize generation dispatch across a broader footprint, and help participants and the states they serve achieve clean energy policy objectives.

The proposed tariff revisions are the product of two intensive stakeholder efforts to evolve and expand access to wholesale energy markets in the West. First, the CAISO proposes a suite of tariff revisions resulting from its Day-Ahead Market Enhancements (DAME) initiative.[2] Second, pursuant to its Extended

[1] The CAISO submits this filing pursuant to section 205 of the Federal Power Act (FPA), 16 U.S.C. § 824d, and Part 35 of the Commission's regulations, 18 C.F.R. Part 35. Capitalized terms not otherwise defined herein have the meanings set forth in appendix A to the CAISO tariff, and references herein to specific tariff sections are references to sections of the CAISO tariff unless otherwise specified.

[2] Materials related to the DAME initiative are available on the CAISO website at https://stakeholdercenter.caiso.com/StakeholderInitiatives/Day-ahead-market-enhancements.

Day-Ahead Market (EDAM) initiative, the CAISO proposes tariff revisions to extend access to its day-ahead market to balancing areas[3] in the WEIM that elect to participate in the day-ahead market.[4] After an extensive public process, both sets of tariff revisions received wide support from stakeholders.

The CAISO respectfully requests the Commission issue an order by December 21, 2023 accepting the DAME and EDAM tariff revisions, with the effective dates described in section VI of this transmittal letter.[5] The CAISO further requests that the Commission extend the time period for comments to 30 days from the date of this filing and the time period for responses to comments to 20 days from the date comments are due given the significance and scope of this filing.

I. Introduction

The CAISO proposes two significant sets of tariff revisions that implemented together will transform and extend its day-ahead market framework, producing significant economic, environmental, and reliability benefits. First, the CAISO proposes critical enhancements to its day-ahead market to address the challenges caused by increasing system variability and uncertainty, which will improve market efficiency and reliability. Second, the CAISO proposes to extend participation in its enhanced day-ahead market to other balancing authorities in the West, expanding access to the broad benefits of a day-ahead market.

The DAME tariff changes establish two new products – imbalance reserves and reliability capacity – to address the growing magnitude of imbalances in the supply and load forecast between the day-ahead and real-time market. These net load imbalances have grown significantly in recent years because of the rapid growth in variable energy resource capacity, significant

---

[3] By balancing area, the CAISO means a balancing authority area as defined in the North American Electric Reliability Corporation (NERC) Glossary of Terms. The CAISO also uses the term balancing authority, consistent with its meaning in the NERC Glossary of Terms, when referring to the responsible entity that maintains balance within the balancing area.

[4] Materials regarding the EDAM initiative are available on the CAISO website at https://stakeholdercenter.caiso.com/StakeholderInitiatives/Extended-day-ahead-market.

[5] Specifically, the CAISO requests that the Commission accept the DAME tariff revisions effective May 1, 2025. The CAISO also requests that the Commission accept the EDAM tariff revisions that address initial EDAM onboarding and implementation effective December 21, 2023, and it requests an effective date of May 1, 2025 for the balance of the EDAM tariff revisions. The CAISO respectfully requests waiver of the 120-day notice requirement for the DAME tariff revisions and the latter EDAM tariff revisions.

As discussed in section V.G of this transmittal letter, conforming tariff changes may be required for each affected balancing authority, and the tariff-owning entities within each affected balancing area. The CAISO intends to submit a separate filing to identify conforming tariff changes within its own balancing area.

weather-related uncertainty, and extreme climate events. This has increased the risk that the CAISO will not have sufficient online capacity and ramp capability available to efficiently and reliably balance the system in real-time.

To avoid these potential adverse outcomes, the CAISO currently uses the residual unit commitment (RUC) process to adjust the load forecast upon which the CAISO bases its capacity procurement. The DAME tariff changes will mitigate the need for this adjustment in RUC. They will help increase overall efficiency through the procurement of imbalance reserves and reliability capacity based on modeled variability and uncertainty.

The CAISO will procure imbalance reserves up and imbalance reserves down to meet the range of expected imbalances between the day-ahead and real-time net load forecasts. The extreme upper and lower range of expected imbalance reflects the range of upward and downward flexible reserves the CAISO foreseeably would need for a given interval if either extreme outcome occurred. The integrated forward market will co-optimize bids for energy, ancillary services, and both types of imbalance reserves in determining market awards, thus enhancing market efficiency.

Similarly, the CAISO will procure reliability capacity up much the same as the capacity it procures today in the RUC process, with the addition of reliability capacity down that represents new capacity procured to address scenarios where the day-ahead market awards too much energy relative to the day-ahead net load forecast. Procuring capacity in the downward direction works together with the upward capacity to inject more flexibility into the process and address potential oversupply in the real-time market.

The second major set of proposed tariff revisions establish the extended day-ahead market framework, which will enable other balancing areas in the West to participate in a day-ahead market. These tariff changes incorporate the enhancements proposed in the DAME initiative. Together, the tariff changes will support the optimal day-ahead commitment of a geographically diverse set of resources across an expanded market area, maximizing the use of available transmission capability and providing broad reliability, economic, and environmental benefits.

WEIM participants can join EDAM and settle all balancing area loads and resources in the day-ahead timeframe in addition to all imbalances between day-ahead positions and the real-time market. Similar to the WEIM, participants in EDAM must demonstrate they have sufficient supply to meet balancing and flexibility tests so the market can more efficiently optimize resources over the expanded market area. Before the day-ahead market runs, this test will review the demand obligations and supply options for each balancing area and determine whether each has sufficient supply to satisfy its forecasted demand

commitments, forecasted imbalance reserve requirements, and forecasted ancillary service obligations. The DAME tariff revisions will enhance the ability to assess whether participating balancing authorities have met these requirements. Balancing areas that pass the test will benefit from the diversity of resources available across the expanded market area, and be pooled together for purposes of the real-time resource sufficiency evaluation in the WEIM. Balancing areas that do not pass the test will be subject to a surcharge for such failure and may not be included in the real-time resource sufficiency evaluation pool.

The extended day-ahead market will optimize the transmission available to the market, including unsold and unscheduled firm transmission capability. It will also identify the most efficient resource commitments, energy transfers, and capacity transfers to meet forecasted demand and uncertainty across the footprint. A transmission customer that submits a self-schedule before the start of the day-ahead market will be assigned a higher market clearing scheduling priority to allow these transactions to clear ahead of other self-schedules and economic bids. Alternatively, the transmission customer may release its transmission rights to the market and receive transfer revenue that accrues across an interface due to energy or capacity price differences between balancing areas.[6] In any case, a transmission provider will adhere to the requirements of its open access transmission tariff (OATT) to release, hold, or prioritize transmission service.

If a transmission customer neither releases nor self-schedules use of its transmission rights by the close of the day-ahead market, the capacity will be available to the day-ahead market, with the transmission customer nonetheless retaining its right to later submit an intra-day self-schedule. The market will afford such balanced intra-day self-schedules a higher priority when so instructed by a transmission provider acting in accordance with its tariff. The market will seek to accommodate all self-schedules submitted after the close of the day-ahead market through re-dispatch. If the market is unable to accommodate the customer's schedule, the market will provide the balancing authority information regarding the infeasibility. The balancing authority would, in accordance with the transmission provider tariff, attempt to address the infeasibility. This approach to transmission availability respects transmission customers' rights under legacy contracts and the Commission's *pro forma* OATT, promotes market efficiency, and provides confidence in transfers between balancing areas that are relied upon to deliver energy, imbalance reserves, and reliability capacity in both normal and stressed system conditions. These market rules can help operationalize resource adequacy programs throughout the West in a coordinated and efficient manner.

---

[6] Transfer revenue represents the separation in the marginal energy costs, imbalance reserve costs, and reliability capacity costs between two balancing areas participating in EDAM when a scheduling limit is reached.

The EDAM tariff changes will also account for greenhouse gas (GHG) emissions by adopting the WEIM resource-specific design to identify which resources serve demand in a GHG regulation area. This approach, with some enhancements, will allow resource scheduling coordinators to recover their cost of compliance with a state's carbon pricing policy. Using this method does not embed the cost of that state's policies in the price for demand outside of a GHG regulation area, and provides a mechanism to identify which scheduling coordinators are electricity importers into a GHG regulation area. In addition, EDAM will introduce a special market run in the day-ahead market – the GHG reference pass – that will use energy bids to optimize dispatch to serve the demand and create a reference schedule that informs the value of a GHG transfer that the market may attribute to a resource. Finally, the CAISO will include a GHG net export constraint to reduce the potential for secondary dispatch, preventing the attribution of GHG transfers to resources in balancing areas that are net importers and reducing the potential for secondary dispatch.

The CAISO will use its existing post-market settlements timelines and procedures with a few modifications to account for unique features of the extended day-ahead market settlement requiring additional provision. In particular, congestion revenue will be allocated to the balancing area where the constraint is modeled. Transfer revenue, similar to congestion revenue at an interface between two balancing areas, will be allocated equally between the two balancing authorities that made the transmission available or to the transmission customer. This ensures congestion revenue is allocated based on the location of the constraint and that transfer revenue is allocated to the market participant that made the transmission available to the market. The allocation of congestion and transfer revenue to the balancing authority will enable each transmission service provider to determine the most appropriate distribution of costs and benefits from participation in the day-ahead market.

With DAME providing an innovative pathway to manage rapidly changing system conditions, integrate incremental resources, and address significant operational challenges, and EDAM providing an enhanced day-ahead market that builds on the successful platform of the WEIM, a diverse array of Western market participants and customers will be well-positioned to realize significant and measurable reliability, economic, and environmental benefits.

II. Executive Summary

A. Day-Ahead Market Enhancements

1. Addressing Energy Imbalances Today

The CAISO's day-ahead market enhancements are driven by the need to address increasing differences in the load forecast net of variable energy resource production (*i.e.*, the net load forecast) used to optimize the day-ahead market compared to the net load forecast used for the real-time market. Two sets of forecasts drive the net load forecast – the gross forecast of load and the production forecast from wind and solar resources. Unless these forecasts for the day-ahead market perfectly match the forecasts for the real-time market, an imbalance is unavoidable. The net load forecast is a key value for the CAISO markets because it represents how much energy the market must procure from firm dispatchable resources to meet system needs.

Although some net load imbalances are to be expected, the magnitude of deviations has grown significantly in recent years. Major causes have been the rapid growth in variable energy resource capacity on the CAISO grid and more extreme weather-related uncertainty throughout the Western grid. Demand for electricity is highly dependent on weather and increasingly impacted by extreme weather conditions, which make it harder to forecast demand accurately. These same extreme weather conditions can also make it harder to forecast production from wind and solar resources.

Large net load imbalances have placed the CAISO at risk of not having sufficient online capacity or insufficient ramp capability in the real-time market. To avoid load shedding or and other harmful outcomes, the CAISO has had to increase its RUC load forecast, which results in RUC procuring more capacity than it otherwise would. These manual interventions have maintained system reliability, but they have come at the cost of inefficient unit commitment and higher overall costs.

2. Description of Imbalance Reserves and Reliability Capacity

To address the net load variability and uncertainty challenges facing the CAISO, the CAISO is proposing a suite of day-ahead market enhancements. The core of these day-ahead market enhancements is the establishment of two new bi-directional products called imbalance reserves and reliability capacity. Essentially, the CAISO will be procuring four new products: (1) imbalance reserves up; (2) imbalance reserves down; (3) reliability capacity down; and (4) reliability capacity up.

Imbalance reserves are a flexible reserve product to cover: (1) uncertainty in the net load forecast between the day-ahead market and real-time market; and (2) real-time ramping needs not covered by hourly day-ahead market schedules. The CAISO separately will procure imbalance reserves to address upward uncertainty and downward uncertainty, respectively. A resource receiving an imbalance reserves award must submit economic bids to the real-time market for its awarded capacity range. The imbalance reserve product will ensure the integrated forward market (IFM) schedules sufficient dispatch capability to meet net load imbalances between the day-ahead and real-time markets.

The CAISO will procure reliability capacity to meet positive or negative differences between cleared physical supply in the integrated forward market and the load forecast. This is similar to the existing RUC capacity product except the CAISO only procures RUC capacity to the extent the forecast exceeds cleared physical supply. With the bi-directional reliability capacity product, the CAISO will replace the existing uni-directional RUC capacity product it procures today with the reliability capacity up product as well as the ability to procure decremental capacity with the reliability capacity down product. As with imbalance reserves, a reliability capacity award will obligate the resource to provide economic energy bids in the real-time market for the quantity of its awards.

3. Bidding for the New Products and Eligibility for Awards

For both imbalance reserves and reliability capacity, suppliers will provide separate bids for the upward and downward products. Each bid will have a single price/quantity pair, subject to a $55/MWh cap on imbalance reserves bids and a $250/MWh cap on reliability capacity bids. Resources that are dispatchable on a 15-minute basis will be eligible to provide imbalance reserves. Eligibility for reliability capacity awards will be broader, with only certain types of demand response resources excluded from eligibility.

For each interval a resource submits bids for imbalance reserves, it must also provide an energy bid in the day-ahead market and must economically bid the portion of the energy bid that overlaps with the imbalance reserve bid. To facilitate the EDAM resource sufficiency evaluation, all eligible resources offering energy bids to the day-ahead market must submit bids for reliability capacity up at the same quantity as their energy bid plus ancillary service self-provision. This ensures all resources in the EDAM resource sufficiency evaluation are available for RUC, including excess supply that participants offered above their resource sufficiency evaluation requirements.

4. Impacts on Resource Adequacy Bidding

The CAISO will amend the bidding obligations for resource adequacy capacity to align with the new products. All resource adequacy capacity (system, local, and flexible) must bid for reliability capacity up, consistent with the existing obligation to submit RUC availability bids. A significant change is that resource adequacy capacity will be able to submit economic bids for reliability capacity up to the $250 bid cap and retain its RUC compensation.

The CAISO will not require resource adequacy capacity to submit bids for reliability capacity down. If resource adequacy capacity self-schedules its energy into the day-ahead market, it will have no obligation to submit imbalance reserves bids. If, however, resource adequacy capacity submits an economic bid to the day-ahead market and is eligible to provide imbalance reserves, it must submit bids for imbalance reserves up and imbalance reserves down for the capacity range covered by the economic bid for energy. Flexible resource adequacy capacity must bid its full range of resource adequacy capacity as imbalance reserves up and imbalance reserves down.

5. Enhancements to the Integrated Forward Market

The integrated forward market currently co-optimizes bids for energy and ancillary services. The CAISO will update the integrated forward market to co-optimize bids for energy, ancillary services, and imbalance reserves. This will enhance the efficiency of unit commitment by securing flexible reserves in a way that effectively considers the trade-offs between providing energy, ancillary services, and imbalance reserves.

The CAISO will model imbalance reserves procurement on how it procures the flexible ramping product. The CAISO will aim to meet the full procurement targets subject to a stepped demand curve, with each step of the demand curve based on the trade-offs between procuring the product and incurring the $247/MWh penalty price of violating contingency reserve requirements. The $247 penalty price, however, will only inform how each step of the curve is set. The CAISO separately will set a penalty price on imbalance reserves of $55, meaning the CAISO will not procure imbalance reserves above that price. This value is roughly comparable to the 80th percentile of historical operating reserves bids submitted to the CAISO and represents the costs the CAISO likely would incur if it had to procure additional spinning reserves instead of imbalance reserves.

The CAISO will consider potential transmission constraints when awarding imbalance reserves. The CAISO proposes two mechanisms to provide

flexibility in how it evaluates imbalance reserves for deliverability. The first is a configurable deployment factor that will determine what percent of imbalance reserves must be deliverable. The second is the flexible activation/deactivation of individual transmission constraints in the deployment scenarios. Under this approach, the CAISO would not necessarily enforce all constraints used in the energy market to evaluate deliverability of imbalance reserves. Instead, it would identify in the business practice manual the constraints it would enforce. Both measures address stakeholder concerns about potential excessive costs of procuring imbalance reserves and market performance. By including these implementation details in the business practice manual, the CAISO can adjust these parameters based on market simulation and observed market results.

The CAISO proposes to extend its current day-ahead market local market power mitigation measures to cover imbalance reserves because it will price the imbalance reserves product on a locational basis and imbalance reserves are fungible with energy. Consistent with the current design, the CAISO would mitigate imbalance reserves bids that provide counter-flow to an uncompetitive constraint (whether that constraint is in the base case or the cases of meeting modeled upward or downward uncertainty) to the higher of the competitive locational price or the default availability bid. The default availability bid is a simplified version of the existing default energy bid set, which is set at a static value of $55.

6. Enhancements to the Residual Unit Commitment Process

The CAISO proposes three major changes to the RUC. First, creating a reliability capacity down product makes RUC a bi-directional process that also can also address oversupply conditions. Second, the CAISO is changing the nature of a RUC award. Today, a RUC award is a firm obligation to provide energy in real-time. To provide greater flexibility for the real-time market to optimize procurement, a RUC award now would obligate the resource to submit an economic bid to the real-time market for its range of awarded reliability capacity. The real-time market would then consider all bids to determine the most economical solution. Third, the CAISO will implement a new RUC market power mitigation pass before procuring reliability capacity. The existing requirement for resource adequacy capacity to submit $0 bids into RUC has obviated the need for explicit RUC market power mitigation provisions. As discussed herein, the $0 bidding requirement is problematic for EDAM, and therefore the CAISO is proposing to remove it. However, its removal raises new possibilities for resources to exercise local market power in RUC, supporting the adoption of the new market power mitigation pass.

7. Financial Settlements Issues

a. Basic Settlement of Imbalance Reserves and Reliability Capacity

The CAISO will pay resources receiving an imbalance reserves or reliability capacity award the locational price of the product for the awarded direction. The CAISO also will amend the bid cost recovery mechanism to account for both the bid costs and revenues from the new products. This will ensure resources remain eligible for uplift payments where warranted. Resources will be subject to unavailability charges for both products. Reliability capacity will be subject to direct rescission for undispatchable capacity.

The CAISO will allocate the costs of procuring the new products based on a two-tiered cost allocation similar to existing allocation processes for other market products. The allocation processes follow cost-causation principles and assign costs to the maximum extent possible to the parties driving the incurrence of those costs, while protecting parties from unreasonably large cost allocations in extreme cases.

b. Transitional Measures for Resource Adequacy Capacity

During the stakeholder process, several parties noted that existing bilateral resource adequacy contracts might not specify the allocation of market payments for imbalance reserves and reliability capacity between the load serving entity and the resource adequacy resource. Resource adequacy capacity has been ineligible for RUC capacity payments since the beginning of the CAISO's nodal market, so resource adequacy contracts likely would not address RUC payments. Further, imbalance reserves are a new product so contracts may not have anticipated creation of such a product. To facilitate the transition to the new products, the CAISO will implement a three-year period during which the CAISO will apply optional gap-filling rules to allocate the market revenue from imbalance reserves and reliability capacity. However, for these transitional rules to apply, both the load serving entity and supplier must mutually elect for this settlement treatment.

c. Ramp Deviation Settlement

There is significant overlap between imbalance reserves awards in the day-ahead market and flexible ramping product awards in the real-time market. They both provide the CAISO with flexible ramping capabilities. In developing imbalance reserves, the CAISO was concerned about the potential for double payments to resources receiving an award for both products. To address this

concern, the CAISO will implement a ramp deviation settlement for imbalance reserves. Under this mechanism, the CAISO will settle a flexible ramping product award in the real-time market based on imbalances with the corresponding five-minute portion of the overlapping imbalance reserves award or forecasted movement embedded in the day-ahead market energy. Including the forecasted movement payments that are already embedded in the day-ahead price provides as complete an imbalance settlement as possible because forecasted movement is already settled explicitly in the flexible ramping product.

The CAISO also addresses a similar existing issue with convergence bidding and with the WEIM in which parties potentially can receive double payment between the forecasted movement payments embedded in the day-ahead market price and the explicit payment for forecasted movement through the flexible ramping product.

d. Impact on Congestion Revenue Rights from Displaced Congestion Revenue

Some stakeholders expressed concern regarding the impact imbalance reserves might have on congestion revenue rights settlement. These parties noted that both energy and imbalance reserves can compete for scarce transmission revenue. Sometimes, the IFM co-optimization could hold back scarce transmission capacity to ensure the deliverability of imbalance reserves instead of using that capacity for energy. In these scenarios, the locational price for imbalance reserves would reflect the scarcity of that transmission, but the locational price of energy in the day-ahead market would not. Because the CAISO settles congestion revenue rights based on congestion revenue from energy in the day-ahead market, some stakeholders expressed concern that implementing imbalance reserves could harm congestion revenue rights holders.

To address this potential concern, the CAISO proposes to calculate the energy market congestion revenue displaced from procuring imbalance reserves and add it to the total costs recovered through the two-tiered imbalance reserves cost allocation. For the CAISO balancing authority, the CAISO would add the displaced congestion funds recovered through that allocation to the pool of funds paid to parties holding congestion revenue rights with modeled flow over the transmission constraints with displaced congestion revenue. The CAISO also will adjust the notional value of congestion revenue rights, which represent the maximum settlement value, to reflect differences in the marginal cost of congestion for energy and imbalance reserves between the source and sink of the congestion revenue right. For EDAM entities, the CAISO would allocate the displaced congestion funds to the EDAM entity for further allocation pursuant to its OATT.

B. Extended Day-Ahead Market

Extending the CAISO's day-ahead market to other balancing areas results from a multi-year stakeholder effort across the West to build on the benefits of the WEIM, the CAISO's real-time market.[7] Since the Commission approved the WEIM framework in 2014, participating utilities have realized $4.20 billion in gross benefits from the WEIM.[8] Anticipating even greater benefits from a two-day optimization, WEIM participants requested the CAISO explore the feasibility of providing similar optimization services in the day-ahead timeframe.[9] The CAISO anticipates annual economic benefits of EDAM will range from $100 million to more than $1 billion in addition to the economic and other benefits the WEIM will continue to provide.[10]

The EDAM design offers a voluntary regional day-ahead market by leveraging the existing features of the CAISO's day-ahead market with targeted adjustments that recognize the unique challenges and needs of the balancing authorities that may participate in EDAM and their embedded and their associated utilities, generators, and customers, in the Western Interconnection.

---

[7] *See, e.g.*, *Cal. Indep. Sys. Operator Corp.*, 147 FERC ¶ 61,231 (WEIM Authorization Order), *order on reh'g, clarification, & compliance*, 149 FERC ¶ 61,058 (2014) (WEIM Rehearing Order). These Commission orders authorized the implementation of the original WEIM market design, which the CAISO has subsequently enhanced and updated in a series of tariff amendments.

[8] Quarterly benefit metrics are available at https://www.westerneim.com/Pages/About/QuarterlyBenefits.aspx.

[9] Fifteen balancing areas that have already joined the WEIM participated in an EDAM feasibility assessment. *See* Public Comment Letter from WEIM Entities re Extended Day-Ahead Market (Sept. 16, 2019), available at http://www.caiso.com/InitiativeDocuments/PublicCommentLetter-EIMEntites-EDAM-Sep16-2019.pdf; Briefing on Extended Day-Ahead Market Initiative (Sept. 18, 2019), available at http://www.caiso.com/InitiativeDocuments/Briefing-ExtendedDay-AheadMarketInitiative-Presentation-Sep2019.pdf.

[10] *Extended Day-Ahead Market: Feasibility Assessment Update from EIM Entities* (2019) (EIM Entities Study), available at http://www.caiso.com/InitiativeDocuments/Presentation-ExtendedDay-AheadMarketFeasibilityAssessmentUpdate-EIMEntities-Oct3-2019.pdf; *The State-Led Market Study* (July 30, 2021), available at https://static1.squarespace.com/static/59b97b188fd4d2645224448b/t/6148a012aa210300cbc4b863/1632149526416/Final+Roadmap+-+Technical+Report+210730.pdf; *CAISO EDAM Benefits Study: Estimating Savings for California and the West Under EDAM Market Scenarios* (Nov. 4, 2022) (CAISO EDAM Benefits Study), available at http://www.caiso.com/Documents/Presentation-CAISO-Extended-Day-Ahead-Market-Benefits-Study.pdf; *Brattle EDAM Simulations: PacifiCorp Results* (Apr. 2023) (Brattle EDAM Study), available at https://www.brattle.com/wp-content/uploads/2023/04/Brattle-EDAM-Simulations-PacifiCorp-Results.pdf. The CAISO EDAM Benefits Study quantified capacity savings of $652 million annually in addition to operational savings of $543 million. The EIM Entities Study quantified operational savings but did not attempt to quantify capacity savings. The Brattle EDAM Study quantified gross benefits of $810 million and net benefits of $438 million annually.

The EDAM design also incorporates the enhancements to the day-ahead market proposed in the DAME initiative. Together these enhancements support the optimal commitment of a geographically diverse set of resources across the footprint of all balancing areas participating in EDAM, optimizing the use of available transmission capability, building upon the WEIM, and providing broad economic, reliability, and environmental benefits.

Under the extended day-ahead market design, load serving entities and resources in balancing areas that participate in the WEIM can choose to extend their participation to the day-ahead market processes. For the balancing authorities that join, the extended day-ahead market will settle all loads and resources in the day-ahead timeframe and all imbalances between day-ahead positions and the real-time market. The extended day-ahead market will optimize the transmission and resources offered into the market to identify the most efficient resource commitments and energy transfers to meet forecasted demand across the footprint. It will also optimize imbalance reserves and reliability capacity across a broader footprint. EDAM will also include mechanisms to harmonize greenhouse gas accounting methods and address the potential under-recovery of historical transmission revenues.

The CAISO will include readiness criteria in the EDAM implementation process that parallel those in the WEIM today to ensure the CAISO and participants are prepared for operation of the day-ahead market in each balancing area. Further, the CAISO will include transitional measures in EDAM similar to those in the WEIM to insulate participants from adverse reliability or market outcomes during the implementation process and throughout their initial participation. The CAISO will also monitor and issue public reports on the performance of the EDAM design and work with stakeholders to refine the design.

EDAM-related activities occur primarily within three timeframes – before the day-ahead market, within the day-ahead market, and after the day-ahead market.

1. Pre-Market Timeframe

In the pre-market timeframe, the extended day-ahead market activities center on preparing for the optimization of the day-ahead market by addressing transmission availability, accounting for legacy transmission contracts and transmission ownership rights, and ensuring each balancing area is resource sufficient. By accounting for the transmission capability necessary to support optimized unit commitment, and ensuring each balancing area has sufficient resources to support its own obligations, the pre-market activities support an effective and efficient resolution in the day-ahead market.

a. Transmission Availability

The proposed EDAM tariff revisions balance stakeholder interests by seeking to maximize the transmission capability each balancing authority makes available to the day-ahead market while respecting existing transmission service rights within all balancing areas. This allows participants to enjoy the widespread benefits of the extended day-ahead market. Specifically, the extended day-ahead market will allow each balancing authority that joins to transfer supply, imbalance reserves, and reliability capacity based upon the economic optimization of resources across a broad geographic area, without threatening the balancing authority's ability to meet its own obligations.[11] The extended day-ahead market accounts for the transmission capability necessary to support this optimized unit commitment in the day-ahead market and transfers between balancing areas, while recognizing that balancing authorities and the associated transmission service providers must conform to the terms and conditions of their amended OATTs required to accommodate this day-ahead market. With the support and input of stakeholders, the CAISO has developed a tailored approach that optimizes the transmission capability made available to support transfers to other balancing areas, while allowing all participating balancing authorities to continue fulfilling their obligations.

Prior to the start of the day-ahead market, a balancing authority participating in EDAM will provide information to the CAISO, including information gathered from transmission service providers in its balancing area, regarding any applicable transfer constraints and the amount of transmission capacity available. This will allow the extended day-ahead market to utilize all transmission capacity available to it, including unsold firm transmission capability available, while honoring existing and legacy rights. This allows transmission service providers and their customers located within the balancing areas to maintain their open access transmission offerings and rights, respectively, including network integration transmission service and firm point-to-point rights.

Recognizing and honoring existing and legacy transmission rights, EDAM provides three different avenues for customers to exercise their transmission rights prior to the day-ahead market. First, the transmission customer can use its rights for its own purposes by submitting a balanced self-schedule associated with registered transmission service rights into the day-ahead market, *i.e.*, a self-schedule representing the same quantity at the source location and the sink location. Self-schedules submitted before the day-ahead market are assigned a scheduling priority in the day-ahead market higher than a self-schedule not

---

[11] Balancing authorities in the West are diverse and include large and small investor owned utilities, federal entities and power marketers, large and small municipal utilities, joint power agencies, and generation-only balancing areas. The EDAM design will accommodate all balancing authorities much as the WEIM does today.

associated with registered transmission service, and these transactions will clear ahead of self-schedules not exercising transmission rights and economic bids. The market will not adjust a self-schedule associated with registered rights to accommodate a self-schedule that is not associated with registered rights, maintaining this market scheduling priority.

Second, as an alternative to using its transmission rights, the transmission customer may elect to release its transmission rights for the market to optimize. In exchange for releasing its rights for optimization, transmission customers are eligible to receive an allocation of transfer revenues if transfer revenue accrues across an interface due to energy or capacity price differences between balancing areas, *i.e.*, revenue from price differences between balancing areas at the transfer location, the transmission customer would receive the transfer revenue. A transmission customer's release of transmission rights to the market allows the market to support optimized transfers in the day-ahead and real-time markets, thus lowering the risk of infeasibilities and the need to redispatch. A transmission customer that pursues this avenue cannot self-schedule released transmission in the day-ahead market or after the close of the day-ahead market.

Finally, a transmission customer may choose neither to release nor self-schedules its transmission rights by the close of the day-ahead market. In this case, the capacity will be available to the day-ahead market for optimization, allowing the market to establish a transfer that delivers lower cost resources from one balancing area to serve demand in another balancing area. Under this third option, however, the transmission customer does not lose its rights that it has today under the OATT. Instead, the transmission customer would retain its right to submit later an intra-day self-schedule associated with its rights. If instructed by an EDAM transmission service provider, the market will afford balanced intra-day self-schedules or schedule changes for registered firm OATT transmission rights a priority in the real-time market higher than other real-time self-schedules and higher than cleared day-ahead self-schedules not based on specific firm OATT transmission rights, including transfer schedules between balancing areas serving load. For all intra-day self-schedules associated with firm OATT rights, the market would redispatch the system through the real-time market to accommodate both the transfer optimized in the day-ahead market and the transmission customer's self-schedule submitted after the close of the day-ahead market. In the rare case the market cannot accommodate both transactions through redispatch, then the market will afford the transfer equal priority with demand in the balancing area and notify the applicable balancing authority of the associated schedules and the infeasibility. The balancing authority would, acting in accordance with applicable tariffs and good utility practice, determine how to resolve the infeasibility. Given the diversity of the Western Interconnection, each balancing authority and transmission service provider may have different means by which they can resolve the infeasibility, including by dispatching additional resources manually, adjusting the transfer limit, adjusting the transmission

customer schedule, or taking any combination of actions or additional actions to resolve the infeasibility and fulfill its balancing authority obligations.

By providing three avenues to honor customers' transmission rights, the extended day-ahead market provides an innovative pathway to promote regional markets and advance the interconnectedness of balancing areas in the West in a non-discriminatory manner. Together, and separately, these avenues respect transmission customers' rights, promote market efficiency and provide confidence in transfers between balancing areas that rely on them to deliver energy, imbalance reserves and reliability capacity in both normal and stressed system conditions. The CAISO and stakeholders discussed other options, including imposing restrictions on the availability of transmission to the market, but deemed these to be less efficient. For example, restrictions can create conditions where congestion occurs even though sufficient transmission capability is available; *i.e.*, transmission automatically "carved out" from the market may not be fully utilized. Similarly, providing a market scheduling priority for late-submitted transmission schedules higher than transfers between balancing areas would undermine confidence in market outcomes. If there is frequent use and exercise of particular transmission rights across discrete and specific paths or flowgates, the EDAM design does allow transmission service providers to request an adjustment to "carve out" the transmission right from the market under limited circumstances specified in the transmission service provider's tariff, which the CAISO would implement. Due to the inefficiencies created by such arrangements that limit the benefits of EDAM to ratepayers, the CAISO expects such carveouts to be rare. This transmission availability framework balances the interests of all stakeholders and is superior to the *pro forma* OATT because it enables balancing areas that may not be in a position to join an independent system operator (ISO) or regional transmission organization (RTO) at this time to nonetheless capture the benefits of a day-ahead market on behalf of their customers.

The extended day-ahead market will provide legacy transmission contracts and transmission ownership rights in an EDAM balancing area a market scheduling priority and settlement consistent with existing mechanisms in the CAISO tariff for these types of arrangements, while making the flow capability available to the market in the CAISO balancing area. These mechanisms provide legacy contract and ownership rights holders with a scheduling priority above other transmission through the day-ahead and real-time markets consistent with their contractual rights. They also provide financial protection from congestion costs associated with the scheduling of legacy rights in the market. Legacy contracts and ownership rights will also be eligible to make their transmission capacity available at interties between EDAM balancing areas in exchange for transfer revenue.

b. Resource Sufficiency

In addition to accounting for available transmission, activities in the pre-market timeframe will include evaluating the resource sufficiency of each balancing authority in the EDAM area, *i.e.*, the aggregate of the CAISO balancing area and all balancing areas participating in EDAM. This is a fundamental underpinning of the extended day-ahead market. Similar to the WEIM, the CAISO expects balancing areas participating in the day-ahead market to demonstrate they have sufficient supply to meet specific balancing and flexibility needs. The EDAM resource sufficiency evaluation or "RSE" embodies these same principles.

Under this structure, before the first run of the day-ahead market, each balancing authority participating in EDAM will be evaluated to determine if its resources are sufficient to meet its projected needs in the day-ahead time horizon. Specifically, the EDAM resource sufficiency evaluation will review the demand obligations and supply options for each balancing area and determine whether there is sufficient supply shown to satisfy the forecasted demand obligations in both the upward and downward directions, the forecasted imbalance reserve requirements, and the forecasted ancillary service obligations. Results of the resource sufficiency evaluation will be provided to each balancing authority on an advisory basis to assist the balancing authority in establishing its resource sufficiency prior to the binding run of the resource sufficiency evaluation at approximately 10:00 a.m. (*i.e.*, immediately prior to running the day-ahead market).

Balancing areas that pass the final binding EDAM resource sufficiency evaluation will be "pooled" together for the WEIM resource sufficiency evaluation. Balancing areas that do not pass the EDAM resource sufficiency evaluation will be assessed the applicable failure surcharge depending on the direction of the failure (upward or downward) and the time period (off-peak or on-peak). If the integrated forward market can satisfy the forecasted needs of the balancing authority to cure its deficiency, then the balancing area will be included in the pool. This pooling approach benefits eligible balancing areas by accounting for the diversity benefit of procured imbalance reserves through the day-ahead market. If balancing authorities do not cure the failure through the integrated forward market, then the balancing area will not be included in the pool, and the balancing authority will be evaluated individually for purposes of the WEIM resource sufficiency evaluation.

The proposed resource sufficiency evaluation recognizes and respects the unique circumstances of balancing areas in the Western Interconnection. In addition, the proposal includes an additional measure that will allow each balancing authority to manage its internal supply if there is a risk of resource

insufficiency approaching real-time. For example, if a balancing area is experiencing resource constraints due to a significant generation outage, the balancing authority needs a mechanism to allow it the balancing authority to limit the supply offered to support transfers to another balancing area. The net export transfer constraint is a voluntary, and optional, mechanism that will enable each balancing authority to preserve sufficient supply to meet its projected needs, while managing reliability conditions based on conditions and circumstances anticipated within its balancing area. A uniform assessment of resource sufficiency, together with a configurable mechanism for each balancing authority to manage its supply, provides an innovative pathway for incrementally advancing the interconnectedness of balancing areas in a just, reasonable, and non-discriminatory manner.

2. Within the Extended Day-Ahead Market

a. Optimizing Available Resources Across the EDAM Footprint

In the extended day-ahead market, all resources and load in a participating balancing area will submit an economic bid or a self-schedule in the day-ahead market based on their availability and operational circumstances. This differs from the load and resource participation model in the WEIM where a base schedule reflects the planned operation of loads and resources in the day-ahead timeframe, and only participating resources submit economic bids, *i.e.*, a base schedule in the WEIM is accounted for but not optimized by the market. The extended day-ahead market will optimize the transmission and resources offered into the market to identify efficient resource commitments and energy transfers to meet scheduled and forecasted load across the footprint.

Resources located outside of the EDAM area may fully participate in the day-ahead market if they are pseudo-tied or dynamically scheduled into an EDAM balancing area or are a designated network resource to serve load in an EDAM balancing area. Other contracted supply located outside of the EDAM area can self-schedule at the interties of non-CAISO balancing areas but cannot economically bid. This restriction results from non-CAISO balancing area concerns that non-source-specific supply that is not deliverable could displace physical internal generation. If the economically bid intertie supply does not perform, it could cause or exacerbate reliability conditions within the balancing area. This paradigm is consistent with the WEIM today. At the same time, designating a network resource to serve an EDAM balancing area enables economic bidding of the resource because the designation imposes specific transmission requirements that increase confidence the supply is deliverable. The added requirement that a designated network resource be source-specific

provides further confidence that the supply is not speculative. In any event, the CAISO will continue to allow economic bidding at its interties with non-EDAM balancing areas as it does today.

At the start of EDAM, the CAISO proposes to apply market power mitigation at the balancing area level similar to the WEIM today. The CAISO intends to evaluate the overall market power mitigation design in its existing Price Formation Enhancements stakeholder initiative. The CAISO will consider extending any design changes arising from that initiative to the extended day-ahead market and the WEIM.

The CAISO proposes to make virtual bidding an optional election for each balancing authority participating in EDAM, allowing the EDAM participant to choose whether to enable convergence bidding within its balancing area. This allows participants to gain day-ahead market experience prior to introducing convergence bidding, which is an unfamiliar and significant market design feature.

b. Harmonizing Greenhouse Gas Accounting in the Day-Ahead and Real-Time Markets

Under the existing Commission-approved greenhouse gas accounting design, the WEIM uses bid adders to identify which resources serve demand in California balancing areas using a resource-specific approach. When offering bids to serve demand in a state with carbon pricing policies, scheduling coordinators for resources located in balancing areas outside of California submit a two-part bid adder consisting of a GHG bid capacity (MW) quantity and a GHG price ($/MWh). The bid adder reflects the willingness of the resource's scheduling coordinator to make the resource's output available to serve demand within California balancing areas and reflects the cost of compliance with California's carbon pricing policies. The market optimization considers the GHG bid adder in addition to the energy bid to determine efficient dispatch and identify which resources serve demand in California balancing areas. If a resource outside of a California balancing area does not submit a GHG bid adder, the market does not consider or attribute the resource to serve California demand.

Resources within California balancing areas recover their costs of compliance with California's carbon policies through their energy bids. The CAISO will extend the current WEIM GHG accounting framework and resource-specific approach to EDAM with several enhancements. This design allows resources' scheduling coordinators to recover their cost of compliance with a state's carbon pricing policy, does not embed the cost of that state's policies in locational marginal prices for demand outside of a GHG regulation area, and

provides a mechanism to identify which scheduling coordinators are electricity importers into a GHG regulation area.

The geographic boundary used for GHG accounting purposes under EDAM (and the WEIM) will reflect state-level boundaries rather than balancing area boundaries. A state-level boundary more accurately represents the boundary of the GHG regulation area because carbon pricing policies are established on a state-by-state basis. Further, the GHG accounting framework for EDAM will allow resources to submit GHG bid adders associated with serving demand in multiple GHG regulation areas, which will allow EDAM to optimize procurement of resources to serve demand in California, Washington, and other states that adopt carbon pricing policies. The market will consider the bid adders in addition to energy bids in determining the most efficient dispatch for serving demand in a specific GHG regulation area. This framework will allow for resource-specific attribution of resources in one GHG regulation area to serve load in another GHG regulation area.

Attributing resources to a GHG regulation area creates the potential for secondary dispatch where resources outside of a GHG regulation area attributed to serve demand within a GHG regulation area are backfilled by higher emitting resources. EDAM will introduce a special market run in the day-ahead market – the GHG reference pass – that will use energy bids to optimize dispatch to serve the demand without GHG transfers into a GHG regulation area. This market run will create a reference schedule to inform the MW value of a GHG transfer the integrated forward market may attribute to a resource. The reference used in real-time will be the difference between the day-ahead market energy schedule and day-ahead market GHG award. The reference schedule for a balancing area that participates in the WEIM and not in EDAM will continue to be a self-submitted base schedule.

The CAISO also proposes to include a GHG net export constraint in EDAM to reduce the potential for secondary dispatch. Participating balancing authorities will not elect whether or not use this constraint. Instead, the CAISO will enforce this constraint to limit the attribution of GHG transfers to resources within a balancing area so they do not exceed a net export transfer schedule from the balancing area. This prevents the attribution of GHG transfers to resources in balancing areas that are net importers. Arguably, these balancing areas are relying on their internal supply (as well as some imports) to meet demand and do not have surplus supply to support demand in a GHG regulation area. This constraint will reduce the potential for secondary dispatch by not attributing a GHG transfer to a resource for which another resource will need to backfill to serve demand in the balancing area that is a net importer. The CAISO will not enforce this constraint when resources located in a balancing area outside of a specific GHG regulation area register that they have a commitment to serve demand within that GHG regulation area. EDAM will recognize this

supply as available to serve demand in the GHG regulation area. When an EDAM balancing area that overlaps with a GHG regulation area fails the resource sufficiency evaluation in the upward direction, EDAM will deactivate all net export constraints for that hour. By deactivating net export constraints, the CAISO will provide a greater pool of resources that may receive an attribution to serve demand in GHG regulation areas. Failure of the resource sufficiency evaluation in the upward direction signals a potential concern about scarcity of supply the CAISO believes outweighs the potential for secondary dispatch. Under the tariff revisions proposed in this filing, the CAISO will also enforce similar rules in the WEIM for each fifteen-minute market interval.

3. Post-Market Timeframe

The CAISO proposes to use its existing post-market settlements timelines and procedures for the extended day-ahead market, except when unique feature of the extended day-ahead market settlement necessitates an additional provision. Such EDAM-specific features primarily arise in the context of transfer and congestion revenues, GHG bid adders, and the RSE failure surcharge.

The extended day-ahead market will model internal transmission constraints, internal transmission limits, and transmission transfer limits between balancing areas. Congestion revenue will accrue when the market model identifies transmission constraints or limits internal to a balancing area. Differences in the marginal cost of congestion component of locational marginal prices will reflect the incremental costs to serve demand. The CAISO proposes to allocate the revenues from these binding transmission constraints to the balancing authority for the balancing area where the constraint is modeled, including any adjustment for congestion differences that may be associated with registered legacy contract rights or transmission ownership rights.

Transmission capability at interfaces between balancing areas will accrue transfer revenue when the transmission scheduling limit binds at a transfer location. The CAISO proposes to share this transfer revenue equally between the two balancing authorities that made the transmission available to facilitate energy transfers, unless a commercial arrangement dictates a different revenue sharing. In addition, the CAISO will allocate a portion of transfer revenue to transmission customers that make transmission available at an interface when they release transmission rights in advance of the day-ahead market. This allocation ensures market participants that make the transmission available to the market receive an allocation appropriate for the transmission made available to the market.

EDAM will use GHG bid adders to establish a marginal GHG cost and identify which resources outside of a specific GHG regulation area support demand in that GHG regulation area. The marginal GHG cost for a GHG

regulation area will be a component of the locational marginal price paid by load within that GHG regulation area. Scheduling coordinators for resources that receive an attribution to serve demand in the GHG regulation area based on their submitted GHG bid adder will receive a payment reflecting the product of their MW attribution and the relevant marginal GHG cost.

Balancing authorities that experience a resource sufficiency failure, other than a *de minimis* failure, will incur EDAM RSE failure surcharges. The CAISO will apply a tiered surcharge based on the amount of the failure. Having a tiered surcharge structure aligns the magnitude of failure with a reasonable charge for the shortfall. The CAISO will distribute the revenue arising from the EDAM RSE surcharges to the balancing areas that satisfied all of the resource sufficiency components in the applicable period. If no balancing area satisfied the resource sufficiency components in the applicable period, the CAISO will not assess the surcharge.

4. Offering Access to the Extended Day-Ahead Market

a. Onboarding and Costs Related to EDAM

As with the WEIM, the decision to participate in EDAM will be made at the balancing area level. Balancing authorities must be WEIM participants to join the extended day-ahead market. The balancing area-level functionality allows the CAISO to utilize available interchange information for operating the extended day-ahead market, while allowing each balancing authority to remain responsible for maintaining reliability and providing open access tariff transmission service in its balancing area.

The processes for joining the extended day-ahead market will largely resemble the processes for joining the WEIM, including initial implementation agreements, onboarding cost recovery mechanisms, and onboarding processes well before participation begins.

The CAISO proposes to recover its EDAM implementation costs through a deposit and incremental charge/refund true-up mechanism based on its cost of similar services. EDAM participants thereafter will pay an administrative fee based upon the CAISO's cost of service and the services necessary to support EDAM participation.

b. Historical Revenue Recovery for EDAM Transmission Service Providers

Although the extended day-ahead market will not affect EDAM transmission owners' costs to construct, own, and maintain transmission

facilities, releasing transmission to the EDAM markets may affect their recovery of such costs. To avoid under-recovery or any misallocation of costs, the extended day-ahead market will provide EDAM transmission service providers an opportunity to recover transmission revenues comparable to their historical cost recovery prior to participation in EDAM.

Specifically, the CAISO will allow transmission service providers to recover:

1. Historical transmission revenues from sales of short-term firm and non-firm transmission products under the transmission service provider's tariff, and for historical Wheeling Access Charge revenues for CAISO transmission owners;

2. A portion of revenues associated with new approved transmission builds that increase the transfer capability between EDAM balancing areas based on the proportional ratio of historical short-term sales to the overall historical transmission revenues; and

3. Revenues for use of the transmission system when wheeling through transfer volumes in an EDAM balancing area are greater than total import and export transfer volumes for the balancing area.

Recovery of these revenues will compensate transmission service providers for their expected, reduced transmission service revenues under EDAM compared to the revenues they earned prior to participation in EDAM. Due to their participation in EDAM, participating transmission owners are expected to sell fewer short-term transmission services because such services will be displaced by EDAM transfers. The CAISO will allocate these costs to gross load across the EDAM area because load, ultimately and primarily benefits from the optimized transfers that will occur with EDAM participation.

C. Severability

The CAISO developed the tariff changes associated with the DAME and EDAM initiatives through separate stakeholder processes and has combined them in this filing. The CAISO will also implement the DAME and EDAM changes through a combined software release. This promotes both efficiency and consistency because the CAISO will be able to test the software functionality fully and resolve any implementation issues that may arise from any of the proposed changes. Also, this allows market participants to manage a single implementation and participate in EDAM with the benefits of the day-ahead market enhancements, providing them similar efficiency.

The DAME and EDAM tariff revisions are severable from each other such that the Commission can approve DAME without approving EDAM; however, it cannot approve EDAM without approving DAME.[12] The DAME tariff changes represent the foundation of needed improvements to the efficiency of the day-ahead market in the face of increasing system variability and uncertainty. The EDAM design is built on this foundation. In other words, the extended day-ahead market requires the enhancements to the day-ahead market. However, DAME is not dependent on EDAM. The Commission can accept DAME as just and reasonable whether or not it is implemented concurrently with EDAM. The CAISO and market participants have worked collaboratively for several years, and the enhancements to the day-ahead market represent significant improvements that should move forward even if the Commission does not accept the EDAM tariff changes. Approving DAME will provide CAISO market participants with significant day-ahead market benefits regardless of the Commission's actions on EDAM.

Within the proposed DAME tariff changes, there are six individual elements that are severable from each other and the remaining DAME tariff revisions.

The first severable element of the DAME tariff provisions is the proposed local market power mitigation for imbalance reserves up. As explained in section IV.D.2.d of this transmittal letter, the mitigation measures established in new tariff section 31.2.3.2 will not trigger initially because of the interaction with the $55 penalty price embedded in the imbalance reserves procurement curve. For this reason, the CAISO could implement the new DAME structure with no mitigation for imbalance reserves. Therefore, the authority to conduct mitigation under this proposed section is severable from the core DAME tariff changes.

The second severable element of DAME is the proposed flexible implementation of imbalance reserves procurement, which allows the CAISO flexibility to enforce transmission constraints and apply a deployment factor, respectively, in the imbalance reserves deployment scenarios. This feature, discussed in section IV.D.3.d of this transmittal letter, permits the CAISO to fine-tune elements of imbalance reserves procurement based on its review of system and market performance. If the Commission were to find it is not just and reasonable for the CAISO to adjust these values in the business practice manual, then the CAISO proposal remains feasible because it: (a) enforces in the deployment scenarios all transmission constraints enforced in the integrated forward market; and (b) requires 100 percent deliverability in the deployment scenarios. The statements in new tariff section 31.3.1.6.3.1 that deployment scenario flows cannot exceed limits "on transmission facilities identified in the Business Practice Manual" and in new tariff section 31.3.1.6.3.1 that deliverability

[12] *NRG Power Marketing, LLC v. FERC*, 862 F.3d 108 (D.C. Cir. 2017).

in deployment scenarios is assessed "as adjusted by the applicable Deployment Factor" are severable.

The third severable element of DAME is the transition period for negotiated availability bids described in section IV.D.2.c of this transmittal letter and new tariff section 39.7.4.2. The CAISO is concerned it initially will not have a sufficient basis to approve a scheduling coordinator's cost-based justifications for a negotiated availability bid value. Without the transition period, the CAISO and scheduling coordinators would be unlikely to reach agreement during the initial period in which DAME is effective. The result would be scheduling coordinators filing for approval of their proposed value with the Commission through a filing under section 205 of the FPA, as outlined in new tariff section 39.7.4.1. The CAISO prefers not to offer the negotiated option when it is unlikely to address a scheduling coordinator's concerns. Nevertheless, if the Commission were to find the proposed transition period is not just and reasonable, the CAISO's overall DAME proposal remains feasible because scheduling coordinators would still have the option, outlined in new tariff section 39.7.4.1, to seek Commission approval of an individualized availability bid.

The fourth severable element is the proposed DAME transitional measures described in section IV.E.1 of this transmittal letter and new tariff section 11.2.6. The transitional measures provide load serving entities and resource adequacy resources a backstop if they cannot agree through bilateral negotiations on how to share the market revenue from imbalance reserves and reliability capacity. The transitional measures provide these entities additional time to resolve their contractual issues. Resolving those contractual issues, however, is neither a necessary nor indispensable element of the CAISO's DAME proposal, and thus the transitional measures are severable from other elements of the proposal.

The fifth severable element of DAME is the CAISO's proposal to charge congestion revenue "displaced" from the deployment scenarios as part of the imbalance reserves cost allocation process and allocate those displaced revenues to parties holding congestion revenue rights with modeled flow over the transmission lines. The CAISO discusses this element in section IV.E.3 of the transmittal letter. The CAISO does not expect imbalance reserves to displace significant congestion revenue from energy in the integrated forward market, but it included this element to assure holders of congestion revenue rights they would be unaffected by creation of the imbalance reserve product. Therefore, the tariff revisions in sections 11.2.1.9, 11.2.4, and 31.3.1.6.4, and the tariff appendix A definition of Notional CRR Value, to effectuate the displaced congestion revenue element of the proposal are severable.

The sixth severable element of DAME is the update implemented in proposed edits to tariff sections 4.12.2 and 4.17.4 regarding registration of

resource information in the CAISO master file for resource-specific system resources and distributed energy resources, respectively. As described in section IV.E.5 of this transmittal letter, these edits conform the master file rules for these resource types with the existing generally-applicable master file rules in Commission-approved tariff section 4.6.4. The CAISO should have updated these two sections when it last revised section 4.6.4. The CAISO now seeks to address that oversight. In any event, these revisions are not directly related to the DAME proposal and are severable from the other DAME tariff revisions.

There are four individual elements of the proposed EDAM tariff revisions that are severable from each other and from the remaining EDAM elements.

The first severable element of the proposed EDAM tariff revisions is the transmission revenue recovery mechanism described in proposed tariff section 33.26 and section V.D.1 of this transmittal letter. These changes constitute a way for transmission owners to recover revenue that may not otherwise accrue under EDAM because the transmission service provider will make transmission available to support transfers between balancing areas. There is no direct association between these changes and the other EDAM changes; although, this element is important to ensure transmission service providers continue to recover expected revenue from their transmission system, particularly in the early years of EDAM participation.

The second severable EDAM element pertains to cost recovery from transmission customers who receive a day-ahead schedule exceeding the transmission rights they hold. This proposal is found in proposed tariff section 33.23 and described in section V.B.1.c of this transmittal letter. This change is not directly related to other EDAM changes and is not a precondition to approval of EDAM, however, it is important to ensure transmission service providers recover the costs of using their transmission system.

The third severable element concerns the option to enable virtual bidding within a balancing area participating in EDAM. This proposal is found in proposed tariff section 33.30.7 and described in section V.C.1 of this transmittal letter. The CAISO recognizes both Commission precedent on virtual or convergence bidding and the desire of interested balancing authorities to gain experience with the day-ahead market first, before undertaking the major step of implementing a completely foreign feature like convergence bidding.

The fourth severable EDAM element concerns the GHG net export constraint, which is found in proposed tariff section 33.32.5 and described in section V.C.3.e of this transmittal letter. The net export GHG constraint limits the attribution of resources serving a GHG regulation area and reduces the risk of secondary dispatch. However, these tariff revisions are not necessary for EDAM and the other GHG enhancements to go into effect.

In conclusion, the CAISO respectfully requests the Commission accept all of the DAME and EDAM tariff changes. The Commission should sever DAME and EDAM only if it is unprepared to accept both in a concurrent order. The best path forward is to approve both DAME and EDAM concurrently because ratepayers across the West will benefit from sets of tariff improvements and because significant uncertainty and additional software configuration changes may result if approval of EDAM is delayed.

III. Background

A. The CAISO's Current Market Structure

The CAISO administers day-ahead and real-time wholesale electricity markets.[13] The day-ahead and real-time markets both operate inside the CAISO balancing area, while the WEIM provides other balancing areas in the Western Interconnection an opportunity to participate in the real-time market.[14] The day-ahead market (DAM) includes four sequential processes: (1) bid submission; (2) market power mitigation for the integrated forward market (IFM); (3) IFM; and (4) residual unit commitment (RUC) process. The real-time market (RTM) includes six sequential processes: (1) bid submission; (2) hour-ahead scheduling process (HASP); (3) real-time unit commitment (RTUC); (4) short-term unit commitment (STUC); (5) fifteen-minute market (FMM); and (6) real-time dispatch (RTD).

The CAISO below summarizes the key features of these market processes as currently in effect, focusing primarily on the day-ahead market.

1. Day-Ahead Market

a. Bidding to the Day-Ahead Market

Bidding to the day-ahead market opens a week ahead of the trading day and closes at 10:00 a.m. on the day before the trading day. Bids can be either

---

[13] Tariff appendix A, existing definition of CAISO Markets. For the sake of clarity, this transmittal letter distinguishes between existing tariff provisions (*i.e.*, provisions in the current CAISO tariff), new tariff provisions (*i.e.*, tariff provisions the CAISO proposes to add in this filing), and revised tariff provisions (*i.e.*, existing tariff provisions the CAISO proposes to revise in this filing).

[14] The WEIM, formerly called the energy imbalance market (EIM), is generally addressed in existing tariff section 29, *et seq.* The term EIM is still used in WEIM-related provisions and defined terms in the existing tariff, the business practice manual, and in some other documents.

economic bids or self-schedules.[15] Scheduling coordinators for generation resources offer separate bids to supply energy, ancillary services, and RUC capacity.[16] Scheduling coordinators for load serving entities submit bids for load in the day-ahead market. This bid submission process includes the opportunity to bid for exports, imports, and wheeling through transactions.

Under the California resource adequacy (RA) program, described in more detail in section II.A.3.d of this transmittal letter, qualified RA capacity has specific obligations to submit market bids. Resources providing generic RA capacity within the CAISO balancing area must economically bid or self-schedule that capacity into the integrated forward market for energy and ancillary services. This applies all hours of the month the resource is physically available. Resources providing flexible RA capacity must submit economic bids for energy and ancillary services in the must-offer hours applicable to the category of flexible RA the resource is providing.[17] Resources providing generic or flexible RA capacity must participate in RUC, but their capacity is optimized at a zero dollar bid.

b. Integrated Forward Market and Associated Market Power Mitigation

The IFM is a financial market where bid-in supply clears against bid-in load and ancillary service requirements. The IFM co-optimizes procurement of energy and ancillary services for each operating hour of the trading day seeking to minimize overall procurement costs, while respecting transmission constraints and inter-temporal resource constraints such as minimum run time and start-up time.

Before issuing schedules for energy and ancillary services in the IFM, the CAISO first conducts a market power mitigation pass. This process identifies

---

[15] A self-schedule is a market bid that indicates a quantity in megawatt-hours (MWh) but does not specify a price. This indicates the scheduling coordinator is a price-taker. Tariff appendix A, existing definition of Self-Schedule. Effectively, self-schedules are requests that the market schedule the transaction irrespective of the market price.

[16] As described further below, RUC capacity currently is capacity the CAISO procures to meet gaps between supply cleared in the market and forecast demand.

[17] Existing tariff section 40.10.3 describes the three categories of flexible RA capacity: base (category 1); peak (category 2); and super peak (category 3). Category 1 flexible RA must submit bids for all days in the 17-hour period running from 5:00 a.m. to 10:00 p.m. Category 2 flexible RA must submit bids for all days in a five-hour period specified in the relevant business practice manual. Category 3 flexible RA must submit bids in the five-hour period applicable for Category 2 on non-holiday weekdays.

and mitigates potentially uncompetitive energy bids to ensure market prices remain competitive. Nothing is scheduled or committed in the IFM market power mitigation pass. The IFM uses bids mitigated in the IFM MPM pass. The CAISO mitigates energy offers to the greater of *default energy bids* or the *competitive locational marginal price*.[18] Default energy bids are the CAISO's estimate of a resource's marginal cost. The competitive locational marginal price is the marginal price of energy minus the non-competitive congestion components at the location of the mitigated resource. The competitive locational marginal price represents the going rate for competitive energy at the relevant location, and it ensures resources are mitigated only to the extent needed to resolve market power for higher-priced bids.

After the mitigation run, the CAISO issues schedules for energy and ancillary services in the IFM using a clean bid set, which consists of bids mitigated through the market power mitigation process and the submitted bids that were not flagged for mitigation. The market optimization software schedules and prices resources in two successive runs. First, the scheduling run produces resource schedules. This involves clearing bids, enforcing the priorities of self-schedules, and potentially relaxing constraints.[19] Second, the pricing run follows the scheduling run and produces the locational marginal prices (LMPs) utilized in settlements. The LMP at each pricing node in the day-ahead market – including the pricing nodes at the interties that connect the CAISO balancing area with other balancing areas – equals the sum of a system marginal energy cost (SMEC) component,[20] a marginal cost of losses (MLC) component, and a marginal cost of congestion (MCC) component of the LMP calculation.[21]

There may be cases where effective economic bids are insufficient to produce a feasible solution, and the market must adjust submitted self-schedules.[22] In these cases, the market uses configurable market scheduling

---

[18] If the resource's unmitigated energy bid were less than the default energy bid or the competitive locational marginal price, there would be no modification to the resource's bid.

[19] Existing tariff section 31.4 specifies the scheduling priorities in the day-ahead market. Existing tariff section 34.12 specifies the real-time market scheduling priorities.

[20] As described in section V.C.4 of this transmittal letter, the implementation of EDAM will include a marginal energy cost for each participating balancing area, which for the CAISO balancing area would be the same as the system marginal energy cost component noted here.

[21] Tariff appendix C, existing section A.

[22] Existing tariff section 27.4.3, *et seq.*; tariff appendix A, existing definition of Real-Time Dispatch; Business Practice Manual for Market Operations, section 6.6.5 (listing market parameter values that are calibrated based on values set forth in the tariff) (Market Operations BPM).

and pricing parameters to reach a feasible solution and set LMPs. The market parameters used in the market include penalty prices that apply when constraints enforced by the CAISO market are relaxed (or violated). The constraints include system energy-balance (*i.e.*, power balance) and intertie constraints.[23] The various constraints have different price values that represent the cost at which the software will relax a constraint if it cannot reach a feasible solution while enforcing the constraint. When the CAISO relaxes a constraint, the CAISO's market software calculates the scheduling run LMPs based on administratively determined relaxation prices, *i.e.*, the penalty prices.[24] Using penalty prices sets the priority level of the power balance constraint relative to other priorities within the market optimization.[25]

c. Residual Unit Commitment Process

The CAISO conducts the residual unit commitment after the IFM. RUC is a physical market process that procures capacity to meet the CAISO balancing area's load forecast.[26] RUC bases the need for incremental capacity on the difference between physical supply that clears the integrated forward market and

---

[23] The system energy-balance constraint ensures that the physical law of conservation of energy (*i.e.*, the sum of generation and imports equals the sum of demand, including exports and transmission losses) is accounted for in the market solution. The shadow price of the system energy-balance constraint establishes the SMEC, which as explained above is a component of the calculation used to determine LMPs. Tariff appendix C, existing sections B-C; Market Operations BPM, section 6.6.5.4. The shadow price is the marginal value of relieving a particular constraint. Tariff appendix A, existing definition of Shadow Price.

[24] *See* existing tariff sections 27.4.3.2.2 – 27.4.3.2.2.4 and 27.4.3.3.2 – 27.4.3.3.4. The penalty prices are set forth in separate tables for the IFM, the RUC, and the real-time market, and reflect the hierarchical priority order in which the constraint associated with each penalty price may be relaxed in the IFM, RUC, or real-time market by the SCUC and SCED software. Market Operations BPM, section 6.6.5. Self-schedules for existing transmission contracts (ETCs), transmission ownership rights (TORs), and converted CAISO transmission service rights are not subject to adjustment pursuant to price relaxation. Existing tariff section 27.4.3.4.

[25] The market optimization software applies one set of penalty prices when the market parameters related to the soft energy bid cap (set at $1,000/MWh) are in effect, which is the case in most market intervals. The market applies a higher set of penalty prices when the market parameters related to the hard energy bid cap (set at $2,000/MWh) are in effect, which can occur when cost-based offers justify the use of the hard energy bid cap. *See* existing tariff sections 27.4.3.2, 27.4.3.3. 30.5.8, 30.7.12, 30.11, and 39.6.1.1; tariff appendix A, existing definitions of Soft Energy Bid Cap and Hard Energy Bid Cap.

[26] Existing tariff section 31.5, *et seq.*; tariff appendix A, existing definition of Residual Unit Commitment.

physical supply needed to meet the demand forecast.[27] Specifically, the CAISO determines the RUC procurement target for a trading hour based on the next day's hourly CAISO demand forecast less the energy scheduled in the day-ahead schedule, and accounting for other factors, as appropriate.[28] The CAISO subsequently selects and commits RUC capacity not already scheduled in the IFM based on the merit order of submitted RUC availability bids.[29] This process bridge any gaps between the IFM – a financial market – and the forecasted needs for physical resources in real-time. Without RUC, the CAISO could be short of power in the real-time or be forced to rely on less economic resources that can start up in the real-time market time horizon.

Resources receiving a RUC availability award generally receive a RUC availability payment in addition to payment for any energy awarded through the real-time market; although, the CAISO can rescind the RUC payment if the capacity is not delivered.[30] These general compensation rules do not apply to RA capacity. RA Capacity receiving a RUC award is ineligible for RUC compensation because being available in RUC is part of the resource's RA obligation. Any RUC compensation would thus be considered double compensation.

RUC has one significant limitation – it only procures capacity incremental to the capacity procured in the IFM. If the physical supply the CAISO procures in the IFM exceeds the demand forecast, RUC has no mechanism to address that potential oversupply. Instead, the CAISO's operators must issue exceptional dispatches to address these issues if they are not confident the oversupply concern is small enough to be managed in the real-time market processes. This issue particularly can occur with multi-stage generating resources. Multi-stage generating resource configurations committed in the IFM carry over to RUC as a fixed input. RUC can them commit the unit to a higher configuration but not a lower configuration. If there is an operational need for that resource to be in a lower configuration, the RUC process currently has no way of meeting that need.

Aside from exceptional dispatches, CAISO operators have one additional tool to manage RUC – in certain scenarios, they can manually adjust the load

---

[27] RUC does not consider virtual supply, described in more detail in section III.A.3.b of this transmittal letter, in evaluating whether the IFM has procured sufficient resources to meet the demand forecast.

[28] Existing tariff section 31.5.3.

[29] Existing tariff section 31.5.5.

[30] Existing tariff section 31.5.7, *et seq.* For each settlement period and resource, scheduling coordinators receive RUC compensation equal to the sum of the RUC availability payment and a RUC bid cost recovery amount (if applicable). Existing tariff section 11.2.2.

forecast that is the basis of RUC procurement.[31] This covers adjustments relating to: anticipated demand response deployment; metered sub-system self-schedules; anticipated deviations from day-ahead forecasted output from variable energy resources; expected resource self-schedules in the real-time market; and expected deficiencies in ancillary services procurement.

2. Real-Time Market

a. Real-Time Market Processes

The fundamental mechanics of the real-time market are substantially similar to the day-ahead market. The first step of the real-time market is bid submission. Real-time market bidding opens after the posting of day-ahead market results at approximately 1:00 p.m. on the day before the trading day. Scheduling coordinators submit bids to supply energy and ancillary services. Load serving entities, however, do not bid to procure load in the real-time market. Instead, the CAISO procures supply to meet forecasted load.

The next step is the HASP, in which the CAISO provides MW hourly-block schedules to intertie resources. The RTUC and STUC are the next two processes, both of which involve providing commitment instructions to units.

The FMM and RTD are the two market optimization processes within the real-time market. Both the FMM and RTD include market power mitigation procedures. The FMM procures energy and ancillary services for a given fifteen-minute period and runs approximately 37.5 minutes prior to that period. The RTD procures energy and ancillary services for a given five-minute period and runs approximately 7.5 minutes prior to that period.

b. Imbalance Settlement

The CAISO settles awards between the real-time market processes and between day-ahead and real-time, on an imbalance basis. Effectively, an award in the FMM only settles at the FMM price to the extent it does not overlap with an existing day-ahead market award. Similarly, an award in RTD only settles at the RTD price if it does not overlap with an FMM award.

c. Differences in Market Optimization – LMP Formulation & Flexible Ramping Product

[31] Existing tariff sections 31.5.3.1 – 31.5.3.6.

Aside from the timing difference relative to the operating hour, the day-ahead market and real-time market optimization have two noteworthy differences.

First, the LMP for each pricing node in the real-time market includes the elements in the day-ahead market plus a component that represents marginal greenhouse gas cost and, for each pricing node within a WEIM balancing area. The real-time LMP also includes a WEIM bid adder component.[32]

Second, in addition to energy and ancillary services, the FMM and RTD also procure the flexible ramping product. The flexible ramping product allows the CAISO to enforce flexible ramping constraints in the real-time market.[33] The CAISO developed the flexible ramping product to manage the ramping capability needed to meet changes in net demand – both forecasted net demand changes and unexpected net demand changes – that had become more challenging with increased variability in demand and increased participation of variable energy resources in the real-time market. The flexible ramping product has two components.

The first component of the flexible ramping product compensates resources for ramping capability provided through the real-time market's energy scheduling and dispatch process, which the tariff designates as forecasted movement.[34] Each run of the real-time market's multi-interval optimization provides a binding energy dispatch or schedule for the upcoming market interval and nonbinding advisory schedules for the subsequent intervals. The forecasted movement represents the change in energy schedule between the binding schedule or dispatch for the upcoming market interval and the advisory schedule for the following market interval of that market run. The CAISO compensates or charges resources and demand for this forecasted movement at a flexible ramping up price or flexible ramping down price, depending on the direction of the total ramp relative to the forecasted movement of the resource or demand.

The second component of the flexible ramping product is the uncertainty awards component, which the CAISO issues to address the potential for errors in the advisory interval demand or supply forecasts that may materialize in a

---

32 Tariff appendix C, existing section B.

33 Existing tariff section 44, *et seq.* In addition, existing tariff section 11.5.9 includes the flexible ramping product in the list of real-time market settlements and specifies that the CAISO will settle it pursuant to existing tariff section 11.25.

34 Existing tariff section 44.3; tariff appendix A, existing definition of Forecasted Movement.

subsequent market run when the interval is financially binding.[35] The CAISO determines the uncertainty requirement based on a statistical estimation of load and supply forecast error. The CAISO then procures additional ramping capacity through uncertainty awards in an amount up to the uncertainty requirement using a procurement curve that weighs the quantity procured against the costs. This procurement curve ensures the total cost of uncertainty awards will not exceed the expected cost of a power balance violation absent the uncertainty awards.

d. Managing Energy Imbalances from the Day-Ahead Market

The purpose of the real-time market is to update procurement based on how events unfold closer to real-time. Changes between the day-ahead market and real-time market in terms of both supply and demand are inevitable. The inevitable differences between the amount of energy scheduled in the day-ahead market compared to the real-time market are commonly called energy imbalances. A key function of the real-time market is to manage the energy imbalances that occur between the day-ahead and real-time markets.

Two key factors drive energy imbalances. The first is the difference in scheduling granularity between markets. The day-ahead market schedules energy in hourly time increments compared to 15-minute and five-minute energy schedules in the real-time market. These granularity differences cause imbalances because the real-time market schedules fluctuate within the hour, whereas the day-ahead market schedules are fixed for the hour. Because of these granularity differences, the real-time market can require faster, more granular intra-hour ramping capability compared to the ramp rate needed to simply transition from one hourly schedule to the next.

The second factor driving energy imbalances is net load forecast uncertainty. There is unavoidable uncertainty in the day-ahead net load forecast. The day-ahead net load forecast cannot predict with certainty the actual net load during the operating day. Any differences between the day-ahead forecast and what actually occurs results in imbalances.

3. Other Noteworthy Market Design Elements

---

[35] Existing tariff section 44.2; tariff appendix A, existing definition of Uncertainty Award. Market participants do not submit separate bids for the flexible ramping product, but instead the CAISO uses energy bids for optimizing procurement of uncertainty awards. Existing tariff sections 27.4.1 and 34.9.

a. Congestion Revenue Rights

The CAISO market design includes congestion revenue rights (CRRs), which are financial instruments that market participants can acquire through a CAISO-administered allocation and auction process or through a secondary registration system.[36] The primary purpose of CRRs is to hedge day-ahead market congestion costs, allowing market participants to address congestion risk.[37] When transmission demand exceeds capacity, locational marginal prices vary depending on congestion levels. Congestion charges can change dramatically based on system conditions and patterns of supply and demand. As the Commission has recognized, CRRs give market participants a level of financial protection against the risks associated with unpredictable congestion charges.[38]

b. Convergence Bidding

In addition to submitting bids or schedules for physical resources and demand, CAISO market participants can also hedge their market positions and manage their exposure to the differences between day-ahead and real-time prices by submitting purely financial bids – called convergence bids or virtual bids in the CAISO tariff.[39] Such bids can be for virtual supply or virtual demand. If a market participant's convergence bid is cleared in the day-ahead market, it is automatically liquidated with the opposite buy/sell position at the real-time price. Convergence bids at both internal nodes and the interties settle at the average of the four fifteen-minute market prices for the hour.[40]

c. Bid Cost Recovery

---

[36] *See* existing tariff section 36, *et seq.*; *see also* http://www.caiso.com/market/Pages/ProductsServices/CongestionRevenueRights/Default.aspx.

[37] Congestion revenue rights are not included in the extended day-ahead market. Instead, a transmission service provider participating in EDAM will account for congestion revenue under its tariff, which provides the opportunity for each transmission service provider to determine the appropriate settlement treatment for its transmission customers. *See* section IV.D.3 of this transmittal letter.

[38] *See, e.g.*, *Cal. Indep. Sys. Operator Corp.*, 149 FERC ¶ 61,093, at P 2 (2014) (citations omitted) ("CRRs are financial instruments that enable their holders to hedge variability in congestion costs. Entities acquire CRRs primarily to offset integrated forward market congestion costs reflected in the congestion component of locational marginal prices (LMPs).").

[39] *See* existing tariff section 30.9. The CAISO uses the terms "virtual" and "convergence" interchangeably in this filing to refer to those types of bids and transactions. Convergence bidding is an optional feature available for balancing authorities to enable in their balancing area as part of the extended day-ahead market. *See* section V.C.1 of this transmittal letter.

[40] Existing tariff section 11.3.

Scheduling coordinators for resources submit three-part bids, which include start-up, minimum load, and energy bid costs.[41] The CAISO's market optimization software considers the start-up and minimum load costs in optimizing for the least-cost commitment or dispatch of all resources. The market only uses the energy bid cost to set the LMP for a given market interval. This creates a risk that the rents from the difference between the LMP and the resource's energy bid costs will provide insufficient revenue to compensate its start-up and minimum load costs. Also, because of inter-temporal constraints (*e.g*., ramping rates, minimum run times, and minimum up times), a resource's energy or ancillary services bid may set the price in one interval in which the CAISO commits the resource, but that resource may not be infra-marginal in other intervals when it must run to meet those inter-temporal constraints because its bid is higher than the market clearing price.

The existing bid cost recovery mechanism addresses these risks by providing uplift payments to resources if their bid costs exceed their market revenue over the day. The CAISO recovers the costs of bid cost recovery payments through uplift payments allocated to a combination of measured demand and virtual bidders.[42] The CAISO calculates bid cost recovery separately for the day-ahead and real-time market.[43] Resources committed in the residual unit commitment process are eligible to receive real-time bid cost recovery.[44]

d. Resource Adequacy

California's RA program, which the CAISO administers in coordination with the California Public Utilities Commission (CPUC) and other local regulatory authorities in the CAISO balancing area, seeks to secure sufficient capacity to support the safe and reliable operation of the CAISO grid.[45] The CAISO's resource adequacy tariff provisions require load serving entities in the CAISO

---

41 Existing tariff section 30, *et seq.*

42 Existing tariff section 11.8 *et seq.*

43 Bid cost recovery will be extended to other balancing areas as a feature of the existing day-ahead market.

44 Units committed in RUC are included in the real-time market bid cost recovery (as opposed to day-ahead market bid cost recovery) because (1) many commitments made in RUC are non-binding so the real-time market makes the binding commitment decision and (2) long-start and extra-long-start resources that do receive binding commitments in RUC are only committed to their minimum sustained operating level (PMin) so they can participate in the real-time market.

45 Existing tariff section 39, *et seq.*

balancing area to submit a year-ahead forward showing and month-ahead showings of the capacity procured to meet their share of the peak load plus any applicable reserve margin, as well as local and flexible capacity requirements. The extended day-ahead market includes a resource sufficiency test, not a resource adequacy requirement. Participating balancing authorities will remain responsible for meeting their own resource adequacy needs and will have the opportunity to participate in a resource adequacy program of their choosing. In other words, the CAISO will continue to administer its resource adequacy program, while other balancing authorities may or may not participate in a resource adequacy program.[46]

B. The Western Energy Imbalance Market Today

The existing WEIM is similar to the proposed EDAM in that it is not a separate market, but a broader geographical scope of the CAISO's real-time market.[47] Accessing the CAISO's real-time market through the WEIM enables balancing areas, transmission service providers, and resource owners in the West to take advantage of the CAISO's real-time optimization to balance supply and demand. WEIM participants have realized $4.20 billion in benefits since the WEIM began operation in 2014.[48] The WEIM enables an exponentially more efficient use of the Western transmission system. It also reduces costs for participants by lowering the amount of costly reserves utilities need to carry.[49] Utilities also can reduce their carbon emissions and use intermittent renewable resources more efficiently. When one utility area has excess hydroelectric, solar, or wind power, the CAISO can deliver it to customers elsewhere. Likewise, when the CAISO balancing area has excess solar energy, it can help meet demand outside of California that otherwise would be met by more expensive resources. The WEIM also increases reliability by providing additional operational visibility across electricity grids and improving the ability to manage transmission line congestion across the region's high-voltage transmission system.

Participation in the WEIM is voluntary both for balancing authorities and for individual resource owners within a participating balancing area.[50] A balancing authority's decision to participate in the WEIM does not involve

---

[46] The interoperability of EDAM and the Western Resource Adequacy Program is discussed further in sections V.B.1 and V.B.2 of this transmittal letter.

[47] *See* WEIM Authorization Order, 147 FERC ¶ 61,231, at PP 7, 74 (establishing the WEIM).

[48] https://www.westerneim.com/pages/default.aspx.

[49] https://www.westerneim.com/Pages/About/HowItWorks.aspx.

[50] WEIM Authorization Order, 147 FERC ¶ 61,231, at PP 2, 8.

"joining" CAISO on either a full or limited basis, and the CAISO does not assume operational control over the transmission facilities in the participating balancing area. Likewise, the CAISO does not manage generator interconnections or transmission planning for WEIM participants. WEIM participants simply participate in the CAISO's real-time market to manage the dispatch of generation to meet demand in the real-time horizons and provide imbalance energy services to their customers.

The WEIM is governed by a five-member Governing Body that shares authority with the CAISO Board of Governors on rules specific to participation to the WEIM.[51] A committee of Western stakeholders nominates members of the WEIM Governing Body. These members bring a high level of professionalism and leadership, with work experience in public power, investor-owned utilities, and as state regulators. Additionally, the WEIM governance design includes a Body of State Regulators (BOSR) which provides a forum for Western state regulators to learn about the WEIM and WEIM-related stakeholder initiatives.[52] This body may express a common position in the CAISO stakeholder process or to the WEIM Governing Body on WEIM issues; and its members are not restricted from taking any position before the Commission or any other forum concerning WEIM matters or the CAISO.

C. DAME Stakeholder Process

The day-ahead market enhancements initiative began in 2019 and comprised six major phases leading to joint approval from the WEIM Governing Body and the CAISO Board of Governors on May 17, 2023. The six phases of the DAME stakeholder process were:

1. Initial Working Groups (June 2019 - August 2019)
2. Straw Proposal Development (February 2020 - July 2021)
3. Analysis and Straw Proposal Revisions (January 2022 - October 2022)
4. Final Proposal Development (December 2022 - January 2023)
5. Postponement and Extended Workshops (February 2023 - March 2023)
6. Revised Final Proposal and Addendum (April 2023)

The first phase of the process involved initial working group discussions with stakeholders about how to enhance the day-ahead market. This process began with a workshop on June 20, 2019, where the CAISO presented two potential revised day-ahead market formulations. Both proposed formulations would procure the imbalance reserves product, but one would procure the product based on the existing day-ahead structure in which IFM and RUC are

---

51 https://www.westerneim.com/Pages/Governance/GoverningBody.aspx.

52 https://www.westerneim.com/Pages/Governance/EIMBodyofStateRegulators.aspx.

sequential market passes, and the other would procure the product in a revised market framework in which the IFM and RUC are integrated into a single market pass. The CAISO requested stakeholder feedback on the advantages and disadvantages of each formulation and discussed the options in a working group meeting on August 13, 2019.

The second phase involved developing a day-ahead market enhancement straw proposal. Based on feedback from the working group process, the CAISO published an initial straw proposal on February 7, 2020 that reflected the integrated IFM/RUC approach. Stakeholder opinion diverged on the initial design of the integrated IFM/RUC, and the CAISO published a revised straw proposal on June 9, 2020 that discussed alternative approaches to integrating the IFM/RUC. The CAISO continued to probe the feasibility of integrating the IFM and RUC, but it ultimately concluded this revised approach was not feasible. Accordingly, the CAISO published a second revised straw proposal on July 21, 2021 that maintained the sequential IFM and RUC passes.

The third phase involved further analysis of the problems spurring the initiative and incremental improvement of the imbalance reserve design. This step began with the CAISO's publication of an analysis report on January 24, 2022 that quantified market imbalances between the day-ahead market and real-time market. Based on this analysis the CAISO released revised straw proposals on April 27, 2022, and October 26, 2022, each of which refined the design of the imbalance reserve product.

The fourth phase involved developing the final proposal. The CAISO published a draft final proposal on December 1, 2022, followed by the final proposal on January 11, 2023. CAISO management intended to present the proposal for approval by the WEIM Governing Body and CAISO Board at their joint February 2023 meeting. However, CAISO management delayed bringing the initiative to the CAISO Board and WEIM Governing Body joint meeting to allow time for further refinements to address continued stakeholder concerns.

The fifth major phase involved a series of workshops held in February and March 2023 and March 2023, with stakeholders and the CAISO's Market Surveillance Committee.

The sixth and final phase involved revising the proposal to account for feedback from the additional workshops. The CAISO published a draft revised final proposal on April 6, 2023. On April 19, 2023, the CAISO posted an addendum to the draft revised final proposal addressing issues regarding the imbalance reserve demand curve. The CAISO published its revised final

proposal on May 1, 2023 (DAME Final Proposal),[53] leading to approval by the WEIM Governing Body and the CAISO Board of Governors on May 17.[54]

After the policy phase of the stakeholder initiative concluded, on June 2, 2023, the CAISO posted initial draft tariff revisions to implement DAME and then posted further revised versions of the DAME tariff language in July. The CAISO provided opportunities for stakeholder comments throughout this lengthy process and made numerous changes to the proposed DAME design and tariff language to address the comments.[55]

D. EDAM Stakeholder Process

Like the DAME stakeholder process, the EDAM stakeholder process started in 2019. The EDAM stakeholder process began with a feasibility assessment to explore extending participation in the day-ahead market to WEIM participants.[56] The CAISO discussed this feasibility assessment with stakeholders in October 2019. That same month, the CAISO published a paper and held a meeting with stakeholders to discuss the issues the CAISO proposed to address through a stakeholder initiative to extend participation in the day-ahead market to WEIM participants.[57] Over the next several months, the CAISO hosted a training course on its day-ahead market[58] and then began a series of

---

[53] The DAME Final Proposal is available at http://www.caiso.com/InitiativeDocuments/RevisedFinalProposal-Day-AheadMarketEnhancements.pdf and is included in attachment C to this filing.

[54] *See* http://www.caiso.com/Documents/DecisiononDay-AheadMarketEnhancements-JointMotion-May2023.pdf. The materials provided to the CAISO Board of Governors and the WEIM Governing Body included a May 10, 2023 memorandum (as amended on May 15, 2023) regarding the DAME design (DAME CAISO Board of Governors and WEIM Governing Body Memo), available at http://www.caiso.com/Documents/DecisiononDay-AheadMarketEnhancements-Memo-May2023.pdf and provided in attachment D to this filing. The vote of the WEIM Governing Body and the CAISO Board of Governors is also included in attachment D.

[55] *See* https://stakeholdercenter.caiso.com/StakeholderInitiatives/Day-ahead-market-enhancements.

[56] See http://www.caiso.com/Documents/ISO-WesternEnergyImbalanceMarketEntities-HosExtendedDayAheadMarketStakeholderCall100319.html. The CAISO Board of Governors was also briefed on this topic during its September 2019 meeting. *See id.*

[57] *See* http://www.caiso.com/Documents/ExtendedDayAheadMarketIssuePaperCall101719.html.

[58] *See* http://www.caiso.com/Documents/TrainingCourseExistingDay-AheadMarketOverview.html.

technical workshops on the EDAM initiative.[59] The CAISO coordinated the schedules for the DAME and EDAM initiatives to help stakeholders manage their resources and engage in both of those interrelated initiatives.[60] After a pause, the CAISO restarted the EDAM stakeholder efforts in November of 2021.

During the next two years, the CAISO issued a series of papers and held a total of nearly 80 public meetings, workshops, and working group meetings to develop and refine the EDAM proposal.[61] The CAISO issued its final proposal for the EDAM design on December 7, 2022 (EDAM Final Proposal).[62] On March 30, 2023, the CAISO posted initial draft tariff revisions to implement the EDAM design and then posted further revised versions of the EDAM tariff language in May and June. The CAISO provided opportunities for stakeholder comments throughout this lengthy process and made numerous changes to the proposed EDAM design and tariff language to address the comments.[63] This high level of CAISO engagement and stakeholder participation indicates the importance of this effort and the willingness of participants in the West to build upon the collaboration and success of the WEIM, increase regional coordination, and support evolving state policy goals.

On February 1, 2023, the CAISO Board of Governors and the WEIM Governing Body approved the proposed design and the filing of the EDAM

---

[59] *See* http://www.caiso.com/Documents/ExtendedDay-AheadMarketInitiativeTechnicalWorkshopSeriesKickingOff211-1220.html.

[60] *See id.*; http://www.caiso.com/Documents/UpdateDay-AheadMarketEnhancements-ExtendedDay-AheadMarketInitiativesCall041620.html.

[61] *See* https://stakeholdercenter.caiso.com/StakeholderInitiatives/Extended-day-ahead-market.

[62] The EDAM Final Proposal is available at http://www.caiso.com/InitiativeDocuments/FinalProposal-ExtendedDay-AheadMarket.pdf and is included in attachment E to this filing.

[63] *See* https://stakeholdercenter.caiso.com/StakeholderInitiatives/Extended-day-ahead-market. The design reflected in the EDAM Final Proposal was informed by over 1,500 pages of diverse stakeholder comments received across different proposal iterations, which are summarized in a stakeholder matrix document available at http://www.caiso.com/Documents/StakeholderCommentsMatrix-ExtendedDay-AheadMarket-FinalProposal-Feb2023.pdf.

proposal reflected in this tariff amendment.[64] Stakeholders, including state public utility commissioners, broadly supported this decision.[65]

IV. Proposed Day-Ahead Market Enhancements

A. Issues with the Current Day-Ahead Market Leading to this Filing

1. Drivers of Increased Net Load Imbalances between the Day-Ahead Market and the Real-Time Market

Some imbalances in the net load between the day-ahead market and real-time market are inevitable. The magnitude of those imbalances, however, has increased materially in recent years. In 2019, the CAISO published a comprehensive report titled *Price Performance Analysis* that summarized and analyzed price formation in the CAISO markets, including factors that contribute to differences in volumes and prices cleared in the day-ahead market compared to the real-time market.[66]

The report identified systematic imbalances in the net load in different market timeframes. The greatest magnitude of imbalance typically occurs between the IFM and fifteen-minute market (as opposed to between the fifteen-minute market and the five-minute market).[67] These imbalances can be as large

---

[64] *See* http://www.caiso.com/Documents/ApprovedDecisiononExtendedDay-AheadMarket-JointMotion-Feb2023.pdf. The materials provided to the CAISO Board of Governors and the WEIM Governing Body included a January 26, 2023 memorandum regarding the EDAM design (EDAM CAISO Board of Governors and WEIM Governing Body Memo), available at http://www.caiso.com/Documents/DecisiononExtendedDay-AheadMarket-Memo-Feb2023.pdf and provided in attachment F to this filing. The vote of the WEIM Governing Body and the CAISO Board of Governors is also included in attachment F.

[65] *See* BOSR Advice to WEIM Governing Body on EDAM Design and Governance (Jan. 26, 2023), available at https://www.westerneim.com/Documents/BOSRAdvicetoWEIMGoverningBodyonEDAMMarketDesignandGovernance-Jan26-2023.pdf.

[66] CAISO Energy Markets Price Performance Report (Sept. 23, 2019) (Price Formation Report), available at http://www.caiso.com/Documents/FinalReport-PricePerformanceAnalysis.pdf.

[67] Although the residual unit commitment process occurs between the IFM and the FMM, RUC procures capacity, not energy, and RUC compensation is not based on imbalances from a prior market process. Thus, assessing day-ahead to real-time imbalances involves comparing IFM prices/volumes to FMM prices/volumes. For ease of reference, however, the CAISO's generally refers to imbalances between the day-ahead market and the real-time market rather than imbalances between the IFM and the real-time market.

as 6,000 MW in a single hour, which is significant considering the CAISO's historical peak demand is approximately 52,000 MW.[68] The report also explained these "net load differences . . . can be expected to lead to price divergence between markets" because "[f]or instance, when conditions of load and [variable energy resource] generation lead to higher net load levels in real-time, the market may be more exposed to higher prices since there is a need to dispatch higher levels of generation, which may come with higher bid prices."[69]

Several factors contribute to the large imbalances between the day-ahead and real-time market. The Price Formation Report explained large imbalances between the day-ahead and real-time market occur because of deviations between the forecast and actual events for both the net load and variable energy resource output. From the CAISO's start-up in 1998 through recent years, the CAISO balancing area consisted of a predictable generation fleet and a predictable load. Resources scheduled hourly in the day-ahead market had relatively predictable real-time load and ramping needs. Over the last 10 years, however, variable energy resources (*i.e.*, wind and solar resources) have become more prevalent. Although these resources are critical in meeting renewable energy and greenhouse gas emission goals, output from these resources is highly dependent on environmental conditions, making their output less predictable than traditional resources. Solar resources also have a daily production cycle that does not necessarily align with electricity demand patterns, requiring other resources to ramp up quickly when solar production is ramping down. These factors can create challenging conditions for system operators. Rather than the relatively predictable load conditions, system operators must manage the more unpredictable and variable net load differences.

Significant weather-related uncertainty throughout the Western grid has also exacerbated net load imbalances. Electricity demand is weather-sensitive, and extreme temperatures can lead to spikes in energy use due to cooling needs, which are difficult to forecast accurately. On the supply side, extreme weather impacts the generation of wind and solar power, and the output can change rapidly based on cloud cover, wind speed, and other atmospheric conditions. These factors create challenges for accurate forecasting and thus contribute to imbalances between the day-ahead market and the real-time market.

2. CAISO Operators Rely on Manual Interventions in RUC to Address Net Load Imbalances

---

68 Price Formation Report at 54.

69 *Id.* at 59.

The persistent net load imbalances between the IFM and real-time market has required the CAISO to intervene manually in the RUC to ensure there is sufficient online capacity to mitigate the risk of load shedding or insufficient ramp capability in the real-time market. Specifically, the CAISO operators exercise the authority granted in existing tariff section 31.5.3 to adjust the RUC load forecast up to ensure that RUC procures additional capacity to address uncertainty between the day-ahead and real-time markets. These operator adjustments to the RUC forecast have increased in frequency and magnitude over the last several years in parallel with the increased magnitude of net load imbalances between the day-ahead and real-time markets.

The CAISO previously adopted both the flexible ramping product and flexible RA capacity constructs in part to address some of the same underlying factors driving net load imbalances. However, they have contributed only minimally to avoiding the need for manual interventions in RUC. In that regard, both the flexible ramping product and flexible RA capacity can only address net load imbalances with resources already committed in the IFM or that can be committed within the real-time market. Additionally, for much of its existence the flexible ramping product has been problematic because the market procured it on a WEIM balancing area level rather than at a nodal level. When transmission constraints materialized in real-time either within a WEIM balancing area or across WEIM balancing area boundaries, the capacity frequently would be stranded behind those transmission limitations.[70]

Thus, absent larger market changes, the CAISO expects its reliance on operator interventions to mitigate the operational risks posed by net load imbalances to increase.

[70] Both the Price Formation Report and the CAISO Department of Market Monitoring's 2021 Annual Report on Market Issues and Performance documented the phenomenon of stranding the flexible ramping product behind constraints. The CAISO largely addressed this deliverability concern through its February 1, 2023 implementation of nodal flexible ramping product procurement. *See* transmittal letter for CAISO tariff amendment to refine flexible ramping product, Docket No. ER22-2661-000, at 5-13 (Aug. 15, 2022). The Commission subsequently approved the CAISO proposal. *Cal. Indep. Sys. Operator Corp.*, 181 FERC ¶ 61,034 (2022).

3. The Status Quo of Relying on Out-of-Market Operator Interventions is Problematic and Unsustainable

Persistent and systematic out-of-market actions taken by CAISO operators signal a gap in the CAISO's market design. These out-of-market actions lead to less efficient and accurate pricing in the real-time market relative to the day-ahead market. The CAISO has identified at least three concrete ways relying on RUC adjustments to the demand forecast is problematic for the market.

a. Sub-optimal and Inefficient Unit Commitment

Addressing imbalances through manual grid operators' adjustments to the RUC demand forecast means the IFM does not co-optimize the procurement of this additional capacity with energy. Without such co-optimization, the CAISO is not sending a clear price signal regarding the value that capacity is providing to address uncertainty. Further, the energy price in the IFM is not reflecting the trade-off of providing energy as opposed to capacity to meet uncertainty. The result is the CAISO day-ahead market has sub-optimal and inefficient unit commitment and is producing less efficient economic energy and ancillary services schedules. The status quo thus presents higher system costs than would occur if the CAISO could specifically value resource commitments needed to meet net load uncertainty and co-optimize procurement of those resources with energy and ancillary services.

b. RUC Availability Payments Provide Insufficient Compensation

In theory, the current CAISO market sends needed price signals through the RUC clearing price and the RUC availability payment. In reality, however, the current RUC compensation methodology does not send effective price signals in many cases because the RUC clearing price does not account for the trade-off for the unit being scheduled for energy in the IFM compared to being scheduled as RUC capacity to address net load imbalances. Regardless of the price, the vast majority of capacity committed in RUC is RA capacity optimized at a zero dollar bid and ineligible for a RUC availability payment. However, there are costs to make resources available in the real-time market, *e.g.*, gas-scheduling costs, costs to set up a hydro system, opportunity costs from other market opportunities, and transmission costs for imports. RA resources do not recover these costs through market payments; they must recover these costs through their bilateral RA contract payments. This is an inefficient way to

compensate RA units for the unique availability obligations a RUC commitment imposes on them.

c. RUC Capacity is Not Necessarily Deliverable Where it is Needed

The RUC process does not ensure the incremental capacity procured to address net load imbalances between IFM and the real-time market will be deliverable to the locations driving the need for the manual adjustments in the RUC forecast. The CAISO's experience with the real-time market's flexible ramping product shows that procuring a product that is undeliverable where needed imposes additional system costs without addressing the underlying concerns driving procurement of that product.

B. Description of Key Day-Ahead Market Enhancements

To address the issues described above, the CAISO proposes to enhance its day-ahead market by introducing two new products – imbalance reserves and reliability capacity. Both products will be bi-directional, meaning there are four new products: (1) imbalance reserves up; (2) imbalance reserves down; (3) reliability capacity down; and (4) reliability capacity up. The CAISO will procure the new products based on bids submitted under the existing day-ahead market bid submission timeline.

The CAISO will update the IFM to co-optimize procurement of energy, ancillary services, and imbalance reserves. The CAISO also will expand the existing market power mitigation process preceding the IFM to include imbalance reserves. The RUC process will now procure the new bi-directional reliability capacity product instead of the uni-directional RUC capacity it procures today. The CAISO also will add a new market power mitigation process for RUC between the IFM and RUC. In summary, the new day-ahead market will involve these five sequential processes:

1. *Bid submission* – Includes bids for energy, ancillary services, imbalance reserves, and reliability capacity.
2. *IFM market power mitigation* – Mitigates potential exercise of market power in procurement of energy and imbalance reserves.
3. *IFM* – Issues day-ahead schedules for energy, ancillary services, and imbalance reserves.
4. *RUC market power mitigation* – Mitigates potential exercise of market power in procurement of reliability capacity.
5. *RUC* – Issues awards for reliability capacity.

1. Overview of Imbalance Reserves

a. Description of the Product – Imbalance Reserves

Imbalance reserves are a flexible reserve product designed to cover both uncertainty in the net load forecast between the day-ahead market and real-time market and real-time ramping needs not covered by hourly day-ahead market schedules. A resource receiving an imbalance reserves award must submit economic bids to the real-time market for its awarded capacity range. Imbalance reserves will ensure the integrated forward market schedules sufficient dispatch capability to meet net load imbalances between the day-ahead and real-time markets.

b. Procuring the Product – Imbalance Reserves

The CAISO will target the procurement of imbalance reserves up and imbalance reserves down to address, respectively, an uncertainty range above and below cleared physical supply. The CAISO will calculate the uncertainty range imbalance reserves will fill based on the historical uncertainty in the day-ahead load, solar, and wind forecasts. That uncertainty range only sets the procurement target. The CAISO will procure a quantity of imbalance reserves for any interval based on a demand curve that decreases procurement as costs increase by assessing the trade-off between the incremental cost and operational value of the reserves. Procurement also will evaluate whether a resource's reserve capacity would be deliverable if upward- modeled uncertainty occurs (called the upward deployment scenario) and if downward- modeled uncertainty occurs (called the downward deployment scenario).

c. Financially Settling the Product – Imbalance Reserves

Resources awarded imbalance reserves will receive a day-ahead payment at the product's locational price. The CAISO will settle ramping capability provided by imbalance reserve awards in the day-ahead market, subject to the ramp deviation settlement described in section IV.E.2.a of this transmittal letter, which considers how an imbalance reserves award in the day-ahead market relates to a flexible ramping product award in the real-time market. The market will recover the costs of imbalance reserves, including displaced congestion

revenue from the deployment scenarios,[71] from entities based on their contribution to the need for procuring the product.[72] The current bid cost recovery process will also account for imbalance reserves bid costs and revenues.

2. Overview of Reliability Capacity

a. Description of the Product – Reliability Capacity

The CAISO will procure reliability capacity to meet positive or negative differences between cleared physical supply in the IFM and the load forecast. This is similar to the existing RUC process except the CAISO currently only procures uni-directional RUC capacity if the forecast exceeds cleared physical supply. With the bi-directional reliability capacity product, the CAISO can also procure decremental capacity in RUC. As with imbalance reserves, a reliability capacity award will obligate the resource to provide economic energy bids in the real-time market for the quantity of its awards.

b. Procuring the Product – Reliability Capacity

Like today, RUC will continue to consider transmission constraints to ensure reliability capacity is deliverable to the aggregate of load in the day-ahead timeframe. Unlike imbalance reserves, RUC will not use a demand curve to procure reliability capacity for procurement. Instead, it will seek to procure the full reliability requirement subject to penalty prices when there is extreme scarcity.

c. Financially Settling the Product – Reliability Capacity

Settlement of reliability capacity will be similar to settlement of imbalance reserves but simpler. Resources awarded reliability capacity would receive a day-ahead payment at the product's locational price. The market will allocate the costs of reliability capacity to demand. As with imbalance reserves, the current bid cost recovery process will account for reliability capacity bid costs and revenues. Unlike the settlement of imbalance reserves, settlement of reliability capacity will not involve deviation settlements with any real-time market awards, nor will it account for displaced congestion revenues.

---

[71] In section IV.E.3 of this transmittal letter, the CAISO describes in detail the concept of displaced congestion revenue.

[72] In section IV.D.7.c of this transmittal letter, the CAISO describes in detail the cost allocation methodology and its rationale.

3. Relationship Between the Products

Although imbalance reserves and reliability capacity are distinct products, they are related. The following figures provide a simple graphical representation of how imbalance reserves and reliability capacity relate to each other and to energy cleared in the day-ahead market.

Figure 1 below illustrates a simple scenario where IFM clears physical supply equal to the load forecast. The market will procure imbalance reserves to cover upward and downward uncertainty requirements. The day-ahead market would not need to procure reliability capacity in the RUC process.

Figure 1

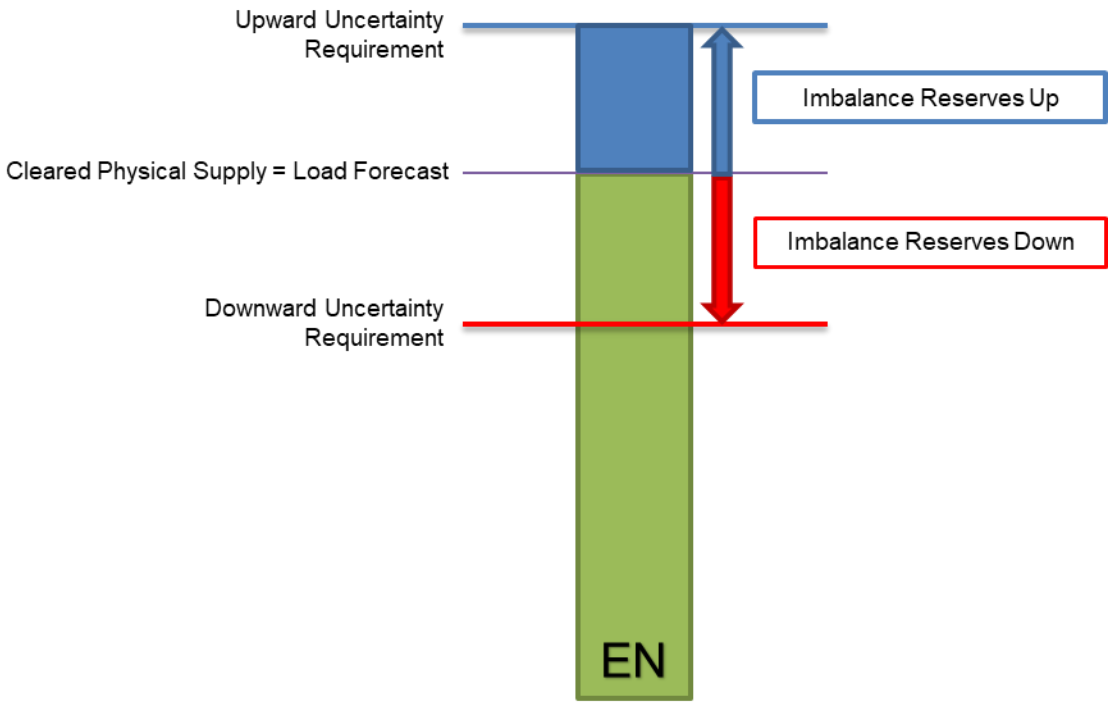


Figure 2 and

Figure 3 below represent more realistic scenarios where physical supply cleared in the IFM is greater than or less than the load forecast. As seen in Figure 2, when the IFM clears less physical supply than the load forecast, the CAISO would procure reliability capacity up to meet that difference.

Figure 3 presents a scenario where the IFM clears more physical supply than the load forecast, which results in procuring reliability capacity down to ensure the real-time market has sufficient decremental capacity.

Figure 2

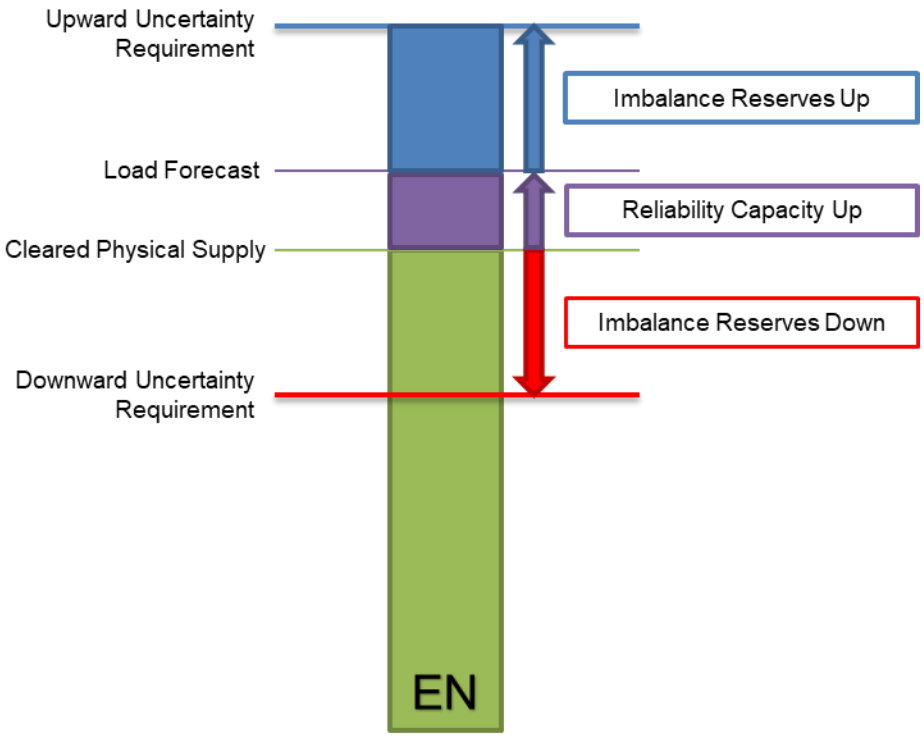
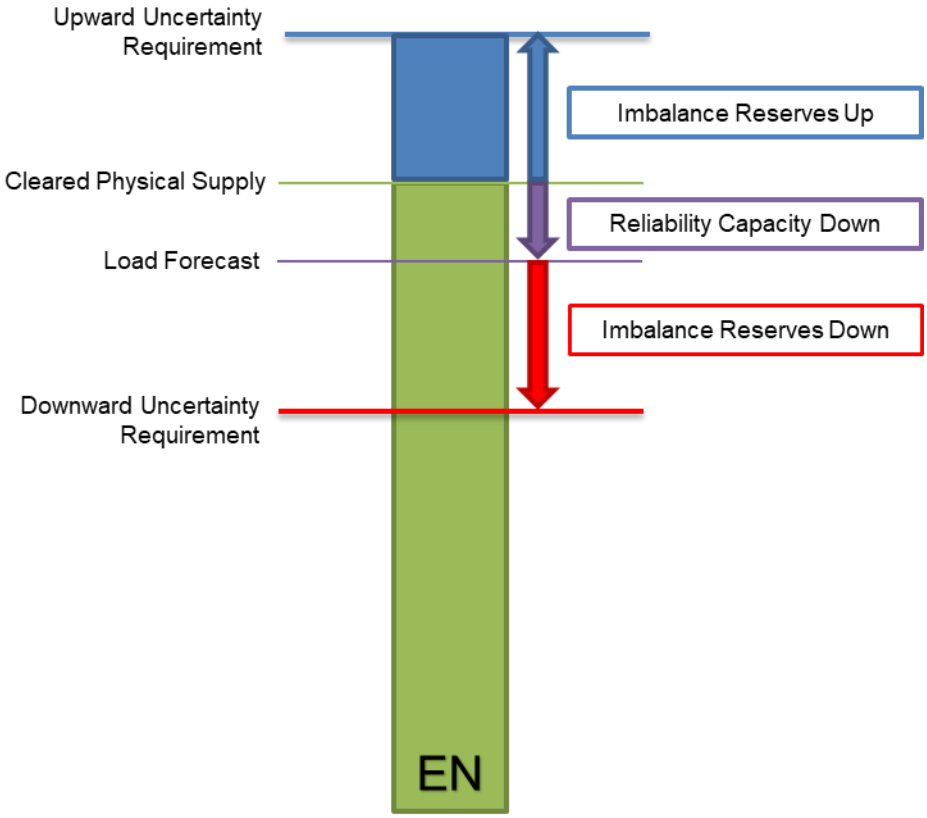

Upward Uncertainty Requirement
Load Forecast
Cleared Physical Supply
Downward Uncertainty Requirement
Imbalance Reserves Up
Reliability Capacity Up
Imbalance Reserves Down
EN

Figure 3



C. New Products Will Address Key Gaps in the Existing Market Structure

The new day-ahead market products will address the major issues the CAISO has identified with the existing day-ahead market and result in a day-ahead market that economically and reliably addresses net load variability and uncertainty. A more effective day-ahead market will in turn promote a more effective and efficient real-time market.

1. Benefits of the Imbalance Reserves Product

The CAISO expects the largest benefits will accrue from the new imbalance reserves product. Procuring flexible ramping capacity in the IFM through a co-optimization that also considers energy and ancillary services will obviate the need for CAISO system operators to adjust the RUC procurement

targets manually on a routine basis.[73] Resources receiving imbalance reserves awards will have capacity ranges capable of meeting all but the most extreme net load imbalances and ramping needs that materialize between the day-ahead and real-time markets. The CAISO expects co-optimized procurement of imbalance reserves will provide these specific benefits compared to relying on the status quo day-ahead market design:

1. Improved unit commitment
2. Enhanced market efficiency
3. Ability to meet real-time operational needs more effectively
4. Increased feasibility of IFM exports

a. Improved Unit Commitment

Co-optimizing imbalance reserve procurement with energy and ancillary services will help maximize the value of these reserves, resulting in more optimal unit commitment decisions and more optimal allocation of system ramping capability. In addition, marginal prices will now consider the opportunity costs of not providing the other products. For example, if a resource is economic for energy but instead is held back to provide imbalance reserves, the marginal prices will ensure that resource earns sufficient revenue from providing imbalance reserves to cover the opportunity cost of not selling energy. Thus, the resource will be indifferent to receiving an incremental energy schedule or imbalance reserve award.

b. Enhanced Market Efficiency

For both the overall system and individual resources, procuring flexible reserves using bids is more efficient. It compensates resources for these flexible reserves directly through market payments, rather than indirectly compensating them through an RA capacity contract.[74] Using bids allows the market optimization to consider costs when scheduling and committing units, leading to better economic outcomes. Marginal prices are a more appropriate mechanism to compensate resources for their availability than fixed contract payments because marginal prices will reflect when and where the reserves are most valued.

---

[73] The CAISO will maintain the authority to make such adjustments but expects that its use of this authority will reduce significantly with implementation of imbalance reserves.

[74] As mentioned above, incremental capacity procured from manual RUC adjustments largely comes from RA capacity that receives no additional compensation from receiving the RUC award.

c. Meeting Real-Time Needs More Effectively

Co-optimizing imbalance reserves with energy will allow the market optimization to consider the full ramping needs of the system (for energy and uncertainty). Currently, there is no assurance the supply committed or scheduled in RUC is sufficient to meet faster ramping needs because RUC involves an hourly award. Imbalance reserves awards, however, require a resource to be 15-minute dispatchable, so the awarded capacity will meet the ramping needs that materialize in real-time more effectively.

Assessing imbalance reserves' deliverability to the nodes that drive the need for procurement of the product will help ensure the procured capacity is usable. This is an enhancement to the status quo because an adjustment in the RUC procurement target does not ensure the incremental RUC capacity procured will be deliverable to the location driving the need for that adjustment. The CAISO's experience with the flexible ramping product highlights the need to ensure ramping products evaluate a resource's ability to provide the product and meet the specific need based on resource characteristics and limitations on the transmission system.

d. Increased Feasibility of IFM Export Schedules

Procuring flexible reserves in the IFM will better ensure IFM export schedules are feasible. Relying on RUC adjustments to procure supply pushes the RUC procurement target farther away from the IFM solution, which can lead to export schedules cleared in the IFM not being feasible in RUC and thus being curtailed in the RUC process. Because the CAISO expects imbalance reserves will significantly reduce the use of RUC adjustments, it is less likely RUC will curtail export schedules cleared in the IFM.

2. Benefits of the Reliability Capacity Product

Imbalance reserves will improve RUC by allowing it to return to its intended purpose of ensuring the day-ahead market schedules sufficient supply to meet a balancing area's demand forecast. The reliability capacity up product, which effectively is the current version of RUC capacity, will accomplish this. Separately, the reliability capacity down product will inject more flexibility into the process by addressing potential oversupply in the real-time market. Oversupply may require manual operator interventions. Oversupply has increased during certain hours due to increasing system variability and uncertainty. Reliability capacity down will obviate the need for operator intervention in many cases. With the creation of the downward product, RUC can serve a broader purpose. It

will help the day-ahead market match demand forecasts rather than merely ensure sufficient supply.

3. Why are Both Products Necessary?

Imbalance reserves and reliability capacity have some similarities. The CAISO will procure both in the day-ahead market to augment energy and ancillary services awards, and awards of both products will trigger real-time market economic bidding obligations. However, the similarities end there. The CAISO needs both products because they serve different purposes, and the CAISO will procure them based on different resource characteristics and system needs. Imbalance reserves address circumstances where the actual real-time net load and associated ramping needs are different than the capacity the day-ahead market procured because net load is above or below the day-ahead forecast. RUC is necessary to procure incremental or decremental capacity to match the outcome of the day-ahead market to the forecast load. Reliability capacity is necessary regardless of any load or variable energy resource forecast uncertainty. RUC assumes the forecast is correct and procures to fill in the gaps. If the CAISO only procured imbalance reserves, there would be no assurance the day-ahead market would procure sufficient physical resources to meet expected demand in real-time. If the CAISO only procured reliability capacity, the CAISO would not have a market product to address net load imbalances and would continue to rely on manual adjustments in the RUC process.

D. Detailed Description of Major Day-Ahead Market Rule Changes

This section IV.D of the transmittal letter describes the principal design elements of the day-ahead market enhancements following the general market timeline from bidding through financial settlement.

1. Bidding for the New Day-Ahead Products and Limitations on Awards

a. Imbalance Reserves – Eligibility to Bid and Limitations on Awards

Suppliers will submit separate single segment price and quantity bids for imbalance reserves up and imbalance reserves down, and the market would use these bids to determine optimal imbalance reserve awards.[75] The imbalance

[75] New tariff section 30.5.2.9 addresses the rules on bidding for imbalance reserves.

reserve bid quantity (MW) must be greater than zero and will be limited to the resource's maximum 30-minute ramp capability.[76] Bids for imbalance reserve up and down bid are capped at $55/MWh. As detailed below in section IV.D.2.c of this transmittal letter, $55 is slightly above the 80th percentile of historical spinning reserve bids (based on data from the first half of 2022). If the CAISO did not procure imbalance reserves, it likely would have to meet the operational need by procuring spinning reserves. The $55 value represents the costs the CAISO likely would procure in foregoing procurement of a MW of imbalance reserves. The bid cap on bidding for imbalance reserves thus represents a design decision to limit bidding at the value of the product to the CAISO. The separate bids for the up and down product could have different hourly price/quantity pairs but only a single price/quantity pair in each hour. The market may schedule a resource to provide both forms of imbalance reserves in the same interval if the resource feasibly can provide both products.

Resource types identified in Figure 4 below are eligible to bid for imbalance reserves.[77] Generally, any resource type capable of adjusting their energy output on a 15-minute basis is eligible to provide imbalance reserves. In general, resources must be scheduled to be online to receive an imbalance reserves award. In addition, the CAISO can award imbalance reserves to offline resources if the resource has a start-up time of 15 minutes or less.[78] Even if a resource is eligible to bid for imbalance reserves, for each interval it submits imbalance reserve bids it must also submit an energy bid in the day-ahead market, and it must economically bid the portion of the energy bid that overlaps with the imbalance reserve bid.

After implementation, the CAISO intends to monitor the various eligibility rules to determine how they affect performance of the imbalance reserve product. Based on that analysis, the CAISO may reassess the rules. For example, the CAISO may consider allowing for longer start-up times or longer ramp horizons.

b. Reliability Capacity – Eligibility to Bid and Limitations on Awards

---

[76] The market will enforce dynamic ramp capability constraints for resources with dynamic ramp rates.

[77] New tariff section 31.3.1.5.1 states the resource types eligible to submit imbalance reserves bids.

[78] New tariff section 31.3.1.5.2.

Suppliers would provide separate single segment price and quantity bids for reliability capacity up and down, and the market would use those bids to determine optimal awards. The reliability capacity bid quantity (MW) must be greater than zero and is limited to the resource's 60-minute ramp capability.[79] This ramp capability limitation matches the time granularity of the product and ensures awarded reliability capacity can meet the operational needs for which the CAISO has procured it. The CAISO will cap bid prices for the reliability capacity up and down product at $250/MWh, which matches the existing limits on RUC bids. The separate bids for the up and down product could have different hourly price/quantity pairs but only a single price/quantity pair in each hour. An award for the upward product and the downward product are mutually exclusive in the same interval.

The resource types identified in Figure 4 are eligible to bid for reliability capacity. In general, all resources currently eligible for the RUC process would be eligible to provide reliability capacity, including storage. Resources must submit an energy bid in the day-ahead market to qualify for reliability capacity, but the bids need not overlap with the economically bid portion of the energy bid.

Under EDAM, all eligible resources offering energy bids in the IFM (and thus included in the EDAM resource sufficiency evaluation) must submit bids for reliability capacity up at the same quantity as their energy bid plus ancillary service self-provision. This ensures all resources in the EDAM resource sufficiency evaluation are available for RUC, including excess supply that participants offered above their resource sufficiency requirements.

c. Special Considerations – Intertie Resources

Hourly intertie resources would be ineligible for imbalance reserves awards because they are not 15-minute dispatchable, but they would be eligible to provide reliability capacity because that product is based on a resource's 60-minute capabilities. Intertie resource that are 15-minute dispatchable and dynamic intertie resources would be eligible to provide both imbalance reserves and reliability capacity. Further, hourly exports to non-EDAM balancing areas can also provide reliability capacity up at CAISO interties, with the obligation to provide a decremental energy bid to dispatch down the export schedule in the FMM if needed.

Intertie resources also must register a resource identification code in the CAISO master file to be eligible to provide imbalance reserves or reliability capacity. Under existing rules, intertie resources can participate in the CAISO market with a transaction identification code. The additional information provided

[79] Revised tariff section 30.5.2.8 contains rules on bidding for reliability capacity.

through master file registration, such as ramp rates and capacity constraints, is necessary for the market to determine what imbalance reserves and reliability capacity awards are feasible. Any intertie resources receiving imbalance reserves and reliability capacity awards would have to submit an e-tag after RUC with a transmission profile equal to the sum of the IFM energy schedule, plus the imbalance reserves up and reliability capacity up awards.

The CAISO also proposes special rules for high-priority (PT) self-scheduled exports, which are exports supported by a resource with non-RA capacity bid into the day-ahead market. To maintain the tariff-defined export priorities and to indicate that economically-bid exports and lower-priority (LPT) export self-schedules that clear the IFM are at risk of curtailment in the real-time market, the CAISO proposes that the resources supporting PT exports must submit bids for reliability capacity up for the self-scheduled quantity.[80] The reliability capacity up bids will provide the market an economic signal of the PT export resource's willingness to be decremented. If the CAISO converts reliability capacity up on an export, it has the same effect as if the CAISO dispatched up a generator serving CAISO balancing area load.

If PT exports do not explicitly bid for reliability capacity up, the CAISO will consider them in the RUC scheduling run with reliability capacity up bids at penalty prices that maintain the merit order of their energy bids in the IFM. If there is no available physical supply capacity in the RUC above energy schedules to meet both the demand forecast and the economic and LPT exports that cleared the IFM, the CAISO will issue economic and LPT exports a curtailment indication for the real-time market in the form of reliability capacity up awards. Such awards created from these inserted bids will be ineligible for RUC compensation because the inserted bid is a tool the CAISO will use to enforce export priorities. The failure of PT exports to submit an actual bid for reliability capacity up reflects an unwillingness to have the export cut based on economics as reflected through the RUC optimization process. This unwillingness justifies its ineligibility for compensation.

d. Special Considerations – Variable Energy Resources

In general, variable energy resources (VERs) will be eligible to provide imbalance reserves and reliability capacity in both directions based on their registered resource parameters. Additionally, VERs will have to bid for a

[80] Revised tariff section 30.5.1(bb).

reliability capacity up quantity equal to their VER forecasted output.[81] This requirement is necessary to align with the EDAM resource sufficiency evaluation rule that all resource capacity in the EDAM resource sufficiency evaluation must bid into RUC as reliability capacity up. Because the EDAM resource sufficiency evaluation will consider VERs at their forecast, VERs must bid reliability capacity up into RUC at their forecast MW output.[82] If VERs do not bid reliability capacity up to their VER forecast, the CAISO will either: (1) generate a bid price of $0 (if the resource submitted no bid for reliability capacity up); or (2) apply the submitted bid price to the additional required quantity.[83]

To ensure day-ahead market awards to VERs are feasible, the CAISO proposes to apply a capacity constraint in the IFM such that the combined awards of energy, upward ancillary services, and imbalance reserves up cannot exceed the resource's VER forecast. The CAISO proposes to apply a similar constraint in RUC such that the IFM awards plus the reliability capacity up award would not exceed the resource's VER forecast.

---

[81] Revised tariff section 31.5.1.2 ("A Scheduling Coordinator representing an Eligible Intermittent Resource must submit RUC Availability Bids for RCU at a quantity equal to their forecasted output based on the forecast referenced in Section 34.1.6").

[82] Independent of EDAM, this rule would be necessary to ensure RUC can consider all physical supply, including the supply forecasted for VERs not bid into the IFM.

[83] Revised tariff section 30.7.3.1 (stating that "to the extent the Scheduling Coordinator for an Eligible Intermittent Resource fails to submit a Bid for RCU up to the quantity of its forecasted output based on the forecast referenced in Section 34.1.6 the CAISO generates a bid for RCU up to the forecasted output").

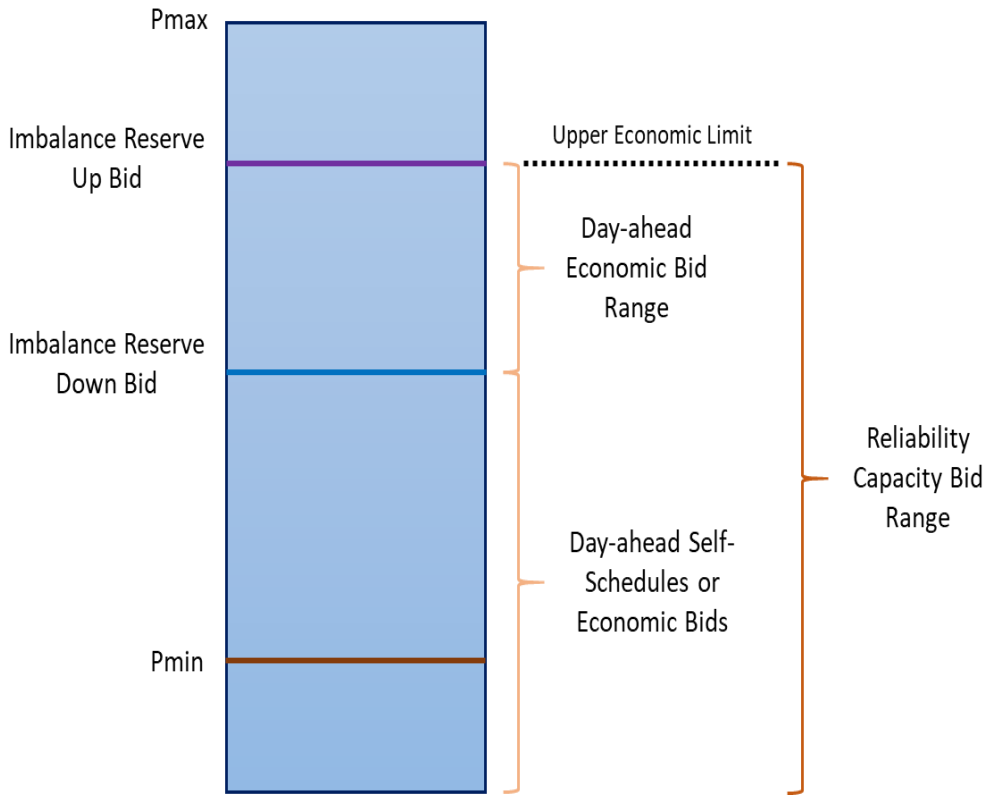
Figure 4

	EN	RCU	RCD	IRU	IRD
Non-Participating Load	Yes	Not Eligible	Not Eligible	Not Eligible	Not Eligible
Virtual Supply	Yes	Not Eligible	Not Eligible	Not Eligible	Not Eligible
Virtual Demand	Yes	Not Eligible	Not Eligible	Not Eligible	Not Eligible
Hourly Block Import	Yes	Eligible	Eligible	Not Eligible	Not Eligible
Hourly Block Export	Yes	Eligible	Eligible	Not Eligible	Not Eligible
15-Min Import	Yes	Eligible	Eligible	Eligible	Eligible
15-Min Export	Yes	Eligible	Eligible	Eligible	Eligible
Dynamic Import	Yes	Eligible	Eligible	Eligible	Eligible
Long-Start Generator	Yes	Eligible	Eligible	Eligible	Eligible
Short-Start Generator	Yes	Eligible	Eligible	Eligible	Eligible
Participating Load w/ 15-Min dispatch capability	Yes	Eligible	Eligible	Eligible	Eligible
Participating Load w/ Hourly dispatch capability	Yes	Eligible	Eligible	Not Eligible	Not Eligible
Variable Energy Resources (Wind/Solar)	Yes	Eligible	Eligible	Eligible	Eligible
Non-Generator Resources (Storage)	Yes	Eligible	Eligible	Eligible	Eligible
Hybrid Resource	Yes	Eligible	Eligible	Eligible	Eligible
Energy Storage Resource	Yes	Eligible	Eligible	Eligible	Eligible
60-Minute Proxy Demand Resource	Yes	Eligible	Eligible	Not Eligible	Not Eligible
15-Minute Proxy Demand Resource	Yes	Eligible	Eligible	Eligible	Eligible
5-Minute Proxy Demand Resource	Yes	Eligible	Eligible	Eligible	Eligible
Reliability Demand Response Resource	Yes	Not Eligible	Not Eligible	Not Eligible	Not Eligible

e. Combining the Rule Sets

To ensure a given resource's combined awards of energy, imbalance reserves, and reliability capacity are all feasible, the CAISO will limit the total awards to the resource's upper economic limit. This term refers to the highest operating level submitted in the resource's energy bid. Figure 5 below graphically demonstrates how these rules interact.

Figure 5



f. Obligations of Resource Adequacy Capacity to Bid for New Products

The CAISO also had to consider whether RA capacity obligations needed adjustment given the new day-ahead market products.

i. Imbalance Reserves Must-Offer Obligation

The CAISO concluded system and local RA capacity would not have an explicit must-offer obligation to submit bids for imbalance reserves. Specifically, system and local RA capacity self-scheduled to the CAISO market would have no obligation to submit imbalance reserves bids. However, RA capacity from resources eligible to provide imbalance reserves (*i.e.*, 15-minute dispatchable resources) would have a must-offer obligation to submit bids for imbalance

reserves on any portion of RA capacity that is bid economically to the IFM.[84] Flexible RA capacity is 15-minute dispatchable and must economically bid to the IFM under existing market rules. The CAISO is not changing these requirements. Therefore flexible RA capacity must submit imbalance reserves bids for their entire range of flexible RA capacity.[85]

These imbalance reserves must-offer requirements will maximize participation of RA capacity in the procurement of imbalance reserve, increasing the competitiveness of the product, improving congestion management, reducing concerns about physical withholding, and helping the CAISO balancing area pass the EDAM resource sufficiency evaluation. Importantly, the proposed tariff revisions will maintain the option for resources to self-schedule energy from generic RA. Without maintaining this option, the CAISO might lose RA capacity from resources that cannot and would not participate by bidding economically.

ii. Reliability Capacity Must-Offer Obligation

RA capacity currently required to provide RUC availability bids will now be required to bid RA capacity for reliability capacity up. Bidding for reliability capacity down will be optional for RA capacity.[86] Marking a change from current practice, the CAISO will not require RA capacity to offer into RUC with $0 availability bids. Instead, RA capacity can bid into RUC at any price between the bid floor and bid cap. This is an important change to facilitate the implementation of EDAM. It would distort the operation of EDAM if all RA capacity were bidding RUC capacity at zero dollars, while resources in the rest of the EDAM area were able to bid economically. Arguably, such a result would suppress RUC prices to the detriment of resources in those other balancing areas that lack an RA capacity contract to compensate them for the zero dollar RUC bidding obligation. Having RA capacity bidding to RUC at zero dollars also would allow EDAM balancing areas to benefit from RA capacity without having contributed to the costs of procuring that capacity.

iii. Real-Time Market Resource Adequacy Bidding

The CAISO proposes maintaining the existing real-time market must-offer obligation. Today, certain RA resources have an obligation to bid or self-schedule in the real-time market even if they do not receive an IFM schedule or binding RUC commitment. The CAISO enforces these obligations through its

---

[84] Revised tariff section 40.6.1(5). RA capacity from resources that are ineligible to submit bids for imbalance reserves would retain their existing must-offer obligations and could not submit imbalance reserve bids.

[85] Revised tariff section 40.10.6.1(a).

[86] Revised tariff sections 40.6.1(4) and 40.10.6.1(d).

tariff and through mechanisms like bid insertion, which enable the market to generate real-time bids for eligible RA capacity that does not submit bids and is not on an outage. After operational experience with the new market products, the CAISO will review the existing real-time market obligations.

iv. Resource Adequacy Bid Generation

RA capacity with an obligation to submit reliability capacity up bids will be subject to bid insertion for reliability capacity up.[87] If the scheduling coordinator for RA capacity does not offer it as reliability capacity into the day-ahead market, the CAISO will extend the bid quantity to the required amount using the submitted bid price. If the resource submits no reliability capacity up bid, then the CAISO will insert a bid at the $55 default availability bid.[88] The RUC bid is a single segment, so the CAISO cannot fill the missing quantity at $55 when there is already a submitted bid price for part of the RA capacity.

The CAISO proposes not to generate bids for imbalance reserves even where RA capacity has a must-offer obligation to submit those bids. The CAISO made this decision to minimize the scope of rule changes for RA resources. RA capacity has an existing obligation to bid into for RUC, and the CAISO can generate such bids if necessary. For this reason, generating bids for reliability capacity up merely modifies an existing rule. The same, however, is not the case for imbalance reserves that are an entirely new product. The CAISO did not want to go this far in revising the existing RA rules outside of a dedicated RA policy stakeholder initiative. Implementing generated bids for imbalance reserves also would raise technical complications. For system and local RA, a resource's obligation to bid imbalance reserves is contingent on how it bids energy into the IFM. Generating imbalance reserves bids for flexible RA capacity would be more straightforward, but would it would constitute a notable change in RA policy because flexible RA capacity is not subject to generated bids today.[89]

---

87 New tariff section 40.6.8(f).

88 The CAISO discusses the default availability bid in section IV.D.2.c of this transmittal letter.

89 The CAISO has identified a multi-year effort to develop enhancements to the RA program. *See* Policy Initiatives Roadmap 2023-2025, available at http://www.caiso.com/InitiativeDocuments/Final2023-2025PolicyInitiativesRoadmap.pdf.

2. Integrated Forward Market – Market Power Mitigation Pass

a. Rationale for Imbalance Reserves Mitigation

The current market power mitigation process in the day-ahead market applies only to the IFM. The CAISO proposes to adjust this existing mitigation run to consider imbalance reserves up. The CAISO would not mitigate bids for imbalance reserves down or bids for imbalance reserves up from non-EDAM intertie resources certified to provide imbalance reserves, consistent with current policy for energy bid mitigation under which interties are presumed competitive and thus not mitigated.[90]

The CAISO is proposing imbalance reserves mitigation because the existing energy bid mitigation alone would not mitigate potential exercises of market power for imbalance reserves bids.[91] Energy and the new imbalance reserve up product are fungible; the market will attempt to reorient energy and imbalance reserve schedules to avoid awarding resources with high priced imbalance reserve bids in favor of awarding them energy schedules. However, this reaction would force the market to schedule energy from a resource with higher bid costs, which would drive up the total production cost. In this way, suppliers could utilize their position on the grid to exercise local market power, driving up costs to the system and increasing their market payments above competitive levels. The CAISO is not proposing to mitigate bids for imbalance reserves down because mitigation applies in cases where a modeled injection would provide counterflow to a binding constraint. In the case of imbalance reserves down, there is no injection of energy to model.

b. Mechanics of the IFM Market Power Mitigation Pass

The CAISO will adjust the existing IFM market power mitigation pass to account for new ways energy bids may require mitigation and to mitigate bids for imbalance reserves up directly.

---

90 *See, e.g.*, System Market Power Mitigation: Revised Draft Final Proposal, at 17 (Sept. 18, 2020) (stating that "the CAISO has not applied a system-level market power mitigation process to its market because it generally has access to large amounts of presumably competitive west-wide power through economic offers at its interties"). This CAISO document is available at http://www.caiso.com/InitiativeDocuments/RevisedDraftFinalProposal-SystemMarketPowerMitigation.pdf.

91 http://www.caiso.com/InitiativeDocuments/Appendix-C-Third-Revised-Straw-Proposal-Day-Ahead-Market-Enhancements.pdf.

In the market power mitigation pass for the IFM, the market will use unmitigated bids to clear bid-in load, bid-in supply, imports, exports, ancillary services requirements, and the imbalance reserve requirements. The market power mitigation pass would evaluate binding transmission constraints in the base scenario (cleared bid-in load), the imbalance reserve up deployment scenario, and the imbalance reserve down deployment scenario for competitiveness.[92]

The CAISO will mitigate energy bids from resources that can provide counter-flow to an uncompetitive constraint in the base scenario or the deployment scenarios. This is because energy marginal prices would have congestion contributions from binding constraints in the deployment scenarios.[93] Resources able to provide counter-flow to an uncompetitive constraint in the upward deployment scenario would also have their imbalance reserve up bid mitigated.

c. Consequences of Mitigation and Default Availability Bids

The CAISO will extend the existing mitigation consequences to imbalance reserve up bids requiring mitigation except that the CAISO will mitigate to the higher of the *Competitive Locational IRU Price* or *IRU Default Availability Bid* (instead of the higher of the default energy bid or competitive locational marginal price).[94]

The *Competitive Locational IRU Price* is the locational price without the non-competitive congestion components from binding constraints in the imbalance reserve up deployment scenario at the location of the mitigated resource.[95]

---

92 Revised tariff section 39.7.2.2(B).

93 See the DAME technical description available at http://www.caiso.com/InitiativeDocuments/DraftTechnicalDescription-Day-AheadMarketEnhancements.pdf. The terms $\sum SF_{i,m,t}\,\mu^{(u)}{}_{m,t}$ and $\sum SF_{i,m,t}\,\mu^{(d)}{}_{m,t}$ represent how transmission constraints in the upward and downward deployment scenarios contribute to the LMP.

94 New tariff section 31.2.3.2. "IRU" refers to imbalance reserve up. "IRD" refers to imbalance reserve down.

95 Tariff appendix A, new definition of Competitive Locational IRU Price.

The *IRU Default Availability Bid* is a new concept similar to the well-established default energy bid but with several key differences.[96] Default energy bids are specific to each resource and designed to approximate a resource's variable costs of providing energy. The CASO offers five methodology options for calculating default energy bids.[97] The variable costs of providing energy can be approximated based on generally understood criteria such as generator performance data, fuel costs, and opportunity costs. However, costs related to a resource's ability to provide reserves are more nebulous. Estimating the variable costs of each resource to provide reserves is subject to significant uncertainty. For this reason, the CAISO proposes a static system-wide default availability bid for imbalance reserve mitigation that applies for all resources across all market intervals. This will provide a mitigation "floor" that balances the need to protect consumers against market power, while also protecting suppliers against excessive mitigation by forcing offers below their costs.

The default availability bid will be set at $55/MWh. As with the $55 cap on imbalance reserves offers, this value represents a high percentile value of historical spinning reserve bids, which is the existing market product most similar to imbalance reserves. Specifically, as reflected in Figure 6 below, the $55/MWh value is set at slightly above the 80th percentile of historical spinning reserve bids using historical data from the first half of 2022. Spinning reserve bids are a reasonable approximation of a resource's cost to provide imbalance reserves because the obligations of providing the product are similar. The CAISO will investigate the relationship between spinning reserve bid prices and prevailing gas prices to consider making the default bid scalable by gas prices. After the CAISO and its stakeholders gain experience with imbalance reserves, the CAISO will re-evaluate the default availability bid value with stakeholders and update it as appropriate.

Figure 6: Spinning Reserve Bid Prices (Jan. – June 2022)

Type	Spinning Reserve Bid Price ($/MWh)
50 Percentile	$1.90
60 Percentile	$5.00
70 Percentile	$21.70
80 Percentile	$50.00
90 Percentile	$100.00

96 New tariff section 39.7.4 covers Default Availability Bids.

97 Existing tariff section 39.7.1 describes Default Energy Bids.

As with energy bid mitigation, the CAISO will offer an option for scheduling coordinators to negotiate the default availability bid.[98] Under the negotiated rate option, the scheduling coordinator begins the process by proposing a value that "must represent the costs of providing the underlying product."[99] Once the parties agree on a rate, the CAISO will submit the negotiated default availability bid rate to the Commission through an informational filing. To ensure the negotiated rate remains reasonable, the CAISO will have authority to require a scheduling coordinator to renegotiate availability bids "that have become outdated, are possibly erroneous, or for which the Scheduling Coordinator has changed."[100]

The negotiated rate option initially will not be available to scheduling coordinators because imbalance reserves and reliability capacity are new products, and the CAISO would not have sufficient experience with the products to validate the reasonableness of proposed rates. The option will be available once the "the CAISO certifies through a market notice it has gained sufficient operational experience with Imbalance Reserves and Reliability Capacity to validate that proposed Negotiated Availability Bids correspond reasonably to the underlying costs of providing the products."[101] The CAISO expects this will take approximately one year but, to provide market participants certainty, the negotiated default availability bid option will be available no more than 18 months after the CAISO implements the new products.[102]

d. The CAISO Will Not Trigger Mitigation Initially

The combined effect of the $55 bid cap on imbalance reserves, and the $55 limit on the imbalance reserves procurement, described below in section IV.D.3.b of this transmittal letter, means that the proposed mitigation procedures initially would not be triggered. However, the CAISO anticipates adjusting those parameters over time. The CAISO concluded it was preferable to propose mitigation provisions now to provide a complete market design rather than wait to develop and propose mitigation when the CAISO adjusts those parameters.

---

98 New tariff section 39.7.4.1.

99 *Id.*

100 *Id.*

101 New tariff section 39.7.4.2.

102 *Id.*

3. Integrated Forward Market – Procurement

a. Establishing Procurement Targets for Imbalance Reserves

The CAISO has modeled calculation of the imbalance reserve requirement based on its existing methodology for establishing flexible ramping product requirements in the real-time market.[103] The CAISO will use historical data to identify the load, wind, and solar forecast error between the day-ahead market and fifteen-minute market. Based on this historical data, the CAISO will conduct a set of statistical regressions to identify connections between the prevailing load, wind, and solar forecasts for each hour and the historical forecast errors. From this set of regressions, the CAISO will establish the imbalance reserves up and down requirements, which it sometimes refers to as the uncertainty requirements. The CAISO initially intends to set the up and down uncertainty requirements at the 97.5 percentile and 2.5 percentile levels of error, respectively. Consistent with the current approach for the flexible ramping product, the CAISO will specify these percentile levels in the business practice manual.[104] This will permit the CAISO to adjust these parameters after gaining operational experience. The CAISO will calculate a separate requirement for each EDAM balancing area based on historical data for the balancing area, but those separate requirements would reflect the EDAM diversity benefit.

b. Demand Curve for Procuring Imbalance Reserves

The regressions will only establish the imbalance reserves procurement targets, meaning that the targets represent the maximum potential procurement. The actual quantities the CAISO procures will be determined using a stepped demand curve with a penalty price set at $55/MW.[105] The CAISO will calculate separate demand curves for each hour in both directions for each EDAM balancing area.[106]

---

[103] Procurement targets for the new imbalance reserve product are addressed in new tariff section 31.3.1.6.1.

[104] Existing tariff section 44.2.4.2; Market Operations BPM, section 7.1.3.1.5 and appendix N.

[105] New tariff section 31.3.1.6.2 describes the procurement curve. The $55 penalty price is the same as the default availability bid**.**

[106] The EDAM resource sufficiency evaluation would not use the imbalance reserve demand curve. Instead, the resource sufficiency evaluation will penalize any imbalance reserve

Using a demand curve represents the relationship between the price of imbalance reserves and the quantity the market will procure. It shows how the market's willingness to procure imbalance reserves changes with the price, helping to establish an appropriate price level while considering the expected cost of not procuring imbalance reserves. The imbalance reserve demand curve thus establishes the price of not fulfilling the imbalance reserve requirement for an hour. This allows the market to determine whether to meet all or some of the upward and downward uncertainty requirements. The market makes this determination by assessing the trade-off between the cost and the value of an incremental unit of imbalance reserves. If the imbalance reserve price is lower than the expected cost of not meeting the uncertainty requirement, the market will continue to procure imbalance reserves. Conversely, if the imbalance reserve price is higher than the expected cost of not meeting the uncertainty requirement, then the CAISO will not procure additional imbalance reserves to meet the balance of the uncertainty requirement.

The market uses penalty prices to establish the priority of different schedules and constraints and to set market prices when schedules or constraints need to be relaxed when there is insufficient supply to satisfy requirements. Setting the penalty price at $55 means the CAISO will not procure the product at any price above that level.[107] The CAISO bases this hard limit on procurement on the same rationale supporting a $55 bid cap for the product – it reflects the high end of the range of spinning reserves offers, which the CAISO would otherwise procure in lieu of imbalance reserves. The CAISO determined imbalance reserves are not worth more than $55/MWh because the CAISO likely will be able to meet the need at that price by procuring an alternative, *i.e.*, spinning reserves. The penalty price thus reflects the CAISO's decision that it should not procure imbalance reserves at a price greater than their value as determined based on historical spinning reserve prices. This procurement curve design also represents a comparable methodological approach to the approach the Commission approved for the Midcontinent Independent System Operator (MISO) for a similar product.[108]

The steps of the procurement curve below the penalty price will resemble that of the flexible ramping product demand curve. The CAISO will calculate demand curves by determining the imbalance reserve requirement that should be

---

requirement relaxation at a high penalty price to ensure all economic imbalance reserve bids are used before incurring an imbalance reserve shortfall, which would result in failing the resource sufficiency evaluation in that direction.

[107] On its own, the $55 bid cap would not limit procurement above this level because congestion can drive the price above the bid cap, as seen in the energy market.

[108] *Midcontinent Indep. Sys. Operator, Inc.*, 181 FERC ¶ 61,160 (2022) (approving multi-step procurement curve capped at $31/MWh for MISO's ramp capability product).

relaxed at different price levels to ensure the cost of imbalance reserve awards does not exceed the expected cost of foregoing them. However, instead of the $1,000/MWh cost used in the flexible ramping product calculation, the avoidance cost of imbalance reserves will be set to $247/MWh, which is the lowest penalty price for violating contingency reserve requirements. Basing the curve on a penalty price lower than the flexible ramping product is appropriate because, unlike with the flexible ramping product, a 1 MW shortage of imbalance reserves is far less likely to violate the power balance constraint by 1 MW and trigger the $1,000 penalty prices, which is the basis for procuring the flexible ramping product. This design framework is consistent with the approach approved by the Commission for MISO.

By capping the demand curve values at the default bid price for imbalance reserve mitigation, market participants can gradually gain experience and adapt to the new market design. This approach allows for a smoother transition, reducing unforeseen issues or price spikes that could arise from a more aggressive initial implementation. However, the CAISO will monitor initial market performance closely to ensure the demand curve cap unintentionally does not stifle market efficiency or suppress price signals essential for maintaining system reliability. As the market matures and the CAISO gains more operational experience, it will be crucial for the CAISO periodically to review and adjust the demand curves to reflect the true value of imbalance reserves and the associated scarcity conditions. This process will involve ongoing collaboration between the CAISO, market participants, and other stakeholders to identify potential improvements and to make informed adjustments to market rules based on empirical evidence.[109]

c. Deployment Scenarios and Deliverability

The market will consider transmission constraints when awarding imbalance reserves in the IFM to ensure they are deliverable if deployed in real-time.[110] The CAISO proposes to use upward and downward deployment scenarios similar to those it currently uses to procure flexible ramping product at a nodal level. Under this approach, the CAISO awards the product only if it is simultaneously deliverable in three scenarios: (1) "base scenario" of meeting demand; (2) upward deployment scenario in which imbalance reserves up capacity is converted to incremental energy in real-time to meet net load; and (3) downward deployment scenario in which imbalance reserves down capacity is converted to decremental energy in the real-time to meet net load. Optimizing

---

[109] If the CAISO revises the $55 penalty price upward, the market power mitigation measures would spring into effect under their own terms. Thus, any revision of the penalty price would impact the procurement curve and mitigation.

[110] New tariff section 31.3.1.6.3.1.

imbalance reserve procurement based on these three scenarios will help ensure scheduled day-ahead physical supply can meet the uncertainty requirements if deployed without violating transmission constraints.

The upward deployment scenario would ensure supply and imbalance reserves up awards are deliverable to the location where upward net load uncertainty may materialize. Conversely, the downward deployment scenario would ensure supply less imbalance reserves down awards are deliverable to the location where the downward net load uncertainty may materialize. The CAISO will model where uncertainty may materialize in both scenarios by using allocation factors derived from historical data about uncertainty at individual load and VER nodes.[111]

d. Flexible Enforcement of Transmission Constraints

The CAISO does not propose to enforce all transmission constraints in the deployment scenarios. Instead, it proposes a flexible activation/deactivation of individual transmission constraints in deployment scenarios in response to optimization performance, market performance, or operational experience. The CAISO will define the specific constraints it will enforce in the deployment scenarios in the relevant business practice manual.[112]

Simultaneously solving three distinct scenarios will place new computational stresses on the market optimization engine. This stress could increase the time needed to achieve a market solution. Limiting the transmission constraints considered in the deployment scenarios can limit the problems posed by delays in the software reaching a solution.

Market simulation or initial operational experience also may uncover that a few transmission constraints are disproportionately driving clearing prices for the products to unreasonably high levels. Where other market products or manual operation actions could address the operational needs posed on the other side of those constraints in a more cost-effective manner, it may be more prudent not to enforce those constraints in the deployment scenario.

Finally, this flexibility will enable CAISO to collaborate with EDAM balancing areas that may have lower frequency and quantity of binding constraints than the CAISO balancing area to identify the most critical constraints to enforce, so as not to reduce computational performance with negligible market impact.

---

111 New tariff section 31.3.1.6.3.2.

112 New tariff section 31.3.1.6.3.1 (deployment scenario flows cannot exceed limits "on transmission facilities identified in the Business Practice Manual").

e. Deployment Factor

The CAISO proposes to implement a tunable parameter – the deployment factor – to control the proportion of imbalance reserve awards that must be deliverable in the deployment scenarios.[113] The deployment factor will be a percentage defined in the business practice manual separately for imbalance reserves up and imbalance reserves down.[114] This parameter will not adjust the percentage of the imbalance reserves procured. The CAISO would still look to procure 100 percent of imbalance reserves target, subject to the procurement curve. Instead, the deployment factor will determine how much of the reserves procured would have to be feasible in the scenario. For example, if the parameter is set to 50 percent, the market will still target procurement of the full imbalance reserve requirement subject to the demand curve, but only half of each imbalance reserve award supply injection and demand withdrawal will be modeled against transmission constraints in the deployment scenarios at each location.

The CAISO intends to set the deployment factor for both the up and down deployment scenarios at 100 percent, meaning the optimization would ensure all imbalance reserves are deliverable in the deployment scenarios. A value of 100 percent also will maximize the robustness of EDAM transfers. A deployment factor set at this value imposes a rigorous deliverability test and maximizes the feasibility of EDAM transfers because they are fully evaluated against physical transmission limits in the deployment scenarios. However, using a configurable parameter, rather than hardcoding the tariff to require 100 percent deliverability, addresses stakeholder concerns about potential excessive congestion costs resulting from the deployment scenarios and virtual arbitrage. Additionally, market simulation or operational experience might indicate a lower value is advisable. For these reasons, the CAISO has concluded that it needs flexibility to consider possible changes to the deployment factor. Any such changes will require changes to the applicable business practice manual and will be subject to stakeholder input through the Commission-approved business practice manual change management process.[115]

---

113 *Id.* (deliverability in deployment scenarios assessed "as adjusted by the applicable Deployment Factor").

114 Tariff appendix A, new definition of Deployment Factor.

115 *See* existing tariff sections 22.4.3 and 22.11.

f. Specialized Constraint for Awarding Imbalance Reserves to Storage Resources

The CAISO's tariff and market systems contain various specific rules for storage resources given their unique operating characteristics. That approach is necessary to implement the proposed market products. Specifically, the CAISO proposes to enforce a constraint to ensure ancillary services schedules and imbalance reserves award for storage resources are feasible given the resource's expected state of charge.[116]

4. Residual Unit Commitment – Market Power Mitigation Pass

a. General Description of Reliability Capacity Mitigation

The CAISO currently does not have a RUC market power mitigation pass. The CAISO has reconsidered that approach and now proposes to include a distinct local market power mitigation pass before RUC to assess the competitiveness of bids for reliability capacity up.[117] If that mitigation pass detects the potential for exercising market power, it would mitigate reliability capacity bids. Any mitigated bids would be inputs to the process. Further, to aid in the performance and solution time of the overall day-ahead market, the CAISO proposes to limit the RUC market power mitigation pass to a 24-hour horizon, rather than RUC's optimization horizon that may extend past the trading day.

b. Need to Apply Mitigation

The CAISO and its stakeholders largely have considered RUC mitigation unnecessary in the past given the significant amount of RA capacity bidding into RUC at $0. Market power is not a concern under these circumstances RA resources cannot physically or economically withhold capacity. However, concerns about market impacts in EDAM have lead the CAISO to propose removing the current $0 bidding obligation in RUC for RA capacity. Even if that RA obligation remained in place, the CAISO is concerned that expanding RUC to other balancing areas whose RUC requirements would not be met by RA capacity with a $0 bidding obligation would undermine the current assumption that the RA bidding rules for RUC forestall attempts to exercise market power in RUC.

---

116 New tariff section 31.3.1.6.5.

117 The new RUC market power mitigation pass is addressed in new tariff section 31.9.

c. Triggering Mitigation

The market power mitigation pass for RUC will use unmitigated reliability capacity bids to procure reliability capacity to meet the CAISO demand forecast. The CAISO will use load distribution factors to distribute the demand forecast to load nodes in the market footprint. The CAISO will enforce transmission constraints using the same shift factors from IFM. It will model reliability capacity awards as energy flows, and the market would evaluate whether binding transmission constraints are uncompetitive using a dynamic competitive path assessment. Resources that could provide counter-flow to uncompetitive constraints would have their reliability capacity up bids mitigated. The market power mitigation pass will not mitigate reliability capacity down bids. The market also will not mitigate the reliability capacity up bids of non-EDAM intertie resources certified to provide reliability capacity, consistent with existing rules that do not apply mitigation to imports because of the presumed competitiveness of interties.

d. Default Availability Bids as Consequence if Mitigation

The market power mitigation pass will mitigate a bid for reliability capacity up subject to mitigation to the higher of the: (1) default/negotiated availability bid; or (2) competitive locational reliability capacity up price.[118] The competitive locational price for reliability capacity is the locational price of reliability capacity minus the non-competitive congestion components at the location of the mitigated resource. The non-competitive congestion components will be derived as the marginal price of reliability capacity up minus the non-competitive congestion components from binding constraints in RUC at the location of the mitigated resource. The reliability capacity up mitigation will provide a mitigation "floor" to balance the need to protect consumers against market power while also protecting suppliers against excessive mitigation forcing offers below a resource's costs.

The CAISO proposes to apply the same $55 default availability bid it proposes for imbalance reserves to reliability capacity. Scheduling coordinators will have the same opportunity to request a negotiated default availability bid once the CAISO certifies it is prepared to enter into such negotiations. The $55 value is designed to be a conservative value that represents a high percentile value of historical non-RA RUC availability offers. Because the obligations of reliability capacity up are similar to the obligations of a RUC capacity awards, the CAISO believes it likely represents the upper bounds of resource's bid costs for the new product. Having the same default availability bid also produces system benefits and implementation efficiencies. Having the same value also makes it

[118] New tariff section 31.9.2.

easier for market participants to track the functioning of these new market power mitigation measures. The CAISO concluded because it can independently justify the $55 value for both imbalance reserves and reliability capacity, then it is appropriate to use the same value as the default bid for both products.

5. Residual Unit Commitment – Procurement

a. Establishing Procurement Targets for Reliability Capacity

If a balancing area's demand forecast is greater than the physical supply that clears the integrated forward market, the RUC process will procure reliability capacity up to provide upward dispatch capability and/or commit additional units.[119] If the balancing area's demand forecast is less than the physical supply that clears the integrated forward market, the RUC process will procure reliability capacity down to provide downward dispatch capability (but would not de-commit units).

b. Meeting the Procurement Targets

Energy schedules, imbalance reserve awards, and ancillary services awards will be fixed in RUC at their IFM schedules. With those fixed inputs, the RUC optimization will procure the needed reliability capacity for each balancing area based on submitted bids and subject to transmission constraints within balancing areas and across EDAM and external interties.[120]

---

[119] The RUC procurement target is addressed in new tariff section 31.5.3.

[120] The processes the CAISO will use to procure to the RUC procurement target are addressed in new tariff section 31.5.5.

c. Transitioning Multi-Stage Generators Down

As part of incorporating the reliability capacity downward product in RUC, the CAISO proposes enhancing RUC regarding how it optimizes awards to multi-stage generating units, which are combined-cycle units and similar resources that operate in distinct operating configurations. Currently, RUC can keep a multi-stage generator in the same configuration or transition the unit to a configuration with a higher output range. When a multi-stage unit needs to be transitioned down to a configuration with a lower output range either to manage congestion or oversupply conditions, the CAISO must issue the resource an exceptional dispatch to move it to that lower configuration. Without creating an explicit functionality for RUC to transition a multi-stage resource to a lower configuration, the resource's ability to provide reliability capacity down would be limited to the downward range within the IFM-committed configuration. That outcome would unreasonably limit the ability of reliability capacity down to provide its anticipated benefits. Accordingly, the CAISO proposes to update RUC to allow it to issue reliability capacity down awards to a multi-stage generating unit that, if converted to energy in the real-time market, would require the resource to transition to a lower configuration than awarded in the IFM.[121] This change to RUC will help manage congestion in the RUC and avoid out-of-market actions by system operators.

d. CAISO Will Maintain Authority for Operators to Manually Adjust RUC Demand Forecast

Implementing imbalance reserves in the day-ahead market should greatly reduce RUC adjustments going forward. However, market operators will still have the authority to use RUC adjustments as needed. Although net load uncertainty is the main reason market operators use RUC adjustments, RUC adjustments can cover other operational risks, such as wildfire risks. Therefore, market participants should not expect the use of operator RUC adjustments in the CAISO balancing area to disappear.[122] The CAISO publishes the RUC load adjustment (MW) and RUC load adjustment reason on OASIS to provide

---

[121] The CAISO is not, however, proposing that a reliability capacity down award would result in decommitting a multi-stage generating unit entirely. New tariff section 31.5.5.3 ("RUC shall not Shut Down resources scheduled through the IFM.").

[122] As reflected in new tariff section 33.31.2.4, the CAISO does not propose that its operators have this same authority over RUC load forecasts submitted by EDAM entities.

transparency. The CAISO will continue to do so after implementation of the proposed day-ahead market enhancements.[123]

6. Real-Time Market Obligations from Imbalance Reserves and Reliability Capacity Awards

Resources with awards for imbalance reserves and reliability capacity must provide economic energy bids for the full range of their reliability capacity and imbalance reserve awards in the real-time market.[124] For resources that fail to meet this requirement, the CAISO will insert generated bids in the real-time market at resources' existing default energy bids. Economic bids enable the real-time market to re-dispatch resources to meet real-time system conditions and imbalances. Real-time self-schedules do not allow the real-time market to re-dispatch the resource. Real-time must-offer obligations apply in the hours that a resource has a reliability capacity or imbalance reserve award.

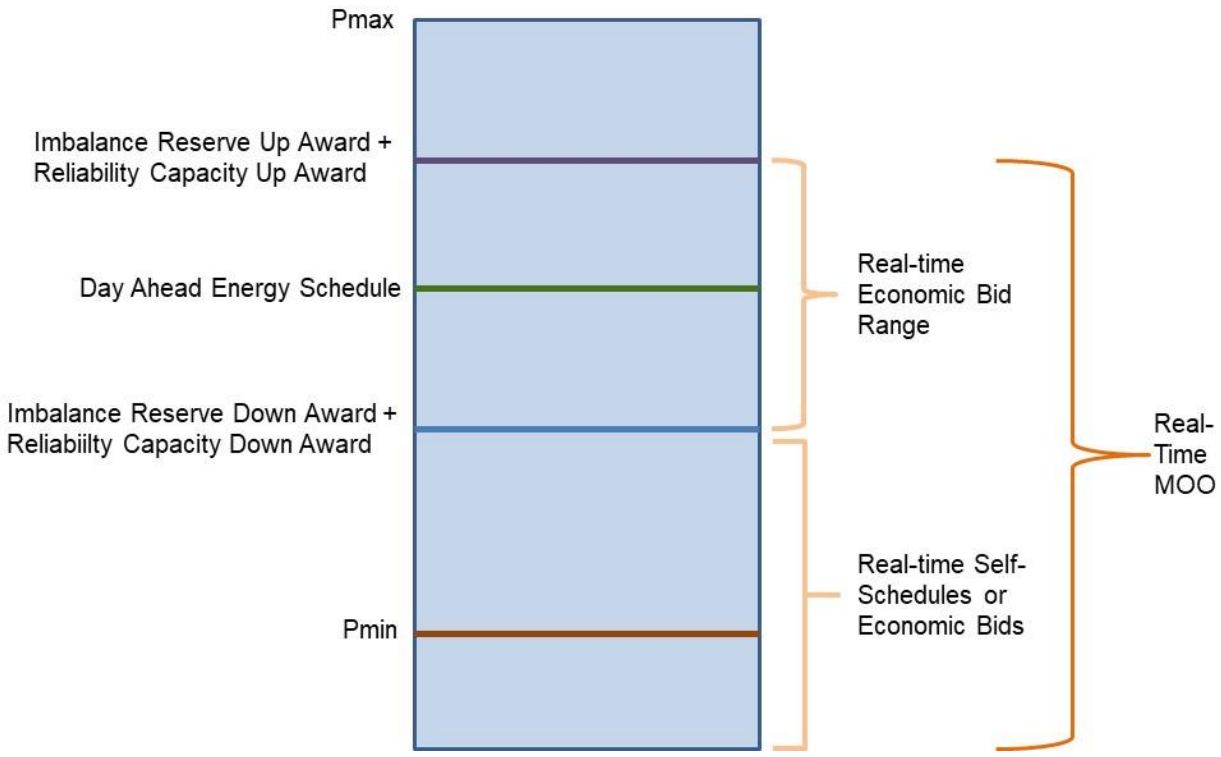
Figure 7 below graphically illustrates the specifics of the minimum real-time market bidding obligations for such resources. A resource must submit economic bids above its day-ahead energy schedule by the amount of imbalance reserves up and reliability capacity up awarded. The resource need not submit additional bids up to its Pmax but may elect to do so. This ensures there are sufficient economic offers to allow the real-time market to dispatch the resource above or below its day-ahead energy schedule. Any portion of a resource's day-ahead energy schedule below the imbalance reserves down and reliability capacity down awards can be self-scheduled or economically bid. A resource cannot submit a self-schedule that exceeds its energy schedule less its imbalance reserves down and reliability capacity down awards. This ensures there are sufficient economic offers to allow the real-time market to dispatch the resource below its day-ahead energy schedule. A resource that can be committed in the real-time market can submit start up and minimum load bids to enable the market to re-optimize the unit commitment decision. This is not a requirement because the resource can elect to self-schedule a portion of its output.

---

123 The CAISO thus proposes only conforming edits to section 31.5.3.1.

124 New tariff sections 31.3.4 (addressing bidding obligations from imbalance reserves) and 31.5.8 (addressing bidding obligations from reliability capacity).

Figure 7



7. Major Financial Settlements Design Elements of the Day-Ahead Market Enhancements

a. Paying Awards and Assessing Unavailability Charges

The CAISO will pay resources that receive an award for imbalance reserves or reliability capacity the locational price for the direction of the product. These payments are separate from settlement of any energy awards in the real-time market. For both products, the CAISO proposes unavailability charges if a resource is unavailable to provide the product. For both products, the CAISO will also subtract such charges from the procurement costs allocated as specified in section IV.D.7.c of this transmittal letter.[125]

---

[125] New tariff section 11.2.1.8 and revised tariff section 11.2.2.2.3 ("RUC Availability Payments rescinded due to Undispatchable Capacity are subtracted from the RUC Compensation Costs allocated per Section 11.8.6.5.3").

The CAISO will consider a resource as having *unavailable imbalance reserves up* capacity to the extent the sum of its energy scheduled in the IFM, upward ancillary services, and imbalance reserves up award minus the five-minute portion of the imbalance reserves award exceeds the upper economic limit as adjusted for outages.[126] The CAISO will consider a resource as having *unavailable imbalance reserves down* capacity to the extent the resource's lower economic limit exceeds the energy scheduled in the IFM minus downward ancillary services minus the imbalance reserves down award plus the five-minute portion of the imbalance reserves award.[127] For both directions, the CAISO will charge back to the resource unavailable imbalance reserves capacity at the higher of the relevant locational price for reliability capacity or the FMM flexible ramping product price in the relevant direction.[128] The "higher of" formulation is necessary to create a stronger performance incentive than would be created through a straight payment rescission rule. Capacity that is not available in real-time reduces the available supply of real-time energy and flexible ramping product and drives up their price. This formulation charges resources commensurate with the harm they cause to the system when unavailable to provide their awarded imbalance reserves.

Unlike the proposed treatment for imbalance reserves, for reliability capacity the CAISO proposes a straight payment rescission for RUC capacity that goes on outage between day-ahead and real-time or fails to submit the required real-time market bids (*i.e.*, the resource has undispatchable capacity for reliability capacity).[129] Unavailable reliability capacity does not affect the price of the flexible ramping price in the same way as unavailable imbalance reserves. A resource will have *undispatchable reliability capacity up* in a settlement interval to the extent the resource's upper economic limit (as adjusted for outages) is less than the sum of energy from IFM, upward ancillary services, imbalance reserves

---

126 New tariff section 11.2.1.8.1. The formulation includes the five-minute imbalance reserve quantity because the unavailability of that portion of the imbalance reserves award is accounted for in the proposed deviation settlement covered in section IV.E.2.a of this transmittal letter.

127 New tariff section 11.2.1.8.2.

128 New tariff sections 11.2.1.8.1 (for imbalance reserves up) and 11.2.1.8.2 (for imbalance reserves down).

129 The term "Undispatchable Capacity" is an existing defined term in the CAISO tariff and refers to ancillary services or RUC capacity "that is not available for use due to a derate or Outage of the resource." Tariff appendix A, existing definition of Undispatchable Capacity. The CAISO proposes to amend this definition for RUC capacity also to include cases where the resource fails to submit the required real-time market bids.

up award, and the reliability capacity up award.[130] A resource will have *undispatchable reliability capacity down* in a settlement interval to the extent the resource's lower economic limit exceeds the award for energy from the IFM minus downward ancillary services minus the imbalance reserves down award minus the reliability capacity down award.[131] For both directions, the CAISO will charge undispatchable capacity back to the resource at the relevant locational price for reliability capacity.

Resources that receive both a reliability capacity and imbalance reserve award in the same direction and are not available or only bid a portion of their combined award will have the unavailable capacity allocated first to reliability capacity and then to imbalance reserves.[132] This is consistent with the CAISO's existing approach of assessing unavailability charges in reverse order of the quality of the product. Because reliability capacity is a lower-priority product, the CAISO assesses its unavailability first before deeming any portion of imbalance reserves unavailable.

b. Bid Cost Recovery Implications of the New Products

The CAISO will include in the existing bid cost recovery mechanism the revenue and bid costs from both imbalance reserves and reliability capacity awards.[133] Similarly, the CAISO will continue to keep cost recovery payments from the integrated forward market and the RUC/real-time market separate because they have different cost allocations. The CAISO will include revenue and bid costs from imbalance reserve awards in the calculation of day-ahead bid cost recovery.[134] The CAISO will continue netting surplus revenues from the RUC process against revenue shortfalls in the real-time market. A revenue surplus would occur in the RUC when the marginal price of reliability capacity exceeds a resource's reliability capacity bid cost. Conversely, the CAISO will net any surplus revenues from the real-time market against revenue shortfalls in the residual unit commitment process.

---

130 New tariff section 11.2.2.2.1. The reference to the "upper operating limit" in the phrase "the lower of the resource's Upper Economic Limit or upper operating limit," is how the proposed tariff revisions will account for the impact of outages.

131 *Id.* 11.2.2.2.1.

132 New tariff section 11.2.1.8.3

133 The exception for this is the bid cost recovery provisions for RA resources will not consider reliability capacity payments and bids because the CAISO presumes unrecovered RUC costs are accounted for in the bilateral RA capacity contract. Revised tariff section 11.8.3.

134 New tariff section 11.8.2.1.8 (accounting for imbalance reserves bid costs) and revised tariff section 11.8.2.2.1 (accounting for imbalance reserves revenue).

c. Allocating Costs of Procuring the New Products

The CAISO's proposed rules for settling the costs of procuring the new products follow cost causation principles and the two-tiered cost allocation approach the CAISO already uses for several settlements processes.[135] For the four products (imbalance reserves up, imbalance reserves down, reliability capacity up, and reliability capacity down), the CAISO separately calculates a tier 1 quantity and tier 1 price appropriate for each of the four products. A scheduling coordinator's allocation in tier 1 for the product is the product of the tier 1 quantity and the tier 1 price. Specifically, the CAISO proposes to allocate tier 1 imbalance reserves costs to generation resources, load, import, and exports based on deviations between their day-ahead market schedules and their availability in the real-time market.[136] Separately, the CAISO proposes to allocate tier 1 reliability capacity up costs to under-scheduled load and net virtual supply,[137] with tier 1 reliability capacity down costs allocated to over-scheduled load and net virtual demand.[138] The CAISO separately will calculate a tier 1 price calculated at the lower of two possible formulations of the average price of the product. These tier 1 prices are designed to avoid allocating excessive costs to individual parties in tier 1 and is consistent with the CAISO's other two-tier cost allocations. This design decision, however, means some costs potentially would not be recovered from tier 1. The CAISO will allocate any costs from procurement of that product not recovered through the tier 1 allocation in the second tier. For all four products, the CAISO will allocate tier 2 costs *pro rata* to metered demand in that hour.

The tier 1 allocation seeks to allocate as much of the total cost of procuring the product as possible to those entities driving the need for the procurement. The MW quantity in each calculation is a proxy of how much of that product the participant caused the CAISO to procure. Ideally, the CAISO would pursue a direct settlement by allocating the MWs of the procured product directly on a one-to-one basis based on how much of the product each entity in tier 1 caused the CAISO to procure. However, it is impossible to make a direct one-to-one connection. For example, the CAISO cannot make a direct connection between a specific wind resource's deviation between its day-ahead forecast and real-time output and the CAISO's need to procure a specific quantity

---

135 Existing tariff sections 11.8.6.4 (IFM Bid Cost uplift allocation) and 11.8.6.5.3 (RUC Compensation Costs).

136 New tariff sections 11.2.1.9.1 (imbalance reserves up tier 1 quantity) and 11.2.1.9.2 (imbalance reserves down tier 1 quantity).

137 New tariff sections 11.8.6.5.3.1 (reliability capacity up tier 1 quantity) and 11.8.6.5.3.3 (reliability capacity up tier 1 price).

138 New tariff sections 11.8.6.5.3.2 (reliability capacity down tier 1 quantity) and 11.8.6.5.3.4 (reliability capacity down tier 1 price).

of imbalance reserves at a specific location because the requirements are determined on a system level, not an individual nodal basis. Also, it would be challenging to determine which loads and resources the CAISO should charge at each nodal location and in what proportion because demand does not bid to buy imbalance reserves. The CAISO therefore proposes the tiered cost allocation process so it can allocate costs in the first tier as closely as possible to what drove the need for the CAISO to incur those costs, while relying on the second tier to ensure full cost recovery for the new products in recognition that all metered demand receives system benefits from those new products.

d. Resource Adequacy Availability Incentive Mechanism

Although resources providing RA capacity will have specific obligations to submit bids for imbalance reserves and reliability capacity, the CAISO does not propose to amend the tariff provisions regarding the Resource Adequacy Availability Incentive Mechanism (RAAIM) to account for these obligations explicitly.

The CAISO considered updating RAAIM to assess bid submission for required amounts of imbalance reserves or reliability capacity. For example, consider a resource providing 100 MW of system RA capacity that submits an economic bid for energy to the day-ahead market for its 100 MW of RA capacity but does not submit the imbalance reserves bid that must accompany the economic bid. The CAISO could have treated the resource as unavailable for purposes of RAAIM because of that failure to submit the imbalance reserves bid.

The CAISO also could have proposed to update RAAIM to assess bidding in the real-time market in compliance with the performance obligations of an imbalance reserves or reliability capacity award. For example, consider a system RA resource that today has a real-time market must-offer obligation that can meet it by submitting either economic bids or self-schedules. Under the day-ahead market enhancements, if that resource submits its required bid for reliability capacity up and RUC awards the bid, then the resource would have a real-time market obligation to submit an economic offer that exists independent of its RA must-offer obligation in the real-time market. If that resource submits a self-schedule to the real-time market, the CAISO could treat the resource as unavailable in the real-time for purposes of RAAIM because the economic offer obligation was triggered indirectly from its status as RA capacity.

Updating RAAIM to account for either of these scenarios, however, would raise undue complication. In the first example, the unavailability is contingent on how the resource meets it must-offer obligation to the day-ahead market. If that same resource, instead had submitted a self-schedule for energy to the day-ahead market, there would be no imbalance reserves bidding obligation.

Configuring RAAIM to account for these conditional obligations would be challenging. In the second example, the CAISO would have to extend RAAIM to assess bidding not directly connected to the RA must-offer obligation, which is a notable expansion of RAAIM's scope. The CAISO concluded there was insufficient justification to confront these complexities and propose these changes.

Most RA capacity is already subject to generated bids that apply where the capacity fails to bid. Thus, RA resources typically incur RAAIM charges only because a unit is on outage, rather than because it failed to bid. Under the proposed new rules, if a resource is on outage it would be on outage for all products, not just for energy or reliability capacity. Also, receiving an imbalance reserves or reliability capacity award creates a must-offer obligation to submit economic bids to the real-time market. However, most RA capacity already has a must-offer obligation to bid into the real-time market, although generic RA can satisfy that obligation with self-schedules. Bid insertion applies to that must-offer obligation. Thus, most RA capacity already participates in the real-time market. Finally, RA capacity will be subject to generated bids for reliability capacity up under proposed revisions to tariff section 40.6.8(f). A reliability capacity up award based on such a generated bid will obligate the resource to submit economic bids to the real-time market. If the resource fails to meet the real-time market must-offer obligation triggered by the awarded reliability capacity, then the CAISO would generate real-time market bids at the default energy bid. Failure to deliver on those bids would subject the resource to the new no-pay rules. This generated bid framework further reduces the likelihood that RA capacity will be unavailable for the new products absent an outage, which still could cause the RA capacity to incur RAAIM. For all these reasons, the CAISO determined there was not justification for updating RAAIM to account for the new products.

e. Grid Management Charge

The market services charge of the grid management charge covers the cost of bidding and clearing the market. Because resources submit bids for imbalance reserves and reliability capacity, the market services charge would apply for imbalance reserve awards, and the CAISO would update the rate to account for these factors.[139] Suppliers may include this cost in their bids.

E. Additional Design Elements

This section IV.E of the transmittal letter describes additional design elements and other aspects of the proposed DAME tariff provisions.

---

[139] Tariff appendix F, revised schedule 1.

1. Measures to Accommodate Long-Term Contracts

The CAISO proposes to introduce an optional three-year transitional settlement mechanism, referred to as DAME Transitional Measures, to allocate payments for imbalance reserves and reliability capacity that overlap with RA capacity between the scheduling coordinators for the resource and the load serving entity.[140] Both scheduling coordinators would need to opt in to the transitional measures. Absent the mutual election of the scheduling coordinators for the resource and load serving entity to opt in, the CAISO will allocate all market payments for imbalance reserves or reliability capacity that overlap with RA capacity to the resource's scheduling coordinator.

The optional transitional measures address the concern that existing bilateral RA contracts between load serving entities and suppliers may not address in explicit detail how to allocate the revenue paid to a resource for the new products between the load serving entity and supplier. RA capacity currently is ineligible for RUC capacity payments, so contracts would not necessarily address RUC payments under the market design enhancements. Further, imbalance reserves are a new product, so no contract would have addressed that directly. The transitional measures can serve as a backstop or "gap-filling" mechanism to navigate the transition if the parties cannot agree on implementing their existing contracts or would prefer the CAISO to handle the issue through these temporary settlement provisions.

To opt in, the scheduling coordinators for both the load serving entity and resource must notify the CAISO of their mutual election. If the scheduling coordinators complete the election within the first 60 days of the effective date of new market products, then the CAISO will apply the settlement measures retroactive to the tariff effective date. For elections completed after the 60-day period, the CAISO will apply transitional measures prospectively starting with the first day of the month after the month in which the election was complete. Performing retroactive settlement adjustments creates logistical and administrative burdens on the CAISO that increase as the period of retroactivity increases. The CAISO therefore needed to limit the retroactive settlement period to 60 days. For elections made after this period, the CAISO must wait to begin applying the transitional measures until the start of the next month because of the monthly nature of the RA program and the challenges to the CAISO systems of making mid-month adjustments to how it handles a given quantity of RA capacity. Once under the transitional measures, the scheduling coordinators can mutually agree to terminate the treatment before expiration of the three-year period and may opt back into transitional measures later in the three-year transition period. The election would apply to all RA capacity on RA supply plans

[140] New tariff section 11.2.6.

relating to that resource/load serving entity pair during the transitional period. The CAISO determined it was infeasible to tie the election to a specific MW quantity because the RA capacity shown for a resource/load serving entity pair can fluctuate over time for various commercial reasons, even if the quantity under contract remains static.

The CAISO will calculate an RA resource's overlapping RA capacity by comparing the resource's shown RA capacity against the resource's "stacked" awards for energy, ancillary services, imbalance reserves, and reliability capacity, arranged in priority of quality.[141] The CAISO creates separate "stacks" of the awards for all upward and downward products, with the lowest-quality products on the bottom. The CAISO will consider any portion of RA capacity that overlaps with either the imbalance reserve awards or reliability capacity awards, which are on the bottom of the stack, overlapping RA capacity potentially subject to the RA true-up mechanism. The CAISO will separately calculate the quantity of overlapping imbalance reserves up, imbalance reserves down, reliability capacity up, and reliability capacity down, and it will separately allocate payments for each of the four overlapping types of capacity.

The CAISO will calculate overlapping *imbalance reserves up* as RA capacity shown for the resource in the month minus energy and upward ancillary services.[142] The overlapping *reliability capacity up* is the same value minus any imbalance reserves up award. The CAISO will calculate overlapping *imbalance reserves down* as the energy schedule minus regulation down minus the RA capacity shown for the resource in the month. The overlapping *reliability capacity down* is the same value minus any imbalance reserves down award. If a resource has multiple RA contracts shown on multiple load serving entities' monthly RA plans, the CAISO will determine the portion of overlapping RA capacity associated with load serving entities that "opt in" to the RA true-up mechanism versus the load serving entities that opt out of the transitional measures by distributing the overlapping RA capacity to the contracted load serving entity in proration to the load serving entity's RA showing as compared to the resource's total RA showings.

For both types of imbalance reserves, the CAISO will calculate an opportunity cost component of the market payments for the overlapping capacity and provide that payment to the resource. The CAISO will allocate the balance

---

141 The order for the upward product in descending order of quality after energy is regulation up, spinning reserves, non-spinning reserves, imbalance reserves up, and reliability capacity up (*i.e.*, reliability capacity up is the lowest-quality product). The descending order of quality for the downward products is regulation down, imbalance reserves down, and reliability capacity down.

142 The value is capped at the imbalance reserves up award. It would be illogical to determine the amount of overlapping capacity that exceeded the actual imbalance reserves award. The CAISO will perform this same capping in determining the overlap for the other three new products as well.

of the payments from the overlapping capacity to the load serving entity. For both types of reliability capacity, the CAISO will allocate payments to the load serving entity with the resource retaining no payment.

Finally, to facilitate bilateral settlement between suppliers and load serving entities, the CAISO will provide load serving entities information about the opportunity cost calculations for RA capacity not subject to the transitional measures. It will be up to the individual parties to determine how to use that information. The CAISO will continue to provide this information even after expiration of the three-year transition period.

2. Real-Time Market Ramp Deviation Settlement

The CAISO proposes a real-time market ramp deviation settlement to avoid potential double-paying resources receiving both an imbalance reserves award in the day-ahead market and a flexible ramping product award in the real-time market. The CAISO also proposes to extend this deviation settlement to WEIM market participants and convergence bidding entities to address a minor existing double payment concern in the current market design that could become more significant under the day-ahead market enhancements and the extended day-ahead market.

a. Ramp Deviation for Imbalance Reserves

Both imbalance reserves and the flexible ramping product compensate resources for providing the CAISO markets with flexible ramping capabilities. The two products have significant overlap in terms of what they seek to accomplish and how resources meet the requirements of the product. This overlap creates double-payment concerns regarding both opportunity costs and forecasted movement.

i. Double-Payment of Opportunity Costs

The CAISO pays resources that receive an imbalance reserve award in the day-ahead market the locational price of imbalance reserves for the corresponding interval. The locational price of imbalance reserves is based on two factors: imbalance reserve bids and any opportunity costs. Opportunity costs for imbalance reserves occur when the market holds a resource out of merit for energy or ancillary services to preserve its ramp capability to provide upward capacity to meet the uncertainty requirements in an interval. Similarly, opportunity costs for energy can occur when the market holds a resource out of merit for energy to preserve its downward capability to provide sufficient ramping to meet the load in a subsequent interval. However, the marginal clearing price

of the flexible ramping product is based only on opportunity costs; there are no bids associated with this product. Thus, a resource awarded both imbalance reserves and flexible ramping product could be paid opportunity costs from both products, even if its energy and ancillary service schedules did not change. This can be seen as an unwarranted double payment.

ii. Double-Payment of Forecasted Movement

In the day-ahead market, all hourly schedules are financially binding across the 24-hour horizon. There are no un-settled "advisory" intervals in the day-ahead market. There is no need to settle forecasted movement in the day-ahead market because the energy prices already reflect the opportunity cost of resources scheduled out-of-merit in previous hourly intervals. However, in the real-time market, only one market interval is financially binding over the optimization horizon. The market produces unsettled "advisory" prices for the remaining market intervals. If the market dispatches a resource for energy in the binding interval to provide ramp capability to meet the energy dispatch of an advisory interval, the resource can incur an opportunity cost if the binding interval price is less than its energy bid. If in this market run the resource incurs an opportunity cost, the advisory interval energy price will increase to reflect this tradeoff. However, the advisory interval energy price is not settled, and when it becomes binding in the next market run, the out-of-merit dispatch is unknown and the opportunity cost is not embedded in the binding energy price. To compensate the resource, the CAISO provides it a separate payment for forecasted movement at the marginal price of ramp capability.[143] This incentivizes the resource to follow its energy dispatch because the resource is indifferent to receiving an incremental energy schedule or a forecasted movement payment because it earns the same profit under both scenarios. However, a resource may receive compensation for forecasted movement both in the day-ahead market (embedded in the energy prices) and in the real-time market (as a side payment). This also can be seen as a double payment.

[143] *See* section 7.1.3.1.4 of the Market Operations BPM for numerical examples.

iii. Ramp Deviation Settlement to Address Double-Payment Concerns

Although imbalance reserves and the flexible ramping product are similar, the products are not identical. Thus, developing an imbalance settlement between the two products is not straightforward like the way the CAISO settles imbalance energy between day-ahead and real-time. In the case of energy, an energy award in the day-ahead market is effectively the same product as an energy award in real-time market. If a resource receives a day-ahead award for energy with the same MW award in the FMM and then RTD, it would be inappropriate to pay that resource once at the day-ahead price, a second time at the FMM price, and third time at the RTD price for the same MW quantity. To address this concern, the existing market only pays or charges that resource for energy in the real-time market that is incremental or decremental to the day-ahead market award. The proposed real-time market ramp deviation settlement aims to address this same concern for imbalance reserves and the flexible ramping product based on the principle that the five-minute ramp capability awarded to a resource in the IFM as either energy movement or imbalance reserves should only receive payment for flexible ramping product in the FMM to the extent the FMM award is incremental to the five-minute capacity procured in the IFM.

The mechanics of the ramp deviation settlement are as follows. The CAISO first must determine the portion of an imbalance reserves award that corresponds to a five-minute interval. The CAISO will determine this *five-minute imbalance reserve quantity* as "the five-minute ramp capable portion of the award measured as the MW quantity of the resource's ramp capability above the Day-Ahead hourly Energy schedule, in the case of IRU, or below that schedule, in the case of IRD."[144] This process breaks down the imbalance reserve award into the same time granularity as a flexible ramping product award. For resources with an award for imbalance reserves up and an upward award of flexible ramping product, the CAISO only settles any incremental award of upward flexible ramping product at the flexible ramp up price.[145] The CAISO compensates the non-incremental quantity of upward flexibility (*i.e.*, the quantity that corresponds to the award for imbalance reserves up) only as imbalance reserves up.

For forecasted movement, the CAISO also will apply a deviation settlement but through a different formulation. Forecasted movement in the FMM

---

144 Tariff appendix A, new definition of Five-Minute Imbalance Reserve Quantity.

145 New tariff section 11.25.2.1.1 (for uncertainty awards).

will be subject to a deviation settlement with forecasted movement in the IFM.[146] This requires the CAISO to include in the tariff a concept of IFM forecasted movement, which does not exist today. The CAISO proposes this to be "the algebraic difference of the Day-Ahead Schedule between consecutive hours."[147] The CAISO will pay forecasted movement in the FMM the flexible ramp up price and charge the flexible ramp down price. Therefore, the CAISO proposes to pay a resource with an upward deviation in forecasted movement at the flexible ramp up price for the quantity of the deviation because the deviation provided additional ramping capability. However, the CAISO will also charge it at the flexible ramp down price for the quantity of the deviation because the upward movement increased flexible ramp down requirements. For similar reasons, the CAISO will pay a resource a downward deviation the flexible ramping down price for the quantity of the deviation and charged it at the flexible ramp up price for the quantity of the deviation. This aligns with the deviation settlement between the FMM forecasted movement and the RTD forecasted movement.

b. Ramp Deviation for WEIM Participants

The WEIM also procures flexible ramping product to commit and position resources to meet future load and supply variability and uncertainty. For WEIM participants, forecasted movement from base schedules equals forecasted movement in the integrated forward market. If resources already are scheduled to ramp in WEIM base schedules, then paying them an additional forecasted movement payment in the FMM for the same ramp constitutes a double payment. This is not a significant concern, but the CAISO nevertheless finds it prudent to address it. Therefore, WEIM participants would also be subject to a forecasted movement deviation settlement in the FMM. The CAISO will establish baseline forecasted movement for each resource on WEIM base schedules.[148] The CAISO will charge any deviations using the forecasted movement deviations rules described above for non-WEIM participants.

c. Ramp Deviation for Convergence Bidding

The existing IFM energy price, which is one-half of how virtual bids are settled, has embedded within it the settlement of forecasted movement. The real-time market price, which is the price at which virtual bids are liquidated, does not include forecasted movement. The CAISO settles forecasted movement separately through the flexible ramping product.

---

146 New tariff section 11.25.1.2.

147 New tariff section 44.3.4.

148 New tariff section 44.3.6.

There is an existing discrepancy because the two prices that are the basis of settling virtual bids (*i.e.,* convergence bids) are based on different factors. In practice, this discrepancy has not had a material impact. However, the CAISO expects that is likely to change with the new market products. Creating an imbalance reserves product co-optimized with energy in the IFM means that the energy price now implicitly will include an opportunity cost of not providing imbalance reserves. This change in the energy price formation will increase the discrepancy in how convergence bidding is settled. Thus, the CAISO proposes to implement a forecasted movement deviation settlement at the FMM flexible ramping product prices for virtual supply and demand similar to what it has proposed for the WEIM.[149] The forecasted movement would be based on the difference in virtual awards between consecutive hours,[150] and then the CAISO would settle it in the same way it settles deviations for the physical market.[151]

3. Congestion Revenue from Deployment Scenarios

a. The Issue of Displaced Congestion Revenue

The CAISO's election to pursue the two-tiered cost allocation for imbalance reserves has implications on how the CAISO collects congestion revenue and settles congestion revenue rights. The CAISO's cost allocation proposal for imbalance reserves means that the CAISO will not collect congestion revenues to cover the marginal cost of congestion in the imbalance reserve deployment scenarios. Whenever a constraint is binding in the deployment scenarios, there could be a shortfall of congestion revenue collected on that constraint because the CAISO would not otherwise collect congestion revenue on the imbalance reserve flow. The imbalance reserve deployment scenario flow can "displace" energy flows over constrained transmission paths. The energy flow that does not occur would "displace" energy market congestion revenues that would otherwise go to those entities with a claim on congestion revenues, which for the CAISO balancing area would be CRR holders and measured demand. For EDAM balancing areas, the determination of the entities with a claim on congestion revenues would depend on individual transmission service provider tariffs.

The CAISO can demonstrate this dynamic of displaced congestion revenue with a simple example where the system has a single locally congested

---

149 New tariff section 11.25.1.2.

150 New tariff section 44.3.5.

151 New tariff section 11.25.1.2.

area with the rest of the system facing no congestion. Consider the marginal energy offer is $20 outside the congested area and $90 inside, while the marginal imbalance reserves up offer is $5 outside the congested area and $7 inside. If the CAISO market uses a MW of the scarce transmission capacity into the congested area for imbalance reserves up instead of energy, then the CAISO would not collect the $70 in congestion revenue. That foregone $70 represents funds that, at least in the CAISO balancing area, would have gone first to CRR holders with CRRs over the transmission line leading into the congested area and then potentially to measured demand if those CRR holders were whole up to their notional value.

The CAISO does not expect the imbalance reserves deployment scenarios to displace significant energy congestion revenues because the trade-off between using transmission for energy as opposed to imbalance reserves depends on the relative difference between the marginal energy and marginal imbalance reserve offers inside and outside the constrained area. In the example above, the market would not use the scarce transmission for imbalance reserves because that would not minimize overall procurement costs. Using the transmission for energy avoids the $70 difference ($90-$20) in energy price, whereas using the transmission for imbalance reserves only avoids incurring the $2 difference ($7-$5) in imbalance reserves up price.

The CAISO expects that the relative differences in marginal energy offers inside versus outside congested areas generally will be higher than the relative differences in marginal imbalance reserves offers inside versus outside congested areas. The CAISO bases this expectation on the lower cost of providing imbalance reserves compared to providing energy. The CAISO expects to use the constrained transmission mostly used for energy. These are reasons the CAISO does not expect the imbalance reserves deployment scenarios will displace significant congestion revenue. Nevertheless, the CAISO acknowledges the possibility this will occur occasionally to the detriment of CRR holders (and congestion revenue recipients in EDAM balancing areas). The CAISO therefore proposes measures to address such displaced congestion revenues.

b. CAISO Proposal to Address Displaced Congestion Revenue

To address the concern of displaced congestion revenue, the CAISO proposes a mechanism to collect congestion revenue rent on imbalance reserve flows and redistribute it to entities entitled to congestion revenues, such as CRR holders. This mechanism will involve calculating displaced congestion revenue from imbalance reserve flows and redistributing it according to existing processes.

The CAISO will determine the displaced congestion revenue from imbalance reserve up flows by calculating and summing (Imbalance Reserve Up Flow) * (Shift Factor) * (Shadow Price of Transmission Constraint) for all binding constraints in the upward deployment scenario.[152] Similarly, the CAISO will determine the "displaced" congestion revenue from imbalance reserve down flows by calculating and summing (Imbalance Reserve Down Flow) * (Shift Factor) * (Shadow Price of Transmission Constraint) for all binding constraints in the downward deployment scenario. This methodology is the formalized way of identifying the $70 of congestion revenue displaced in the example above.

The CAISO will collect this displaced revenue through the proposed imbalance reserve cost allocation.[153] Congestion contributions to transmission constraints in an EDAM balancing area from resources in other EDAM balancing areas contribute to the congestion offset of the EDAM balancing area where the congestion occurs. For EDAM balancing areas, the CAISO will return the revenues to the balancing area to re-distribute to its participants according to its tariff processes. For the CAISO balancing area, the CAISO will allocate the revenue through its existing CRR settlements process.[154] The CAISO proposes to redefine the congestion revenue collection to fund the CRR notional value to include imbalance reserve deployment scenarios.[155] Additionally, the CAISO proposes to redefine the notional value of CRRs to add the marginal cost of congestion differences between source and sink for imbalance reserve deployment scenarios.[156] CRRs represent an entitlement to congestion revenue and, therefore, it is appropriate for their settlement to reflect this new form of congestion revenue.

4. Treatment of Metered Subsystems, Existing Transmission Contracts, and Transmission Ownerships Rights

The CAISO proposes to maintain its existing approach for existing transmission contracts (ETCs) and transmission ownership rights (TORs) in which the valid and balanced portion of ETC or TOR self-schedules will be

---

152 New tariff section 31.3.1.6.4.

153 New tariff section 11.2.1.9.

154 Revised tariff section 11.2.4.

155 This proposal will not change the existing CRR nomination and auction processes to account for imbalance reserves. The CRR model will not withhold transmission capacity for the CRR nomination and auction processes.

156 Tariff appendix A, revised definition of Notional CRR Value.

exempt from cost allocations.[157] For both imbalances reserves and reliability capacity, as long as the ETC and TOR self-schedules meet the market participant's demand, the CAISO need not procure additional imbalance reserves or reliability capacity to serve load of those rights holders. Thus, the CAISO proposes to exclude the ETC and TOR self-schedules from the tier 1 and tier 2 cost allocations for both products up to the valid and balanced portion of ETC and TOR self-schedules.

The CAISO proposes one significant change for settlements with metered sub-systems (MSSs).[158] Currently, MSS operators must make an election on four options that govern how the MSS participates in the markets, including whether it opts in or out of participating in RUC.[159] The CAISO proposes to eliminate the RUC opt-out option because creation of a downward product in RUC would not align with an opt-out, and it would create difficulty in designing a resource sufficiency evaluation for EDAM if the CAISO had an internal load bubble that was not part of RUC. Otherwise, the new provisions largely will apply to MSSs just as they apply to other classes of market participant. The CAISO proposes to maintain the current settlement of MSS operator day-ahead energy schedules for those MSSs that have elected gross settlement or net settlement. For both reliability capacity tier 1 and reliability capacity tier 2 cost allocations, MSS operators will settle in a similar manner as non-metered subsystem resources, regardless of their net versus gross selection. An MSS operator that has elected to load follow to manage its own load variability will not receive a reliability capacity tier 1 or a reliability capacity tier 2 cost allocation because it faces other cost exposure in cases where the energy they provide to the market does not closely match its load. For both imbalance reserve tier 1 and imbalance reserve tier 2 cost allocations, MSS operators will settle in a similar manner as non-metered subsystem resources, regardless of their net versus gross selection. Finally, an MSS operator that has elected to load follow to manage its own load variability will receive imbalance reserve tier 1 and imbalance reserve tier 2 cost allocations based on its net portfolio uninstructed deviations.

---

[157] ETCs and TORs represent scheduling and transmission rights that pre-dated CAISO operations and entitle holders to a "perfect hedge" against congestion charges, exempting such self-schedules from any congestion charges stemming from the congestion component of the locational marginal price. *See* existing tariff sections 11.2.1.5 and 11.5.7.1.

[158] MSSs are geographically contiguous systems that operate as a standalone electric utility within the CAISO balancing area and balance the own loads and resources within their territory.

[159] An MSS operator also can elect between: net settlements or gross settlements; to load follow or not to load follow with its generating resources; and to charge or not to charge the CAISO for their emissions costs.

5. Clarifying and Conforming Changes to Existing Tariff Provisions

Along with the tariff amendments directly necessary to implement the day-ahead market enhancement design elements discussed above, the CAISO proposes a range of clarifying and conforming changes to existing tariff language.

The CAISO proposes a set of conforming amendments to acknowledge the new products created through this filing. For example, the CAISO is amending existing tariff sections that refer generally to submission of bids for energy and ancillary services to include references to bids for imbalance reserves and reliability capacity or refer generically to "Bids" without reference to the specific products.

The tariff provisions for the day-ahead market heavily use the current defined term "CAISO Forecast of CAISO Demand." To acknowledge that other balancing areas will participate in the day-ahead market by virtue of the EDAM initiative, the CAISO is amending this defined term to refer generally to the "CAISO Forecast of BAA Demand" and updates existing references in tariff section 31.

In several instances, amending the tariff to implement the day-ahead market enhancements highlighted ambiguity and lack of clarity in certain existing tariff provisions that the CAISO is not changing substantively in this initiative. The CAISO is using this filing to improve and clarify these tariff provisions. For example, in tariff section 31.5.4, the CAISO proposes to delete references to several RUC procurement constraints that are no longer necessary. Similarly, in tariff sections 4.12.2 and 4.17.4, the CAISO proposes to reconcile tariff language covering master file parameter registration rules for resource-specific system resources and distributed energy resources, respectively, with the generally applicable master file rules stated in tariff section 4.6.4.[160]

---

[160] *Cal. Indep. Sys. Operator Corp.*, 167 FERC ¶ 61,001, at PP 4, 35 (2019) (approving proposal requiring master file parameters to reflect "design capabilities . . . when operating at maximum sustainable performance over Minimum Run Time, recognizing that resource performance may degrade over time").

F. Major Issues of Stakeholder Discussion

1. Demonstrating the Need for the Imbalance Reserves Product

Throughout the stakeholder process, several parties questioned the need for the imbalance reserves product. These parties noted the proposed imbalance reserves product is unlike products in other ISO/RTO markets and suggested the CAISO does not need it. Some parties also commented that the role imbalance reserves are meant to play could be fulfilled by additional ancillary services procurement.

The CAISO does not agree imbalance reserves are an unnecessary product. As an initial matter, the CAISO notes that there is no single just and reasonable market design. All the CAISO needs to do is to demonstrate that its proposal is just and reasonable.[161] There is ample evidence this is the case. In its recent technical conference process on modernizing electricity market design, the Commission itself acknowledged ISO/RTO markets already have faced, or soon may face, challenges from net load variability and directed each ISO/RTO to submit informational reports on this issue.[162]

In response to these challenges, other ISO/RTO markets already have developed similar products. In 2014, MISO "developed the Ramp Capability products to provide additional operational and interchange scheduling flexibility to respond to variations in Load, and the variability of supply from intermittent sources, by maintaining sufficient ramp capability to respond to Load variations by ramping up or down to adapt to system changes."[163] Similarly, on July 6, 2023, the Southwest Power Pool (SPP) implemented its uncertainty reserve

---

161 *See, e.g., New Eng. Power Co.*, 52 FERC ¶ 61,090, at 61,336 (1990), *aff'd sub nom. Town of Norwood v. FERC*, 962 F.2d 20 (D.C. Cir. 1992) (proposed rate design need not be perfect, it merely needs to be just and reasonable); *City of Bethany v. FERC*, 727 F.2d 1131, 1136 (D.C. Cir. 1984) (*City of Bethany*) (finding that, when determining whether a proposed rate was "just and reasonable" as required by the FPA, the Commission properly did not consider "whether a proposed rate schedule is more or less reasonable than the alternative rate designs"); *Louisville Gas & Elec. Co.*, 114 FERC ¶ 61,282, at P 29 (2006) (the just and reasonable standard under the FPA is not so rigid as to limit rates to a "best rate" or "most efficient rate" standard, but rather a range of different approaches often may be just and reasonable).

162 *Modernizing Wholesale Electricity Market Design*, 179 FERC ¶ 61,029, at PP 14-15, 19-20 (2022).

163 Transmittal letter for MISO tariff amendment on enhancement of demand curves for up ramp capability and short-term reserve products, Docket No. ER22-2891-000, at 2 (Sept. 20, 2022).

product,[164] which is designed to "determine the amount of upward flexibility that is necessary to address both forecasted and unforeseen changes in system needs, and to procure and price that upward flexibility consistent with least-cost principles . . . ."[165] In approving SPP's proposed product, the Commission identified that procuring a flexibility reserves product in the market "will more appropriately procure and price the system's projected flexibility needs within the market and reduce both the need for SPP operators to rely on out-of-market actions and the resulting make-whole payments."[166] Notably, the products in both SPP and MISO are procured in both the day-ahead market and real-time market timeframes. In this sense, the CAISO's proposal to add the day-ahead market imbalance reserves product to its existing real-time market flexible ramping product brings the CAISO into alignment with SPP and MISO. Further, the Commission has already recognized that products procured specifically to meet ramping needs are distinct from ancillary services.[167]

The CAISO acknowledges imbalance reserves would have attributes that are distinct from those existing products. For example, the CAISO understands those other products are not priced at a nodal level and are not procured based on distinct offers but instead are priced on opportunity costs. These incremental updates on designing a flexible reserves product, however, reflect the specific issues and opportunities seen in Western markets with the accelerated penetration, magnitude and diversity of variable energy resources, not yet seen to the same degree in other markets.

2. Granularity of Procurement and Accounting for Transmission Constraints

Throughout the stakeholder process and especially in the 2023 workshop process, some stakeholders urged the CAISO to adopt a zonal approach to imbalance reserves procurement, similar to the CAISO's procurement of ancillary services. Specifically, stakeholders raised concerns that the CAISO's proposal to model transmission constraints to ensure deliverability at a nodal level would increase costs and add unnecessary complexity because, among other reasons,

---

164 *Sw. Power Pool, Inc.*, Notice of Effective Date, FERC Docket No. ER22-914-000 (June 6, 2023).

165 *Sw. Power Pool, Inc.* 180 FERC ¶ 61,088, at P 3 (2022).

166 *Id.* at P 17.

167 *Midcontinent Indep. Sys. Operator, Inc.*, 149 FERC ¶ 61,095, at P 45 (2014) (rejecting protest that a ramp capability product requires comparable settlement treatment to ancillary services).

nodal procurement would require accompanying market power mitigation measures.

Driven by the experience with undeliverable flexible ramping product in the real-time market, the CAISO's preferred approach throughout the process was to procure imbalance reserves nodally to maximize the product's deliverability. During the 2023 workshop process, the CAISO re-examined this question, and published a comparative matrix[168] highlighting the tradeoffs between two major options summarized as follows:

- Nodal approach – procuring imbalance reserves within the IFM (co-optimized with energy and ancillary services) using deployment scenarios to ensure the awards are transmission feasible if deployed as energy.
- Zonal approach – procuring imbalance reserves within the IFM (co-optimized with energy and ancillary services) using zonal procurement similar to ancillary services.

The CAISO considered these concerns, but it maintained the nodal procurement approach with two key revisions to address concerns about the relative costs of imbalance reserves procurement. First, the CAISO proposed the flexible activation/deactivation of individual transmission constraints in deployment scenarios. This will permit the CAISO to consider how specific constraints might drive clearing prices for imbalance reserves and would permit the CAISO and EDAM balancing areas to strike tradeoffs between deliverability and price impacts. Second, the CAISO proposed to use the deployment factor in the imbalance reserves deployment scenarios. Similar to the flexible activation of constraints, the ability to set the deployment factor and reevaluate it on an ongoing basis will permit the CAISO and parties to balance these competing factors.

3. Which Day-Ahead Market Process Should Procure Imbalance Reserves?

Some participants questioned if the CAISO should co-optimize imbalance reserves with energy in the IFM or procure them separately in RUC with reliability capacity. One stakeholder offered a specific counter-proposal for the CAISO to co-optimize procurement of imbalance reserves and reliability capacity in RUC using nodal procurement to respect transmission constraints. The stakeholder's rationale for shifting imbalance reserves to RUC was that it would limit price impacts of imbalance reserve procurement because imbalance reserves would not compete for scarce transmission with energy procured in the IFM.

---

168 http://www.caiso.com/InitiativeDocuments/ComparisonMatrix-Day-AheadMarketEnhancements.pdf.

The CAISO considered this approach but determined it was inadvisable. By not co-optimizing energy and imbalance reserves procurement, the CAISO would lose efficiencies in unit commitment, scheduling, and prices. The stakeholder's suggested approach also would lead to inconsistent price formation and modeling with flexible ramping product because imbalance reserves bids no longer would be co-optimized with energy, whereas flexible ramping product awards would still be co-optimized with energy in the real-time market. For these reasons, the CAISO's proposal to co-optimize imbalance reserves with energy in the IFM is reasonable.

4. Hybrid Demand Curve

The stakeholder process considered multiple options on how to procure imbalance reserves to meet the requirements. At different points, the CAISO proposed various penalty price structures, ranging from demand curves to graduated penalty prices that relax the imbalance reserve requirement as the cost increases, as well as strict penalty prices that protect the full imbalance reserve requirement at higher costs.

The CAISO initially proposed in its Draft Revised Final Proposal published on April 6, 2023, to use the demand curve approach described in section IV.D.3.b of this transmittal letter for EDAM balancing areas, but it instead proposed a hybrid imbalance reserve demand curve for the CAISO balancing authority. Under this approach, the CAISO balancing authority would have met half of its the imbalance reserve requirements by a demand curve and half through high penalty prices. This was intended to address concerns of prioritizing low-priority exports over imbalance reserves for the CAISO balancing area's net load uncertainty, ensuring more predictable export volumes.

In the final stages of the stakeholder process, however, the CAISO updated its proposal and decided not to adopt a hybrid approach for the CAISO balancing area. Further evaluation of this approach revealed that a hybrid approach could lead to high prices that exceed the operational benefit of the product. Given the changes to the CAISO's proposal on export bidding in RUC that allow the market to signal exports may not be feasible in real-time, the demand curve approach reduces the CAISO balancing area's exposure to unpredictable export volumes. The CAISO thus concluded the demand curve approach was appropriate for the CAISO balancing area.

5. Limitations on Imbalance Reserves Awards

Early in the stakeholder process, the CAISO initially proposed limiting a resource's imbalance reserves award to its 15-minute ramp capability because imbalance reserves are procured to meet the actual potential 15-minute ramping needs in responding to uncertainty or ramping needs that materialize intra-hour in real-time. Because the CAISO procures to meet a 15-minute need, it seemed reasonable to limit awards to what a resource can provide in 15 minutes.

This proposal, however, raised stakeholder concerns that limiting awards in this way was too restrictive and would raise costs unnecessarily. In that regard, a portion of real-time net load uncertainty arises with longer lead times and correlate between hours.[169] Consequently, the CAISO concluded a 15-minute ramping restriction might be unnecessarily costly because it would require more resources to provide imbalance reserves and may excessively commit 15-minute resources relative to the real-time operational need. Accordingly, the CAISO extended the limitations on imbalance reserves awards from a resource's 15-minute ramp capability to its 30-minute ramp capability. The CAISO, however, maintained the proposed requirement that resources providing imbalance reserves must be 15-minute dispatchable.

6. Tunable/Configurable Parameters

As the stakeholder process came to completion, multiple stakeholders expressed concern about the difficulty of reaching final positions on the proposal without firmly understanding how the new products would work in practice. The CAISO runs a robust market simulation program before implementing most market design changes. However, these participants were concerned the market simulation would occur after finalization of the policy proposal and its submission to the Commission.

In general, the CAISO notes that significant market design changes often are presented for Commission approval prior to market simulation. Nonetheless, in response to these concerns, the CAISO committed to implementing a range of configurable market parameters in the software design to permit the CAISO to make implementation refinements readily based on experience during the market simulation period and during the initial stages of implementation. The CAISO developed this configurability to provide off-ramps and the ability to pivot quickly should the CAISO identify unintended adverse consequences. Consistent with

---

169 http://www.caiso.com/InitiativeDocuments/DraftRevisedFinalProposal-Day-AheadMarketEnhancements.pdf, p. 10.

the FPA and the Commission's rule of reason, those parameters that significantly affect the CAISO's rates, terms, and conditions of service are defined in the proposed tariff provisions.[170] Examples include the imbalance reserve demand curve cap and the default availability bid values. Although the CAISO's systems would provide flexibility to adjust these values on short notice, the CAISO would specify the values in the tariff and would not be permitted to change them without Commission approval. The CAISO concluded the specific value of other parameters, such as the deployment factor, do not significantly affect terms of service and need not be included in the tariff under the rule of reason. To maintain the capability to address unforeseen issues quickly, the CAISO elected to define these other values in the business practice manual where possible.

These steps towards greater flexibility addressed stakeholder concerns but raised new concerns regarding transparency on how the parameters will be set at the start, how they will evolve or change, and what discretion the CAISO will have in making those changes. To address these issues comprehensively the CAISO provided a detailed matrix identifying how it intends to tune the configurable elements of the design, both during testing, implementation and after go-live.[171] The CAISO also committed to work with the Department of Market Monitoring to report on the performance of alternative parameters and settings before and after implementation of the day-ahead market enhancements.

G. Connection to EDAM

The proposed day-ahead market enhancements are critical not only for the CAISO balancing authority standing on its own but also for the expanded day-ahead market. The new products will help ensure EDAM participants, including the CAISO, can benefit from the footprint-wide diversity in the day-ahead market's optimization. The CAISO's proposed day-ahead market enhancements address the unique requirements of an EDAM design to ensure transfers determined in that market are both economic and reliable.

1. Benefits of Imbalance Reserves in EDAM

One of the key benefits of EDAM is to utilize diverse resources across multiple balancing areas to meet load and operational needs across the West more efficiently. Imbalance reserves will be an important component of EDAM's ability to drive these efficiency benefits. The CAISO commissioned the CAISO EDAM Benefits Study to estimate EDAM benefits. As part of that study, CAISO

---

170 *See, e.g.*, *Cal. Indep. Sys. Operator Corp.*, 119 FERC ¶ 61,053, at P 38 (2007).

171 http://www.caiso.com/InitiativeDocuments/FlexibleParameterMatrix-Day-AheadMarketEnhancements.pdf.

requested a sensitivity study to elaborate on the role of imbalance reserves in the EDAM benefit calculation. The CAISO published this study on November 15, 2022 and held a public webinar to discuss the study results on November 18, 2022. The study results showed imbalance reserve are an important component in realizing the inter-regional dispatch efficiency goals of EDAM. The study found that without the imbalance reserve component, the potential EDAM benefit for the State of California would be about 60 percent lower, with other Western states benefits similarly compromised.

Several factors explain these benefits. First, pooled procurement of imbalance reserves across a wider geographic footprint will reduce EDAM balancing authorities' individual net load uncertainty requirements through the EDAM diversity benefit because the CAISO expects net load uncertainty would not materialize coincidently across the larger and more geographically diverse EDAM market footprint. Second, creating a common market product to address forecast net load uncertainty in the day-ahead timeframe will build confidence in energy transfers between balancing authorities scheduled in the day-ahead market through EDAM. EDAM participants can rely on other EDAM balancing areas to support their transfer obligations because of a common day-ahead market imbalance reserve product. Third, imbalance reserves will more efficiently utilize resource capacity by providing balancing areas access to resources across EDAM.

Besides reducing the overall reserves needed to address net load forecast uncertainty in the day-ahead timeframe, imbalance reserves will more efficiently select the resources to provide these reserves. It will provide EDAM balancing areas access to resources that can provide these reserves at a cost lower than their own resources. In addition, it will provide additional revenue opportunities to balancing areas with more efficient and flexible resources. Imbalance reserve transfers will be firm, ensuring a balancing area can access any imbalance reserves that come from another balancing area to meet its net load uncertainty. Finally, imbalance reserves establish a consistent treatment of uncertainty in the EDAM resource sufficiency evaluation. This ensures equitable evaluation of each balancing area's uncertainty needs.

2. Benefits of Reliability Capacity in EDAM

The rationale for procuring reliability capacity – ensuring physical supply is committed to cover differences in cleared physical supply and net load forecast – applies as much to the CAISO as it applies to other EDAM balancing areas. Using the reliability capacity product, as opposed to some other mechanism, to meet this need across the EDAM footprint will be beneficial. As with imbalance reserves, using a common product across the footprint can minimize the cost of ensuring there is enough physical supply in the EDAM footprint to meet each

balancing area's load forecast by creating a deeper pool of resources from which each balancing area can meet its obligations.

V. The Extended Day-Ahead Market

As a balancing authority, the CAISO fulfills its responsibility to balance supply and demand through its day-ahead and real-time markets, which are synchronized to determine the most economical commitment and dispatch of resources, taking into account system constraints. In recent years, the CAISO has worked to integrate resources located within its balancing authority, including variable energy resources, into the markets the CAISO utilizes to allocate transmission capacity, manage congestion, enable spot energy trading, and reliably operate the CAISO controlled grid. Consistent with the Commission's long-standing direction, the CAISO has worked to remove barriers to entry to the day-ahead market to allow participation by all resources technically capable of providing the products it procures. Simultaneously, the CAISO has advanced numerous initiatives to enhance its markets to reflect price formation and product enhancements, including extending its real-time market to other balancing authorities in the Western Interconnection.

In 2014, the Commission issued the WEIM Authorization Order[172] and the WEIM Rehearing Order,[173] accepting the CAISO's proposal to allow balancing authorities outside of the CAISO to participate voluntarily in the imbalance energy portion of the LMP-based real-time market the CAISO operates within its own balancing area. Since initiation of the WEIM, the CAISO has integrated 22 balancing areas into the advanced market system and established governance structure.[174] The extension of the day-ahead market through EDAM is a direct descendent of these collective efforts.[175]

---

172 *See* WEIM Authorization Order 147 FERC ¶ 61,231 (accepting the CAISO tariff amendment to extend its real-time market to other balancing areas).

173 *See* WEIM Rehearing Order, 149 FERC ¶ 61,058 (order denying requests for rehearing and granting in part and denying in part requests for clarification of the WEIM Authorization Order).

174 *See* https://www.westerneim.com/Pages/About/default.aspx.

175 The EDAM framework builds upon the tariff framework developed for the WEIM. In essence, new tariff section 33 houses the majority of the EDAM-specific provisions and is numbered in parallel with other sections of the existing tariff, while other sections have been amended to account for EDAM-specific elements. This tariff framework supports the "applies to" test that defines the scope of joint authority by the WEIM Governing Body and the Board of Governors. *See* EDAM Governance Review Committee Final Proposal, available at https://www.westerneim.com/Documents/EDAM-Governance-Final-Proposal-WEIM-Governance-Review-Committee-Phase-3.pdf.

A. Participation Framework

A balancing authority can elect to provide customers the substantial economic, reliability, and environmental benefits of participating in the CAISO day-ahead market through the EDAM framework proposed in this filing.[176] Notably, the extended day-ahead market is not a new market, rather it takes advantage of the existing CAISO day-ahead market by adding new procedures to accommodate the voluntary participation of other balancing authorities by extension of that market without disrupting the current market structure.[177] Like the WEIM, participation in the extended day-ahead market is projected to provide substantial economic benefits for customers, stemming primarily from the availability of a broad pool of resources to serve load and savings from replacing existing manual processes with the CAISO's automated process. Also, EDAM will provide a wider range of tools to maintain reliability in the face of changing system conditions in the Western U.S. and significant public policy benefits by better integrating variable energy resources.[178]

Participation in the extended day-ahead market by a balancing authority is entirely voluntary, and a balancing authority can terminate its participation with a six-month notice and without exit fees. The primary prerequisite for balancing authorities to join the extended day-ahead market is that the balancing authority already be participating in the WEIM, or that the balancing authority will join the WEIM concurrently with its joining EDAM. A balancing authority that chooses to participate in EDAM will remain responsible for maintaining the reliability of its balancing area, including meeting operating reserve and capacity requirements, scheduling, curtailment of the transmission facilities under its operational control, and manually dispatching resources to maintain reliability. Functional separation of participants in EDAM will remain similar to the WEIM, with each participating entity retaining its associated functions.[179] Retaining these functions empowers participating entities to continue their organizational operation as they do today, while at the same time supporting their participation in a multi-balancing-area day-ahead market. This participation model complements the WEIM and allows

---

176 See section II.B of this transmittal letter for a discussion of the benefits of EDAM participation, including estimated financial benefits ranging between $100 million and more than $1 billion annually.

177 In this regard, EDAM builds on the principles of the Commission-approved WEIM. *See* WEIM Authorization Order, 147 FERC ¶ 61,231, at P 74.

178 *See id.* at P 75.

179 *See* new tariff section 33.4(a)-(c); *compare* existing tariff section 29.4(a)-(b) *with* existing tariff appendix B.17, *pro forma* EIM Entity Agreement, section 2.1; *see also* new tariff sections 33.4.3, 33.4.5, and 33.4.7 (requiring the scheduling coordinator to be separate from other functions unless subject to the standards of conduct, similar to the WEIM).

for coordinated participation, including a balancing authority's continued participation only in the WEIM. This means the CAISO will simultaneously support three approaches for interested parties to participate in CAISO markets – real-time market participation only (*i.e.*, WEIM participation), day-ahead market and real-time market participation together[180] (*i.e*., WEIM and EDAM participation), and full participation with the CAISO when a transmission-owning utility becomes a participating transmission owner in the CAISO balancing area.[181]

1. Implementation, Procedures, and Fees

As in the WEIM, a balancing authority seeking to participate in the extended day-ahead market must enter into an implementation agreement that establishes the implementation date for participation in the day-ahead market, requires the CAISO to perform changes to its systems, and specifies the implementation fee so the CAISO can recover its costs.[182] These implementation activities include planning and project management, administrative management, full network modeling of resources, system integration and functional testing, metering and settlements, and operations readiness and training. The implementation process also includes system integration and functional testing, as well as training, market simulation and a period of parallel operations. The proposal also extends the readiness framework established for the WEIM with tailored updates and modifications for EDAM. This will allow the CAISO and participants to confirm they are prepared

---

180 *See* new tariff sections 29.1(a)-(c), 29.11(a), 29.29, 29.31, and 29.34 (integrating participation in the day-ahead and real-time markets by establishing the relationship between tariff section 29 and tariff section 33).

181 The Valley Electric Association was the most recent utility outside the CAISO balancing area to join the CAISO as a participating transmission owner. *See Cal. Indep. Sys. Operator Corp.*, 137 FERC ¶ 61,194 (2011) (accepting Transition Agreement that sets forth the terms and conditions for Valley Electric Association to make the transition from the Nevada Power Company balancing area to the CAISO balancing area); *Cal. Indep. Sys. Operator Corp.*, Commission Letter Order, Docket No. ER12-2623-000 (Oct. 15, 2012) (accepting amendment to Transition Agreement); *Cal. Indep. Sys. Operator Corp.*, Commission Letter Order, Docket No. ER13-586-000 (Jan. 17, 2013) (accepting second amendment to Transition Agreement); *Cal. Indep. Sys. Operator Corp.*, Commission Letter Order, Docket No. ER17-695-000 (Feb. 15, 2017) (accepting third amendment to Transition Agreement).

182 *See* section V.A.1 of this transmittal letter.

for production operations, with the option for extension of the implementation date if necessary.[183]

The CAISO will recover its costs to facilitate a new balancing authority joining EDAM through an implementation fee based on its cost of service. For example, the CAISO will apply its fully burdened rate calculated based on the most recently published triennial cost of service study.[184] The actual onboarding costs per balancing area will vary depending on the size and complexity of the effort, and the CAISO will collect a $300,000 deposit from a balancing authority seeking to join the extended day-ahead market to cover the actual start-up costs incurred.[185] The CAISO estimated this deposit amount based on the costs the CAISO anticipates incurring for an average sized balancing area interested in EDAM participation. If the deposit exceeds the actual cost incurred, the CAISO will refund the excess amount including any interest accrued on the remaining deposit. If the actual cost of performing the onboarding service exceeds the deposit, additional deposits in $300,000 increments will be required. Any invoice payment past due will accrue interest, per annum, calculated in accordance with 5 C.F.R. § 1315.10.[186] If the balancing authority joining EDAM fails timely to pay any undisputed costs, the CAISO will not be obligated to continue implementation activities unless and until all undisputed amounts are paid. After successful implementation, the CAISO will provide a report that details deposit(s) received, actual costs incurred, and applicable interest earnings (on deposit balance).

The ongoing administrative costs of EDAM services will consist of the CAISO's existing market services charge and a new system operations charge, both volumetric charges.[187] The market services charge represents fees for the real-time market and the day-ahead market services that the extended day-ahead market offers, and applies to awarded MWh of energy and MW of capacity.[188] The systems operations charge represents the fees for real-time dispatch services, and it applies to metered flows in MWh of supply and demand. Assessing an EDAM systems operations charge in this manner is reasonable

---

183 *See generally* new tariff section 33.2 (establishing a readiness framework for EDAM with requirements appropriate for a participant already in the WEIM); *compare with* existing tariff section 29.2(b) (including readiness procedures for the WEIM).

184 http://www.caiso.com/InitiativeDocuments/2023-Cost-of-Service-Study-and-2023-GMC-Update-Jul122023.pdf.

185 New tariff section 33.11.5.

186 This interest rate is also used in comparable circumstances in existing tariff section 29.11(g)(2).

187 New tariff section 33.11.6 and existing tariff appendix F, schedule 1, revised part A.

188 As noted above, balancing authorities participating in EDAM must also participate in the WEIM.

because it represents the CAISO's efforts to manage the dispatch of energy and capacity in day-ahead and real-time. Once a balancing authority begins participating in the extended day-ahead market, it will no longer pay WEIM administrative fees, it will only pay EDAM administrative fees, which cover services provided by the CAISO for both the day-ahead and real-time markets. As participation in EDAM grows, the increase in volumes will contribute to lower market services and system operations rates, all other factors held constant. Under this approach, all customers receiving market services (including customers in the existing CAISO footprint) will benefit from a lower market services rate because the market service charge will be calculated using the incremental day-ahead MWh volumes from participants.

2. Transitional Protective Measures

Several WEIM transitional measures guard against potential adverse reliability or market outcomes in a particular balancing area or in the broader market area. The CAISO proposes to extend these existing measures to the extended day-ahead market for a similar timeframe so the CAISO and participants have mechanisms to respond to potentially adverse reliability or market outcomes during the initial period these participants are integrating into the CAISO's day-ahead market.

Prior to implementation, each balancing authority joining EDAM can request a change in its implementation date if it determines it cannot proceed on that date.[189] Over the years, WEIM entities have exercised their right to change their implementation date during WEIM implementation due to lack of readiness or other factors, such as the timing of requisite approvals. The CAISO believes prospective EDAM entities should have a similar opportunity.[190]

The CAISO may, within 60 days following the implementation date, temporarily suspend participation of a balancing authority in EDAM for a period not to exceed 60 days.[191] Although the CAISO as market operator institutes the suspension, the CAISO would make the decision for any temporary suspension only after consultation and coordination with the participant. This suspension feature has yet to be exercised following a balancing area implementation in the

---

189 New tariff section 33.2.7; *compare with* existing tariff section 29.2(b)(6)(B).

190 The CAISO tariff refers to a balancing authority participating in EDAM as an EDAM entity. This transmittal letter generally refers to balancing authorities participating in EDAM.

191 New tariff section 33.1.1; *compare with* existing tariff section 29.1(d). This provision has never been invoked in the WEIM during its nearly ten years of operation, and would likely only be invoked in response to a significant and unexpected market or system operational issue arising at the onset of EDAM participation.

WEIM but remains a prudent safeguard at the outset of EDAM operation within a balancing area.

The WEIM transitional price discovery mechanism is another important, existing feature that will also apply to the extended day-ahead market. For a period of six months following implementation, the CAISO will not apply certain transmission constraints and will relax certain transmission and/or power balance constraints.[192] In these circumstances, the CAISO will determine prices consistent with other provisions of its tariff, effectively substituting the last economic bid for what would otherwise be a parameter price. This feature has helped balancing areas transition to the WEIM and will similarly help them transition to EDAM.

The CAISO also proposes to extend its day-ahead price correction authority from five business days to 10 business days for a three-month period following an EDAM implementation date.[193] Extending the window for the CAISO to assess and implement EDAM price corrections will help facilitate resolution of any lingering implementation-related issues. This protective measure has been a valuable tool in the WEIM, and it should apply to EDAM for the same reason.

Finally, the CAISO will apply an EDAM measure that is available beyond initial participation in the WEIM. Specifically, the CAISO may interrupt balancing area participation in the market when operational circumstances have caused or are likely to cause abnormal system conditions that require immediate action to maintain system reliability or there is a communication failure that prevents access to CAISO systems.[194] In these instances, although the CAISO makes the ultimate decision, the CAISO will coordinate closely with the affected balancing authority to assess whether interruption is necessary to ensure reliable operation. This measure will also extend to the CAISO balancing area if isolation from the rest of the EDAM market area is necessary, while supporting the separate running of the day-ahead market within its own balancing area. This is important in limited circumstances to allow interruption of the market and maintain system reliability.

---

192 New tariff section 33.27.1; *compare with* existing tariff section 29.27(b). This transitional pricing period may be extended beyond six months, but an extension requires Commission approval.

193 New tariff section 33.35; *compare with* existing tariff section 29.35.

194 New tariff section 33.7.4; *compare with* existing tariff section 29.7(j). The CAISO has exercised this WEIM authority. For example, the CAISO has temporarily interrupted a balancing area's participation in the WEIM due to transmission outages that electrically separated two participating balancing areas.

3. Participants in Each Balancing Area

There will be four identified participant roles in each balancing area participating in the extended day-ahead market, versus two in the WEIM. A scheduling coordinator must represent each participant undertaking these roles.[195]

The first role is a balancing authority that enables EDAM participation in its balancing area. The CAISO refers to this participant as an EDAM entity, and it will be an extension of the WEIM entity function.[196] Accordingly, during the implementation process, an existing or new WEIM entity will enter into EDAM Entity Implementation Agreement with the CAISO and an addendum to its EIM Entity Agreement with the CAISO, which agreements will be effective within the balancing area according to their terms. The second participant role is a resource owner or operator, referred to as an EDAM resource, which will enter into an addendum to its EIM Participating Resource Agreement with the CAISO. The third participant role is an EDAM transmission service provider, which will enter into an EDAM Transmission Service Provider Agreement with the CAISO.[197] The fourth participant role is an EDAM load serving entity, which will enter into an EDAM Load Serving Entity Agreement with the CAISO.[198]

4. Participation in EDAM Versus the WEIM

The CAISO's day-ahead market optimizes all supply and demand within a balancing area, including settlement of the day-ahead market awards, operation of the real-time market based on the day-ahead market schedules, and settlement of differences between day-ahead market schedules and real-time market results. This structure requires participation in the day-ahead market, through either an economic bid or a self-schedule, by all supply and demand within a balancing area; otherwise, the day-ahead market would not clear as designed. The extended day-ahead market accounts for these differences through additional roles and responsibilities applicable to the associated day-ahead market functions and integration of these roles and responsibilities with the associated real-time market functions.

---

195 New tariff section 33.4; *compare with* existing tariff section 29.4.

196 Tariff appendix A, new definition of EDAM Entity.

197 In many cases, a balancing authority participating in EDAM as an EDAM entity will be the only or most significant transmission service provider in a balancing area. Due to the need to coordinate with transmission service providers on certain aspects of EDAM, including transmission availability issues, the CAISO has determined it will be beneficial to have a direct contractual relationship with all transmission service providers in the EDAM area.

198 The various types of agreements described above (including scheduling coordinator agreements) are discussed in greater detail in section V.A.6 of this transmittal letter.

In the WEIM, resources are divided into: (1) those eligible to participate and registered to submit bids into the real-time market, referred to as EIM participating resources, and (2) all other resources within the WEIM balancing area, which are registered as non-participating resources and are ineligible to submit bids in the real-time market. The balancing authority submits base schedules to represent the expected operation of non-participating resources and intertie transactions in the WEIM. The CAISO issues dispatch instructions only to WEIM participating resources and settles the delivered energy and imbalance energy directly with the scheduling coordinator for the participating resource. The CAISO settles imbalances from the base schedule with the balancing authority, not the resource owner or operator. Base schedules also reflect forecasted demand within the balancing area, and the CAISO settles imbalance between the base schedule and the metered quantity with the balancing authority. This structure is reasonable for operating a real-time market providing imbalance energy service within a participating balancing area within the operational hour, but this structure is insufficient for extension of the day-ahead market because all supply and demand resources must be available for the optimization to clear within each balancing area and establish feasible energy and capacity transfers between balancing areas.

All resources in a balancing area participating in EDAM will be required to submit bids or self-schedules into the day-ahead market,[199] which will produce an end result similar to base scheduling in the WEIM through self-scheduling of supply and the exercise of existing transmission rights. A self-schedule in the market indicates the resource does not have an economic offer expressing a willingness for the market to optimize and commit it. Self-scheduled resources are price-takers that want their output to flow irrespective of market prices, and they typically have existing transmission rights to deliver that generation. Because they will be obligated to submit bids or self-schedules, resources that are non-participating resources in the WEIM and currently represented by a WEIM entity scheduling coordinator must establish a resource-specific scheduling coordinator relationship with the CAISO as an EDAM resource. It will be unnecessary for EDAM participants to submit base schedules in the real-time market because EDAM participants will have day-ahead market schedules that serve the purpose of WEIM base schedules.

---

[199] As a result of this different market design approach, the tariff terminology for resources in EDAM is different from the tariff terminology for resources in the WEIM. An EDAM Resource Facility is the physical resource in an EDAM balancing area that can deliver energy, curtailable demand, demand response services, or similar services. An EDAM Resource is the owner or operator of an EDAM Resource Facility. Tariff appendix A, new definitions of EDAM Resource and EDAM Resource Facility. By comparison, an EIM Resource is the physical resource, and an EIM Participating Resource is the owner or operator of an EIM Resource. Tariff appendix A, existing definitions of EIM Resource and EIM Participating Resource.

The CAISO day-ahead market clears supply and demand bids, which requires load serving entities within a balancing area in EDAM to self-schedule or economically bid their demand. As the starting point, the CAISO would look to the scheduling coordinator for demand within the balancing area to bid and self-schedule. Having a single scheduling coordinator for all load serving entities in a balancing area participating in EDAM supports the coordination required for forecasting and the reliable and efficient submission of demand bids into the day-ahead market. The CAISO can also enable individual load serving entities within a balancing area to represent their demand in the market separately from the rest of the demand in the balancing area. The individual load serving entity would need to work with the balancing authority and the CAISO through the implementation process to model its demand separately. Separating individual load serving entities within a balancing area will require separate metering that satisfies the standards of the balancing authority and the CAISO tariff. Additionally, the load serving entities would need a scheduling coordinator to represent their demand in the market and submit additional information, including meter data, to support settlement, and enter into an EDAM Load Serving Entity Agreement.[200]

5. Stakeholder Positions on the EDAM Participation Framework

Stakeholders broadly supported a voluntary participation model for EDAM, similar to the WEIM, which does not bind participants to lengthy participation timeframes and provides for ease of exit if necessary. Stakeholders also sought and broadly supported application of WEIM transitional protective measures to the extended day-ahead market that allow entities and the CAISO to manage participation risk to the extent there are unexpected operational, reliability, or financial impacts from participation. Stakeholders largely supported the role and function of the day-ahead market and incorporation of the proposed enhancements to EDAM.

Some stakeholders sought the ability to use base schedules for resources in EDAM, similar to the WEIM. The CAISO explained, however, that to provide the benefits of the day-ahead market to customers in EDAM, unlike the WEIM, all resources in a balancing area must be represented in the market and submit bids (economic bids or self-schedules). Such resources will receive a day-ahead schedule that serves the same purpose as a base schedule, thus removing the option to submit base schedules. Base schedules are reasonable for operating a

[200] As discussed in section V.C.1 of this transmittal letter, balancing authorities participating in EDAM may elect at the onset of their participation whether to implement virtual bidding in their balancing areas in the day-ahead market. In balancing areas that implement convergence bidding, load-serving entities will also be able to engage in convergence bidding to promote convergence between day-ahead and real-time prices.

real-time market providing imbalance energy service within a participating balancing area, but this structure is not sufficient for extension of the day-ahead market because all supply and demand resources must be available for the optimization to clear within each balancing area and establish feasible energy and capacity transfers between balancing areas.

Other stakeholders sought assurance resources can participate in the extended day-ahead market without modification of underlying supply contracts. The CAISO clarified that, assuming the entity holding the supply contract and the entity serving its load with the resource have the same scheduling coordinator, the CAISO settlement with the single scheduling coordinator for any energy payments can be managed and redistributed in accordance with the terms of its contract to ensure the contract price is covered. The scheduling coordinator can then manage redistribution of those payments in accordance with the terms of its contract to ensure the contract price is covered. This treatment would support EDAM settlements for resources subject to regulatory must-take requirements or with special contractual provisions regulated by state commissions without the need for fundamental contractual changes.[201]

[201] Some contracting parties may wish to enter into supporting arrangements to settle differences between the market price and the contract price, which are relatively common and will not alter fundamental supply arrangements. The CAISO will continue to support inter-scheduling coordinator trades within its balancing area but will not extend this feature to EDAM, which is consistent with the WEIM.

6. EDAM Agreements

a. Overview of EDAM *Pro Forma* Agreements and Addenda

The CAISO is including in Appendix B of the CAISO tariff various *pro forma* agreements, and addenda to existing *pro forma* agreements, that will enable participation in the day-ahead market by an EDAM entity, EDAM entity scheduling coordinator, EDAM resource, EDAM resource scheduling coordinator, EDAM transmission service provider, and EDAM load serving entity as follows.

b. EDAM Entity Implementation Agreement

A balancing authority that currently participates in the WEIM and seeks to participate in EDAM, or that seeks to begin participating in the WEIM and EDAM concurrently, must first execute a *pro forma* EDAM Entity Implementation Agreement with the CAISO. This agreement requires the prospective EDAM entity to compensate the CAISO for the start-up costs necessary to incorporate the EDAM entity into the day-ahead market. The agreement also establishes the implementation date for participation in the day-ahead market and requires the prospective EDAM entity to sign a *pro forma* addendum to its existing EIM Entity Agreement no later than 90 days prior to its implementation date.

Under the agreement, the CAISO will perform changes to its systems, in accordance with the CAISO tariff, to allow the prospective EDAM entity to participate in the day-ahead market. In addition to the implementation activities and fee discussed in section V.A.1 of this transmittal letter, the prospective EDAM entity must satisfy the CAISO tariff requirements applicable to an EDAM entity prior to its implementation date. Specifically, the prospective EDAM entity must: (1) demonstrate that it satisfies all qualifications for participation as an EDAM entity, (2) show that it is authorized to make transmission available in its balancing area consistent with the CAISO tariff and the applicable transmission service tariffs, contracts, rules, procedures, or other arrangements, (3) enter into an EDAM addendum to its EIM Entity Agreement with the CAISO governing its participation in the day-ahead market, and (4) secure representation by an EDAM entity scheduling coordinator.

The CAISO may terminate the EDAM Entity Implementation Agreement: (1) under new tariff section 33.1; or (2) if the prospective EDAM entity materially defaults under the agreement or new tariff section 33 and does not cure the default within 30 days. The prospective EDAM entity may terminate the agreement, without penalty, by giving at least 30 days written notice to the CAISO. In the event of termination, the CAISO will make every effort to halt work and related costs on the implementation as soon as practical and refund any

payments provided by the prospective EDAM entity in excess of costs already incurred by the CAISO.

The agreement also discusses how the CAISO will settle deposits, including any refunds owed. In addition, the EDAM Entity Implementation Agreement provides the prospective EDAM entity the right to audit the CAISO's records to verify costs incurred by the CAISO. The remaining terms are identical to or consistent with those contained in other similar *pro forma* agreements.

c. EDAM Addendum to EIM Entity Agreement

Any balancing authority that seeks to participate in EDAM must enter into a *pro forma* EDAM Addendum to its EIM Entity Agreement with the CAISO. The prospective EDAM entity must execute the addendum no later than 90 days before its designated implementation date.

This addendum incorporates new tariff section 33 into the underlying EIM Entity Agreement, thereby allowing an EIM entity also to participate as an EDAM entity in the day-ahead market. The addendum obligates the CAISO to provide open access to the day-ahead market under the terms of the CAISO tariff. In turn, the EDAM entity will have provisions in its tariff to enable operation of the day-ahead market in accordance with the CAISO tariff. The balancing authority also must ensure it is represented by an EDAM entity scheduling coordinator, which may be the EDAM entity, itself, or another entity certified by the CAISO to perform the requisite function. Under the addendum, the EDAM entity must provide information to the CAISO regarding the network topology of its facilities within its balancing area, non-participating resources, and loads in accordance with the CAISO tariff. The EDAM entity also will make available for use in the day-ahead market transmission capacity on its system that is not otherwise encumbered, reserved, scheduled, or being used by its transmission customers or others. Finally, the EDAM entity will provide the CAISO with real-time information regarding the availability of transmission capacity for use in the day-ahead market.

Once the addendum becomes effective, it will remain in effect until terminated with 180 days' notice pursuant to the same process set forth in the underlying EIM Entity Agreement. If the balancing authority terminates its participation in EDAM, it may continue to participate in the WEIM under the terms of the underlying EIM Entity Agreement.

d. EDAM Addendum to EIM Entity Scheduling Coordinator Agreement

Any EIM entity scheduling coordinator that wishes to act as an EDAM entity scheduling coordinator must execute a *pro forma* Addendum to the EIM Entity Scheduling Coordinator Agreement.

The addendum, which incorporates new tariff section 33, requires the EDAM entity scheduling coordinator to perform all associated obligations under tariff section 33, including obligations related to scheduling, settlement, system security policy and procedures, billing and payments, confidentiality, compliance with standards of conduct requirements, and dispute resolution.[202] The EDAM entity scheduling coordinator also will have primary responsibility to the CAISO for the payment obligations under tariff section 33. In addition, it will ensure that each EDAM entity that it represents executes an EDAM Addendum to the EIM Entity Agreement.

Once effective, the addendum to the EIM entity scheduling coordinator agreement will remain in effect until terminated under the same process set forth in the underlying agreement. If the EDAM entity scheduling coordinator terminates its participation in EDAM, it may continue to participate in the WEIM as an EIM entity scheduling coordinator under the terms of the underlying EIM Entity Scheduling Coordinator Agreement.

An EDAM entity scheduling coordinator must have the same scope for its EDAM participation as it has for its corresponding relationship with the EIM entity it represents in the WEIM, *i.e.*, the scope of participation in the day-ahead market must correlate directly with the scope of participation in the real-time market to ensure alignment of scheduling coordinator's responsibilities in the CAISO markets.

e. EDAM Addendum to EIM Participating Resource Agreement

The owner or operator of each resource in an EDAM balancing area must participate in the day-ahead market as an EDAM resource and execute a *pro forma* Addendum to the EIM Participating Resource Agreement. The addendum, which incorporates new tariff section 33, requires the EDAM resource to perform

---

[202] Consistent with new tariff section 33.4.3, a governmental entity that is an EDAM entity scheduling coordinator may also be an EDAM resource scheduling coordinator or a scheduling coordinator for one or more EDAM load serving entities if it agrees to comply with standards of conduct equivalent to those set forth in 18 C.F.R. § 358.

all associated obligations under tariff section 33. Owners and operators of resources in a balancing area participating in EDAM that are not currently EIM participating resources are required to enter into both a new EIM Participating Resource Agreement and an EDAM addendum to that agreement. Each EDAM resource also must be represented by an EDAM resource scheduling coordinator.

The EDAM resource also must register its resource with the CAISO as an EDAM resource facility. Accordingly, the addendum requires listing in Schedule 1 of the underlying EIM Participating Resource Agreement all EDAM resource facilities not already included in Schedule 1 as an EIM participating resource, including the operating characteristics of the EDAM resource facilities. The EDAM Resource must notify the CAISO of any changes to these characteristics under the requirements in the underlying EIM Participating Resource Agreement.

Once the addendum becomes effective, it will remain in effect until terminated under the same process set forth in the underlying EIM Participating Resource Agreement. If the EDAM resource terminates its participation in EDAM, it may continue to participate as an EIM participating resource under the terms of the underlying EIM Participating Resource Agreement.

f. EDAM Addendum to EIM Participating Resource Scheduling Coordinator Agreement

Any EIM participating resource scheduling coordinator that wishes to act as an EDAM resource scheduling coordinator must execute a *pro forma* Addendum to the EIM Participating Resource Scheduling Coordinator Agreement. A new entity that wishes to act as an EDAM resource scheduling coordinator must execute both the EIM Participating Resource Scheduling Coordinator Agreement and the addendum.

The addendum, which incorporates new tariff section 33, requires the EDAM resource scheduling coordinator to perform all associated obligations under tariff section 33, including obligations related to scheduling, settlement, system security policy and procedures, billing and payments, confidentiality, compliance with standards of conduct requirements, and dispute resolution. The EDAM resource scheduling coordinator also will have primary responsibility to the CAISO for the payment obligations under tariff section 33. In addition, it will ensure that each EDAM resource for which it submits bids enters into an Addendum to the EIM Participating Resource Agreement and an EIM Participating Resource Agreement where one is not already in effect.

Once effective, the addendum will remain in effect until terminated pursuant to the same process set forth in the underlying EIM Participating Resource Scheduling Coordinator Agreement. If the EDAM resource scheduling coordinator terminates its participation in EDAM, it may continue to act as a scheduling coordinator in the WEIM as an EIM participating resource scheduling coordinator under the terms of the underlying EIM Participating Resource Scheduling Coordinator Agreement. An EDAM resource scheduling coordinator must also have the same scope for its EDAM participation as it has for its corresponding relationship with the EIM participating resources it represents in the WEIM.

g. EDAM Transmission Service Provider Agreement

An EDAM transmission service provider is a transmission service provider that owns transmission or has rights to provide transmission service on an EDAM intertie or within an EDAM balancing area that makes transmission service available for use in the day-ahead market through an EDAM entity.[203] This is a new role for EDAM to clarify the roles associated with transmission available in the day-ahead market. An entity may not qualify as an EDAM transmission service provider solely by being a network integration transmission service customer of another entity. All transmission service providers within a balancing authority participating in EDAM must execute the *pro forma* EDAM Transmission Service Provider Agreement.

The agreement, which incorporates new tariff section 33, requires the EDAM transmission service provider to perform all associated obligations under tariff section 33, including having provisions in its tariff to enable operation of the day-ahead market and ensuring scheduling coordinators represent its transmission customers that submit schedules in the day-ahead market. The EDAM transmission service provider must use the EDAM entity scheduling coordinator for its balancing area as its sole scheduling coordinator. This ensures a single EDAM point of contact for balancing authority and transmission owner communications regarding operation of the transmission facilities in the balancing area. The EDAM transmission service provider must provide information about transmission capacity available to the day-ahead market to its EDAM entity scheduling coordinator and the CAISO.

The CAISO may terminate the agreement if (1) the EDAM transmission service provider materially defaults under the agreement or tariff section 33 and does not cure the default within 30 days, or (2) the EDAM entity for the balancing area where the EDAM transmission service provider is located terminates participation in EDAM. If the EDAM transmission service provider will no longer

---

[203] Tariff appendix A, new definition of EDAM Transmission Service Provider.

make transmission service available in the day-ahead market, it may terminate the agreement on 180 days' notice, provided the termination must be concurrent with the termination of participation in the day-ahead market by the EDAM entity for the balancing area where the EDAM transmission service provider operates. The remaining terms in the agreement are identical to or consistent with those contained in other similar *pro forma* agreements.

h. EDAM Load Serving Entity Agreement

The EDAM load serving entity is responsible for demand within an EDAM balancing area, and the EDAM entity must authorize it to represent its load in the day-ahead and real-time markets.[204] This is a new role for EDAM to clarify the roles associated with representation of demand in the day-ahead market. A load serving entity that wishes to act as an EDAM load serving entity must execute the *pro forma* EDAM Load Serving Entity Agreement.

The agreement requires the EDAM load serving entity to provide information regarding its load, including the submission of bids and settlement of demand, to the CAISO for day-ahead market purposes in accordance with the CAISO tariff. The EDAM load serving entity also is required to be represented by an EDAM load serving entity scheduling coordinator, which may be the EDAM load serving entity or another entity certified by the CAISO to be an EDAM load serving entity scheduling coordinator.[205]

The CAISO may terminate the agreement if (1) the EDAM transmission service provider materially defaults under the agreement or new tariff section 33 and does not cure the default within 30 days, or (2) the EDAM entity for the balancing area in which the EDAM load serving entity is located terminates participation in EDAM. If the EDAM load serving entity no longer wishes to participate in EDAM, it may terminate the agreement by giving at least 90 days written notice to the CAISO. The remaining terms in the agreement are identical to or consistent with those contained in other similar *pro forma* agreements.

7. Metering and Telemetry

---

[204] It may be that an EDAM entity represents most or a load in a balancing area, in which case the EDAM entity would execute the EDAM Load Serving Entity Agreement in addition to the other agreements the EDAM entity must execute, as described above.

[205] The CAISO has not proposed a new agreement for an EDAM load serving entity scheduling coordinator, which may use the services of any scheduling coordinator, including the EDAM entity scheduling coordinator.

The CAISO's metering provisions are set forth in existing tariff section 10. This section will apply in the EDAM area, meaning that scheduling coordinators in new EDAM balancing areas must ensure compliance with the CAISO's metering provisions.[206] All EDAM resources – including generating units, storage resources, distributed energy resource providers, demand response providers, and interties – must be CAISO metered entities or scheduling coordinator metered entities and comply with existing tariff section 10.[207] This broad application of existing rules ensures fairness and accurate metering across a new market footprint.

To ensure compliance across new tie points, the CAISO proposes to require each balancing authority participating in EDAM to ensure the separate metering of any load aggregation point in its balancing area not represented by the EDAM entity scheduling coordinator so that associated demand can be settled.[208]

The CAISO also proposes to accommodate any legacy generating units in the EDAM footprint that have not updated to modern metering systems. In the CAISO, this has not been an issue because local regulators' metering requirements have kept pace with the CAISO's metering requirements. However, other states may have units where the costs to update the metering of legacy units would outweigh the benefits, principally with units soon to retire. As such, the CAISO proposes to provide a narrow exception to its metering granularity requirement: scheduling coordinators for EDAM resources that cannot meter the EDAM resource facility's energy every 15 minutes or faster may not submit economic bids or provide ancillary services, and they must submit self-schedules in the day-ahead and real-time markets.[209] This will allow legacy units to participate in the day-ahead market without requiring expensive upgrades, and it will protect against inaccurate price signals and settlement data.

---

206 New tariff section 33.10.

207 New tariff sections 33.10 and 33.10.3. *See also* transmittal letter for tariff amendment to implement metering rules enhancements initiative, Docket No. ER17-949-000, at 1-6 (Feb. 8, 2017) (discussing the CAISO's metering rules comprehensively). This tariff amendment was accepted by a letter order issued on March 31, 2017. For CAISO metered entities, the CAISO directly polls the meters and performs the validation, estimation, and editing procedures to produce settlement qualify meter data. For scheduling coordinator metered entities, the scheduling coordinator polls the meters, performs the validation, estimation, and editing procedures, and submits the resulting settlement qualify meter data to the CAISO. Resources elect to be one or the other based on myriad factors, including the complexity of their metering configuration, local regulatory authority requirements, and power purchase agreement requirements.

208 New tariff section 33.10.1.

209 New tariff section 33.10.2.

Based on the CAISO's discussions with EDAM transmission service providers and resource owners, the CAISO expects very few resources will need to use this provision, and even those few will likely only need to use the provision for a few years before retirement or retrofitting.

As with metering, the CAISO proposes to apply the same telemetry requirements in the CAISO and the WEIM to EDAM.[210] This means EDAM resources must satisfy communications, telemetry, and control requirements in a manner that ensures that the CAISO and EDAM entities are able to monitor the operations of the EDAM resource as necessary to maintain reliability.[211] Just as in the CAISO, an EDAM resource facility will be exempt from telemetry requirements if it has a rated capacity of less than 10 MW, unless it seeks to provide ancillary services.[212] EDAM resource facilities must comply with any EDAM entity or local regulatory authority requirements in addition to the EDAM telemetry provisions in the CAISO tariff.[213]

B. Pre-Market Activities

1. Ensuring Sufficient Transmission Is Available for the Day-Ahead Market

There must be sufficient available transmission capability to support optimized unit commitment in the day-ahead market and transfers between balancing areas for the extended day-ahead market to deliver the expected benefits to a diverse group of participants over a broad area. The CAISO proposal achieves this goal by balancing two key objectives regarding transmission availability. First, the design seeks to maximize the transmission capability available to the day-ahead market and thus improve efficiency and outcomes for all participants. Second, the design respects the rights of transmission customers and facilitates their participation in the market. The extended day-ahead market does not include financial transmission rights and thus not require or permit transmission service providers or their customers to

---

210 New tariff section 33.10.4. This provision mirrors the CAISO's own telemetry provision in existing tariff section 7.6.1, but with minor edits for clarity. For example, tariff section 7.6.1 refers to a "direct communication link," but modern Inter-control Center Communications Protocols enable data exchange over wide area networks between control centers, so tariff section 33.10.4 instead refers more generally to "control requirements."

211 New tariff section 33.10.4.

212 *Id*.

213 *Id.*

convert any transmission rights under existing tariffs to tradable or financial rights.[214]

The changes proposed to the CAISO tariff will establish an EDAM market transmission availability framework that: (1) prescribes the use of transmission across the broader EDAM area in the day-ahead market to improve outcomes for customers, and (2) harmonizes an ISO/RTO day-ahead market structure with transmission customer rights utilized to serve load and engage in the bilateral market under a *pro forma* OATT structure.

The transmission availability framework will require each transmission service provider in an EDAM balancing area to amend its tariff to account for the market availability of transmission on its system consistent with the EDAM provisions of the CAISO tariff, thus enabling operation of the day-ahead market within a balancing area that joins EDAM.[215] Transmission service providers in EDAM balancing areas will continue to maintain and administer their approved OATTs, as the CAISO will due under its tariff within the CAISO balancing area. At the same time, the CAISO and other transmission service providers will continue to sell transmission on their systems and will consider the specific matters their individual tariffs will need to address for their balancing area.[216] Although amendments to individual transmission service provider tariffs are beyond the scope of the instant filing, the CAISO expects the Commission will consider such amendments as targeted variations from the *pro forma* OATTs of such transmission service providers to be evaluated under the "consistent with or

---

[214] As such, the EDAM proposal is consistent with section 218 of the FPA, 16 U.S.C. § 824r(b) (stating that "nothing in [the FPA] confers on the Commission the authority to require an electric utility or person to convert to tradable or financial rights (1) firm transmission rights" pursuant to contract or by reason of ownership of transmission facilities as of the date the Energy Policy Act of 2005 was enacted, or "(2) firm transmission rights obtained by exercising contract or tariff rights associated with" such firm transmission rights).

[215] *See* new tariff section 33.4.2 (clarifying that each transmission service provider individually remains responsible for providing transmission service to its customers); *see also* existing tariff section 29.4(b)(3)(B). Transmission service outside the EDAM area will continue in the WEIM and in other balancing areas as it does today.

[216] https://stakeholdercenter.caiso.com/StakeholderInitiatives/Extended-day-ahead-market-ISO-balancing-authority-area-participation-rules.

superior to" standard,[217] just as the Commission did when it accepted tariff amendments of transmission service providers participating in the WEIM.[218]

The WEIM utilizes available transmission capability in the real-time market to support optimization of economic offers, meet imbalance energy needs and facilitate energy transfers between balancing areas.[219] The extended day-ahead market will similarly maximize available transmission capability in the day-ahead market to support optimization of demand and supply[220] and facilitate transfers of energy and capacity between balancing areas.[221] However, some additional complexity arises in the context of EDAM compared to the WEIM because under EDAM, all generation will be committed or dispatched by the market (in the day-ahead and real-time timeframes, respectively). Moreover, the transmission

---

[217] In both Order Nos. 888 and 890, the Commission found that transmission service providers may submit FPA section 205 filings proposing rates, terms and conditions in their tariffs that are "consistent with or superior to" the *pro forma* OATT. *Promoting Wholesale Competition Through Open Access Non-Discriminatory Transmission Servs. by Pub. Utils.; Recovery of Stranded Costs by Pub. Utils. & Transmitting Utils.*, Order No. 888, FERC Stats. & Regs. ¶ 31,036, at 31,770 (1996) (cross-referenced at 75 FERC ¶ 61,080) (Order No. 888); *Preventing Undue Discrimination & Preference in Transmission Serv.*, Order No. 890, 118 FERC ¶ 61,119, at P 135 (2007) (Order No. 890). The Commission's application of this standard can take into account the unique tariff structure or market design of an ISO or RTO. *See, e.g., N.Y. Indep. Sys. Operator Corp., Inc*., 123 FERC ¶ 61,134, at P 13 (2008) ("[W]e recognize that NYISO's [New York Independent System Operator, Inc.] proposed deviations from the *pro forma* OATT reflect the actual market design used by NYISO, and find these deviations to be consistent with or superior to the *pro forma* OATT, except as otherwise addressed below.").

[218] *See, e.g., Nev. Power Co.,* 151 FERC ¶ 61,131, at P 85 (2015) ("Except as discussed below, we find that NV Energy has met its burden of proof to demonstrate that the proposed OATT revisions are just and reasonable and not unduly discriminatory or preferential pursuant to section 205 of the FPA, and that the proposed revisions to NV Energy's OATT are consistent with or superior to the Commission's pro forma OATT. Moreover, we find that NV Energy has adequately demonstrated that participation in the EIM can be expected to provide NV Energy's customers with economic benefits, in the form of interregional dispatch savings, reduced flexibility reserve, and reduced renewable energy curtailment, as well as reliability benefits through improved situational awareness, responsiveness, and ability to optimize transmission capacities previously left unused."). The Commission affirmed this finding on rehearing. *Nev. Power Co.,* 153 FERC ¶ 61,306, at P 30 (2015).

[219] *See* existing tariff section 29.17 (providing for the transmission made available to the WEIM); *see also* existing tariff appendix B.17, *pro forma* EIM Entity Agreement, section 2.5 (providing for the transmission made available to the WEIM).

[220] Supply in EDAM will include energy, imbalance reserves, and reliability capacity, but not ancillary services. Ancillary services will be self-provided by transmission service providers participating in EDAM and not optimized by EDAM, although the ancillary services will count as supply in the resource sufficiency evaluation.

[221] *See* new tariff sections 33.16 – 33.18 (providing for the transmission made available to the WEIM), *see also* tariff appendix B.27, new *pro forma* Addendum to EIM Entity Agreement (incorporating new tariff section 33).

capability supporting schedules resulting from the day-ahead market must remain available through the real-time market, whereas transmission that supports the real-time market represents whatever is available in real-time after customers have exercised transmission service rights under the Commission's *pro forma* OATT.

The day-ahead market schedules resulting from EDAM are the basis for the operation of the real-time market, and market participants need a substantial degree of assurance they can rely upon the results, particularly given that energy and capacity transfers between balancing areas will be part of resource sufficiency evaluation and ultimately support reliability. Having the transmission capability backing day-ahead market awards available through the real-time is essential for the integrity of the day-ahead market results and providing confidence in the transfers between balancing areas. A high degree of confidence in transfers between balancing areas is critical as each balancing authority maintains its reliability and transmission-related functions in EDAM. Having invested in EDAM, the participating balancing authority will be relying on the market to determine reliable, economic, and feasible schedules to serve load on its system. Establishing this level of confidence is the primary reason the market will afford a transfer between balancing areas equal priority with demand in the balancing area and defer to the balancing authority to determine what operator action will fulfill its balancing authority obligations under its tariff to address any underlying infeasibility.[222] As discussed below, the CAISO will work with EDAM transmission service providers to accommodate balanced intra-day schedule changes associated with the exercise of specific firm OATT transmission rights in real-time.[223]

The CAISO carefully conceived this balanced transmission availability framework with the input of stakeholders throughout the Western Interconnection, recognizing that transmission service providers participating in EDAM would need targeted variations from the terms and conditions of the *pro forma* OATT as permitted under Commission policy. All stakeholders involved in developing the widely supported transmission availability framework understand

---

222 *See generally* new tariff section 33.7.5 (establishing the equal priority and providing each balancing area the discretion to determine appropriate operator action to resolve a market infeasibility); *see also* section V.B.1.f of this transmittal letter (discussing stakeholder positions on this provision).

223 Firm rights under OATT service may vary but are normally in the form of Network Integrated Transmission Service, Firm Point-to-Point, or Conditional Firm rights. The permissible uses for the higher priority will be as prescribed by the transmission service provider and consistent with their OATT provisions. In this transmittal letter the reference to specific firm OATT transmission rights is intended to capture the transmission service providers' firm rights offerings under their tariffs.

that some discrete enhancements to the terms and conditions of transmission service are a prerequisite to enable a day-ahead market across a multi-balancing area market, and the CAISO presents the EDAM transmission availability framework as an approach that will lead to changes consistent with or superior to the *pro forma* OATT.[224]

For these reasons, the CAISO respectfully requests that the Commission accept the proposed CAISO tariff provisions implementing the EDAM transmission availability framework.

a. Transmission System Information

The first step to account for available transmission capability is administrative and requires all balancing authorities participating in the extended day-ahead market to provide the CAISO with all of the necessary information from the transmission service providers in its balancing area. This necessary information includes updates to the network model, outages, and day-ahead scheduling limits that allow the CAISO to utilize flow-based congestion management respecting intra-balancing area physical transmission constraints, balancing area-specific power balance constraints, and intertie constraints between balancing areas (both physical limits and scheduling limits).[225] The CAISO will integrate this information into the associated systems.

As noted above, EDAM is structured to provide day-ahead market opportunities and benefits to balancing areas outside the CAISO balancing area. Unlike circumstances where a transmission owner is joining an ISO or RTO by transferring operational control of its transmission to the ISO/RTO, the balancing authorities participating in EDAM remain responsible for the reliability of the transmission system in their area. Accordingly, the CAISO will look to the balancing authority participating in EDAM, in coordination with its transmission service providers, to provide system information required to support market operations and ensure the CAISO has the latest most accurate current state of the system, look-ahead scheduling, and forecasting information from different sources so the day-ahead market operates effectively within a multi-balancing area environment.

---

224 The Commission has recognized that its open access requirements are not intended to prevent customers from realizing the benefits of ISO or RTO markets under terms and conditions that differ from the *pro forma* OATT. For example, In Order No. 890, the Commission held that "the purpose of this rulemaking is not to redesign approved, fully-functioning RTO or ISO markets." Order No. 890 at P 143; *see also id.* at PP 157-58.

225 *See generally* new tariff section 33.4 (establishing the roles and responsibilities of EDAM market participants at a high level).

The CAISO will receive all updates of transmission system and other pertinent operational information directly from the entities responsible for reliability of their transmission systems and the associated transmission services within the participating balancing area, and the CAISO will incorporate that information directly into the optimization to produce a quality solution. This enables EDAM, like the WEIM, to function in a multi-balancing area environment in a compatible and complementary manner with the transmission services, while respecting the transmission rights of third parties and scheduled transactions in the market.

The day-ahead market relies upon the flow-based capability of the transmission system within a balancing area participating in EDAM to support optimized unit commitment, including the internal flows supporting the transfers between participating balancing areas. Availability of internal flow capability in EDAM aligns with the availability of internal transmission in the WEIM and will ensure that the day-ahead market model relies upon the same transmission capability available in the real-time market.[226] This internal flow-based capability is broken down by three categories of available transmission – legacy contract rights, transmission ownership rights, and OATT service rights[227] – as well as the distinction of whether the transmission capability supports a transfer between balancing areas in the EDAM footprint or imports, exports, or wheels through a balancing area in the EDAM footprint.[228]

b. Legacy Contracts and Third-Party Ownership Rights

---

[226] *See Tri-State Transmission & Generation Ass'n*, 174 FERC ¶ 61,250, at P 3 (2021) ("Each EIM Entity must ensure that transmission service providers in its BAA [balancing authority area] make unused transmission capacity available for use in the CAISO real-time market.") (citing existing tariff section 29.17(c)).

[227] *See* tariff appendix A, new definitions of EDAM Legacy Contract and EDAM Transmission Ownership Right (defining an EDAM Legacy Contract as a transmission service contract entered into prior to the effective date of the EDAM transmission service provider's tariff or otherwise not governed by the terms of the applicable tariff, and defining an EDAM Transmission Ownership Right as a third-party right on transmission facilities within an EDAM balancing area that is not subject to the applicable tariff).

[228] The CAISO refers to interfaces between balancing areas in the EDAM area as "internal interties", and it refers to interfaces with non-EDAM balancing areas, including balancing areas in the WEIM, as "external interties." This distinction matters because transmission capability between balancing areas in EDAM supports the transfer of energy and capacity in the day-ahead market and this requires unique modeling capabilities and specific provision on the availability and resulting revenue. *See* tariff appendix A, new definitions of EDAM Internal Intertie and EDAM External Intertie.

The CAISO honors existing contracts in existence on the CAISO operations date and transmission ownership rights in the CAISO balancing area through a series of requirements and procedures designed to maintain the existing and underlying rights within the CAISO markets, including the day-ahead market construct.[229] Similarly, the CAISO will honor legacy transmission contracts, *i.e.*, non-OATT transmission service agreements entered into with an EDAM transmission service provider prior to the effective date of the applicable OATT or otherwise not governed by the terms of that tariff, and will also honor transmission ownership rights by third parties on transmission facilities within an EDAM balancing area not subject to the applicable OATT according to these existing mechanisms by cross-referencing the requirements and procedures in the EDAM provisions of the CAISO tariff.[230] These procedures require the balancing authority and transmission service provider to register the underlying rights with the CAISO so the CAISO can establish an associated contract reference number in its systems. The assigned contract reference number will afford scheduled use of the rights treatment in the day-ahead and real-time markets consistent with the rights, which may include both a scheduling priority and a financial right.

The CAISO will provide a balanced legacy contract or ownership right self-schedule associated with a contract reference number a priority higher than other self-schedules in the market, and will allow changes to the associated self-schedule through real-time while maintaining the higher scheduling priority throughout this timeframe. In addition, the CAISO will afford the balanced portion of a legacy contract or ownership right schedule associated with a contract reference number financial protection from congestion charges and losses, sometimes called the "perfect hedge," to the extent the underlying contract rights support such financial protections. Alternatively, holders of legacy contracts and ownership rights will be eligible voluntarily to make their transmission capacity available at interties between balancing areas in exchange for the accrued transfer revenue as discussed below in section V.B.1.d of this transmittal letter. This transfer revenue opportunity will extend to existing contracts and ownership rights in the CAISO balancing area, thereby establishing comparable and equitable treatment of all non-tariff based transmission rights in the broader EDAM footprint.

---

[229] *See* existing tariff section 16 (establishing the requirements and procedures that honor existing contracts), and existing tariff section 17 (establishing the requirements and procedures that honor ownership rights); *see also Cal. Indep. Sys. Operator Corp.*, 116 FERC ¶ 61,274, at PP 901-1003 (2006), *order on clarification & reh'g*, 119 FERC ¶ 61,076, at PP 442-85 (2007) (accepting tariff sections 16 and 17).

[230] *See* new tariff section 33.16 (establishing the requirements and procedures that honor legacy contracts), and new tariff section 33.17 (establishing the requirements and procedures that honor ownership rights).

c. Exercising OATT Rights in the Market

The CAISO proposes several provisions to allow customers with transmission service rights under *pro forma* OATTs to elect how to exercise those rights in EDAM. The first step requires registration with the CAISO of all firm and conditional firm transmission service customer rights within a balancing area, including point-to-point service rights and network transmission service rights.[231] The registration process for customer rights will be a simplified version of the legacy contract registration process, and likely will include some degree of automation so the process can account for both long-term and short-term rights under an existing OATT as well as ongoing transmission sales under the transmission service provider's tariff. This registration process will associate: (a) a source in an EDAM balancing area to an external location, (b) an EDAM external location to a sink within the EDAM balancing area, (c) a wheel through an EDAM balancing area from/to an external location, (d) a source to a sink within an EDAM balancing area, or (e) a source or sink to/from an intertie with another balancing area participating in EDAM. Registration of network integration transmission service rights will allow for the identification of multiple sources and sinks. Again, similar to the process for legacy contracts and ownership rights, the CAISO will assign each registered transmission customer right a contract reference number for schedules that indicate the customer's use of the associated transmission capability in the day-ahead market.[232]

The scheduling coordinator for the transmission customer will reference the assigned contract reference number to provide the CAISO information about scheduled use of a customer's rights.[233] The customer will use its OATT rights by submitting a balanced self-schedule in the markets associated with the contract reference number, and the market scheduling priority assigned to the self-schedule will depend upon the terms of the applicable EDAM transmission provider tariff and whether the use was communicated to the market prior to the close of the day-ahead market at 10:00 a.m. The deadline of 10:00 a.m. on the

---

231 *See* new tariff sections 33.18.1 (providing a registration process for internal rights and rights at external interties), and 33.18.2 (requiring registration of rights at internal interties using a similar process).

232 *See* new tariff section 33.18.3 (providing for issuance of a CRN and the associated treatment in the day-ahead market).

233 Contract reference numbers registered for firm OATT transmission rights (*i.e.*, not legacy contract rights or transmission ownership rights) are not eligible to receive a financial protection against congestion costs. This is consistent with how the CAISO tariff currently treats transmission service not under existing contracts or transmission ownership rights.

day prior to the operating day follows the firm deadline established in section 13.8 of the *pro forma* OATT, which provides, "Schedules for the Transmission Customer's Firm Point-To-Point Transmission Service must be submitted to the Transmission Provider no later than 10:00 a.m. of the day prior to commencement of such service."

Balanced self-schedules associated with a contract reference number submitted prior to the close of the day-ahead market will receive a priority higher than a self-schedule not associated with a contract reference number, and below a self-schedule associated with a contract reference number representing a legacy contract or ownership right.[234] The CAISO will not adjust a self-schedule associated with registered OATT rights communicated to the CAISO by 10:00 a.m. (before the close of the day-ahead market) to accommodate a self-schedule that is not associated with registered OATT rights. This provides a priority higher than other self-schedules not exercising registered transmission rights and higher than economic bids, affording the transmission customer near certainty that the self-schedule will clear the day-ahead market. The timely exercise of rights and associated market priority allows the market to identify the volume/magnitude of transfers between areas and attribute the associated confidence for all balancing authorities to serve load reliably through the market.

To the extent a transmission customer does not exercise its firm OATT rights through balanced self-schedules in the day-ahead market, *i.e.,* by 10:00 a.m. on the day prior to the operating day, the scheduling coordinator for the transmission service customer can still use them in the real-time market by submitting a balanced self-schedule associated with the transmission rights.[235] The requirement that customers exercise their specific firm OATT transmission rights by submitting balanced supply and demand self-schedules is a key feature of the EDAM design and is consistent the contract path point requirements for scheduling firm point-to-point service under the *pro forma* OATT.[236] The CAISO expects these intra-day balanced schedule changes associated with specific OATT transmission rights will be accommodated by the EDAM transmission service provider if practicable, consistent with section 13.8 of the *pro forma* OATT, which provides "Schedules submitted after 10:00 a.m. will be accommodated, if practicable." Transmission customers must submit such intra-day schedule changes associated with specific OATT transmission rights prior to

---

234 *See* new tariff sections 33.18.3.1 and 33.18.3.2 (providing for the relative priority of a self-schedule associated with a CRN).

235 If a self-schedule or intra-day schedule change is not balanced, it will be treated as a regular real-time self-schedule and will not have the priority afforded to balanced schedules exercising OATT rights.

236 EDAM Final Proposal at 33.

the close of the real-time market established in the transmission service provider tariff regarding WEIM.[237] The EDAM transmission provider will make the ultimate determination whether a post-10:00 a.m. schedule using point-to-point service is practicable in accordance with its OATT.

Absent EDAM transmission provider instructions supporting a different priority, the CAISO will afford the intra-day schedule submissions a priority in the real-time market higher than other real-time self-schedules and equal to cleared day-ahead self-schedules, including transfer schedules between balancing areas serving load. Thus, the real-time market will seek to accommodate these self-schedules and, if necessary, redispatch the available supply across all balancing areas based on the transfer capability between balancing areas to support the exercise of the customer's rights. If there is insufficient redispatch capability in the real-time market, the market will manage all schedules through the market clearing process until the solution becomes infeasible and the power balance constraint is relaxed, ultimately affording transfers between balancing areas equal priority to demand within the balancing areas.[238] The CAISO will monitor the frequency and magnitude of the exercise of these transmission rights across different systems and interfaces to understand the market impacts of redispatch and infeasibilities in the market.

In the context of EDAM, transmission service providers will play a role in determining how to accommodate such post-10:00 a.m. point-to-point service schedules. If instructed by an EDAM transmission provider, the CAISO will afford balanced intra-day self-schedules or schedule changes associated with a contract reference number for balanced, specific, firm OATT transmission rights a priority in the real-time market higher than other real-time self-schedules and higher than cleared day-ahead self-schedules not based on specific firm OATT transmission rights, including transfer schedules between balancing areas

---

[237] In the WEIM, the Commission accepted a deadline proposed by each transmission service provider for the submission of base schedules, which generally has been set at T-57.5. This enables each balancing area to account for all schedules prior to the deadline for the WEIM resource sufficiency evaluation at T-40. *See, e.g.*, NV Energy OATT, Attachment P, section 4.2.4.5.2. The Commission found these modified real-time scheduling timelines to be just and reasonable and necessary to implement the WEIM. *Nev. Power Co.,* 151 FERC ¶ 61,131, at PP 161-164.

[238] *See* new tariff section 33.7.5 (establishing an equal priority between day-ahead transfers and demand within a balancing area and requiring coordination and communication concerning any curtailment of transfers between balancing areas to maintain reliability); *see also* EDAM Final Proposal, appendix 3, Mathematical Formulation of the WEIM Power Balance Constraint Relaxation (establishing the formulation for establishing the conditions when the power balance constraint within a balancing area will be relaxed, providing for confidence in transfers, and supporting the equal priority of demand and transfers established in tariff section 33.7.5).

serving load.[239] Such balanced intra-day self-schedules using specific OATT firm transmission rights will have a priority below a self-schedule associated with a contract reference number representing a legacy contract or ownership right. Unlike individual customer legacy contracts or ownership rights, balanced intra-day self-schedules using specific firm OATT transmission rights will not receive a perfect hedge and such schedules will be responsible for congestion or redispatch costs.[240] This approach is consistent with the treatment of non-legacy contracts in ISOs and RTOs with firm OATT transmission rights where such customers pay market congestion charges.[241]

The CAISO believes this higher priority for intra-day self-schedules using balanced, specific, firm OATT transmission rights where instructed by EDAM transmission service providers coupled with the redispatch process across a multi-balancing area market footprint goes beyond the actions that transmission service providers are obligated to undertake today under the *pro forma* OATT or may undertake within their balancing area. Indeed, the proposed actions go even further than existing measures by leveraging the supply capabilities of multiple balancing areas to accommodate these real-time schedule changes by customers with OATT rights that submit such schedule changes after the 10:00 a.m. deadline.[242] In other words, EDAM will afford even greater tools and flexibility than exists today to accommodate, if practicable, such post-deadline schedules seeking to use firm point-to-point service. This approach maintains confidence in EDAM transfers clearing the day-ahead market, which EDAM entities depend upon to serve load, while respecting the late exercise of transmission customer rights in the real-time market. In applying the priorities described above that may result from an infeasibility, the market will inform the

---

239 *See* new tariff sections 33.18.2.2.3 and 33.18.3.1.

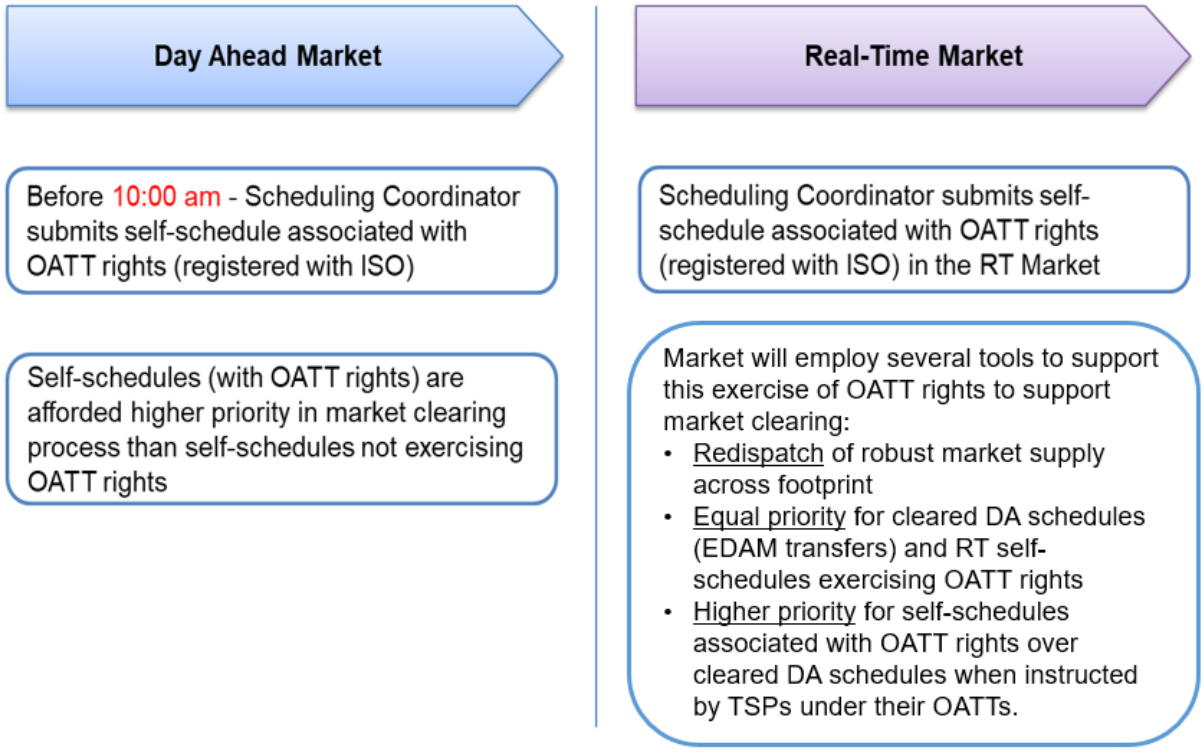
240 In the context of the WEIM, the Commission similarly found that applying imbalance energy charges, including congestion charges to deviations from firm transmission customers' schedules after T-55 (and before T-20) is "just and reasonable and not unduly discriminatory" and "appropriately allocates the costs of imbalances to the customers causing such costs and is necessary to implement the EIM." *Nev. Power Co.,* 151 FERC ¶ 61,131, at P 162; *order on reh'g*, *Nev. Power Co*., 153 FERC ¶ 61,306, at PP 32-40.

241 *See, e.g.,* NYISO OATT, sections 6.7.1.1 and 6.7.1.2.

242 As noted above, section 13.8 of the *pro forma* OATT provides that "Schedules submitted after 10:00 a.m. will be accommodated, if practicable." The Commission has recognized that section 13.8 gives transmission service providers flexibility to "accommodate" transmission service schedules submitted after 10:00 a.m. *See Integration of Variable Energy Res.*, 139 FERC ¶ 61,246, at P 136 (2012) ("The Commission recognizes that transmission providers have flexibility under [section 13.8 of] the *pro forma* OATT to award transmission service based on transmission capability that becomes available when firm transmission service is not scheduled by 10:00 am the day prior to operation."); *Allegheny Power Sys. Inc*., 77 FERC ¶ 61,266, at 62,100 (1996) (finding that the "if practicable" language in section 13.8 "permits flexibility").

balancing authority of the magnitude of the infeasibility and the affected schedules so they may take appropriate action according to their balancing authority obligations and the applicable transmission service provider tariffs.[243]

The following illustration represents the market scheduling priorities established through EDAM as described above. These market scheduling priorities work with the various avenues for transmission to become available in the day-ahead market and together represent the EDAM transmission availability framework. The CAISO carefully conceived this framework with the input from all stakeholders, a substantial majority that support it. The framework effectively harmonizes the extended day-ahead market framework with the *pro forma* OATT framework.



A key policy element of the EDAM market design is that resources in EDAM must have transmission reserved under the respective transmission service provider's tariff. Consequently, the transmission service provider will assess a transmission charge to resources located in an EDAM balancing area

[243] *See, e.g.*, existing tariff sections 29.4(b), 29.7(g), and 29.34(o) (establishing responsibility of the balancing authority to manage conditions within its area based on information provided by the CAISO as the market operator of the WEIM).

that submit a bid in the day-ahead market and have not reserved transmission capacity associated with their schedule under the applicable transmission tariff through firm, conditional firm, or network integration service.[244] This EDAM transmission service provider charge will be based upon the shortest duration of firm transmission service offered under the transmission service provider's tariff, including any applicable penalties. Requiring transmission service reservations or network designations in advance will ensure transmission service providers with revenues normally received to avoid or limit undue cost shifts, support continued transmission planning, avoid free riding, and allow for resources to participate in the market. It is important to recognize the vast majority of resources, are likely to meet the transmission reservation requirement today as they have reserved transmission service or have been designated as a network resource by its off-taker. This proposed design nonetheless ensures that all supply resources will continue to contribute to the costs of the associated transmission system and its availability in EDAM.

d. Transmission Available to Support Transfers Between Balancing Areas

Transmission between balancing areas participating in EDAM will be available to support the transfer of energy, imbalance reserves and reliability capacity within three categories of transmission. Transmission within those three categories can be released for market optimization, respecting any pre-market net export transfer constraints,[245] and this released transmission will provide significant transfer capability and support optimized transfers among balancing areas participating in EDAM. The design models these transfers using transfer system resources with associated scheduling limits based on the transmission capability available at each intertie location, *i.e.*, a transfer location between two balancing areas participating in EDAM, which the balancing authority will communicate to the CAISO prior to the day-ahead market.

The first category of transmission available for transfers is the balancing authority's external supply resources needed to support its day-ahead resource sufficiency evaluation.[246] A balancing area that has load serving entities with contracted supply in another balancing area participating in EDAM will be supported by a transfer system resource. Prior to the day-ahead market, the

---

244 *See* new tariff section 33.23 (establishing a transmission charge for schedules that do not have an associated transmission reservation).

245 *See* section V.B.3 of this transmittal letter.

246 *See* new tariff section 33.18.2.1 (governing transmission to support resource sufficiency provided by the participating balancing authority, *i.e.*, the EDAM entity).

balancing authority will communicate to the CAISO the transmission capability available at each of its internal interties that will be available in the day-ahead market and support a transfer at that intertie location.[247] These transfers support the EDAM resource sufficiency evaluation and the resulting diversity benefit that result from the transfer of energy and capacity in the day-ahead market.

The second category of transmission considered in the extended day-ahead market is transmission customer rights, broken down by type, which may be self-scheduled or released in exchange for transfer revenue. A transmission rights holder, whether receiving with service under a *pro forma* OATT, a legacy contract or ownership right, may submit a self-schedule as described in section V.B.1.b of this transmittal letter. All self-schedules submitted in the day-ahead market will be available to support transfers between balancing areas participating in the day-ahead market, including transfers supporting the EDAM resource sufficiency evaluation.

Alternatively, the transmission customer can elect voluntarily to release its transmission rights to the market. To the extent transfer revenue accrues across the interface due to energy or capacity price differences between balancing areas, the CAISO will settle those transfer revenues with the transmission customer that released its transmission rights for market optimization.[248] Transmission customers releasing their rights cannot self-schedule such rights in the day-ahead market or after the close of the day-ahead market.

Any portion of a registered right that is neither released nor self-scheduled in the day-ahead market (*i.e.,* by 10:00 a.m.) will be available for optimization in the day-ahead market.[249] In essence, the transmission customer would retain its right to submit an intra-day self-schedule later. Absent instructions by the transmission service provider, balanced intra-day self-schedules associated with specific firm OATT transmission service would receive equal priority. The market also would redispatch the system through the real-time market to accommodate both the transfer and all self-schedules submitted after the close of the day-ahead market. If the market cannot accomplish this, the market will afford the transfer an equal priority with demand in the balancing area and provide the balancing authority information regarding the infeasibility. If instructed by the

---

247 The balancing authority must also procure firm, conditional firm or network integration transmission service to deliver the supply to its balancing area (new tariff section 33.18.2.1). With respect to the CAISO balancing area, this would include any procurement or reservation of available transfer capability to deliver the supply to its balancing area (new tariff section 33.18.4).

248 *See* section V.D.3 of this transmittal letter (describing the settlement of transfer revenue).

249 The market will accommodate the transmission customer's later self-schedule through the higher or equal priority for self-schedules in accordance with EDAM transmission service provider instructions coupled with economic redispatch for all later self-schedules, as described in section V.B.1.c of this transmittal letter.

EDAM transmission service provider, balanced intra-day self-schedules associated with firm OATT transmission service would receive a higher priority when submitted in accordance with instructions by EDAM transmission service provider. The balancing authority, in accordance with the applicable transmission service provider tariff(s), would determine whether to dispatch additional resources manually, adjust the transfer or the self-schedule, or take other operator action to fulfill its balancing authority obligations, including firm load shedding as a last resort.

The third category of transmission considered in the extended day-ahead market is the unsold available transmission capability available at transfer locations prior to the day-ahead market. Each transmission service provider in an EDAM balancing area must make unsold firm transmission capability available prior to the day-ahead market for the market to optimize transfers between balancing areas participating in the day-ahead market. The transmission service provider will determine the available transmission capability under its tariff, including transmission capability reserved for reliability, and it will communicate the associated limit to the CAISO.[250] As with the other categories of transmission considered by the day-ahead market, transmission service providers make unsold firm available transmission capability on interfaces available to the market only at the interfaces between two balancing areas participating in the extended day-ahead market and not at the interface of a balancing area not participating in the day-ahead market. After the market close of the day-ahead market through to the publication of day-ahead market results, the transmission service provider cannot make further sales of available transfer capability. After the publication of day-ahead market results, the CAISO will produce a report identifying the amount of unsold transmission capability the day-ahead market optimized, so the transmission service provider can resume sales of the transmission after the day-ahead market results are published.

e. Transmission that May Not Be Available in the Day-Ahead Market

It is necessary for a day-ahead market covering multiple balancing areas and transmission service provider territories to account for the fact each balancing area will have its own transmission service terms and conditions and mechanisms for managing procurement and transmission constraints/limitations

[250] The CAISO expects each transmission service provider in an EDAM balancing area to determine available transmission capability consistent with its tariff-approved methodology to assess available transfer capability based on attachment C to its tariff, including any provisions accounting for transmission reliability margin or TRM.

to maintain reliability. Accordingly, if a balancing authority informs the CAISO through the implementation process, or anytime following its participation in EDAM, that transmission availability should be restricted in the day-ahead market to accommodate the exercise of transmission customer rights, the CAISO will adjust day-ahead market transmission availability of the affected transmission elements.[251] Because the EDAM design depends heavily on making transmission capacity available to the market to reach efficient outcomes, the CAISO expects the transmission service provider will request adjustment of available transmission only under narrow, limited, and specific circumstances as provided in the transmission service provider's tariff. For example, a certain transmission path may consistently produce infeasibilities due to persistent intra-day schedule changes. In such cases, it may be appropriate to limit the availability of the transmission in advance of the day-ahead market, rather than repeatedly facing infeasibilities that are difficult to manage in real-time or produce market outcomes that are inefficient or unreasonable. In these circumstances, the CAISO will defer to these transmission service providers and the terms of their tariffs.

As in the WEIM today, it is appropriate to provide deference to the EDAM balancing authority and its associated transmission service providers to manage these infeasibilities.[252] The balancing authority and other entities with reliability- and transmission-related functional responsibilities in each balancing area are in the best position to make determinations regarding how to administer the terms and conditions of the affected transmission customer's rights based on applicable tariffs and other agreements.[253] This should not be the responsibility of the CAISO as a market operator.

Although the CAISO tariff provides an equal market scheduling priority for transfers and load,[254] each EDAM transmission service provider would retain the ability to manage its system under stressed system conditions within its balancing area, including consideration of any obligations associated with a

---

251 *See* new tariff section 33.18.3.3.

252 The CAISO also provides reliability coordinator services to most balancing authorities and transmission operators in the WEIM area today and will continue to provide such services. *See* existing tariff section 19 (providing for reliability coordinator services).

253 For example, an EDAM transmission provider, in accordance with its tariff, may determines if it is necessary to reduce or suspend EDAM transfers between balancing areas based on operational judgment (*e.g.*, in the event of infeasibility). *See* new tariff section 33.7.5 (establishing an equal priority between day-ahead transfers and demand within a balancing area and requiring coordination and communication concerning any curtailment of transfers between balancing areas to maintain reliability).

254 *See* new tariff section 33.7.5.

resource adequacy program in which it participates.[255] This priority framework retains the value of firm transmission rights by providing the customer with a higher market scheduling priority for intra-day self-schedules associated with specific firm OATT transmission service in accordance with EDAM transmission service provider instructions and a priority equal to cleared day-ahead EDAM transfers for all other intra-day self-schedules as described in section V.B.1.d of this transmittal letter. From there, the CAISO will use the full capability of the real-time market, including utilizing redispatch, to accommodate these rights.

Today, firm transmission rights under the *pro forma* OATT are not guaranteed to flow under any and all conditions. Customers with OATT firm point-to-point transmission service rights can schedule their rights with a certain degree of confidence. If there is a transmission derate in real-time, however, firm point-to-point rights holders are not held harmless relative to other priority schedules. In the event of a derate, both firm point-to-point customers and network integration customers may not be ensured absolute protection for schedules using their rights.

The EDAM self-schedule process will provide at least a comparable high degree of certainty in the ability to schedule firm OATT rights. Due to the transmission contributions across the entire EDAM area and flexibility afforded transmission operators by EDAM, the CAISO does not anticipate a need for the transmission service provider to cut schedules except in the event of a derate. The proposed market scheduling priority and real-time market measures mitigate the concern of affording the transmission customer rights a market scheduling priority higher than transfers and other cleared day-ahead schedules. The EDAM transmission provider, acting in accordance with its tariff, will make the final determination of schedule priorities after the close of the day-ahead market. Granting a higher priority for all intra-day schedule changes submitted after the 10:00 a.m. deadline, even when there is no basis in the EDAM transmission service tariff for such a higher priority, would degrade the confidence in EDAM transfers because the transfer could be unwound to accommodate the exercise of transmission customer rights in the real-time market. Unwinding transfers awarded based on transmission capability that becomes available when firm transmission service is not scheduled by 10:00 a.m. the day prior to operation would adversely affect other entities that depend on such market transfers, including other resource adequacy program participants, further eroding the confidence in transfers, reducing market efficiency, and possibly creating an

[255] *See* new tariff section 33.18.3.3 (establishing a process whereby a balancing authority can inform the CAISO of limited circumstances whereby transmission should not be available to the market).

inappropriate arbitrage opportunity.[256] In any case, each EDAM transmission service provider will remain responsible for managing its system and coordinating with their balancing authority under stressed system conditions.

The CAISO believes limiting the availability of transmission in the day-ahead market must be the exception rather than the rule, recognizing reduced transmission availability could significantly affect the benefits of EDAM and have adverse economic effects on congestion prices settled with the balancing authority and passed down to wholesale customers. Unnecessarily restricting the transmission available in the day-ahead market would reduce efficiency, potentially causing congestion that could have been avoided if the transmission had been available.

[256] The Commission has recognized that "transmission providers have flexibility under the *pro forma* OATT to award transmission service based on transmission capability that becomes available when firm transmission service is not scheduled by 10:00 am the day prior to operation." *Integration of Variable Energy Res.*, 139 FERC ¶ 61,246, at P 136.

f. Stakeholder Positions on Transmission Availability

Stakeholders favored the concept that transmission supporting delivery of resources to meet the EDAM resource sufficiency evaluation be available to the market. This approach promotes mutually beneficial transfers, and to the extent the scheduling limit at the interfaces is binding, transmission revenue could accrue to the benefit of entities making transmission available.

Stakeholders almost universally supported a design to maximize transmission availability in the day-ahead market, including unused or unscheduled firm/conditional firm transmission, to support transfers and derive the benefits of optimized commitment through the day-ahead market. A few stakeholders, primarily transmission customers holding firm or conditional firm transmission rights, sought a design under which they could elect whether to make their transmission rights available to the market or withhold them from the market even if they have not scheduled use of those firm transmission rights by the 10:00 a.m. day-ahead deadline. In particular, these stakeholders contended that, if a transmission customer exercises previously unscheduled firm transmission rights that may support a day-ahead schedule, they should be assured of delivery and not be exposed to additional risks associated with the exercise of their transmission rights by receiving a higher scheduling priority than proposed by the CAISO. At least one stakeholder with firm transmission rights raised concerns that the EDAM design's use of unscheduled firm transmission rights in the day-ahead market could either devalue or undo their firm transmission rights under the *pro forma* OATT.

In response to these concerns, the CAISO clarified that firm transmission rights registered with the CAISO, whether legacy rights or firm OATT rights, remain eligible for scheduling after the close of the day-ahead market, thereby preserving the physical right through the real-time market where WEIM scheduling practices would apply. The use of unscheduled capacity in the day-ahead market after 10:00 a.m. does not change this outcome; rather, the CAISO would accommodate their physical rights through redispatch as required.[257] As noted above, the CAISO believes this accommodation of firm transmission rights not scheduled by 10:00 a.m. day-ahead goes beyond the actions transmission service providers are obligated to undertake today under section 13.8 of the *pro*

[257] As discussed in section V.B.1.d of this transmittal letter, the CAISO would assign equal priority to balanced intra-day self-schedules relative to day-ahead schedules absent instructions by the EDAM transmission service provider that the ISO provide balanced self-schedules associated with specified firm point to point rights higher priority than day-ahead schedules.

forma OATT.[258] Indeed, EDAM will leverage the supply capabilities of multiple balancing areas to accommodate these real-time schedule changes, thus establishing a reasonable limit of what EDAM will accommodate consistent with or superior to the *pro forma* OATT. The Commission has found variations in *pro forma* OATT scheduling timelines are reasonable to allow customers to realize the substantial benefits of the WEIM.[259] EDAM will provide widespread benefits to customers.

As the CAISO further considered how to implement the OATT provision regarding the accommodation of post-10:00 a.m. intra-day schedules if practicable, the CAISO determined it is appropriate to afford such intra-day schedules using firm OATT transmission rights a priority higher than other self-schedules not using firm rights, legacy contracts, or transmission ownership rights when the higher priority is based an instruction from the EDAM transmission provider acting in accordance with its tariff. If the intra-day self-schedules are not submitted consistent with the instructions provided by the EDAM transmission provider, they will receive equal priority to day-ahead schedules as specified in section V.B.1.d of this transmittal letter.

The CAISO further acknowledges that providing transmission customers with firm OATT rights that submit all intra-day schedules after the 10:00 a.m. deadline a market scheduling priority higher than all other self-schedules would ensure delivery of their self-schedules submitted after the close of the day-ahead market; however, the CAISO and most stakeholders disagreed with these stakeholders position, largely because the transmission customers can still schedule their unused rights in real-time, receive a market scheduling priority equal to cleared day-ahead schedules, including transfers, and rely on CAISO redispatching all of the supply available to it from the various participating EDAM balancing areas.[260] Further, firm transmission service will continue to maintain a higher priority through the market scheduling process compared to non-firm

---

[258] Section 13.8 of the *pro forma* OATT provides that firm point-to-point schedules submitted after 10:00 a.m. of the day prior to the commencement of such service "will be accommodated, *if practicable*" (emphasis added).

[259] *See Nev. Power Co*., 151 FERC ¶ 61,131, at PP 161-164. In that order, the Commission found the WEIM would provide numerous economic and reliability benefits to NV Energy's customers and that, in order to effectuate the WEIM, it is necessary for NV Energy and its customers to submit forecast data in advance of T-20. The Commission affirmed this finding on rehearing. *Nev. Power Co.,* 153 FERC ¶ 61,306, at P 40 (". . . in order to realize the numerous benefits from the EIM, participants must submit forecast data consistent with the timelines established by CAISO in order to run the security-constrained economic dispatch.").

[260] Legacy contract rights and transmission ownership rights will receive a higher priority through the real-time market if supported by the contract, similar to existing contract rights and ownership rights in the CAISO balancing area. *See* new tariff sections 33.16 and 33.17.

transmission rights, similar to today under an EDAM transmission service provider's tariff.[261] The CAISO believes that the EDAM transmission availability framework strikes the right balance between the need for transmission to be available in the market and the certainty market participants expect from the market.[262] As explained above, affording all self-schedules submitted after the close of the day-ahead market a higher scheduling priority where an EDAM transmission provider has not instructed the CAISO that a higher priority is needed to implement its OATT, would create inefficiency and undermine the objective to maintain confidence in all transfers.[263]

Some stakeholders also expressed a desire to have the right to carve their OATT transmission rights out of the market to the extent they are rights that support exports from or wheels through an EDAM balancing area to a non-EDAM balancing area to meet resource adequacy or other obligations. The proposed EDAM tariff provisions do not "carve out" such transmission rights (*i.e.*, take such capacity away from the market), but they allow customers to exercise their rights associated with such transactions. To do so, transmission customers would submit a market self-schedule associated with firm transmission rights registered with the CAISO. The market would respect these rights through an appropriate scheduling priority. This recognizes the need of parties to export from or wheel through another balancing area to serve their load and meet reliability needs. The design also provides for the physical "carveout" of certain transmission rights across particular frequently scheduled paths where the applicable transmission service provider determines such carveouts are necessary in accordance with its tariff. If there is frequent use and exercise of particular transmission rights across discrete and specific paths or flowgates, the transmission service provider could request an adjustment to "carve out" the transmission right from the market

---

[261] Transmission service in the CAISO balancing area will follow the CAISO tariff market priorities to facilitate the equivalent priority of transfers and load. *See, e.g.,* existing tariff sections 31.4 and 34.12 (establishing the real-time market and day-ahead market scheduling priorities).

[262] The CAISO further notes that the market is constantly re-dispatching resources to account for changed conditions, including changes to load, topology due to outages, generation availability, or other factors. As such, the CAISO does not propose to assign any direct costs of redispatch to any specific scheduling coordinator representing transmission customer rights. Instead, each balancing area and transmission service provider in the extended day-ahead market will have the opportunity to work with their transmission customers and load serving entities to determine an equitable allocation of any shortfall or excess from the pool of congestion revenues and transfer revenues the CAISO allocates to the balancing area for sub-allocation according to its tariff. *See infra* Section V.D.3 of this transmittal letter (describing the allocation of transfer revenues and congestion revenues).

[263] Transmission service in the CAISO balancing area will follow the CAISO tariff export and wheeling priorities to facilitate the equivalent priority of transfers and load. *See, e.g.,* existing tariff section 23 and tariff appendix L (establishing the scheduling priorities for wheeling through and export schedules).

under limited circumstances specified in the transmission service provider's tariff, which the CAISO would implement.[264]

The CAISO and stakeholders considered a variety of options to balance the objectives of maximizing the transmission capability available to the market (and the associated customer benefits) with the firm transmission rights of OATT customers, understanding that EDAM does not interfere with the obligation of each balancing authority to maintain reliability and does not disrupt the authority of any transmission service provider to manage transmission schedules under stressed system conditions. As explained above, the decision was to provide: (i) transmission customer schedules using specific firm OATT transmission rights submitted after the day-ahead market a higher priority than cleared day-ahead schedules in accordance with EDAM transmission provider instructions, (ii) all other transmission customer schedules using firm OATT transmission rights submitted after the day-ahead market an equal priority to cleared day-ahead schedules (iii) the full redispatch capability of the market in an effort to accommodate all intra-day schedules, (iv) the transmission service provider with discretion to determine the relative priority if accommodating these schedules was infeasible, and (v) the option for the transmission service provider to "carve out" the transmission if the end redispatch was not sufficient.

The CAISO appreciates that this approach will require each transmission service provider to amend its OATT to implement these features. Each EDAM transmission service provider's development of these supporting OATT rules and procedures is a significant next step in the evolution of EDAM, and the CAISO is committed to continuing its work with all interested participants through the associated approval processes, while continuing to look for uniform ways to address these issues. Accordingly, this transmission availability framework provides room for further consideration of the concerns expressed by some stakeholders that EDAM could diminish the value of their firm OATT rights, and allows the Commission to consider these important questions further when individual transmission providers participating in EDAM file the requisite OATT changes.[265] The CAISO believes these further efforts will build upon and enhance this framework. Based on these efforts, the CAISO will evaluate whether to enhance its tariff in the future to consider further conforming changes as may be appropriate based on the Commission-accepted OATT changes. The

---

[264] *See* sections II.B.1.a and V.B.1.f of this transmittal letter (describing the option to "carve out" transmission from the market and explaining the inefficiency of doing so except in limited circumstances).

[265] The CAISO recognizes that some potential EDAM transmission service providers may not be subject to the Commission's jurisdiction. Nonetheless, the CAISO expects that the associated tariff provisions to implement EDAM, much the same as in the WEIM today, will coalesce around a standardized approach that supports broader market participation, with variances as needed to account for unique circumstances or existing business practices of individual transmission providers.

fact the CAISO has provided the opportunity for EDAM transmission service providers to develop OATT modifications for future consideration by the Commission does not undermine the CAISO's basic proposal. Indeed, EDAM transmission service providers will depend on the Commission's acceptance of the CAISO's proposal as a foundation for developing their OATT changes and submitting them to the Commission.

Numerous stakeholders expressed concerns about the interaction between EDAM and the Western Resource Adequacy Program (WRAP), a new voluntary resource adequacy planning and compliance program in the Western Interconnection for which the Commission recently approved a tariff framework effective January 1, 2023.[266] The CAISO expects the first binding season for the WRAP to be the summer of 2025, followed by a three-year transition period ending in the summer of 2028.[267] These stakeholders raised several issues and questions early in the EDAM stakeholder process and expressed appreciation for the CAISO's collaboration, which resolved many of the questions and issues. Given the critical role of resource adequacy in the West and the central role WRAP will play, the CAISO worked to ensure that the market design respected WRAP functionality, and believes that its proposed solution will provide a high degree of certainty for WRAP participants to meet their bilateral obligations to one another. In the reviewing the draft tariff language, however, these stakeholders raised two additional related issues related to the WRAP similar to the stakeholder comments addressed above.

First, they contended that real-time self-schedules using firm or conditional firm transmission service rights for WRAP transactions backed by capacity should be given a scheduling priority in the extended day-ahead market higher than the scheduling priority of cleared day-ahead schedules. While the CAISO did not adopt a special resource adequacy scheduling priority, the CAISO clarifies that if instructed by the EDAM transmission provider, certain intra-day self-schedules using specific firm OATT transmission service rights may be afforded higher priority than cleared day-ahead schedules when so instructed by EDAM transmission providers acting in accordance with their OATTs. Such firm OATT transmission rights may support WRAP schedules, and insofar as they do, WRAP schedules would be afforded the same higher priority treatment. However, such treatment should arise out of the EDAM transmission provider's OATT requirements. In practice the CAISO expects it will be able to accommodate most, if not all, such self-schedules using firm and conditional firm rights after the Commission's 10:00 a.m. deadline through redispatch and thus WRAP transactions should have a high degree of confidence of delivery.

---

266 *Nw. Power Pool*, 182 FERC ¶ 61,063 (2023).

267 *Id.* at PP 13-14.

In the alternative, certain stakeholders proposed that the CAISO should address more specifically in the CAISO tariff how EDAM will accommodate transmission service provider "carveouts" to accommodate intra-day self-schedules in the real-time market that were not scheduled day-ahead. Based on this feedback, the CAISO revised new tariff section 33.18.3.3 to clarify the role of the EDAM transmission provider tariff to establish the narrow and limited circumstances when it would request the CAISO restrict the transmission available to the day-ahead market, with the CAISO simply accepting notice from the balancing authority as supported by the transmission service provider's tariff. Further, as the CAISO explained in the EDAM Final Proposal, the CAISO remains committed to continue engaging with EDAM participants and WRAP representatives, and it will strive to ensure and build upon the interoperability between EDAM and the WRAP through future evolution of the market design as needed.[268] Thus, if experience with EDAM and the WRAP over the next few years reveals interoperability issues, the CAISO will work with stakeholders and WRAP representatives to address them at that time.

Lastly, stakeholders generally understood and did not oppose the requirement to require customers that submit a bid in the day-ahead market and have not reserved transmission capacity associated with their schedule under the applicable transmission tariff to pay a transmission charge for use beyond a transmission reservation. Nonetheless, some stakeholders focused instead on the CAISO tariff requiring transmission service providers to offer an hourly firm service option. The CAISO has not proposed this as a requirement in its tariff, understanding that some transmission service providers may not currently provide hourly firm service. However, hourly firm transmission service is not a requirement under the *pro forma* OATT and is not something required to support extension of the day-ahead market. Stakeholders recommending this approach are essentially engaging in a collateral attack on the Commission's prior decisions not to require transmission service providers to offer firm hourly transmission service. Transmission service providers participating in EDAM are in the best position to consider whether to offer hourly firm transmission service to their customers, not the CAISO.

2. Ensuring Resource Sufficiency in the Day-Ahead Time Horizon

Resource sufficiency has been a cornerstone of the expansion of the CAISO's markets. The success of the WEIM is, in part, due to the foundational support of the WEIM resource sufficiency evaluation, which ensures each balancing authority participating in the WEIM can adequately balance its own supply and demand prior to accessing the CAISO's real-time market. The WEIM resource sufficiency evaluation has functioned to ensure each participating

[268] EDAM Final Proposal at 9.

balancing authority retains its respective resource adequacy obligation to ensure long-term availability of sufficient resources within its own area while allowing each balancing authority to accrue benefits through a wider-area optimization in real-time.

In designing EDAM, the CAISO worked with stakeholders to incorporate a uniform resource sufficiency evaluation into the extended day-ahead market design.[269] As further explained below, the structure of the EDAM resource sufficiency evaluation incentivizes each balancing authority to adequately demonstrate a balance of its own supply and demand prior to entering the day-ahead market. Under this structure, balancing authorities in the EDAM footprint that satisfy the EDAM resource sufficiency evaluation will receive the revenue generated by resource sufficiency evaluation failure surcharges paid by those that do not pass the evaluation and will be pooled together for purposes of the WEIM's resource sufficiency evaluation. A balancing authority in EDAM that cannot demonstrate it has sufficient resources to satisfy the obligations for its own balancing area will be allocated an EDAM resource sufficiency evaluation failure surcharge, but it will remain eligible for inclusion in the pool of passing entities if the insufficiency is cured through the day-ahead market. If the failing balancing authority does not cure the insufficiency through the day-ahead market, then it will be evaluated as a stand-alone balancing area for purposes of the WEIM resource sufficiency evaluation. Serving as a uniform standard applicable to all balancing authorities in the EDAM footprint, the EDAM resource sufficiency evaluation is a just and reasonable pillar supporting the EDAM framework.[270]

a. Uniform Assessment of Resource Sufficiency for All EDAM Balancing Areas

A key element of the just and reasonable design of EDAM is the uniform application of the EDAM resource sufficiency evaluation. As proposed, the EDAM RSE respects each balancing authority's existing resource adequacy regime and its control over its own transmission system, but it also sets a minimum expectation of the evaluation of sufficient resources needed to satisfy the day-ahead needs of the balancing authority. Because balancing authorities across the West are not subject to a common resource adequacy or resource planning program, the EDAM RSE serves as the common mechanism to ensure

---

269 New tariff sections 33.11.2, *et seq.* and 33.31.1, *et seq.*; *see also* EDAM Final Proposal, appendix 1: EDAM RSE Application Formulation (proposing a formulation for the EDAM resource sufficiency evaluation).

270 *See, e.g.*, *Cal. Indep. Sys. Operator Corp.*, 153 FERC ¶ 61,087, at P 72 (2015) (explaining that a uniform resource sufficiency evaluation, applied to all entities on an equivalent basis, is just and reasonable).

day-ahead supply sufficiency without duplicating or supplanting the existing resource adequacy regimes. The uniform EDAM RSE ensures each balancing authority can meet its own obligations before it engages in transfers with other balancing authorities in the EDAM area through the day-ahead market. By requiring each balancing authority in the EDAM area (*i.e.*, the aggregate of all balancing areas participating in EDAM) to demonstrate sufficient forward procured supply each day, the EDAM RSE structure discourages entities with insufficient resources from "leaning" on its neighboring balancing authorities. The CAISO designed the EDAM RSE proposed herein to work in harmony with, and not act as a replacement of, the WEIM resource sufficiency evaluation assessments performed in real-time.[271]

Building upon the principles of the WEIM resource sufficiency evaluation, where the resource sufficiency evaluation tests are uniformly applied to a diverse group of balancing authorities, the EDAM RSE will apply uniformly to produce an hour-by-hour assessment of resource sufficiency for the day-ahead timeframe for each balancing authority in EDAM. In particular, the proposed tariff revisions provide the structure for performing the EDAM RSE prior to running the day-ahead market.[272] The CAISO will perform "advisory" runs of the EDAM RSE and provide the results to each balancing authority at 6:00 a.m. and 9:00 a.m. each day. The CAISO will perform the final binding run of the EDAM RSE at approximately 10:00 a.m. In each advisory and binding run, the EDAM RSE application will establish the hourly quantity of upward sufficiency or downward insufficiency for each balancing authority in the EDAM area looking across three components in the day-ahead timeframe: forecasted demand, imbalance reserves, and ancillary services requirements.[273]

This framework is consistent with each balancing authority's individual obligations to meet its individual requirements for forecasted load, uncertainty, and ancillary services. Specifically, each balancing authority in the EDAM area will submit its demand forecast and variable energy supply forecast to establish its daily demand requirement prior to 9:00 a.m., either by using the CAISO's forecasting system or a direct submission from an approved forecasting provider.[274] To recognize the diversity benefit of participating in EDAM, the CAISO will establish daily imbalance reserve requirements for each balancing authority that account for the breadth and depth of the broader geographic EDAM footprint and the supply options made available therein. Each balancing

---

271 Existing tariff sections 29.34(l)-(n). The EDAM resource sufficiency evaluation is also compatible with the resource adequacy program in the CAISO balancing area and other balancing areas that may participate in the Western Resource Adequacy Program (WRAP).

272 New tariff section 33.31.1.

273 New tariff sections 33.31.1.2 and 33.31.1.3.

274 New tariff sections 33.31.1.2.1.1 and 33.31.4.

authority's imbalance reserve requirements will be fixed at 9:00 a.m., thereby providing entities with certainty as to their resource sufficiency obligations that can be addressed prior to the final binding EDAM RSE at 10:00 a.m. Ancillary services requirements for each balancing authority will be submitted prior to 9:00 a.m. and in accordance with the CAISO's ancillary services self-provision requirements in existing tariff section 8.

To perform the evaluation, the EDAM RSE application will model an EDAM balancing area's entire load and supply on a single bus[275] and perform a unit commitment optimization using all of the existing CAISO resource models. The optimal function of the EDAM RSE will be set to minimize the total cost as a means to determine the most efficient use of the varying resource types and capabilities made available to EDAM. By using existing market models in each hour of the day-ahead time horizon, the EDAM RSE application will establish the hourly requirements in the upward and downward direction for each balancing authority in the EDAM area across each of the three component parts and will compare each balancing authority's requirements against the same balancing authority's EDAM RSE-eligible supply. Resources within the CAISO balancing area can be resource sufficiency evaluation-eligible supply without undertaking additional registration requirements; intertie resources and EDAM resources will be designated as resource sufficiency evaluation-eligible supply following completion of the registration process that will be further described in the Business Practice Manual for the Extended Day-Ahead Market.

A balancing authority in the EDAM area will pass the final binding EDAM RSE if it meets all of the component requirements in each hour of the day-ahead time horizon. In recognition of the common demonstration of daily resource sufficiency, the passing EDAM entities will be "pooled" together and evaluated as a single group for purposes of the WEIM resource sufficiency evaluation.[276] Entities that satisfy the upward requirements of the EDAM RSE will be pooled together in the EDAM upward pool. Likewise, entities that satisfy the downward requirements of the EDAM RSE are pooled together in the EDAM downward pool. If a balancing authority in the EDAM area fails to satisfy all component requirements of the EDAM RSE, such entity remains eligible for inclusion in the

---

[275] The EDAM RSE application will not reflect a full security constrained economic dispatch because it does not include transmission constraints or calculate resultant power flows. EDAM Final Proposal at 60.

[276] The EDAM RSE application will test an EDAM entity's ability to meet its balancing area requirements, including demand and ancillary service obligations, in each of the 24 hours of the day-ahead market run, as well as the flexibility to ramp between the requirements in each hour. Accordingly, in the WEIM, the CAISO will only evaluate balancing authorities in the EDAM area pursuant to the WEIM RSE flexibility test and not the capacity or balancing tests. New tariff section 33.31.1.4.1.

EDAM upward pool and/or the EDAM downward pool if it can resolve the insufficiency through supply procured in the IFM. If the entity cannot resolve its failure through the IFM, then the entity will not be placed into the pool, and it will be evaluated individually for purposes of the WEIM resource sufficiency evaluation.[277]

Entities placed in the EDAM upward pool and the EDAM downward pool commit to timely e-tag all imports and exports.[278] Entities that fail to comply with the required e-tagging protocols, and do not otherwise re-supply, will be removed from the pool.[279] Entities removed from the pool(s) also will be evaluated individually for purposes of the WEIM resource sufficiency evaluation.

b. EDAM Resource Sufficiency Evaluation Failure Surcharges

The tariff revisions set forth the framework for the EDAM RSE failure surcharge and provide for three types of surcharges: the *EDAM RSE On-Peak Upward Failure Insufficiency Surcharge*, the *EDAM RSE Off-Peak Upward Failure Insufficiency Surcharge*, and the *EDAM RSE Downward Failure Insufficiency Surcharge*.[280] Scheduling coordinators for EDAM entities that fail the EDAM RSE by more than a *de minimis* amount, including failing entities that cured their failure through the IFM, will be allocated the applicable EDAM RSE failure surcharge.[281] Revenue generated by the EDAM RSE failure surcharges will be allocated to the scheduling coordinators for the balancing authorities in the EDAM area that passed the EDAM RSE.[282] Each entity allocated charges for an EDAM RSE failure, or otherwise allocated revenue associated with such surcharges, will distribute the charge (or revenue) in accordance with the provisions of its applicable tariff(s).[283]

---

277 New tariff section 33.31.1.4.

278 For avoidance of doubt, the EDAM resource sufficiency evaluation does not propose a surcharge for tagging failures.

279 New tariff section 33.31.1.6. The CAISO will evaluate entities that are excluded from the pool as an individual balancing areas in the WEIM resource sufficiency evaluation.

280 New tariff sections 33.11.2 and 33.31.1.5.

281 The tariff defines a *de minimis* failure as the higher of 10 MW or an amount less than or equal to one percent of the balancing area's upward imbalance reserve requirement for the hour. New tariff section 33.31.1.5.1(i).

282 New tariff sections 33.11.2.2.1, 33.11.2.2.2, and 33.11.2.2.3.

283 The CAISO is currently undertaking a separate stakeholder initiative to address CAISO balancing area-specific elements required for EDAM participation, including how to allocate CAISO balancing area EDAM resource sufficiency evaluation failure surcharges and revenues, and options for curing CAISO balancing area EDAM advisory resource sufficiency evaluation

The *EDAM RSE On-Peak Upward Failure Insufficiency Surcharge* addresses deficiencies during on-peak hours, which present the most severe risk associated with identified resource deficiencies in the day-ahead time horizon. To incentivize all balancing authorities in the EDAM area to demonstrate sufficient supply and flexibility in each of the sixteen on-peak hours, the *EDAM RSE On-Peak Upward Failure Insufficiency Surcharge* applies in all sixteen on-peak hours during the day.[284] Stakeholders agreed that identified resource deficiencies during off-peak hours present a lesser degree of risk. Accordingly, the *EDAM RSE Off-Peak Upward Failure Insufficiency Surcharge* will yield a lower failure surcharge and will apply during the failed hour only. Likewise, the *EDAM RSE Downward Failure Surcharge* will be allocated to the applicable scheduling coordinator in the failed hour only.

The proposed tariff revisions govern the *EDAM RSE On-Peak Upward Failure Insufficiency Surcharge*.[285] The CAISO will calculate this surcharge as the product of three values to reflect (1) the failure quantity, (2) the applicable surcharge price, and (3) a scaled adder for severe and persistent failures. First, the CAISO will determine the failure quantity of the *EDAM RSE On-Peak Upward Failure Insufficiency Surcharge* based on the failing entity's highest *EDAM RSE Hourly Upward Deficiency Quantity* during the sixteen on-peak hours.[286] Second, the CAISO will determine the applicable surcharge price by the applicable trading hub price (*e.g.*, Mid-Columbia or Palo-Verde).[287] The CAISO selected these

---

shortfalls. *See* https://stakeholdercenter.caiso.com/StakeholderInitiatives/Extended-day-ahead-market-ISO-balancing-authority-area-participation-rules. That ongoing stakeholder initiative, which is anticipated to result in a future tariff amendment, is separate from the initiatives that resulted in this tariff amendment and need not be completed before the Commission issues an order accepting the tariff revisions contained in this filing.

284 New tariff sections 33.11.2.1.1 and 33.11.2.2.1. In accordance with the tariff, all of the times listed in these tariff sections are Pacific Time. *See* existing tariff section 1.3.2(i). Details on EDAM resource sufficiency evaluation Upward Credit for passing hours.

285 New tariff section 33.11.2.1.1.

286 The EDAM resource sufficiency evaluation Hourly Upward Deficiency Quantity is the MW sum total of the upward failures during any single operating hour. Tariff appendix A, new definition of EDAM RSE Hourly Upward Deficiency Quantity. The EDAM resource sufficiency evaluation On-Peak Upward Failure Insufficiency Surcharge utilizes the highest value during any of the sixteen on-peak hours as the failure quantity. New tariff sections 33.11.2.1.1 and 33.31.1.5.1; tariff appendix A, new definition of EDAM RSE On-Peak Upward Failure Insufficiency Surcharge.

287 Some stakeholders raised concerns regarding the ongoing liquidity of the bilateral hubs following implementation of EDAM. The CAISO commits to monitoring the liquidity of referenced trading hubs. If a hub does not remain sufficiently liquid, it may no longer be appropriate to use it as a reference point and the CAISO will reevaluate the proposed reference.

trading hubs as appropriate price proxies because they represents the cost the failing entity would bear as it sought additional supply to cure the insufficiency. Third, the scaled adder is composed of two parts, with the *EDAM RSE Failure Multiplier* reflecting the severity of the failure and the *EDAM RSE Failure Scaling Factor* reflecting the persistence of failures. The *EDAM RSE Failure Multiplier* reflects the severity of the failures, denoted by three tiers. Tier 1 failures are *de minimis* failures and the *EDAM RSE Failure Multiplier* will be set to zero; tier 2 failures are those of a magnitude less than or equal to fifty percent of the entity's upward imbalance reserve requirement and increase the *EDAM RSE Failure Multiplier* from zero to 1.25; tier 3 failures are those of a magnitude greater than fifty percent of the entity's upward imbalance reserve requirement and increase the *EDAM RSE Failure Multiplier* to 2. The *EDAM RSE Failure Scaling Factor* reflects persistence and adds one percent to the *EDAM RSE Failure Multiplier* for every additional day during the preceding thirty-day period in which the entity had a tier 2 or tier 3 failure in the upward direction.[288] For example, seven failures over the preceding 30 days would increase the *EDAM RSE Failure Multiplier* of a tier 2 failure from 1.25 to 1.31.

The *EDAM RSE On-Peak Upward Failure Insufficiency Surcharge* will also include a credit to account for any of the sixteen on-peak hours in which the entity satisfied the EDAM RSE. The CAISO will determine the *EDAM RSE On-Peak Upward Credit* as the product of the highest *EDAM RSE Hourly Upward Deficiency Quantity* and the load-weighted average LMP of the load aggregation point in the balancing area.[289] If the *EDAM RSE On-Peak Credit* amount exceeds the surcharge amount, the *EDAM RSE On-Peak Upward Failure Insufficiency Surcharge* will be capped at zero.[290]

The tariff also sets forth the *EDAM RSE Off-Peak Upward Failure Insufficiency Surcharge* and the *EDAM RSE Downward Failure Insufficiency Surcharge*.[291] Like the on-peak surcharge described above, the CAISO will calculate the off-peak surcharge as the product of three values to reflect: (1) the failure quantity, (2) the applicable surcharge price, and (3) a scaled adder for severe failures. Unlike the *EDAM RSE On-Peak Upward Failure Insufficiency Surcharge*, which applies in each of the sixteen on-peak hours, the *EDAM RSE Off-Peak Upward Failure Insufficiency Surcharge* applies only in the failed hour.

In addition, the tariff includes the *EDAM RSE Downward Failure Insufficiency Surcharge*.[292] The CAISO will calculate this surcharge as the

---

288 Tariff appendix A, new definition of EDAM RSE Failure Scaling Factor.

289 Tariff appendix A, new definition of EDAM RSE On-Peak Upward Credit.

290 New tariff section 33.11.2.1.1.

291 New tariff section 33.11.2.1.2.

292 New tariff section 33.11.2.1.3.

product of two values to reflect: (1) the failure quantity and (2) the applicable surcharge price. The CAISO will determine the failure quantity based on the *EDAM RSE Hourly Downward RSE Deficiency Quantity* during the failed hour, and the applicable surcharge price will be the marginal energy cost of the failing balancing area. The CAISO does not expect downward insufficiency to present a challenge across a more geographically diverse EDAM footprint and a scaled adder for severity is not included.

Through the stakeholder process, the CAISO evaluated the need for surcharges associated with failures of the resource sufficiency evaluation. The WEIM resource sufficiency evaluation does not include a component for financial penalties, but a physical limitation on transfers is a more appropriate consequence given the near real-time nature of the WEIM's imbalance product. In the day-ahead time horizon, locking transfers may not incentivize resource sufficiency in the same manner as it does in real-time and could significantly reduce the benefits of the extended day-ahead market. Each of the failure surcharge types are uniquely calibrated to address the severity of the failure and the time period in which the failure occurs and establish a targeted proxy price to address the specific risks presented. Each of the surcharges is narrowly tailored to address the unique circumstances presented, and taken together they constitute a just, reasonable, and not unduly discriminatory structure to address resource sufficiency failures.[293]

c. Stakeholder Positions on the Resource Sufficiency Evaluation Proposal

Some stakeholders initially questioned the need for a uniform resource sufficiency evaluation, contending that the resource adequacy programs governing the load serving entities within each balancing authority should be sufficient. In the stakeholder process, the CAISO and interested parties evaluated the various resource adequacy programs that govern entities that may wish to participate in EDAM including, but not limited to, the CPUC's resource adequacy program and the developing WRAP. After recognizing the differences in how the various programs ensure resource adequacy within their jurisdictions, stakeholders agreed that the EDAM resource sufficiency evaluation was an appropriate tool to complement and maximize the value of these programs by ensuring participants can account for the capacity and optimize use of resources they have procured to support reliability within their individual footprints. On

---

[293] *See, e.g.*, *Midcontinent Indep. Sys. Operator Corp.*, 175 FERC ¶ 61,093, at P 12 (2021) (determining that the proposed rate structure, which was narrowly tailored to address the unique circumstances, was just and reasonable); *Cal. Indep. Sys. Operator Corp.*, 145 FERC ¶ 61,135, at PP 9, 14 (2013) (finding that the proposed rate structure was just and reasonable as it was narrowly tailored to address the unique challenges).

balance, stakeholders agreed that using a uniform resource sufficiency evaluation to test the sufficiency of supply procured under different resource adequacy and resource planning programs was the appropriate means to ensure equitable participation optimal resource commitments. Although not all stakeholders agreed with utilizing the EDAM resource sufficiency evaluation, the large majority of stakeholders supported the EDAM resource sufficiency evaluation as a tool to ensure entities in different resource adequacy programs (or without any centralized resource adequacy program) all bring adequate resources to meet their individual balancing authority obligations.

In addition to ensuring each balancing authority in the EDAM area demonstrates resource sufficiency prior to the day-ahead time horizon, some stakeholders requested that the EDAM resource sufficiency evaluation also reflect transmission constraints and power flows over transmission facilities used to facilitate EDAM transfers. Underlying this concern is the potential that a balancing area could pass the EDAM resource sufficiency evaluation yet still have insufficient resources available to clear the day-ahead market. In the course of numerous working groups and stakeholder meetings, the CAISO explored this option and determined that including transmission elements in the EDAM resource sufficiency evaluation would require using the CAISO's full network model, which in turn would require integration with additional market applications. Functionally, the EDAM resource sufficiency evaluation application would become a proxy run of the CAISO's existing day-ahead market. Given the computation time required to run the application in this configuration, implementing such a configuration would require trade-offs between the number and frequency of advisory runs prior to the final binding EDAM resource sufficiency evaluation and the accuracy of the application itself. Stakeholders ultimately agreed that, while modeling transmission constraints would have some benefits, it would not increase the accuracy of the test sufficiently to justify the significant effort and complexity of implementing such modeling at this time. The CAISO will monitor and evaluate the EDAM resource sufficiency evaluation to ensure it performs as expected without the modeling of transmission constraints, and it will continue working with its stakeholders as EDAM is implemented to explore whether and how the EDAM resource sufficiency evaluation can reflect such constraints.

The CAISO worked closely with stakeholders to ensure that the EDAM resource sufficiency evaluation reflects all owned and contracted resources available for the next day. Stakeholders raised questions and concerns about the compatibility of the EDAM resource sufficiency evaluation with the various resource types owned and contracted within the balancing authorities considering joining EDAM, as well as external supply resources. To aid entities participating in EDAM in their gas procurement planning, the CAISO will provide EDAM resource sufficiency evaluation advisory results two days prior to the relevant day-ahead market to facilitate alignment with the gas nomination cycle.

For stakeholders with hydroelectric supply within their balancing area, the EDAM resource sufficiency evaluation permits entities to manage daily energy limits and hourly energy bids, which in turn inform minimum discharge obligations and availability to discharge. To ensure appropriate consideration of variable energy resource supply in the EDAM resource sufficiency evaluation, the EDAM resource sufficiency evaluation application will create default supply bids for the difference between the bid-in quantity and forecast quantity. The CAISO will use default supply bids derived from the variable energy resource forecast for purposes of the EDAM resource sufficiency evaluation only.

Stakeholders requested that the EDAM resource sufficiency evaluation also permit use of resource or load modification programs that EDAM cannot explicitly model, including delivered firm energy contracts commonly executed through arrangements pursuant to Service Schedule C of the Western Systems Power Pool (WSPP) Agreement and demand response programs within their borders. To accommodate delivered firm energy contracts, the CAISO will utilize a registration system to prevent double counting and ensure sufficient resource integrity before classifying such resources or programs as eligible for the EDAM resource sufficiency evaluation. Non-resource-specific intertie bids submitted at the border of the CAISO balancing area will count in the EDAM resource sufficiency evaluation when such bids are associated with a resource adequacy contract, originate from a resource pseudo-tied with the CAISO, originate from resources dynamically scheduled into the CAISO, or otherwise have an executed agreement with the CAISO that specifies performance and operating requirements. Emergency supply or load modification programs that inform a balancing authority's next-day operating plan may be accounted for through manual forecast adjustments, consistent with the methodology under the WEIM resource sufficiency evaluation.[294] The CAISO will monitor and evaluate the EDAM resource sufficiency evaluation-eligible supply construct and continue working with stakeholders as EDAM is implemented to ensure the EDAM resource sufficiency evaluation reflects all owned and contracted resources available for the next day.

3. Ensuring Sufficient Resources to Serve Balancing Area Needs

In evaluating whether a balancing authority has sufficient resources to meet its own obligations prior to the day-ahead market, EDAM will enable each balancing authority to utilize a configurable net export transfer constraint to communicate to the CAISO the amount of its resources that can be optimized to

[294] New tariff section 33.31.4.1; *see also* section V.C.2 of this transmittal letter (describing external resource participation in the EDAM area).

support EDAM transfers to another balancing area.[295] This mechanism enables each balancing authority to control its supply to meet its own needs before assisting the needs of others. The net export transfer constraint, a configurable feature, will allow balancing areas, including the CAISO balancing area, to manage its internal supply more effectively, while also allowing such supply for support regional needs. The constraint will be calculated as the available capacity of supply bids from all resources multiplied by a configurable confidence factor, minus the EDAM RSE requirements and a configurable non-exportable capacity margin. The net export transfer constraint formula is designed to allow each balancing authority to select the appropriate reliability margin and confidence factor input variables to reflect the unique needs of its balancing area. When enabled, the constraint will enforce an hourly limit on the amount of net EDAM transfer exports from a balancing area in the EDAM footprint. No balancing authority can reduce the constraint to impose a limit below the higher of zero or the transmission service made available to support a net export in the EDAM resource sufficiency evaluation. A balancing authority must elect to enable the constraint prior to 9:00 a.m. on the day before the trading day and may choose which hours of the operating day the constraint will apply.

The voluntary adoption of this constraint prior to the trading day will allow each balancing area to maintain excess supply to meet its expected day-ahead needs and manage reliability conditions. In the CAISO balancing area, this feature protects against EDAM export transfers from resource adequacy capacity, similar to the CAISO's ability to recall lower-priority exports that are not contractually obligated to serve external load. In participating balancing areas outside of California, this constraint will also enable non-CAISO balancing authorities to manage reliability and resource adequacy obligations, such as obligations arising from participation in WRAP.

The EDAM resource sufficiency evaluation, in conjunction with the net export transfer constraint, will instill confidence in the availability of supply to realize an optimal market solution and encourage entities to make excess supply available to the market beyond what they need to meet the EDAM resource sufficiency evaluation.

C. Day-Ahead Market Activities

Section III.A.1 of this transmittal letter describes operation of the day-ahead market in the CAISO balancing area, section IV.G of the transmittal letter describes the connection between DAME and EDAM, and section V.B.1 of the transmittal letter describes the scheduling of transmission to support EDAM

---

[295] New tariff section 33.31.3; *see also* EDAM Final Proposal, appendix 4: Mathematical Formulation of the Net EDAM Export Transfer Constraint (proposing a detailed formulation for this constraint).

transfers. This section V.C of the transmittal letter describes additional changes necessary to extend the day-ahead market to other balancing areas. Otherwise, EDAM participants representing supply and demand will submit economic bids and self-schedules according to the same timelines and procedures as other scheduling coordinators in the day-ahead market today.[296]

1. Virtual Bidding

The CAISO day-ahead market currently enables virtual bidding, also known as convergence bidding. Although virtual bidding has benefits, it also has resulted in some of the Commission's most significant enforcement actions for market manipulation.[297]

Based on overwhelming stakeholder input, the CAISO proposes to allow EDAM transmission service providers to enable virtual bidding in their balancing areas at the onset of their participation in EDAM but will not mandate it.[298] EDAM virtual bidding would function just as CAISO virtual bidding functions today, except with a larger footprint.[299] Alternatively, EDAM balancing authorities may elect to forgo allowing virtual bidding in their balancing areas at the outset of EDAM. A large number of external stakeholders staunchly objected to implementing mandatory virtual bidding at the start of EDAM. They stressed their lack of experience with day-ahead markets, their unfamiliarity with virtual bidding, and their concerns about the potential financial and reliability effects impacts of virtual bidding in the early stages of EDAM under these circumstances. Once EDAM participation begins, the CAISO will begin a stakeholder process to determine a permanent virtual bidding design for the EDAM and CAISO footprint. The optional transition period will give EDAM transmission service providers, their customers and market participants, and their local regulators experience and comfort with two-day markets before enabling virtual bidding. During the EDAM stakeholder initiative, many EDAM market participants and regulators stressed the importance of such a transition period in the decision to join the day-ahead market.

---

296 New tariff section 33.30.

297 *See, e.g.*, *Barclays Bank PLC*, 161 FERC ¶ 61,147 (2017).

298 New tariff section 33.30.7.

299 The CAISO will retain its authority to suspend or limit virtual bidding. *See* existing tariff section 7.9. Each EDAM Entity may recommend that the CAISO suspend convergence bidding in its balancing area, provided that the CAISO will make the ultimate determination as to such recommendation. New tariff section 33.30.7.3.

If an EDAM transmission service provider elects to use the optional transitional period, load serving entities within the balancing area may not bid or self-schedule demand above the amount of supply within the balancing area. The CAISO will enforce this demand limitation based on the available supply, accounting for the difference between variable energy resource bids and their forecasts.[300] This limitation will help mitigate the risk of inaccurate price signals in the day-ahead market.

The CAISO and the Department of Market Monitoring will continue to monitor and evaluate the extended day-ahead market's performance with or without convergence bidding in balancing areas to help inform the evolution of the future design and address any unintended consequences of an optional convergence bidding design at the onset of EDAM.

2. External Resource Participation

External resource participation in EDAM refers to the opportunity for resources physically located outside of the EDAM area, *i.e.*, resources located in a non-EDAM balancing area (including WEIM balancing areas that have not joined EDAM) to participate in the day-ahead market operating in the EDAM area. The approach to external resource participation in EDAM provides numerous options for source-specific and non-source-specific resources to bid economically and self-schedule into the day-ahead market consistent with current practices in the CAISO balancing area and balancing areas in the WEIM. Accordingly, the tariff requirements for participation in EDAM by external resources accounts for existing rules at CAISO interties,[301] existing rules at WEIM interties,[302] and the resource sufficiency evaluation.[303] This means the opportunities for participation in the extended day-ahead market may be different depending on whether the external resource is associated with a pseudo-tie, a dynamic schedule, a self-schedule, or an economic bid.[304] Further, the

---

300 New tariff section 33.30.7.2.

301 *See, e.g.*, existing tariff sections 27, 30.5.2.4.1, and 31.8.

302 *See, e.g.*, existing tariff section 29.34(f)(3) (providing for base schedules of imports and exports); *see also* existing tariff section 29.34(i)(2) (providing an EIM balancing authority discretion whether to enable economic intertie bidding, which to date has not been enabled by any participant); *Ariz. Pub. Serv. Co.*, 155 FERC ¶ 61,112, *reh'g denied*, 156 FERC ¶ 61,227 (2016) (accepting proposal to allow external resources to participate in the WEIM via dynamic scheduling).

303 New tariff section 33.30.8.3.

304 Dynamic schedules and pseudo-ties are the two basic categories of dynamic transfers. A dynamic transfer is the transfer of energy or ancillary services from resources interconnected in one balancing area into another balancing area pursuant to a dynamic signal processed in the

resource's participation may depend on whether the intertie is associated with a boundary between a balancing area in the WEIM and a balancing area outside the WEIM, an external intertie in all cases, a boundary between a balancing area in the WEIM and EDAM, an external intertie with the EDAM area, or between a balancing area in the WEIM or EDAM and the CAISO balancing area, an internal EDAM area intertie.[305]

The table below illustrates the different types of external resources and their ability to offer supply into the day-ahead market under EDAM.

Type of External Resource	Market Participation
1. Source-Specific: Pseudo-Tied Resource	Economically bid and self-schedule in the day-ahead market
2. Source-Specific: Dynamically Scheduled Resource	Economically bid and self-schedule in the day-ahead market
3. Off-System Designated Network Resource	Economically bid and self-schedule in the day-ahead market
4. Other Contracted Supply[306]	Self-schedule in the day-ahead market at EDAM external interties
5. Non-Source-Specific, Non-Contracted Supply	Cannot self-schedule or economically bid into the market at EDAM external interties, except at CAISO external interties, unless economic bidding has been enabled according to existing tariff section 29.34(i)(2)

balancing authorities' energy management systems. A dynamic schedule is a dynamic transfer in which the resource supplying the energy or ancillary services remains under the control of the host balancing authority – the balancing authority for the balancing area where the resource is interconnected to the electric system. Under a dynamic schedule, the host balancing authority includes the resource's output in its balancing of supply and demand. A pseudo-tie is a dynamic transfer in which the balancing authority for the balancing area into which a resource's energy and ancillary services are delivered – called the attaining balancing authority – accounts for the resource supplying the energy or ancillary services. The attaining balancing authority also performs other balancing authority functions for the resource, even though the resource is interconnected to the electric system within another balancing area – called its native balancing area. *See* existing tariff sections 4.14 – 4.15; tariff appendix A, existing definitions of Dynamic Schedule and Pseudo-Tie; existing tariff appendices M and N.

305 New tariff section 33.30.8.

306 This type of external resource includes supply under contract to serve load in the balancing area as further described below.

First, external resources delivered through a pseudo-tie are deemed located within the attaining balancing area based on modeling, telemetry, and other requirements. These arrangements and obligations are captured in tariffs and under contractual provisions between the balancing authorities and the resource owner, and they establish obligations related to performance, delivery, and other operational requirements. As such, these resources will continue their association with either a WEIM, EDAM, or CAISO intertie consistent with the arrangements they use today.[307]

Second, external resources delivered through a dynamic schedule represent source-specific supply located outside of a balancing area, but the market treats them as supply available to the balancing area similar to other dispatchable resources inside the balancing area. Because the resource remains outside of the balancing area, the introduction of the WEIM required further modeling that the CAISO implemented through a feature called a "mirror system resource."[308] This modeling feature allows accurate real-time pricing of a resource located in a WEIM balancing area dynamically scheduled into another balancing area in the WEIM area, including the CAISO balancing area, and will continue between a WEIM balancing area that joins EDAM and a balancing area remaining in the WEIM. These resources may continue to participate in EDAM. However, extending the day-ahead market to other balancing areas will mean that dynamic schedules that have been between WEIM balancing areas may become dynamic schedules between two balancing areas in EDAM. In these cases, the transfer of the resource's output through a dynamic schedule will no longer produce accurate modeling and pricing, even with a mirror system resource, so pseudo-ties will be required to implement these dynamic schedules.[309]

This inaccuracy occurs because the resources in the source EDAM balancing area that produce the energy for the dynamic schedule are also optimally scheduled to produce energy for transfers between the EDAM balancing areas. The mirror resource solution works for dynamic schedules into

---

307 New tariff section 33.30.8.1.

308 Existing tariff section 29.27(c).

309 Resources located in an EDAM balancing area that otherwise may have utilized dynamic scheduling in the WEIM will instead be able to submit an economic bid at their location and have the opportunity to transfer their energy or capacity between balancing areas based on the optimal market solution. In addition, these resources can be designated to support the resource sufficiency evaluation of another balancing area, essentially assuring the transfer capability is available and the associated delivery is economic. If these options are insufficient to support the delivery of supply from the resource, pseudo-tying the resource from one EDAM balancing area to another EDAM balancing area, including the CAISO balancing area, will be necessary.

the CAISO balancing area today because it is possible to attribute the import resources with an adder to the bid price of the resources in the source balancing area. This same feature will not work when the day-ahead market is extended to other balancing areas because a source balancing area participating in EDAM will have its resources optimized in the same day-ahead market solution as other balancing areas in the day-ahead market, including the CAISO balancing area. Dynamic schedules without the mirror resource feature working properly would clear only after the EDAM transfers are at their respective scheduling limits, and they may not even clear even when their bid price is lower than the LMP at their location. To address this condition and permit dynamic scheduling among balancing areas in EDAM, it would be necessary to dedicate transmission capacity which is inefficient because that dedicated transmission capacity could support economic transfers but may go unused if the dynamically scheduled resource is not economic. Accordingly, resources within EDAM balancing areas will be required to pseudo-tie if they are dynamically scheduled to another EDAM balancing area; EDAM will not support dynamic schedules between EDAM balancing areas.

Third, the EDAM design will support external resource participation by source-specific off-system designated network resources.[310] An external resource located in a balancing area outside of the EDAM area can economically bid or self-schedule at the intertie where the resource is a network resource to serve load under the terms of the transmission service provider's tariff.[311] Because the resource is a network resource under the terms of the tariff, its supply is deliverable on firm transmission to the intertie of the balancing area where the load is served, and it will be delivered to load using the network integration transmission service across the balancing area where the load is located. Allowing participation by an off-system designated network resource in this way allows the generation the load serving entity has procured and designated to serve its load to bid economically into the market, and in turn it allows the day-ahead market to serve the demand more cost-effectively. Similarly, source-specific supply that is otherwise not associated with a designated network resource, but is owned or under contract to serve load within

---

310 New tariff section 33.30.3. The resource must be located in a WEIM balancing area, not a balancing area outside of the WEIM. This ensures appropriate modeling of the resource in the CAISO full network model, and the resource characteristics and abilities will be registered and visible to the market operator. Similarly, bids must be associated with the intertie of the EDAM balancing area where the load associated with the designated resource is located, and the bid-in quantity cannot exceed the amount of the off-system network resource being designated.

311 An off-system designated network resource under the terms of the Commission's *pro forma* OATT can be designated to serve load based upon: (1) an attestation that the designation is supported by a contract for the supply, (2) the supply will be delivered on firm transmission to the balancing area where the load is located, and (3) the resource is designated to serve that specific load. *See Preventing Undue Discrimination & Preference in Transmission Serv.*, 123 FERC ¶ 61,299, at PP 177-78, 182-85 (2008).

a balancing area participating in EDAM, can self-schedule or economically bid at an intertie with a balancing area in the WEIM or outside of the WEIM. These resources have a contractual relationship with the balancing authority in EDAM or another load serving entity within the balancing area and their delivery is reliable.

However, and fourth in the table above, other contracted supply must be self-scheduled and cannot economically bid at an intertie between a balancing area in EDAM and a balancing area outside of EDAM. This may include non-source-specific import supply under contract, such as firm energy contracts where the source of the generation is unknown ahead of the 10:00 a.m. day-ahead market close (*e.g.*, WSPP Schedule C arrangements), which can self-schedule at an EDAM intertie.[312] These contracted supply arrangements impose certain performance and delivery requirements that instill greater confidence regarding their performance and delivery. Moreover, the balancing authority can better account for these from a reliability perspective by considering them when defining the level of ancillary services procurement necessary to support them.

Finally, non-source-specific supply not under contract to serve demand within an EDAM balancing area cannot offer supply at external interties of balancing authorities participating in EDAM, either by self-scheduling or economic bidding, unless the balancing authority has enabled economic bidding at external interties with respect to the WEIM.[313] This extension of the Commission-accepted WEIM approach to EDAM is premised upon stakeholder reiteration of the same concerns expressed in the development of the WEIM, *i.e.*, a concern that EDAM balancing areas will be exposed to increase uncertainty about the delivery of the supply, which could raise reliability risks if the supply is not delivered as expected.

This exclusion from participation in EDAM is limited to non-source-specific supply resources at EDAM external interties with balancing authorities participating in EDAM, and does not apply at the CAISO external interties.[314] Within the CAISO balancing area, and unlike in other balancing areas

---

312 The tariff revisions also specify that bids from delivered firm energy contracts may participate in the day-ahead market, and that such contracts include but are not limited to WSPP Schedule C arrangements, CAISO resource adequacy imports, and similar forward contracted supply. New tariff section 33.30.8.2. Further, all energy scheduled from such non-resource-specific supply contracts must have a submitted e-tag within three hours following publication of the day-ahead market results and satisfy other e-tagging requirements. New tariff section 33.30.8.3.

313 *See* existing tariff section 29.34(i)(2) (providing balancing areas the option to enable economic external intertie bidding by non-source specific resources). To date, external economic external intertie bidding by non-source-specific resources has not been enabled by any WEIM balancing area.

314 New tariff section 33.30.3.

participating in EDAM, energy and ancillary services procurement is co-optimized through the market and, as such, the CAISO can better manage potential risks associated with economic bidding at its external interties. Moreover, economic bidding at CAISO external interties has been a longstanding and beneficial feature of the CAISO market, providing suppliers an opportunity to offer excess supply into the market, enabling the CAISO to serve demand more economically. Other balancing authorities considering participating in EDAM expressed reliability concerns about these types of transactions at their interties, and the CAISIO is precluding them at the onset of EDAM unless a balancing authority has enabled economic bidding of such supply under its tariff consistent with the WEIM rules. In any event, the CAISO remains committed to working with market participants to evaluate the EDAM and WEIM external resource participation model and evolve the design as participants gain experience in the day-ahead market and supply structures across the West continue to evolve, potentially in a manner that further mitigates the reliability concerns many stakeholders have expressed.

Most stakeholders support or do not oppose continuing to allow economic bidding at CAISO external interties because this historically has provided an opportunity to derive value for excess supply that may bid into the day-ahead market. Stakeholders recognize the value the CAISO balancing area derives from excess supply offered economically by non-source-specific resources in the day-ahead market. Economic bidding at the CAISO external interties has been a longstanding and beneficial feature, allowing suppliers to offer excess supply to the CAISO balancing area and enabling the CAISO to serve demand more economically. Some stakeholders expressed equity concerns about allowing economic intertie bidding at the CAISO external interties but not allowing it to the same extent at the external interties of other EDAM balancing areas. Nonetheless, the CAISO views a continuation of the practices in the WEIM and the CAISO balancing area to be a reasonable starting point for EDAM, namely allowing non-resource specific resources to bid economically at the CAISO interties, but not allowing them to bid economically at the interties of other balancing authorities participating in the WEIM/EDAM unless authorized at the balancing area interties.

3. Greenhouse Gas Accounting

a. State Programs

The CAISO's EDAM design accounts for the costs arising from state GHG accounting and reduction policies that price carbon.[315] This includes factoring in GHG emission costs incurred for GHG regulatory compliance associated with

[315] States other than California and Washington, such as Oregon, have greenhouse gas policies that address emissions without specifically pricing carbon.

power transactions, reflecting those costs in the CAISO's security constrained least-cost dispatch, and facilitating any required GHG reporting and verification processes. The CAISO proposes to extend the WEIM GHG framework to EDAM. This design allows resources' scheduling coordinators to recover their cost of compliance with a state's carbon pricing policy and provides a mechanism to identify which scheduling coordinators are electricity importers into a GHG regulation area. To ensure the market respects state policies, it does not reflect the price of carbon in LMPs paid by demand outside of a GHG regulation area. The CAISO also proposes several enhancements to its current GHG design to reflect new state programs and address stakeholder concerns involving the potential for emissions leakage.[316] The CAISO also proposes enhancements to the WEIM tariff provisions to align the day-ahead market GHG accounting rules with the real-time market.

California's greenhouse gas cap-and-trade program requires covered entities, including electric generators and electricity importers, to retire allowances for the emissions they produce. These allowances are available to emitters via auction or bilateral purchases, ultimately establishing a price on carbon.[317] Within California, emitting resources can reflect the cost of compliance in their energy bids similar to other costs associated with the production of energy. For imports into California associated with the WEIM, California's program utilizes a "first jurisdictional deliverer" structure to identify the compliance entity for imports. California regulations define the WEIM participating resource scheduling coordinator as the entity with the compliance obligation, if its resource receives an attribution to serve load within California.[318] For resources located outside California and participating in the WEIM, the resources may use a bid adder to indicate a willingness to serve demand in California.[319] The CAISO's proposal extends this existing framework into EDAM,

---

316 Revised tariff section 29.32, *et seq.* and new tariff section 33.32, *et seq.* Secondary dispatch can occur if lower-emitting resources dispatched to serve demand in a GHG area require additional higher-emitting resources to backfill to serve demand outside of that GHG regulation area. The CAISO discusses secondary dispatch in more detail below and explains it can lead to "leakage" under state GHG policies.

317 In some cases, limited free allowances may also be allocated to certain entities or industries to ease the transition into a compliance program, reduce leakage, or offset ratepayer costs.

318 California Code of Regulations Title 17, section 95802(a) defines the electricity importer as the WEIM entity scheduling coordinator. The California Air Resources Board (CARB) is currently undertaking a rulemaking to update the rules for EDAM. The CAISO anticipates CARB will update the existing definitions to include reference to EDAM.

319 At the outset of the WEIM, the Commission recognized that WEIM participating resource scheduling coordinators could submit a bid adder to reflect their GHG costs on a daily basis for each of their WEIM participating resources. WEIM Authorization Order at PP 238-40. The

but with some additions specific to the nature of the day-ahead market to recognize the potential for additional GHG regulation areas. Using bid adders allows the market to attribute transfers into a GHG regulation area to EDAM resources and WEIM participating resources. It also allows sellers of power to recover their production costs associated with compliance with a state program that prices carbon.

Beginning in 2023, Washington implemented a cap-and-invest program. Washington's program uses its own compliance instruments, so the price of carbon is different from the price of carbon under California's program. If an emitting resource located in California is supporting an import into Washington or vice versa, the resource would separately have a compliance obligation under both states' regimes.[320] Like California, Washington's program identifies the first jurisdictional deliverer[321] as the compliance entity for electricity imported into the state. Washington defines a "first jurisdictional deliverer" as the owner or operator of an electric generating facility in Washington or an electricity importer. The statue does not define which entity serves as the "importer," and instead defers the determination to a future rulemaking.[322] Washington's Climate Commitment Act requires Washington agencies to define a more specific rule and methodology for addressing imported electricity associated with a centralized electricity market by October 1, 2026.[323] Washington's Department of Ecology has initiated a rulemaking.[324] The CAISO's EDAM framework allows Washington

---

Commission later approved the CAISO's proposal to allow WEIM participating resource scheduling coordinators to submit an hourly bid quantity to express how much of the resource's output could support an EIM transfer serving CAISO demand through the submission of an hourly price in its bid adder. *Cal. Indep. Sys. Operator Corp.*, 153 FERC ¶ 61,087, at PP 57-58 (order on WEIM Year 1 Enhancements).

320 This would occur in a situation where a resource located in either California or Washington receives an attribution through the market to serve demand in the other state.

321 Revised Code of Washington (RCW) section 70A.65.080(1)(c).

322 RCW 70A.65.010(38).

323 Washington's Climate Commitment Act states that by October 1, 2026, the department of ecology, in consultation with the department of commerce and the utilities and transportation commission, shall adopt by rule a methodology for addressing imported electricity associated with a centralized electricity market. RCW 70A.65.080(1)(c). The Washington Department of Ecology has put out a temporary rule that defines the importer as the "energy imbalance market purchasing entity located or operating in Washington that receives the delivery of electricity transacted through the energy imbalance market." *See* https://ecology.wa.gov/Regulations-Permits/Laws-rules-rulemaking/Rulemaking/WAC-173-441-446#CR101; WAC (Washington Administrative Code)-173-446-040(3)(e)(iv), available at https://app.leg.wa.gov/WAC/default.aspx?cite=173-446-040&pdf=true.

324 *See* https://ecology.wa.gov/regulations-permits/laws-rules-rulemaking/rulemaking/wac-173-441-446.

to identify EDAM resource scheduling coordinators and WEIM participating resource scheduling coordinators as the electricity importer based on market optimization results because the market tool will identify these entities as the source of emissions and allow them to report.

b. Proposed GHG Regulation Areas and Bid Adders

To recognize multiple state GHG programs, the CAISO proposes to update the current balancing area boundaries used for modeling and GHG accounting to create separate GHG regulation areas based on the GHG boundary as defined by a state's jurisdiction.[325] This will allow the CAISO to reflect the costs of compliance for resources within those boundaries, but it will not subject resources outside of the GHG regulation areas to these costs unless they offer their supply to support a GHG transfer into a GHG regulation area. The CAISO will model GHG Regulation Areas[326] in EDAM and the WEIM to align the boundaries across all market runs.[327] However, the pricing nodes included in a specific GHG Regulation Area for the fifteen-minute market and real-time dispatch, and that same GHG Regulation Area for EDAM, may differ because some balancing areas may participate in the WEIM but not in EDAM.[328]

Currently, the CAISO will only model California and Washington as GHG regulation areas because they are the only two states in the WEIM area pricing carbon. However, the CAISO's modeling approach allows for flexibility of either including new GHG regulation areas if additional states implement similar programs, or collapsing GHG regulation areas into one GHG regulation area if jurisdictions link their programs. Furthermore, this approach allows the CAISO to model unique circumstances such as special state provisions for entities whose

---

325 New tariff section 33.32.1.2 describes GHG Regulation Areas. These will reflect the pricing nodes of the CAISO balancing area or an EDAM entity balancing area within a state jurisdiction that has priced GHG emissions as part of a state GHG reporting and reduction program. Resources pseudo-tied or dynamically scheduled into a GHG Regulation Area will be considered to be located within that area.

326 The CAISO proposes a new defined term "GHG Regulation Area" that encompasses the nodes of a balancing area within a state jurisdiction that has priced greenhouse gas emissions as part of a state carbon reduction law or regulation. Tariff appendix A, new definition of GHG Regulation Area.

327 New tariff section 29.32(a) amends existing tariff language to incorporate the concept of multiple GHG Regulation Areas.

328 New tariff section 29.32(a)(2).

generation or load is not associated with the state for GHG accounting purposes, such as certain areas under federal jurisdiction.[329]

Using this new framework for modeling GHG regulation areas, the CAISO proposes to extend to EDAM a bid adder structure similar to what the Commission approved for the WEIM. This design allows a scheduling coordinator for a resource located outside of a specific GHG regulation area to submit a voluntary GHG bid adder,[330] which reflects its willingness to serve demand in a GHG regulation area as an electricity importer.[331] GHG bid adders will consist of a price and a MW quantity, consistent with the WEIM GHG design. Scheduling coordinators for resources located outside of GHG regulation areas may submit separate bid adders as an hourly bid component specific to each GHG regulation area. Scheduling coordinators for resources located in a GHG regulation area may submit a separate bid adder as an hourly bid component specific to another GHG regulation area.[332] Similar to the WEIM GHG design, scheduling coordinators for resources located within a specific GHG regulation area will not have to submit a bid adder to serve demand within that GHG regulation area and instead can include any GHG compliance costs in their energy bid.[333] To align the market rules in the WEIM with EDAM, the CAISO proposes to recognize GHG bid adders related to each new GHG regulation area in the fifteen-minute market and real-time dispatch.[334]

Similar to the current WEIM rules, the CAISO proposes to calculate a maximum GHG bid adder price for each resource located outside each GHG regulation area on a daily basis. This calculation reflects a resource's highest average heat rate on its heat rate curve, the applicable GHG allowance price derived from published indices, and the resource's applicable emission rate.[335]

---

329 *See, e.g.*, WAC-446-040(3)(e)(ii)-(iii), available at https://app.leg.wa.gov/WAC/default.aspx?cite=173-446-040&pdf=true.

330 In tariff appendix A, the CAISO proposes a new defined term "GHG Bid Adder" to replace "EIM Bid Adder" and recognize the use of such bid adders in both EDAM and the WEIM.

331 New tariff section 33.32.1.1 describes the use of these bid adders as specific to Washington and California. As relevant to the WEIM, these proposed tariff amendments are in new tariff section 29.32(a)(2)(A).

332 New tariff section 33.32.1.3; new tariff section 29.32(a)(2)(A).

333 New tariff section 29.32(a)(2). This includes resources pseudo-tied or dynamically scheduled into a GHG Regulation Area.

334 New tariff section 29.32(a)(2).

335 The CAISO proposes a minor clarification to the current WEIM GHG rules to reflect that it uses the average heat rate, not the incremental heat rate, on the heat rate curve in calculating the maximum bid adder. Using the average heat rate avoids the potential for cost under-recovery because the heat rate changes throughout the resource's operating range and is typically higher

The CAISO also provides an option for resources to negotiate a maximum GHG bid adder for each regulation area. Bid adders will be not less than $0/MWh and not greater than 110 percent of the resource's maximum GHG bid adder price GHG maximum compliance cost.[336] This is the same cost-based approach currently in place for bid adders in the WEIM. The CAISO has extended this calculation for bid adders to serve demand in both California and Washington. The CAISO also proposes revisions to the WEIM provisions in new tariff section 29.32 to recognize multiple GHG regulation areas.

Consistent with the Commission's Order No. 831,[337] the CAISO will allow resources to bid above the soft energy bid cap when their bids are cost-verified, including the relevant cost-verified GHG bid adder.[338] In cases where a resource utilizes the existing reference level change request process contained in tariff section 30.11, which allows for an adjustment to the default energy bid,[339] the CAISO will allow a resource to bid up to the sum of its adjusted default energy bid and the relevant maximum daily GHG bid adder. However, the sum of the GHG bid adder price and the energy bid price cannot exceed the hard energy bid cap of $2,000/MWh, as is the case today.[340] This rule will treat resources located within a GHG regulation area and outside of a GHG regulation area in the same manner if their energy costs and GHG compliance costs exceed the soft energy bid cap.

c. Enhancements to Address Secondary Dispatch Concerns

(*i.e.*, less efficient) at lower operating points. Resources can submit bids lower to reflect their actual costs.

336 New tariff section 33.32.1.3.

337 *Offer Caps in Markets Operated by Regional Transmission Organizations and Independent System Operators*, Order No. 831, FERC Stats. & Regs. ¶ 31,387 (2016) (Order No. 831), *order on reh'g and clarification*, Order No. 831-A, 161 FERC ¶ 61,156 (2017) (Order No. 831-A).

338 New tariff section 33.32.1.6; *see* new tariff section 29.32(a)(4) for parallel language as applicable to the WEIM. The hard energy bid cap is the maximum energy bid price the CAISO will use for purposes of clearing the CAISO Market Processes, and is set at $2,000/MWh. The soft energy bid cap is the maximum Energy Bid price submitted by Scheduling Coordinators for resources the CAISO will use for purposes of clearing the CAISO Market Processes without cost verification, and is set at $1,000/MWh. Existing tariff section 39.6.1.1; tariff appendix A, existing definitions of Hard Energy Bid Cap and Soft Energy Bid Cap.

339 The CAISO's existing process for evaluating resources that wish to bid above the soft energy bid cap uses a reference level change to the default energy bid as the vehicle for verifying costs. The CAISO does not propose any changes to this process as part of this proposal. *See* existing tariff section 30.11.

340 *Id.*

The integrated forward market will take into account GHG bid adders in selecting energy produced by resources outside of a specific GHG regulation area up to the associated MW quantity included in the bid adder, as is the case in the WEIM.[341] The integrated forward market will not dispatch resources located outside of a GHG regulation area for attribution to serve demand in a GHG regulation area if the MW quantity included in the GHG bid adder is zero or if a scheduling coordinator for a resource does not submit a GHG bid adder.[342] The integrated forward market does not consider GHG bid adders when selecting resources to serve demand outside of a GHG regulation area. This ensures a state's carbon pricing policies do not affect prices for energy outside of the GHG regulation area. The CAISO's restated tariff provisions for the WEIM also reflect these market rules.

A centralized market that seeks to reflect the price of carbon established by a GHG regulation area but spans both GHG regulation areas and non-GHG regulation areas creates the potential for secondary dispatch. Secondary dispatch can occur if lower-emitting resources receive an attribution to serve demand in a GHG regulation area, resulting in higher-emitting resources backfilling.[343] The CAISO proposes two specific enhancements to address secondary dispatch: an optimized counterfactual in the integrated forward market and a GHG net export constraint. The latter will apply in the WEIM and is severable from the remainder of the CAISO's GHG design proposal. The Commission has previously accepted tariff revisions to support modeling enhancements in the WEIM to improve the accuracy of the CAISO's GHG attributions to generation actually dispatched to serve demand in a GHG

---

341 New tariff section 33.32.2.1; *see* new tariff section 29.32(b)(1), which contains the same substance as the existing tariff, but updates the language to reflect new terminology.

342 New tariff section 33.32.2.4; *see also* new tariff section 29.32(b)(3), which reflects the same policy as the existing tariff but updates the language to reflect new terminology.

343 The CAISO does not identify emissions associated with secondary dispatch; rather, CARB accounts for them through its outstanding emissions calculation by calculating total California WEIM emissions at the unspecified source rate and then subtracting emissions of WEIM participating resources outside of California that the market optimization attributes as supporting California demand, subject to a transmission loss factor. CARB then addresses WEIM outstanding emissions through the direct retirement of freely allocated allowances that CARB would otherwise allocate to electric distribution utilities within California. The outstanding emissions calculation is subject to CARB's jurisdiction. The CAISO understands Washington's approach will continue to evolve and may address secondary dispatch. As explained above, Washington's Climate Commitment Act requires the Washington Department of Ecology, in consultation with the Washington Utilities and Transportation Commission and Washington Department of Commerce, to adopt by regulation a methodology in the cap-and-invest context for addressing imported electricity associated with a centralized energy market by October 1, 2026.

regulation area.[344] These modeling changes also addressed the potential for secondary dispatch by reducing GHG attributions to resources that would have generated even without demand in the GHG regulation area, as reflected in WEIM resources' base schedules.

The WEIM nets transfers into and out of balancing authorities in GHG regulation areas to determine the volume of real-time supply to attribute to participating resources based on their bid adders.[345] The same will be true upon implementation of EDAM, except the CAISO will model and attribute transfers into GHG regulation areas instead of balancing areas. Subject to submitted GHG bid adders, the WEIM limits any GHG attributions to a MW value equal to the WEIM participating resource's dispatchable bid range between a resource's base schedule and its upper economic limit.[346]

EDAM balancing areas will not submit resource base schedules.[347] As described in section V.C.3.d of this transmittal letter, EDAM will include a GHG reference pass to inform the MW attribution of GHG transfers to EDAM resources in the integrated forward market based on their bid adders. In the integrated forward market, EDAM would limit an attribution to the lower of: (i) the GHG bid capacity, (ii) the positive difference between a resource's upper economic limit and its GHG reference pass, or (iii) the optimal energy schedule.[348] The CAISO proposes to rely on the EDAM resource's day-ahead schedule to limit the MW value of a real-time GHG transfer. In real-time, this limit will reflect the lower of: (i) the MW value of the GHG bid adder, (ii) the resource's upper economic bid minus the day-ahead energy schedule, plus the resource's total day-ahead attribution to serve demand in a GHG regulation area, or (iii) the resource's real-time market energy schedule.[349] The difference between the day-ahead energy schedule and the day-ahead attribution reflects the MW value associated with serving load outside the GHG regulation area and is analogous to a base schedule in the WEIM. These rules seek to improve the accuracy of the CAISO's GHG attributions to generation actually dispatched to serve demand in a GHG regulation area. They also address concerns with the potential for secondary

---

344 *Cal. Indep. Sys. Operator Corp.*, 165 FERC ¶ 61,050, at P 17 (2018).

345 The CAISO proposes a new defined term "GHG Transfer" that defines the algebraic difference between supply and demand in a GHG regulation area.

346 Existing tariff section 29.32(b)(2). This also considers any applicable derates and ancillary services capacity reservations, for the relevant hour.

347 *See* section V.A.4 of this transmittal letter (describing differences between participation in EDAM and the WEIM).

348 New tariff section 33.32.2.2.

349 New tariff section 29.32(b)(2).

dispatch by reducing GHG attributions to resources that would have generated even without demand in the GHG regulation area, as reflected in the EDAM resources' integrated forward market schedules or WEIM resources' base schedules.

d. The GHG Reference Pass is an Optimized Counterfactual

In EDAM, the CAISO proposes to conduct an optimized GHG counterfactual based on submitted bids after the day-ahead resource sufficiency evaluation but prior to running the integrated forward market.[350] In this process, the CAISO will identify reference schedules to reflect what dispatch would have occurred without GHG transfers. These reference schedules allow the market to identify an eligible MW value for EDAM resources located outside of a GHG regulation area to receive an attribution in the integrated forward market to serve demand in a GHG regulation area. For this limited purpose, the reference schedules are similar to base schedules in the WEIM, The MW value identified in reference schedules will reflect the difference between an EDAM resource's GHG reference schedule and its upper economic limit.

The CAISO calls this counterfactual for the extended day-ahead market the GHG reference pass. It optimizes resources across the EDAM area based on bids but without transfers into GHG regulation areas. The optimized result will create reference schedules that depict what supply will serve demand outside of a GHG regulation area based on submitted bids. Because the reference schedules reflect supply bids from resources across the EDAM footprint, the CAISO expects the results will be more accurate than base schedules submitted in the WEIM. Given this higher level of accuracy, the CAISO anticipates that the GHG reference pass will result in a material reduction in secondary dispatch arising from dispatch of EDAM resources to serve demand in a GHG regulation area. The GHG reference pass will establish a baseline to determine what dispatch would have occurred in the non-GHG regulation area without offers to serve demand in GHG regulation areas and then will limit attributions to the dispatchable bid range between the resource's GHG reference pass schedule and the resource's effective upper economic bid.[351]

In this market process, the CAISO will use day-ahead bids and self-schedules of resources to optimally clear supply and demand bids without allowing GHG transfers into GHG regulation areas. The GHG reference pass will not allow net imports into a GHG regulation area in order to approximate how a

---

350 New tariff section 33.32.2.3.

351 As described above, the integrated forward market will limit attribution to a value equal to the lower of the described value, the MW value in the resource's bid adder, or the resource's day-ahead energy schedule. New tariff section 33.32.2.2.

balancing area outside the GHG regulation areas will meet its own load with its internal generation as well as supply from other balancing areas outside of the GHG regulation area. The goal of the GHG reference pass is to reflect how supply resources can optimally serve demand in the EDAM footprint without net imports into the GHG regulation areas and the associated cost of compliance with GHG regulation. Therefore, the reference pass will find an optimal market solution without net imports into the GHG regulation areas and the associated GHG regulation cost. A feature of the GHG reference pass is that it will not dispatch resources that have registered that their capacity is committed to serve demand in a GHG regulation area. This supply will receive a GHG reference schedule of zero MW, thereby allowing it to receive a full attribution to serve demand in the GHG regulation area through the integrated forward market.[352]

As described above, the results of this GHG reference pass will serve as a counterfactual to limit the attribution of GHG transfers to EDAM resources in the integrated forward market to the difference between a resource's reference schedule and upper economic limit. For resources that participate in both EDAM and the WEIM, the GHG counterfactual used in place of a base schedule during the real-time market is the difference between the resource's day-ahead market energy schedule and day-ahead market GHG award. For resources that participate in the WEIM and not EDAM, the CAISO will continue to use the self-submitted base schedule as the GHG counterfactual.

The simplified example shown in the table immediately below reflects a counterfactual for, and the attribution of, a 100 MW EDAM resource located outside of a specific GHG regulation area that receives a 40 MW energy schedule from the GHG reference pass based on its bid. If the day-ahead upper economic bid is 90 MW, then 50 MW of that resource's capacity is attributable to serving a GHG regulation area (or a combination of GHG regulation areas) in the integrated forward market. For the purpose of this example, assume that the resource received a day-ahead energy award of 90 MW and a GHG award of 50 MW in the integrated forward market. In the real-time market, the counterfactual becomes the difference between the day-ahead energy award and GHG award, *i.e.*, 40 MW. Assuming the upper economic bid in the real-time market is now 100 MW, the resource can then receive a GHG award up to 60 MW. This 60 MW of capacity eligible for a GHG award cannot exceed the resource's real-time energy award and its GHG bid adder quantity.

Market Process	**Upper Economic Bid**	**Energy Award**	**MW Attributable to GHG Areas**

[352] New section 33.32.2.3. EDAM will afford similar treatment to a pseudo-tie or a dynamically scheduled resource into a balancing authority that has demand in a GHG regulation area, if the resource's scheduling coordinator registers that the supply is associated with demand in the GHG regulation area.

DAM: GHG Reference Pass	90 MW	40 MW	n/a
DAM: IFM	90 MW	90 MW	50 MW
RTM	100 MW	100 MW	60 MW

e. GHG Net Export Constraint

To reduce the potential risk of secondary dispatch further, the CAISO also proposes to utilize a new GHG net export constraint that will limit the attribution of resources serving a GHG regulation area.[353] This net export constraint is specific to the CAISO's GHG modeling and should not be confused with the net export transfer constraint discussed in section V.B.3 of this transmittal letter.

This GHG net export constraint in the integrated forward market limits the aggregate attribution of GHG transfers to EDAM resources within a specific EDAM entity balancing area. Under this constraint, attributions of GHG transfers to EDAM resources located in an EDAM balancing authority outside of a GHG regulation area may not exceed the net exports from that EDAM balancing area. If an EDAM balancing area is a net importer during any hour, no EDAM resource located within the EDAM balancing area may receive an attribution of a GHG transfer during that hour. This constraint reflects how these EDAM balancing areas are relying on their internal supply as well as imports to meet demand and do not have surplus supply to support demand in a GHG regulation area. This constraint will reduce the potential for secondary dispatch by not attributing a GHG transfer to a resource for which another resource will need to backfill to serve demand in the balancing area that is a net importer. This constraint will also limit the aggregate attribution of resources within a specific GHG regulation area to serve demand in another GHG regulation area such that the attribution may not exceed the net exports from these resources' native balancing areas.

This modeling feature responds to stakeholder feedback related to the WEIM's GHG design, which permits WEIM participating resources to receive a GHG attribution even if the balancing area in which it is located is a net importer during a market interval. Although this constraint will restrict attributions in EDAM, it will not apply to "committed capacity" located outside a specific GHG regulation area that is obligated to serve demand within that GHG regulation

[353] New tariff sections 29.32.1 and 33.32.5. These two proposed tariff sections are severable from the rest of the GHG design proposed in this filing. The CAISO proposes to implement the GHG net export constraint to limit secondary dispatch and better align attribution with the transfer capabilities of the system.

area, so long as the scheduling coordinator registers that capacity with the CAISO.[354] This allows capacity owned or contracted to serve demand in a GHG regulation area to receive a full attribution to serve that demand consistent with ownership or contractual entitlements. The CAISO also proposes to implement the GHG net export constraint in the WEIM so the same constraints apply in both markets.[355] This enhancement directly addresses stakeholder concerns involving the potential for secondary dispatch.

The CAISO also proposes to relax this constraint in cases where it might pose reliability or pricing risks. If a balancing area that overlaps with a GHG regulation area fails the EDAM resource sufficiency evaluation in the upward direction, the CAISO will not impose the GHG net export constraint in the relevant day-ahead or fifteen-minute market interval. This means resources outside the GHG regulation area may receive an attribution to serve that deficient balancing area even if that attribution is above the net exports of the resource's native balancing area.

The following simplified example reflects three scenarios (labeled A, B, and C) related to the CAISO's approach to limit attribution using the GHG net export constraint. The constraint limits the aggregate GHG attribution to resources in a balancing area in the non-GHG area to the higher of the optimal net transfer or the aggregate available "committed capacity" in that balancing area. In the example, balancing area 1 is in a non-GHG regulation area, and balancing area 2 is entirely within a GHG regulation area.

In scenario A, balancing area 1 is a net importer and therefore resources within its balancing area may not receive an attribution to serve demand in a GHG regulation area. In scenario B, balancing area 1 is a net exporter with an optimal net transfer of 100 MW. As a result, resources within balancing area 1 may receive an attribution to serve demand in a GHG regulation area up to 100 MW. Scenario C outlines how the CAISO will approach committed capacity. In this example, balancing area 1 has a 100 MW resource, of which 20 MW is committed through contracting to balancing area 2. In this case, the resource's GHG reference schedule cannot exceed 80 MW, leaving 20 MW available for attribution unless the optimal net transfer from balancing area 1 is below 20 MW. The resource in balancing area 1 may receive an additional attribution above 20 MW to serve demand in balancing area 2 at a value above the resource's GHG reference schedule but not to exceed balancing area 1's optimal net exports for that hour.

---

354 New tariff sections 29.32.1 and 33.32.5. Committed capacity refers to contracted capacity.

355 New tariff section 29.32.1 describes the GHG Export Constraint as applicable to the WEIM.

Scenario	Description	Total Attribution to Balancing Area 1
A.) Net importer	Balancing area 1 is a net importer for the hour	None
B.) Net exporter	Balancing area 1's net transfer limit is 100 MW	Up to 100 MW
C.) Accounted-for committed capacity	Balancing area 1 has a 100 MW resource, of which 20 MW is committed capacity to balancing area 2	Up to 20 MW of committed capacity The resource in balancing area 1 may receive an additional attribution to serve demand in balancing area 2 at a value above the resource's GHG reference schedule but not to exceed balancing area 1's net exports for that hour.

The CAISO recognizes enforcing this constraint may create market inefficiency in some intervals but it should improve the accuracy of the CAISO's GHG attributions to generation actually dispatched to serve demand in a GHG regulation area. These modeling changes also reduce the potential for secondary dispatch by eliminating GHG attributions to resources that could require other resources to backfill for them in order to serve demand outside of the GHG regulation areas. Finally, the constraint respects forward contracting to access supply to serve demand in the GHG regulation areas.

f. GHG Settlement

As part of the proposed design for GHG accounting payments, the CAISO will calculate a separate marginal GHG cost[356] for each GHG regulation area in both the day-ahead and real-time markets. This approach differs from the current WEIM rules under which the CAISO calculates one marginal GHG cost for the GHG regulation area of the State of California. Today, in the WEIM, resources outside of California that receive an attribution to serve demand within California receive the system marginal energy cost, which includes the marginal

[356] The CAISO proposes a revision to this already defined term to recognize its applicability in both EDAM and the WEIM. Tariff appendix A, revised definition of Marginal Greenhouse Gas Cost.

GHG cost. The CAISO then subtracts this marginal GHG cost at pricing nodes of participating resources located outside of California that serve demand outside of California balancing areas.[357] With the formation of more than one GHG regulation area with different prices for GHG compliance, the CAISO is proposing for both EDAM and the WEIM to calculate separate marginal GHG costs for each GHG regulation area.[358] This will create an additional payment above the marginal energy cost, called Greenhouse Gas Emission Cost Revenue, for resources located outside of a specific GHG regulation area that receive an attribution to serve demand in that GHG regulation area.[359] The applicable marginal GHG cost for a GHG regulation area will reflect the GHG bid adder of the marginal resource selected through the market optimization to serve demand in that GHG regulation area.[360]

As proposed, in the day-ahead market, GHG payments to resource scheduling coordinators will reflect the product of the obligation to serve demand in a specific GHG regulation area and the marginal GHG cost from the integrated forward market for the respective GHG regulation area.[361] EDAM will collect these payments through locational marginal prices paid by load in the GHG regulation areas. Similar to today, scheduling coordinators for resources within a GHG regulation area will recover the cost of GHG compliance through energy payments.[362] These rules will ensure resources that offer their supply and receive an attribution through the day-ahead market to serve demand in a GHG regulation area receive compensation for the compliance obligations they will face as electricity importers into a GHG regulation area. In addition, the rules ensure demand within the GHG regulation area bears the cost of that payment. Locational marginal prices for supply serving demand outside of a GHG regulation area will not reflect these costs.

In the fifteen-minute market and five-minute real-time dispatch, the GHG payments to resource scheduling coordinators will reflect the product of the obligation to serve demand in a specific GHG regulation area and the marginal GHG cost for the respective GHG regulation area.[363] The WEIM will collect

---

357 Tariff appendix C, existing section F.

358 Tariff appendix C, new section A.5.

359 The CAISO proposes to revise the existing term Greenhouse Gas Emission Cost Revenue to make it applicable to both EDAM and the WEIM. Tariff appendix A, revised definition of Greenhouse Gas Emission Cost Revenue.

360 The CAISO also proposes clarifying changes to its bid cost recovery calculations to account for GHG bid costs and revenues in both the integrated forward market and real-time market. *See, e.g.*, revisions to existing tariff sections 11.8.2.1, 11.8.2.2, 11.8.4.1, and 11.8.4.2.

361 New tariff sections 11.2.1.1 and 33.11.3.7.

362 *Id.*

363 New tariff section 11.5.10.

these payments through locational marginal prices paid by load in the GHG regulation areas. Similar to today, scheduling coordinators for resources within a GHG regulation will recover the cost of GHG compliance through energy payments. Real-time market settlements of GHG payments reflect deviations or imbalances from day-ahead market settlements. A resource's fifteen-minute market settlement for GHG payments associated with an attribution to serve demand in a specific GHG regulation area reflects any imbalance from the resource's day-ahead attribution for that GHG regulation area. A resource's real-time dispatch GHG settlement for a specific GHG regulation area reflects any imbalance from the resource's fifteen-minute market GHG attribution for that GHG regulation area. This is similar to how the WEIM works today. Under the proposed approach, these rules will ensure resources that offer their supply and receive an attribution to serve demand in a GHG regulation area receive compensation for the compliance obligations they will face as electricity importers into a GHG regulation area. In addition, the rules ensure demand within the GHG regulation area bears the cost of that payment. Locational marginal prices for supply serving demand outside of a GHG regulation area will not reflect these costs.

The following simplified example reflects the settlement of an EDAM resource located outside of a specific GHG regulation area that receives a 50 MWh attribution to serve demand in that GHG regulation area in the day-ahead market for a particular trading hour and an incremental attribution of 10 MWh in each of the real-time market intervals of that operating hour. In total for that hour, the resource will receive a $500 MWh payment from its day-ahead market attributions and an additional $400 MWh payment from its real-time market attributions. The additional payment reflects the increase in the real-time marginal GHG cost for its 50 MWh day-ahead attribution that rolled over into real-time and for its incremental real-time market attribution of 10MWh at the real-time marginal GHG cost.[364]

Market Interval	**Attribution and Marginal GHG Cost**	**Payment**
Day-ahead market	50 MWh at $10 marginal GHG cost	$500 MWh
Fifteen-minute market intervals	60 MWh at $15 marginal GHG cost	$900 MWh

[364] The resource's imbalance settlement could also be lower if it received less of an attribution in the real-time market than it received in the day-ahead market and/or if the marginal GHG cost is lower in the real-time market than in the day-ahead market.

Five-minute real-time dispatch intervals	60 MWh at $15 marginal GHG cost	$900 MWh[365]

The CAISO also proposes to establish offset accounts in the day-ahead market and real-time market for marginal GHG costs associated with a specific GHG Regulation Area.[366] These offset accounts ensure there is a balance between amounts collected from demand in a GHG regulation area and payments made to compensate resources that receive an attribution to serve that demand. These offset accounts maintain financial neutrality for the CAISO as a market operator and address variances that may arise between dispatch and settlement quantities across market intervals. The Commission has previously accepted using these neutrality adjustments through offset accounts in the CAISO's markets.[367] Conceptually, marginal GHG cost payments made to resources receiving an attribution to serve demand in a GHG regulation area should be equal to amounts charged to demand for that payment. To the extent the sum of the settlement amount for GHG payments and costs for a specific GHG regulation area does not equal zero because of variances in meter data or tagged quantities, the CAISO will assess charges or make payments to metered demand within a specific GHG regulation area through these offset accounts.

Today, the CAISO utilizes its real-time imbalance energy offset account to ensure neutrality between payments made to resources based on their attributions to serve demand in California balancing areas and amounts collected from demand in those balancing areas to make those payments. With the implementation of two GHG regulation areas with separate marginal GHG costs, the CAISO proposes to create a separate offset account for each GHG regulation area. This will ensure the CAISO maintains neutrality within each GHG regulation area. Additionally, the CAISO is proposing tariff revisions to exclude marginal GHG costs from the current real-time imbalance energy offset.[368]

g. Data Availability

365 This $900 MWh amount consists of the $500 MWh attribution from the day-ahead market listed above plus the incremental deviation in real-time (*i.e.*, 50 MWh x $5 increase at the real-time marginal GHG cost plus 10 MWh x $15).

366 New tariff sections 33.11.3.9.2 and 11.5.4.1.4.

367 *See Cal. Indep. Sys. Operator Corp.*, Commission Letter Order, Docket No. ER19-2497-000 (Sept. 6, 2019).

368 Revised tariff section 11.5.4.1.1.

Finally, the CAISO proposes to confirm its data availability and disclosure policies as they relate to GHG in the tariff.[369] For resources located in California and for resources located in EDAM balancing areas, the CAISO will notify scheduling coordinators of their reference pass schedules and their market results reflecting the MW quantity of any energy supporting transfers that served a particular GHG regulation area. The CAISO also will have the authority to disclose information regarding GHG transfers to a governmental authority only so long as the information does not disclose confidential information of any market participant. The CAISO intends to disclose such information to governmental authorities involved in overseeing state GHG policies. The CAISO proposes parallel obligations related to GHG transfers in the WEIM.[370] These tariff provisions will ensure market participants have access to the data needed to meet their compliance obligations, and state agencies can review aggregated data relevant to their programs.

4. Locational Marginal Price Formation Changes

The CAISO calculates the LMP in each market interval for each node using a reference bus that represents the distributed load in the market area (*i.e.*, the full EDAM area of all balancing authorities participating in the extended day-ahead market or the full WEIM area of all balancing areas participating in the extended day-ahead market or the full WEIM area of all balancing areas participating in the real-time market).[371] Today the CAISO derives a marginal energy cost based on the shadow price of the power balance constraint at the optimal solution across the entire market area. Upon implementation of EDAM, the CAISO will derive a marginal energy cost based on the shadow price of the power balance constraint at the optimal solution for each balancing area in the market area.[372]

The textbook definition of the locational marginal price at a given location or pricing node is the cost of serving the next increment of demand at that location. In the CAISO's current market solution, the LMP is synthesized from its components, namely the marginal energy cost (MEC), the marginal cost of losses (MCL), the marginal cost of congestion (MCC), and the marginal cost of greenhouse gas (MCG) – an additional component developed to represent GHG bid adders in the WEIM.[373] In the traditional market with a single balancing area,

---

369 New tariff section 33.32.6.

370 New tariff section 29.32.2.

371 Revised tariff appendix C.

372 *Id.*

373 *See Cal. Ind. Sys. Operator Corp.*, 153 FERC ¶ 61,087, at PP 57-58 (accepting CAISO's proposal regarding the greenhouse gas bid adder enhancements).

like today's day-ahead market for the CAISO balancing area, the MEC is the shadow price of the power balance constraint, which balances supply and demand in the market area.

With the extension of the real-time market to other balancing areas starting in 2014 under the WEIM, a power balance constraint is enforced for each balancing area reflecting the net transfer as the mismatch of supply and demand within each balancing area. This results in a MEC for each balancing area. To maintain the traditional system marginal energy cost (SMEC) as part of the WEIM, which as its name suggests is the same for all locations in the market area, the CAISO balancing area was selected as a reference setting balancing area of the SMEC for the market area. This transformation from the MEC to the SMEC in the WEIM does not alter the LMP, which remains the sum of its components. Instead, the difference among other balancing areas' MEC and the CAISO balancing area MEC is added to the MCC for each balancing area in the WEIM to account and allocate congestion offset costs among balancing areas.[374] With the introduction of transfer settlement in EDAM and the separation between transfer and congestion revenue, however, this transformation is no longer applicable.[375] The difference between the MECs of two balancing areas in EDAM will result in transfer revenue from the transfers between them, and as such it cannot be added to the MCC. The MEC for each balancing area in EDAM must be referenced, and it is no longer necessary or appropriate to reference a SMEC in the LMP formulation.

The market model will include transfer system resources on each side of the market transfer between balancing areas participating in the extended day-ahead market to represent the MEC of the respective balancing area. The MEC may differ between balancing areas in the market area when market transfers between the balancing areas are scheduled at their respective scheduling limits. The MEC difference between the balancing areas on either side of a specific market transfer generates transfer revenue.[376]

The CAISO will calculate the MCC at each node as the net contribution of the shadow prices of the binding constraints at the optimal solution, weighed by

---

374 *See* existing tariff section 11.5.8.4 (accounting for congestion offset costs between balancing areas in the WEIM); *see also Cal. Ind. Sys. Operator Corp.*, 155 FERC ¶ 61,329, at P 35 (2016) (accepting revisions to enhance the calculation and allocation of congestion offset costs between balancing areas in the WEIM).

375 *See* section V.D.3 of this transmittal letter (discussing the calculation and allocation of congestion revenue and transfer revenue between balancing areas).

376 No stakeholders raised specific concerns regarding the proposed changes to tariff appendix C.

the respective power transfer distribution factors. For market area intertie resources at a scheduling point and transfer resources at a transfer location, the power transfer distribution factors to an intertie constraint or intertie scheduling limit at that scheduling point is +1 for an import and –1 for an export.[377] The CAISO does not consider the effect of transmission losses in calculating power transfer distribution factors; they depend only on the network configuration. Furthermore, the difference between the power transfer distribution factors at two nodes with respect to any binding transmission constraint, and thus the difference between the MCC at these nodes, is independent of the selection of the reference bus.

Under the proposed tariff revisions, the CAISO will calculate the MCL at each node as the product of the MEC and the rate for marginal losses at that node, where the rate for marginal losses is the sensitivity (partial derivative) of system losses to an increment of power injected at that node and absorbed by the reference bus.[378] The distributed load reference used in this calculation is the scheduled load in the integrated forward market and the distributed demand forecast in the residual unit commitment process and the real-time market, within the market area. This calculation reflects the area interchange control feature of the alternating current power flow where the net scheduled interchange of a balancing area is kept constant, while the iterative solution distributes loss deviation from the previous iteration to the reference bus. Consequently, the rate for marginal losses of the transfer system resources that model a transfer at a location between two balancing areas in the EDAM market area may be different because these transfer system resources belong to different balancing areas. The CAISO will set the MCL for both of these transfer system resources to the average rate for marginal losses between the two so that there is no MCL difference between the transfer system resources on either side of a specific market transfer. The calculation of the MCL ignores the marginal losses on transmission facilities outside the market area.

To reflect the LMP calculation at scheduling points outside the market area accurately, the CAISO will deploy an aggregated generation pricing node to account for prices at scheduling points separately from the transfer locations between balancing areas in the market area. The CAISO calculates LMPs for intertie resources at scheduling points, which are represented in the network model as nodes or aggregations of nodes external to the market area (*i.e.*, at the boundary of a balancing area inside the market area with a balancing area outside the market area), through the same process it uses to calculate LMPs for nodes within the market area. When system resources are registered at a scheduling point to a balancing area in the EDAM market area to model energy or capacity imports/exports from/to a balancing area outside the market area, the

---

377 Tariff appendix C, revised section A.3.

378 Tariff appendix C, revised section A.4.

CAISO will distribute the import/export energy schedule or capacity award of the system resource to a default generation aggregation point of the balancing area outside the market area that is the source/sink. If the source/sink balancing area is unknown at the time the market runs, the CAISO will distribute the import/export energy schedule or capacity award of the relevant system resource to a generic generation aggregation point for the relevant scheduling point, and the LMP components of the system resource will reflect the marginal losses and congestion at the relevant aggregation point.

In certain cases, system resources are registered at a scheduling point to a balancing area in the market area to model energy imports/exports from/to another balancing area inside the market area. This occurs because of differences in the applicable market area between the day-ahead market and the real-time market when a balancing area is outside the EDAM area in the day-ahead market, but inside the WEIM area in the real-time market. In this case, the CAISO cannot distribute the day-ahead energy schedule of the relevant system resource in the real-time market to the default aggregation point of the source/sink balancing area that is in the WEIM area because the resources in that balancing area are dispatched optimally. Instead, the market distributes the day-ahead energy schedule of the relevant system resource to the node(s) of the relevant scheduling point using a WEIM mirror system resource at the same scheduling point.[379] The WEIM mirror system resource belongs to the source/sink balancing area, and its base energy schedule matches the day-ahead energy schedule of the system resource it mirrors. The WEIM mirror system resource that mirrors a system resource has an export base schedule that matches the day-ahead import schedule of its mirrored system resource, or a base import schedule that matches the day-ahead export schedule of its mirrored system resource.[380] The LMPs of the WEIM mirror system resource and the system resource it mirrors will be different because the MEC, MCC, and MCL differ because the two resources belong to different balancing areas in the market area.

D. Post-Market Processes and Outputs

The CAISO's extended day-ahead market will produce resource commitments and EDAM energy transfers that the CAISO, as the market operator, will ultimately settle and allocate to the appropriate scheduling

---

379 *See* existing tariff section 29.27(c) (providing for WEIM mirror system resources between balancing areas), and new tariff section 33.18.1 (requiring WEIM mirror system resource registration for system resources at transfer locations between a balancing area in the EDAM area and a balancing area in the WEIM area).

380 New tariff section 33.30.9.

coordinator for the balancing authority. The proposed tariff framework addresses the structure for such charges and the creation of any revenue. Individual balancing authorities, including the CAISO, will distribute charges and revenue to the appropriate entities within their balancing area in accordance with the terms and conditions of their individual tariffs.[381]

1. Transmission Revenue Recovery

Participating in the day-ahead market will not affect transmission owners' costs of owning, maintaining, or constructing transmission facilities. For example, the CAISO will not become the transmission planner for balancing areas participating in EDAM. Moreover, EDAM transmission owners will still be subject to the rates accepted by the Commission.[382] Most expected EDAM transmission owners also use formula transmission rates, which help mitigate the risk that joining EDAM will suddenly result in over- or under-recovery of their revenue requirements. Any change in costs or revenues would roll over into the next annual update of the formula rate. Even if a transmission owner has stated rates, it would have the opportunity to file a new rate case soon after joining EDAM to account for any new costs and revenues.

Although participating in the day-ahead market will not affect the transmission owner's transmission costs, it may affect the *allocation* of revenues received for using its transmission system. If transmission owners release transmission to the CAISO's day-ahead market, they will forego some transmission sales established in their rate cases before joining EDAM. Stakeholders, EDAM transmission owners, and EDAM local regulators have raised concerns that changes to transmission owner revenue recovery due to transmission owner participation in EDAM may result in unexpected downstream cost shifts for ratepayers. To avoid unintended cost shifts at the outset of their participation in EDAM, the CAISO proposes to protect EDAM transmission owners against the risk of foregone transmission revenues. The CAISO will assess an EDAM access charge[383] to recover the eligible revenue shortfalls of

---

381 https://stakeholdercenter.caiso.com/StakeholderInitiatives/Extended-day-ahead-market-ISO-balancing-authority-area-participation-rules.

382 EDAM transmission owners can also have rates subject to another applicable regulator or self-regulated by publicly-owned entities; however, the vast majority of potential transmission owners participating in EDAM are Commission jurisdictional public utilities.

383 CAISO regional and local access charges (known collectively as transmission access charges or TAC) are unaffected and apply to the CAISO balancing area only. *See* existing tariff section 26 and tariff appendix F, existing schedule 3.

one EDAM balancing area to the other EDAM balancing areas.[384] The CAISO and prospective EDAM transmission owners expect these access charges will be relatively small at the beginning and potentially become lower as EDAM transmission owners adjust their rates to reflect recovery of their revenue requirements under a model where they use their transmission assets to support participation in the day-ahead market. The CAISO included projected EDAM access charges in appendix 2 to the EDAM Final Proposal (provided in attachment E to this filing).

EDAM transmission owners may include in the EDAM access charges only those shortfalls expected to result from transitioning from bilateral transmission service to day-ahead market service. These include potential shortfalls associated with expected revenues[385] from sales of hourly non-firm point-to-point, daily non-firm point-to-point, weekly non-firm point-to-point, monthly non-firm point-to-point, hourly firm point-to-point, daily firm point-to-point, weekly firm point-to-point, and monthly firm point-to-point transmission service.[386] These are the transmission services most likely to be displaced by EDAM transfers.

Likewise, the CAISO will permit EDAM transmission owners to recover revenue shortfalls associated with wheeling through an EDAM balancing area or the CAISO balancing area associated with an EDAM transmission service provider in excess of the total net transfers of the EDAM balancing area.[387] In periods where this excess occurs, the transmission owner will be compensated for the transmission use that supports the excess wheeling.[388] Recovering these eligible revenue shortfalls through the EDAM access charge will mitigate potential rate shocks among ratepayers of EDAM transmission service providers due to any loss of expected revenues, maintaining just and reasonable rates for EDAM transmission owners.

All costs related to sales with an EDAM transmission owner's merchant/marketing function are ineligible to be recovered through the EDAM

---

384 New tariff sections 33.26 and 33.26.1. All provisions in new tariff section 33.26 are severable from each other, and new tariff section 33.26 as a whole is severable from the rest of this filing.

385 The shortfalls between expected revenues and actual revenues from these services reflect services a transmission service provider may not sell after transmission capacity is released for optimization in the day-ahead market, resulting in EDAM transfers.

386 New tariff section 33.26.2.1.

387 New tariff section 33.26.2.3.

388 *Id*.

access charge because EDAM participation will impact the total cost and revenue equally, negating any impact to ratepayers.[389] To determine the costs eligible for recovery through the EDAM access charge, EDAM transmission owners will first calculate their recoverable revenue based on their average rate approved by the Commission or applicable regulator for the preceding three years. The costs recoverable through the EDAM access charge consist of the difference between the EDAM recoverable revenue and actual transmission recovered revenue eligible for recovery.[390]

Additionally, EDAM transmission owners may recover in the EDAM access charge a portion of costs that would not appear in the three-year lookback, but whose cost recovery EDAM affects.[391] These include certain new network upgrade costs and revenue shortfalls from sales of non-firm and short-term firm transmission associated with the release of transmission capacity resulting from the expiration of EDAM legacy contracts. EDAM transmission owners can only include in the EDAM access charge new network upgrade costs that relate to EDAM transmission service. Eligible new network upgrade costs are those that increase transfer capability between EDAM entity balancing areas or between the CAISO balancing area and an EDAM entity balancing area and are energized after the EDAM entity begins participation in the day-ahead market.[392] Likewise, an EDAM transmission owner cannot shift all of its eligible new network upgrade costs into the EDAM access charge.[393] The EDAM access charge can only include a percentage of the projected revenue[394] from the new network upgrades equal to the EDAM transmission owner's ratio of (a) the non-firm and short-term firm point-to-point historical EDAM recoverable transmission revenues[395] to (b) the EDAM transmission owner's total revenue requirement. For example, if non-firm and short-term firm sales constitute 10 percent of an EDAM transmission owner's total revenue requirement, and a new eligible upgrade costs $100 million, the maximum amount recoverable through the EDAM access charge would be $10 million (*i.e.*, 10 percent of $100 million). This caps the EDAM access charge recovery of eligible new network upgrades by applying the historic ratio of non-firm/short-term firm transmission revenues to

---

389 New tariff section 33.26.2.1.

390 New tariff section 33.26.2.1.1.

391 New tariff section 33.26.2.2.

392 *Id*.

393 *Id*.

394 In other words, the EDAM access charge can only include projected revenues associated with approved new transmission costs.

395 *I.e.*, those described in the previous paragraph.

total transmission revenues to the costs of the new transmission facility. This will mitigate the risk of rate shock from new network upgrades, and dis-incent transmission owners from attempting to construct new network upgrades so they can fully allocate the project costs to the EDAM access charge.

The CAISO also proposes to include a provision requiring the CAISO to include examples of network upgrades that increase transfer capability in the applicable business practice manual.[396] Generally, network upgrades increase transfer capability if they increase total transfer capability, create new interfaces, increase the simultaneous import limits at existing interfaces, result from an interregional transmission project to increase transfer capability, or are identified through the Western Electricity Coordinating Council (WECC) path rating process to do the same.[397] These tariff provisions will help protect transmission ratepayers from undue costs, while ensuring they contribute to network upgrade costs commensurate with the benefits they receive. Absent these tariff provisions, transmission customers of EDAM transmission service providers could free-ride on new network upgrades, unduly shifting costs to other ratepayer classes.

Using the aggregate annual costs described above for each EDAM transmission owner, the CAISO will compute a $/MWh rate specific to each EDAM balancing area.[398] To form the numerator of the rate, the CAISO will divide each EDAM transmission owner's revenue shortfall to the EDAM balancing areas associated with the other EDAM transmission owners by (a) the EDAM transmission service provider's gross load divided by (b) the total EDAM area gross load minus gross load of the EDAM transmission service provider. Accounting for the EDAM transmission service provider's gross load in relation to the overall EDAM gross load helps ensure EDAM access charges do not allocate costs beyond potential benefits, because the EDAM transmission owner's impact on the EDAM access charge will be proportional to its own share of gross load in the EDAM area.

The CAISO will assess the EDAM access charge to gross load in each EDAM balancing area.[399] Gross load represents end-use customer demand (adjusted for distribution losses), including demand served by excess behind-the-

396 New tariff section 33.26.2.2.

397 *Id*.

398 New tariff section 33.26.1.1.

399 New tariff section 33.26.3.

meter production.[400] Each EDAM access charge will recover the projected recoverable revenue shortfalls for the EDAM balancing areas outside the balancing area for that access charge, such that no EDAM balancing area will be assessed its own projected recoverable revenue shortfalls.

Once collected through the EDAM access charges, the CAISO will allocate revenues collected to EDAM entities on behalf of each such EDAM transmission owner, in proportion to its share of EDAM projected recoverable revenue shortfalls.[401] For example if EDAM transmission owner A expected to recover $6 million from the EDAM access charge, and EDAM transmission owner B expected to recover $4 million, and the CAISO collected $12 million through the access charge, transmission owner A would receive $7.2 million and transmission owner B would receive $4.8 million. The same calculation would apply if the CAISO recovered less than expected through the access charge. In either situation, the CAISO tariff requires that any under- or over-recovery be rolled into the next year's forecasted recoverable revenue from the access charge.[402] This true-up will help right-size the EDAM access charge year-to-year based on expected collections and actual collections.

To ensure complete transparency over the EDAM access charges, the CAISO proposes to require EDAM transmission owners to provide the CAISO with all documentation necessary to determine each component of the EDAM access charge.[403] At a minimum, this documentation will include: (a) the final order from the Commission or the local regulatory authority effecting the approved transmission rates; (b) the sums for each recoverable revenue component and true-up; and (c) an authorized affidavit from each EDAM transmission owner attesting to the accuracy of the data provided.[404] The CAISO may include other requirements in its business practice manuals to ensure consistent practices, and to ensure third parties can review and verify the inputs to the EDAM access charges. For each EDAM transmission owner, the CAISO will maintain on its website the current sum of each recoverable revenue component, the total true-up, and total eligible recovery. The CAISO also will maintain on its website each EDAM access charge, including the rate, the gross

---

400 The CAISO proposes to revise the existing definition of gross load to include demand directly connected to the distribution system of an EDAM transmission service provider in an EDAM balancing area. Tariff appendix A, revised definition of Gross Load. Under both the existing and the revised definition, gross load excludes demand served onsite by the customer's own behind-the-meter production, such as rooftop solar generation that serves the customer's own demand. Gross load also is the billing determinant for the CAISO's transmission access charges.

401 New tariff section 33.26.3.

402 New tariff section 33.26.1.2.

403 New tariff section 33.26.4.

404 *Id*.

load, and the total eligible recovery in that balancing area, similar to how the CAISO maintains data for its transmission owners' transmission access charges. This transparency will help ensure load-serving entities, regulators, and ratepayers can review and verify the accuracy of the inputs and assumptions within the EDAM access charges.

2. Settlements Generally

The extended day-ahead market will settle in accordance with the CAISO tariff-based timelines and procedures associated with settlement of all market participant transactions. The CAISO will assess and settle all day-ahead market charges in EDAM in same manner as it does today in the CAISO balancing area today; however, there are some a unique dimensions to EDAM necessitates additional EDAM-specific settlement provisions.[405] Differences in settlement between EDAM and the CAISO day-ahead market in a single balancing area primarily arise in the context of the resource sufficiency evaluation surcharge and the greenhouse gas accounting rules (both discussed above),[406] and in the context of transfer and congestion revenues allocated between balancing areas (discussed immediately below).

3. Transfer and Congestion Revenue Allocation

The CAISO models internal transmission constraints, internal transmission limits, and transmission transfer limits in the WEIM today, and will continue to do so in EDAM. If these internal transmission limitations or constraints are reached, the market will seek to redispatch around them. The marginal cost of congestion accounts for differences between the incremental cost to serve demand in different locations, and the CAISO recovers these costs as congestion revenues. In the WEIM, the CAISO settles this congestion revenue through one charge code on a net basis where the revenues accrued are offset by incurred congestion costs.[407] This calculation accounts for congestion revenue accrued due to binding transfer scheduling constraints among the balancing areas associated with the transfer locations between balancing areas. In EDAM, the CAISO proposes to account for these accrued incremental costs as internal congestion revenues and will allocate them to the balancing authority where the binding constraint is modeled, including any adjustment for congestion

---

405 New tariff section 33.11, *et seq.*

406 *See* sections V.B.2 and V.C.3 of this transmittal letter.

407 *See* existing tariff section 11.5.4.1 (allocating congestion revenue through a single offset calculation).

differences that may be reversed for legacy contracts or ownership rights.[408] Allocating congestion revenues associated with binding internal transmission constraints within a balancing area to the corresponding balancing authority participating in EDAM is appropriate because that is where the constraint is modeled. In the CAISO balancing area, these congestion revenues will fund CRRs.

Entities in the WEIM also make transmission available at interties between balancing areas to support energy transfers. This practice will continue with EDAM along with the addition of capacity transfers.[409] The transmission across interfaces between balancing areas may have limitations or other constraints, largely due to the amount of transmission capability made available to the market across the transfer interfaces, which sets a scheduling limit. When this transfer scheduling limit binds, the market will seek to re-dispatch to meet the obligations across the market footprint (*i.e.*, the EDAM area), and it may award energy or capacity from higher-cost resources within a balancing area to meet demand and uncertainty requirements in another balancing area. This will result in price separation of the marginal energy component between the balancing area where the constraint is located and the rest of the EDAM area (or price differences in imbalance reserves or reliability capacity). This price separation represents the accrual of transfer revenue in the extended day-ahead market. Rather than settling these revenues under the umbrella of congestion revenue using one settlement charge code, the CAISO will separate this revenue component as accrued transfer revenue and settle it independently from accrued congestion revenue.[410] This separation of the congestion revenue from the transfer revenue supports separate accounting, which in turn provides for the allocation of transfer revenue to the rights holders that voluntarily made transmission available to the day-ahead market. Otherwise, EDAM would be unable to account for this revenue these rights holders are entitled to under the EDAM transmission availability framework.[411]

---

408 *See* new tariff section 33.11.1.2 (allocating congestion revenue internal to each balancing area without consideration of transfer scheduling constraints).

409 Transfer revenue will similarly accrue for price differences between the marginal imbalance reserve and reliability capacity products. For simplicity, this discussion focuses on the price difference in energy, which will account for the most transfer revenue.

410 *See* new tariff section 33.11.1 (separating the allocation of day-ahead congestion and transfer revenue into distinct calculations); *see also* new tariff section 11.5.4.1.2(b)(3) (modifying the real-time congestion offset calculation to reflect the changes in the settlement of day-ahead congestion revenue).

411 *See* section V.B.1.b-c of this transmittal letter (describing the option for legacy contract, ownership, and transmission customer rights holders to make transmission available to support EDAM transfers in exchange for transfer revenue).

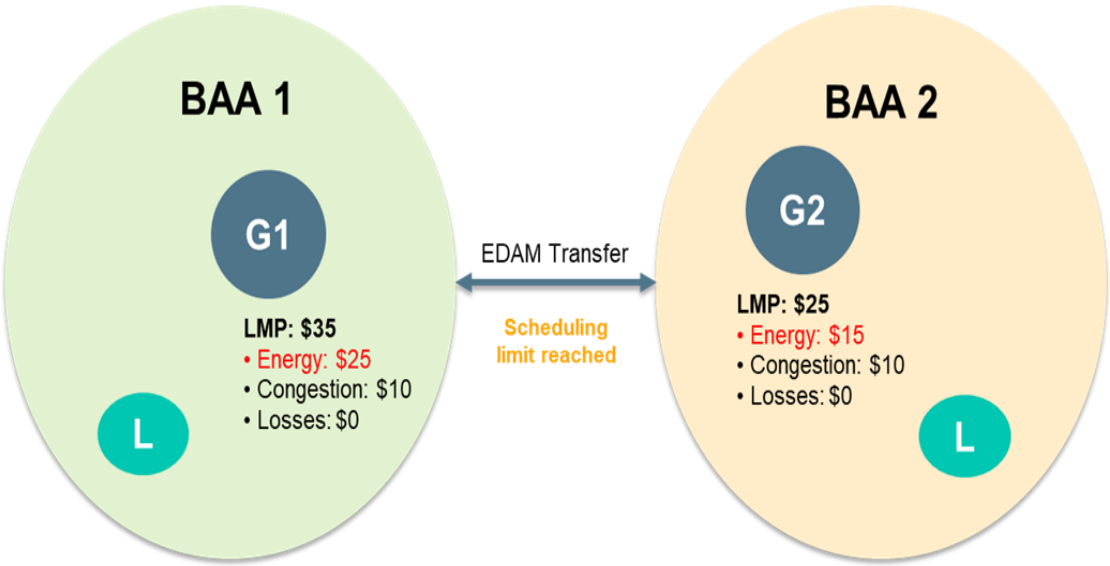
As described in the LMP discussion in section V.C.4 of this transmittal letter, in a single balancing area like today's CAISO day-ahead market, the marginal energy cost (MEC) is the shadow price of the power balance constraint. With the extension of the real-time market in the WEIM, the market enforces a power balance constraint for each balancing area introducing the WEIM net transfer as the mismatch of supply and demand in each balancing area. This results in a MEC for each balancing area. To maintain the traditional system marginal energy costs in the WEIM, which is the same for all locations in the market footprint, the CAISO balancing area was selected as a reference setting for the SMEC, whereas the difference between any other balancing area MEC and the CAISO MEC was added to the MCC for that balancing area.

The settlement of imbalance energy at the relevant nodal MCC determined the real-time congestion revenue or shortfall collected from each balancing area. The real-time balancing area congestion revenue/shortfall in the WEIM does not distinguish between congestion generated by the price separation between balancing areas versus internal transmission constraints. However, to ensure the appropriate balancing area is distributed the appropriate congestion revenue or congestion shortfall, the WEIM also calculates a MCC distribution price. The MCC distribution price is the sum associated the congestion attributed to the enforcement of the net transfer constraint by balancing area based upon the released transmission at transfer location plus the congestion associated with internal transmission constraint. The congestion attributed to enforcing the net transfer constraint is distributed to the balancing area that released unsold transmission at transfer location to facilitate the transfer of energy between the balancing areas. In addition, the internal transmission constraint congestion was retained in the balancing area where the internal transmission constraint is defined. This approach breaks down when transfer revenue, which is essentially congestion revenue at the transfer location between balancing areas in EDAM, is separately calculated and settled.

In EDAM, congestion revenue is collected from the settlement of energy or capacity and MCC difference between source or sink associated with an binding internal transmission constraints. Allocating EDAM congestion revenues in all cases to the balancing area where the constraint is binding is appropriate because that balancing area is responsible for resolving the constraint, whether through re-dispatch or otherwise. Thus, it is reasonable to allocate the congestion revenue to the balancing area where the internal transmission constraint materialized. This rationale applies regardless of the condition that triggers the internal constraint, (*e.g.*, caused by purely internal congestion or imports across multiple interties simultaneously creating congestion on internal transmission constraints). Simultaneous import flow conditions and internal path interactions are a common occurrence across balancing areas, and the host balancing area is responsible for responding to the condition occurring in its

balancing area. Thus, it should receive the congestion revenue from all internal flows and when external flows cause internal constraints to bind.

The following example illustrates the accrual and settlement of transfer revenue between balancing area 1 (BAA1) and balancing area 2 (BAA2) based on the price differences between the two balancing areas. For simplicity, this example reflects a difference in energy prices, with a similar outcome possible when price differences for reliability capacity or imbalance reserves occur and the scheduling limit between balancing areas is reached.



- Transfer revenue accrues when the scheduling limit is reached at a transfer point between two EDAM areas.
- Transfer revenue is the difference in the energy component of the LMP between the two EDAM areas. In this example, transfer revenue is $10 as the difference in energy component of LMP in BAA1 and BAA 2.
- Allocated 50:50 between the two EDAM areas.

The two EDAM entities that made the transmission available to facilitate the energy or capacity transfer generally will share the transfer revenue equally. The exception to the equal sharing of transfer revenue is for situations where a transmission customer releases its transmission rights and makes capacity available at an interface in advance in return for transfer revenue. In such cases, the transmission customer will receive the full allocation of transfer revenue associated with its released right. EDAM will also accommodate any agreed-

upon and documented different sharing arrangement of transfer revenues between two balancing areas.[412]

The separation of congestion revenue settlement from transfer revenue settlement in EDAM is necessary for accurate accounting and settlement of day-ahead energy and capacity transfer costs because transmission customers that make their rights available for the day-ahead market and forgo their right to later scheduled use in exchange for transfer revenue must be settled separately. Transfer revenue represents the market value of transmission made available at internal interties by the service provider or customers holding transmission rights. Congestion revenue, on the other hand, represents the cost to serve demand across the internal transmission system of a balancing area. Without separation of congestion revenue from transfer revenue, the CAISO would be unable to allocate transfer revenue cleanly and directly to transmission customers because the monies would be comingled with congestion revenue that would be allocated to the balancing area supporting the transfer.[413]

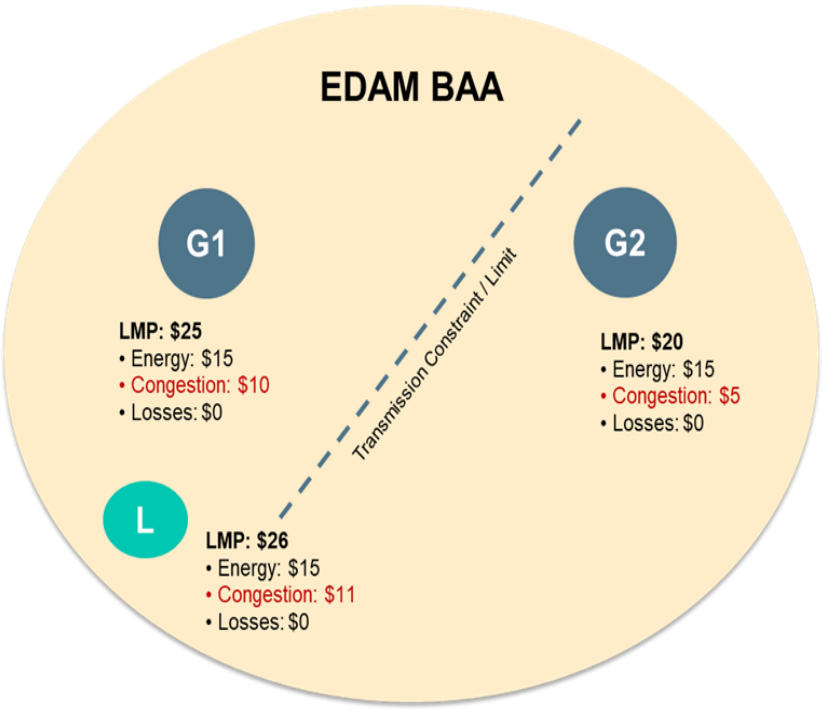
This settlement mechanism supports an important dimension of transmission availability and relates to changes in the formulation of the marginal energy component of the locational marginal price that facilitate extension of the day-ahead market across a multi-balancing area footprint.

The following example illustrates the accrual of congestion revenue within a balancing area based on the locational marginal price differences between generation G1 and G2 and load L due to a binding constraint between locations.

---

412 New tariff section 33.11.1.1.1.

413 *See Real-Time Settlement Review*, Draft Final Proposal, at 8-13 (Oct. 20, 2021) (evaluating potential cost shifting within the WEIM area and between WEIM and non-WEIM balancing areas, and concluding that the CAISO should settle all system resources associated with transfers). This document is available on the CAISO website at http://www.caiso.com/InitiativeDocuments/DraftFinalProposal-Real-TimeSettlementReview.pdf.



- Congestion revenue is the difference in congestion price between Generation and Load across a binding internal constraint.
- The congestion revenue is $1 for every MWh delivered from G1 and $6 from G2 to serve Load.
- Congestion revenue is allocated to the EDAM entity where the constraint materialized.

Settling transfer revenue separately from congestion revenue in EDAM is also necessary to support CRRs in the CAISO balancing area. CRR holders purchase or are allocated rights from a source to a sink to hedge their exposure to internal transmission congestion.[414] CRR holders in the CAISO balancing area have not purchased rights on transmission between two balancing areas. The CAISO thus will still require an allocation of congestion revenue separate from transfer revenue to fund CRRs.

The CAISO will adjust congestion revenue within an EDAM balancing area for legacy contract and ownership rights that receive a hedge against congestion and will settle with the scheduling coordinator for the balancing authority.[415] The transmission service provider would thereafter allocate the congestion revenue under the provisions of its tariff.[416] The allocation of congestion revenue to an EDAM balancing area is consistent with the allocation of congestion offset costs

414 *See generally* existing tariff section 36.

415 New tariff section 33.11.3.8 (providing for settlement of legacy contracts and ownership rights).

416 New tariff section 33.11.1.2 (providing for settlement of congestion revenue).

in the WEIM today.[417] Moreover, the transmission service provider within an EDAM balancing area is in the best position to determine what an equitable allocation among its transmission customers is. An attempt to standardize this allocation under the CAISO tariff would invariably infringe upon the relationship between the EDAM transmission service provider and its customers.[418]

4. Stakeholder Positions on Transfer and Congestion Revenue Allocation

Stakeholders supported an equal distribution of transfer revenue that accrues between balancing areas as equitable. Establishing an equal sharing allocation as the default recognizes that both balancing areas brought transmission to the interface that supported the transfer. Further, two balancing authorities can mutually agree to an allocation different than an equal sharing to support a commercial, agreed upon arrangement that requires a different allocation.

A few stakeholders recommended extending the current WEIM congestion distribution to EDAM with a slight modification. Specifically, they suggested the MCC distribution price should not retain the congestion associated with internal transmission constraint to the balancing area in which it was modeled; rather, it should be shared equally among the balancing areas similar to transfer revenues. The CAISO considered this suggestion and concluded adopting this position would increase the revenue to transmission customers and reduce compensation to the balancing area. This revenue represents an important dimension of the pool of revenue available for each balancing area to sub-allocate under the associated transmission owner tariff. The transmission service provider is in the best position to determine an equitable allocation of these revenues among its transmission customers, and directing the revenue from the CAISO to the transmission customers would undermine this opportunity for each balancing area to pool congestion revenue and determine the most appropriate allocation methodology among its customers.

---

417 Existing tariff section 11.5.8 (providing for settlement of the congestion offset between balancing areas, including the CAISO balancing area).

418 Similarly, although the CAISO balancing area has implemented congestion revenue rights, they may not be appropriate for all balancing authorities to implement at the outset of EDAM, particularly in the Northwest given the limitations on the Commission's ability to require the conversion of firm transmission rights to tradable or financial rights. *See* FPA section 218, 16 U.S.C. § 824r(b) (stating that "nothing in [the FPA] confers on the Commission the authority to require an electric utility or person to convert to tradable or financial rights (1) firm transmission rights" pursuant to contract or by reason of ownership of transmission facilities as of the date the Energy Policy Act of 2005 was enacted, or "(2) firm transmission rights obtained by exercising contract or tariff rights associated with" such firm transmission rights). The CAISO will continue to collaborate with stakeholders to explore the potential for CRRs to be made available as a means to allocate congestion revenue within an EDAM balancing area.

This stakeholder recommendation further highlights the need to distinguish between transfer revenue associated when the net transfer constraint binds and the congestion revenue generated when an internal transmission constraint binds. As explained above, in EDAM transfer revenue is collected from the settlement of transfer schedules of energy or capacity between two balancing areas when the net transfer constraint binds. It is appropriate to allocate transfer revenue to the balancing areas whose transmission service providers or transmission customers schedule either their rights across the transfer or release those rights to the market for optimization the transfer of energy and capacity because: (1) it provides transmission customer the ability to protect against marginal price differences of energy scheduled to flow between balancing areas, and (2) compensation for releasing transmission capacity to market for optimal dispatch of energy or capacity from a resource in one balancing area to meet the need of another balancing area.

E. Additional General Tariff Provisions

1. Creditworthiness

Scheduling coordinators participating in EDAM must comply with the creditworthiness requirements in existing tariff section 12.[419] If a scheduling coordinator fails to satisfy these requirements, the CAISO may implement enforcement actions pursuant to tariff section 12. These are the same creditworthiness requirements that apply to scheduling coordinators taking part in the WEIM.[420]

2. Dispute Resolution, Liability, and Confidentiality

The intake of disputes initiated by EDAM participants in the day-ahead market will be subject to existing tariff section 11.29.8 and managed through the CAISO's customer inquiry, dispute, and information (CIDI) system. EDAM participants must follow the dispute resolution process described in existing tariff section 13, which requires parties to engage in informal good-faith negotiations and mediation before resorting to arbitration.[421]

---

419 New tariff section 33.12.

420 Existing tariff section 29.12.

421 New tariff section 33.13.

The provisions in existing tariff section 14 regarding force majeure events, indemnity, liability, and penalties, as well as the confidentiality provisions in existing tariff section 20, also will apply to EDAM participation.[422] WEIM participants already are obligated to follow these same provisions.[423]

[422] New tariff sections 33.14 and 33.20.

[423] Existing tariff sections 29.14 and 29.20.

3. Miscellaneous

Miscellaneous tariff requirements, such as those regarding assignment, notice, and governing law and forum, are included in existing tariff section 22 and will apply to EDAM.[424] Other miscellaneous tariff requirements specifically relate to EDAM participation. For example, the CAISO may pass on taxes it incurs due to an entity's EDAM participation to the applicable EDAM scheduling coordinator. The tariff requirements also address how title to energy will pass in the day-ahead market and specify that neither the CAISO nor a balancing authority participating in EDAM will be a "Purchasing Selling Entity" under NERC requirements for purposes of e-tags or EDAM transfers.[425]

F. Components of the CAISO's Day-Ahead Market Design Not Included in the EDAM Design

EDAM does not include every market design element of the CAISO's day-ahead market processes that currently applies to the CAISO balancing area. In particular, the CAISO is not proposing at this time to extend existing functionality for inter-scheduling coordinator trades, ancillary services procurement, or congestion revenue rights to EDAM. These features are either discretionary or are addressed through other mechanisms under the open access transmission tariffs that will remain in effect in non-CAISO balancing areas participating in EDAM. The Commission should accept the EDAM design without these features, but it should not foreclose the opportunity for the CAISO and market participants to pursue them as potential future enhancements.

1. Inter-Scheduling Coordinator Trades

At this time, the CAISO is not proposing to support inter-scheduling coordinator trades within non-CAISO balancing areas participating in EDAM. Inter-scheduling coordinator trade functionality is a voluntary settlement service the CAISO makes available to scheduling coordinators to facilitate the market settlement of bilateral transactions that invoice the delivery of energy and ancillary services, as well as a means to allocate market payments and costs between bilateral counterparties.[426] The functionality involves a series of pre-

---

424 New tariff section 33.22.

425 *Id.*

426 Existing tariff section 28. *See also Cal. Ind. Sys. Operator Corp.* 116 FERC ¶ 61,274, at PP 453-54. Inter-scheduling coordinator trading functionality also accommodated the settlement of certain bilateral energy contracts entered into by the State of California during the 2000-2001 energy crisis. These contracts had delivery provisions that could give the seller the choice of delivering power at any node within the CAISO's balancing area. These contracts have since terminated.

market validation and post-market confirmation steps to ensure the accuracy of settlements on which scheduling coordinators rely.

The functionality for inter-scheduling coordinator trades has the potential to assist participants in EDAM, but is unnecessary to ensure extension of the day-ahead market is just and reasonable. Inter-scheduling coordinator trades are not a necessary service for entities to participate in a day-ahead market process. Buyers and sellers can still determine how they will manage the allocation of any market payments and costs under bilateral transactions between them. The CAISO does not offer the functionality for inter-scheduling coordinator trades to participants in the WEIM.[427] Likewise, it is unnecessary for initial implementation of EDAM. However, the CAISO will discuss with stakeholders whether to extend this settlement functionality to participants in EDAM as a future design enhancement.

2. Ancillary Services Procurement

Today, the day-ahead and real-time market processes in the CAISO balancing area provide for the procurement of four ancillary services – regulation up, regulation down, spinning reserve, and non-spinning reserve.[428] The CAISO market procures these services from ancillary service providers whose resources have met specific technical requirements. Scheduling coordinators for these resources submit economic bids to offer their capacity or self-provide these services. The CAISO co-optimizes ancillary services with energy when it procures these ancillary services in the day-ahead and real-time markets. When the CAISO implements imbalance reserves through its day-ahead market enhancements proposed in this filing, it will also co-optimize ancillary services procurement with these commodities in the CAISO balancing area.

The CAISO will not procure or settle ancillary services in other balancing areas at the outset of EDAM.[429] Instead, the CAISO will consider whether a participating balancing area has secured sufficient capacity to meet its own ancillary service requirements and, in the case of reserve sharing groups, whether there is sufficient transmission to support delivery of those ancillary services as part of the resource sufficiency evaluation. Meeting ancillary services requirements will remain a balancing authority function. Transmission providers will continue to secure and allocate the costs of ancillary services such

---

427 Existing tariff section 29.28.

428 In addition, the CAISO contracts for a fifth ancillary service – voltage support – annually (or for such other period as the CAISO may determine is economically advantageous) and on a daily or hourly basis as required to maintain system reliability. *See generally* existing tariff section 8; tariff appendix A, existing definition of Ancillary Services.

429 New tariff section 33.8.

as frequency regulation and operating reserves through various schedules of their open access transmission tariffs.

EDAM will operate based on the input that each balancing area is securing sufficient ancillary services to meet its requirements. Balancing areas participating in EDAM will self-provide their ancillary services in both the day-ahead and real-time market and the market will validate this input through the resource sufficiency evaluation. The integrated forward market will not optimize ancillary service capacity of balancing area participating in EDAM and will not optimize transmission for energy transfers when that transmission is set aside for use by reserve sharing groups to deploy ancillary services. This approach respects each balancing authority's responsibility for securing adequate ancillary services under its respective open access transmission tariffs.

The CAISO plans to continue to work with stakeholders to explore implementing functionality to optimize bids and self-schedules for ancillary services in EDAM as a potential future design enhancement. Prior to implementing any such functionality, the CAISO will propose market rules for consideration by the Commission.

3. Congestion Revenue Rights

Congestion Revenue Rights are financial instruments that market participants can acquire through a CAISO-administered allocation and auction process to provide a level of financial protection against the risks associated with unpredictable congestion charges.[430] The CAISO financially settles CRRs based on the difference in the marginal cost of congestion component of the day-ahead market locational marginal price between two pricing points – a source and a sink – on the CAISO's system, multiplied by the MW quantity of the CRRs a market participant holds between the two points.

The CAISO maintains a CRR model that reflects the most up-to-date direct current full network model. This model includes constraints and network topology that seeks to reflect, as closely as possible, similar constraints and network topology expected in the day-ahead market. The CRR model, however, supports annual and monthly CRR allocations and auctions and runs well before the day-ahead market.[431]

---

[430] *See generally* existing tariff section 36. *See also Cal. Indep. Sys. Operator Corp*., 149 FERC ¶ 61,093, at P 2 (citations omitted) ("CRRs are financial instruments that enable their holders to hedge variability in congestion costs. Entities acquire CRRs primarily to offset integrated forward market congestion costs reflected in the congestion component of locational marginal prices (LMPs).").

[431] Existing tariff section 36.4.1.

The CAISO is not proposing to extend CRRs to balancing areas participating in EDAM as part of its initial design. Transmission customers outside of the CAISO currently manage their risk of congestion by purchasing physical transmission rights under transmission service providers' open access transmission tariffs or by exercising rights under legacy contracts. They will continue to do so under EDAM. These rights reflect existing path-specific transmission. EDAM will compensate transmission customers with any energy transfer payments when they make transmission available to the market. Alternatively, EDAM will provide scheduling priority to transmission customers that self-schedule their physical transmission rights. Like the use of financial transmission rights,[432] using physical transmission rights to hedge the cost of congestion does not insulate transmission customers from all congestion costs. Physical transmission rights do, however, mitigate the risk of congestion costs between a specific source and delivery point on the transmission system.

The CAISO will continue to discuss with stakeholders whether to make CRRs available across the EDAM area as a future design enhancement and how best to align CRRs with physical transmission rights secured under open access transmission tariffs. This effort will include assessing how to ensure the market adequately funds CRRs as well as their value as an effective tool for market participants to hedge the cost of congestion between individual sources and sinks.

G. Conforming Tariff Changes for Entities Participating in EDAM

Each entity that participates in EDAM, including the CAISO, must evaluate what changes to its individual tariff may be required to implement the final Commission-approved EDAM framework, understanding the EDAM framework has been designed with the goal of accommodating each of the diverse group of balancing authorities in the West. For the CAISO balancing area, much of the relationship with its existing tariff provisions and the application of those provisions in EDAM have been included in the framework. For other balancing areas, as stakeholder discussion revealed, although there is commonality in conforming changes that will need to be made, ultimately each potentially impacted balancing area and transmission service provider must address its unique considerations and work through its established stakeholder and governance processes to develop supporting tariff changes for participation in EDAM.

The CAISO has already initiated a stakeholder initiative to identify any portions of its tariff that may require amendment to allow the CAISO to participate in EDAM at the go-live date. This initiative will determine the method to distribute

---

[432] *Cal. Ind. Sys. Operator Corp.* 116 FERC ¶ 61,274, at P 731.

the revenue, and charges, associated with EDAM resource sufficiency evaluation failures of the CAISO balancing area and EDAM transfer revenues. Additionally, the initiative will produce guiding principles for the CAISO balancing area's use of the EDAM net export transfer constraint and a process for recovering historical wheeling access charge revenues. Throughout the course of this initiative, the CAISO is working with its stakeholders to develop a just and reasonable solution to allocate the EDAM benefits, and obligations, among the members of its balancing area. The CAISO anticipates it will complete its stakeholder process and submit its proposed tariff revisions in a separate filing pursuant to section 205 of the FPA shortly after consideration of its initial proposal, and assuming approval, by the CAISO Board of Governors at its September 2023 meeting.[433]

Other balancing authorities, including non-jurisdictional balancing authorities, will use similar processes to address similar concerns and can look to the CAISO's leadership and experience from within its stakeholder process, as well as engagement during the approval processes applicable to the other balancing authorities.[434] Entities that administer jurisdictional tariff services will need to harmonize their existing rules with the EDAM framework, particularly in the context of making transmission available to EDAM, in addition to specific cost allocation considerations within their balancing area.[435]

For example, the EDAM framework will recover congestion and transfer revenues that the CAISO will allocate to the balancing area, and each balancing area will need to develop a methodology to re-allocate the revenue and costs under the associated EDAM transmission service provider tariff. This cost allocation methodology may become somewhat standard across balancing areas through their associated approval processes prior to their participation in EDAM; however, the CAISO tariff does not prescribe it.

---

433 *See* the extended day-ahead market CAISO balancing authority area participation rules stakeholder initiative, located at https://stakeholdercenter.caiso.com/StakeholderInitiatives/Extended-day-ahead-market-ISO-balancing-authority-area-participation-rules. According to the current schedule, the CAISO intends to submit for its Board of Governors consideration this fall a proposal that addresses all of the issues necessary to support its balancing area participation, followed by a filing with the Commission prior to the end of this year. This schedule is designed to ensure the Commission receives all rules associated with participation in EDAM by the CAISO prior to the requested effective date of the EDAM framework presented in this filing, and to support acceptance of the CAISO balancing area specific changes before the first balancing area outside of CAISO participates in EDAM.

434 Generation-only balancing authorities may participate in EDAM similar to participation in the WEIM; however, the nature of a generation only balancing authority may not raise similar cost allocation considerations given the absence of demand within the balancing area.

435 While the CAISO's initiative is limited the tariff changes necessary to allow its own balancing area to participate in EDAM, the public stakeholder initiative allows other entities to benefit by learning about the issues raised and the potential solutions evaluated

During the EDAM stakeholder process, consideration was given to whether it would be appropriate for the CAISO tariff to include applicable cost allocation principles. Some stakeholders would prefer the CAISO tariff included requirements to hold transmission customers harmless from congestion costs to the extent practicable, while others preferred to defer to the EDAM transmission service provider tariff and associated approval process. In the end, the CAISO concluded that establishing criteria for the cost allocation determinations of other balancing areas participating in EDAM was not its proper role. Accordingly, the EDAM framework defers to the EDAM transmission service provider tariff for this determination, as it does with respect to other transmission service provider functional responsibilities.

VI. Effective Dates of DAME and EDAM Tariff Revisions, Request for Waiver of Notice Requirements, and Request for Commission Order

As to the EDAM tariff revisions contained in this filing, the CAISO respectfully requests an effective date of December 21, 2023 for new tariff sections 33.1, 33.2, and 33.4 and the new *pro forma* agreements to be included in appendix B of the CAISO tariff, which constitute the extension of EDAM, the initial onboarding and implementation provisions, the roles and responsibilities of EDAM participants, and the agreements participants need to execute.[436] For all other tariff revisions to implement EDAM and the DAME tariff revisions, the CAISO plans to implement concurrently the day-ahead market enhancements and the extension of the enhanced day-ahead market. This will facilitate and simplify development, testing, and implementation. The CAISO requests an effective date of May 1, 2025 for these other tariff revisions.[437]

The CAISO respectfully requests waiver of the Commission's 120-day notice requirement to permit the requested May 1, 2025 effective date.[438] Good cause exists to grant the requested waiver. The tariff revisions to become effective May 1, 2025 will require significant amounts of time to implement pursuant to a four-step process. First, the EDAM tariff revisions will require the largest and most complex suite of software enhancements since the CAISO's

---

[436] The tariff revisions that will go into effect on December 21, 2023 are incorporated into the clean CAISO tariff sheets contained in attachment A-1 to this filing and are shown in redline format in attachment B-1 to this filing.

[437] The tariff revisions that will go into effect on May 1, 2025 are incorporated into the clean CAISO tariff sheets contained in attachment A-2 to this filing and are shown in redline format in attachment B-2 to this filing.

[438] Specifically, pursuant to section 35.11 of the Commission's regulations, 18 C.F.R. § 35.11, the CAISO respectfully requests waiver of the notice requirement in section 35.3(a)(1) of the Commission's regulations, 18 C.F.R. § 35.3(a)(1), to allow the EDAM tariff revisions to go into effect more than 120 days after submittal of this filing.

market redesign and technology upgrade almost 15 years ago. Second, prospective EDAM participants may need to follow extensive approval processes in each of the jurisdictions where they operate and must understand the market design in which they will participate to obtain such approvals. Third, prospective EDAM participants and transmission owners will need to revise their own tariffs and rate schedules extensively to adopt the new processes, costs, and revenues proposed here. Fourth, prospective EDAM participants and EDAM scheduling coordinators will need significant software enhancements to transition from participation solely in the WEIM to participation in EDAM as well, including a period of market simulation and parallel operations. Therefore, it is appropriate for the Commission to grant the requested waiver.

In addition, for the same reasons described above regarding the effective dates of the DAME and EDAM tariff revisions, the CAISO respectfully requests that the Commission issue an order by December 21, 2023 accepting this tariff amendment in its entirety. It is critical for the CAISO and market participants to have a reasonable degree of certainty regarding the market design so they can make decisions and make commitments accordingly. No one can begin the four steps regarding the EDAM tariff revisions described above without initial regulatory approval from the Commission. Therefore, the CAISO and prospective EDAM participants need Commission approvals in an order on the complete suite of tariff revisions by the date the first subset of them will become effective and well before the vast majority of them become effective. This will provide the CAISO, prospective EDAM participants, and other stakeholders certainty regarding all of the steps needed to participate in EDAM.

Finally, consistent with Commission precedent recognizing the actual implementation date of some market rule changes can depend on variables that cannot be fully predicted in advance and additional time may be needed to implement some market rule changes, the CAISO further requests authorization to inform the Commission of the actual effective date of the proposed May 1, 2025 tariff changes through a subsequent filing within five business days following their implementation.[439]

In support of these requests, and to ensure the Commission has a full record in this proceeding, the CAISO further requests that the Commission set the date for comments 30 days from the date for filing, *i.e.*, September 21, 2023, and set the date for answers to 20 days from the date comments are due, or October 11, 2023. This will help market participants prepare meaningful

---

[439] *See Cal. Indep. Sys. Operator Corp.,* 172 FERC ¶ 61,263, at PP 1, 39 (2020). The CAISO has included an effective date of 12/31/9998 as part of the tariff records submitted in this filing. The CAISO will notify the Commission of the actual effective date of these tariff records within five business days of implementation in an eTariff submittal using Type of Filing code 150 – Report.

comments and provide the CAISO additional time to review intervenors' comments, without eroding the time for the Commission's consideration because the CAISO is submitting this filing 120 days prior to the requested effective date for some of the tariff provisions and an order on all of the tariff provisions.

VII. Filing

A. Communications

John C. Anders
William H. Weaver
Andrew Ulmer
David Zlotlow
John Spomer
Heather Curlee
Sarah Kozal
California Independent System Operator Corporation
250 Outcropping Way
Folsom, CA 95630
Tel: (916) 608-7144
Fax: (916) 608-7296
janders@caiso.com
bweaver@caiso.com
aulmer@caiso.com
dzlotlow@caiso.com
jspomer@caiso.com
hcurlee@caiso.com
skozal@caiso.com

Sean Atkins
Bradley Miliauskas
Davis Wright Tremaine LLP
1301 K Street, NW
Suite 500 East
Washington, DC 20005
Tel: (202) 973-4295
Fax: (202) 973-4495
seanatkins@dwt.com
bradleymiliauskas@dwt.com

B. Service

The CAISO has served copies of this filing on the CPUC, the California Energy Commission, and all parties with scheduling coordinator agreements under the CAISO tariff. In addition, the CAISO has posted a copy of the filing on the CAISO website.

C. Contents of this Filing

Besides this transmittal letter, this filing includes the following attachments:

Attachment A-1	Clean CAISO tariff sheets with an effective date of December 21, 2023
Attachment A-2	Clean CAISO tariff sheets with an effective date of May 1, 2025
Attachment B-1	Redlined CAISO tariff sheets with an effective date of December 21, 2023
Attachment B-2	Redlined CAISO tariff sheets with an effective date of May 1, 2025
Attachment C	DAME Final Proposal
Attachment D	DAME CAISO Board of Governors and WEIM Governing Body Memo/Vote
Attachment E	EDAM Final Proposal
Attachment F	EDAM CAISO Board of Governors and WEIM Governing Body Memo/Vote

VIII. Conclusion

For the reasons set forth in this filing, the CAISO respectfully requests that the Commission issue an order accepting the tariff revisions in this filing by December 21, 2023, effective as of the dates specified herein.

Respectfully submitted,

/s/ John C. Anders
Roger E. Collanton
General Counsel
John C. Anders
Deputy General Counsel
Anthony Ivancovich
Deputy General Counsel
William H. Weaver
Assistant General Counsel
Andrew Ulmer
Assistant General Counsel
David Zlotlow
Lead Counsel
John Spomer
Lead Counsel
Heather Curlee
Senior Counsel
Sarah E. Kozal
Counsel
California Independent System
Operator Corporation
250 Outcropping Way
Folsom, CA 95630

Counsel for the California Independent
System Operator Corporation